BEFORE START

FLIGHT DECK PREPARATION . . . . . COMPLETED
LIGHT TEST . . . . . CHECKED
OXYGEN & INTERPHONE . . . . . CHECKED
YAW DAMPER . . . . . ON
INSTRUMENT TRANSFER SWITCHES . . . . . NORMAL
FUEL . . . . . _____ KGS & PUMPS ON
GALLEY POWER . . . . . ON
EMERGENCY EXIT LIGHTS . . . . . ARMED
PASSENGER SIGNS . . . . . SET
WINDOW HEAT . . . . . ON
HYDRAULICS . . . . . NORMAL
AIR COND & PRESS . . . . . _____ PACK(S), BLEEDS ON, SET
NAV SWITCHES . . . . . AS REQUIRED
AUTOPILOTS . . . . . DISENGAGED
INSTRUMENTS . . . . . X-CHECKED
ANTISKID . . . . . ON
AUTO BRAKE . . . . . RTO
SPEED BRAKE . . . . . DOWN DETENT
PARKING BRAKE . . . . . SET
STABILIZER TRIM CUTOUT SWITCHES . . . . . NORMAL
WHEEL WELL FIRE WARNING . . . . . CHECKED
RADIOS, RADAR & TRANSPONDER . . . . . SET
RUDDER & AILERON TRIM . . . . . FREE & ZERO
PAPERS . . . . . ABOARD
FMC/CDU . . . . . SET

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N1 & IAS BUGS . . . . . SET

– – – – – – – – – – CLEARED FOR START – – – – – – – – –

DOORS . . . . . CLOSED

FLIGHT DECK WINDOWS . . . . . LOCKED

AIR CONDITIONING PACKS . . . . . OFF

START PRESSURE . . . . . _____PSI

ANTICOLLISION LIGHT . . . . . ON

AFTER START

ELECTRICAL . . . . . GENERATORS ON

PITOT HEAT . . . . . ON

ANTI–ICE . . . . . AS REQUIRED

AIR COND & PRESS . . . . . PACKS ON, FLT

ISOLATION VALVE . . . . . AUTO

APU . . . . . AS REQUIRED

START LEVERS . . . . . IDLE DETENT

BEFORE TAKEOFF

RECALL . . . . . CHECKED

FLIGHT CONTROLS . . . . . CHECKED

FLAPS . . . . . _____, GREEN LIGHT

STABILIZER TRIM . . . . . _____UNITS

CABIN DOOR . . . . . LOCKED

TAKEOFF BRIEFING . . . . . REVIEWED

– – – – – – – – – CLEARED FOR TAKEOFF – – – – – – – –

ENGINE START SWITCHES . . . . . ON

TRANSPONDER . . . . . ON

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AFTER TAKEOFF

AIR COND & PRESS . . . . . SET
ENGINE START SWITCHES . . . . . OFF
LANDING GEAR . . . . . UP & OFF
FLAPS . . . . . UP, NO LIGHTS

DESCENT – APPROACH

ANTI–ICE . . . . . AS REQUIRED
AIR COND & PRESS . . . . . SET
ALTIMETER & INSTRUMENTS . . . . . SET & X–CHECKED
N1 & IAS BUGS . . . . . CHECKED & SET

LANDING

ENGINE START SWITCHES . . . . . ON
RECALL . . . . . CHECKED
SPEED BRAKE . . . . . ARMED, GREEN LIGHT
LANDING GEAR . . . . . DOWN, 3 GREEN
FLAPS . . . . . _____, GREEN LIGHT

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SHUTDOWN

FUEL . . . . . PUMPS OFF
GALLEY POWER . . . . . AS REQUIRED
ELECTRICAL . . . . . ON_____
FASTEN BELTS . . . . . OFF
WINDOW HEAT . . . . . OFF
PITOT HEAT . . . . . OFF
ANTI–ICE . . . . . OFF
ELECTRIC HYDRAULIC PUMPS . . . . . OFF
AIR COND & PRESS . . . . _____ PACK(S), BLEEDS ON, GRD
EXTERIOR LIGHTS . . . . . AS REQUIRED
ANTICOLLISION LIGHT . . . . . OFF
ENGINE START SWITCHES . . . . . OFF
AUTO BRAKE . . . . . OFF
SPEED BRAKE . . . . . DOWN DETENT
FLAPS . . . . . UP, NO LIGHTS
PARKING BRAKE . . . . . AS REQUIRED
START LEVERS . . . . . CUTOFF
WEATHER RADAR . . . . . OFF
TRANSPONDER . . . . . AS REQUIRED

SECURE

IRS MODE SELECTORS . . . . . OFF
EMERGENCY EXIT LIGHTS . . . . . OFF
AIR CONDITIONING PACKS . . . . . OFF
APU/GROUND POWER . . . . . OFF
BATTERY . . . . . OFF

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Normal Checklists CI.1.1

Introduction CI.1.1

Normal Checklist Operation CI.1.1

Non-Normal Checklists CI.2.1

Introduction CI.2.1

Non-Normal Checklist Operation CI.2.1

Non-Normal Checklist Use CI.2.3

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Introduction

Checklists contain, in abbreviated form, information required by the trained flight crew to operate the airplane in normal situations and to cope with non-normal situations. This introduction contains guidelines for operation with the printed version of the checklists contained in the Quick Reference Handbook (QRH).

A table of contents is provided for each section except Section 1, Normal Checklists.

Normal Checklist Operation

Normal checklists are organized by phase of flight and are used to verify that certain critical procedural steps have been accomplished. Only procedural steps that, if omitted, would have a direct and adverse impact on normal operations are included.

Normal checklists are used by the flight crew after accomplishing all applicable procedural items. When parked at the gate, the captain calls for checklists. During taxi, the pilot taxiing calls for checklists. After takeoff, the pilot flying calls for checklists. Each crewmember responds for systems and controls in the crewmember's area of responsibility. However, in the Before Takeoff and Landing checklists the pilot flying verifies and responds to checklist items.

The position of the control or indication is visually verified and stated in response to a checklist challenge. When a disagreement between the response and checklist answer occurs, it is mandatory that the checklist be discontinued until the item is resolved. Following the completion of each normal checklist, the crewmember reading the checklist states, "_____ CHECKLIST COMPLETE."

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Introduction

This chapter contains checklists used by the flight crew to cope with non-normal situations. The checklists are grouped in logical sections which match the system description chapter arrangement in Volume 2. The checklists are arranged in alphabetical order within in each section.

Checklists without a visible cue (such as DITCHING) are called unannunciated checklists.

All unannunciated checklists are found in the first section of the Non-Normal checklists chapter. Some unannunciated checklists also appear in an associated systems section (such as INFLIGHT ENGINE START in the Engines, APU section.)

A condition statement is provided for all non-normal procedures.

Checklists may contain both recall and reference items. Recall items are critical steps that must be accomplished from memory. Reference items are actions to be accomplished while reading the checklist. In the Table of Contents for each non-normal checklist section, the titles of checklists containing memory items are printed in bold type.

Certain amplified information is included in brackets [ ] when the reason for the task may not be self-evident. These items are printed in normal font characters and are not bold.

Non-Normal Checklist Operation

Non-normal checklists begin with steps to correct the situation or condition. Information for planning the remainder of the flight is included. When special tasks are required to configure the airplane for landing, the tasks are included as part of the non-normal checklist. Flight patterns for certain non-normal situations are located in Non-Normal Maneuvers chapter, and show the sequence of configuration changes.

While every attempt is made to establish necessary non-normal checklists, it is not possible to develop checklists for all conceivable situations, especially those involving multiple failures. In certain unrelated multiple failure situations, the flight crew may have to combine elements of more than one checklist and/or exercise judgment to determine the safest course of action. The captain must assess the situation and use sound judgment to determine the safest course of action.

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There are some situations which always require landing at the nearest suitable airport. These situations include, but are not limited to, conditions where:

- the non-normal checklist contains the words “Plan to land at the nearest suitable airport.”
- cabin smoke or fire which persists
- one main AC power source remaining (such as engine, or APU)
- one hydraulic system remaining (the standby system is considered a hydraulic system), or
- any other situation determined by the crew to present a significant adverse effect on safety if the flight is continued.

It should be stressed that for persistent smoke or a fire that cannot be positively confirmed to be completely extinguished, the earliest possible descent, landing, and passenger evacuation should be accomplished.

Checklists prescribing an engine shutdown must be evaluated by the captain to ascertain whether an actual shutdown or operation at reduced thrust is the safest course of action. Consideration must be given to probable effects if the engine is left running at minimum required thrust.

Non-normal checklists also assume:

- After engine start and prior to takeoff, the appropriate non-normal checklist is accomplished if a non-normal condition is identified. Upon completion of the checklist, the Dispatch Deviation Guide or airline equivalent is consulted to determine if Minimum Equipment List relief is available.
- System controls are in the normal configuration for the phase of flight prior to the initiation of the non-normal checklists.
- Aural alerts are silenced and the system reset by the flight crew as soon as the cause of the alert is recognized.
- the EMERGENCY position of the oxygen regulator is used when necessary to provide positive pressure in the masks and goggles to evacuate contaminates. The 100% position of the oxygen regulator is used when positive pressure is not required, but contamination of the flight deck air exists. The NORMAL position of the oxygen regulator is used if prolonged use is required and the situation permits. Normal boom mic operation is restored when oxygen use is no longer required.
- Indicator lights should be tested to verify suspected faults.

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- Flight crew reset of tripped fuel pump and fuel pump control circuit breakers in flight is prohibited. Reset of any other tripped circuit breaker is not recommended, unless in the judgement of the captain, the situation resulting from the circuit breaker trip has a significant adverse effect on safety. These other tripped circuit breakers may be reset once, after a short cooling period (approximately 2 minutes.) A ground reset of a tripped circuit breaker by the flight crew should only be accomplished after maintenance has determined that it is safe to reset the circuit breaker.
- Flight crew cycling (pulling and resetting) of circuit breakers to clear non-normal conditions is not recommended.

Non-Normal Checklist Use

Non-normal checklist use commences when the airplane flight path and configuration are properly established. Only a few situations require an immediate response (such as stall warning, ground proximity PULL UP and WINDSHEAR warnings, and rejected takeoff.) Usually, time is available to assess the situation before corrective action is initiated. All actions should then be coordinated under the captain's supervision and performed in a deliberate, systematic manner. Flight path control should never be compromised.

When a non-normal situation is evident, at the direction of the pilot flying, both crewmembers systematically and without delay accomplish all recall items in their areas of responsibility.

The pilot flying calls for the checklist when:

- the flight path is under control
- the airplane is not in a critical stage of flight (such as takeoff or landing)
- all recall items are complete.

For those checklists containing only recall items or a combination of recall and reference items, the pilot not flying first verifies that each recall item has been accomplished. The checklist is normally read aloud during such verification, but the pilot flying is not required to respond except for items that are not in agreement with the checklist

Checklist reference items, including the response or action, are read aloud by the pilot not flying with appropriate action being taken by the crewmember in whose area of responsibility each control is located. After positioning the control, the crewmember taking the action also states the checklist response. In the non-normal landing checklist, the pilot flying also verifies and responds to checklist items.

The pilot flying may also direct reference procedures to be accomplished by recall if no hazard is created by such action, or if the situation does not permit reference to a checklist.

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Notes and information items, except for condition statements, are read aloud. The pilot flying need not repeat these items, but should acknowledge that the items were heard and understood.

The pilot flying is to be made aware when deferred items exist. Accomplishment of such items may be delayed until the appropriate point during approach or landing.

Pilots must be aware that checklists cannot be created for all conceivable situations and are not intended to replace good judgment. In some cases, deviation from checklists may, at the captain's discretion, be necessary.

Following completion of each non-normal checklist, the pilot not flying states: "_____ CHECKLIST COMPLETE." When a non-normal checklist is complete except for the deferred items, the pilot not flying states: "_____ CHECKLIST COMPLETE EXCEPT DEFERRED ITEMS."

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737-300 CFM56-3_22K KG FAA

General PI.10.1

Takeoff Speeds PI.10.1
V1(MCG) PI.10.2
Clearway and Stopway V1 Adjustments PI.10.2
Stab Trim Setting PI.10.2
VREF PI.10.3
Flap Maneuver Speeds PI.10.3
Slush/Standing Water Takeoff PI.10.4
Slippery Runway Takeoff PI.10.6
Takeoff %N1 . . . . . PI.10.8
Assumed Temperature Reduced Thrust PI.10.9
Takeoff Speeds (20K Derate) PI.10.11
V1(MCG) (20K Derate) PI.10.12
Clearway and Stopway V1 Adjustments (20K Derate) PI.10.12
Slush/Standing Water Takeoff (20K Derate) PI.10.13
Slippery Runway Takeoff (20K Derate) PI.10.14
Takeoff %N1 (20K Derate) PI.10.16
Assumed Temperature Reduced Thrust (20K Derate) PI.10.17
Max Climb %N1 . . . . . PI.10.19
Go-around %N1 . . . . . PI.10.19
Flight With Unreliable Airspeed / Turbulent Air Penetration PI.10.20

All Engines PI.11.1

Long Range Cruise Maximum Operating Altitude PI.11.1
Long Range Cruise Control PI.11.2
Long Range Cruise Enroute Fuel and Time - Low Altitudes PI.11.3
Long Range Cruise Enroute Fuel and Time - High Altitudes PI.11.5
Long Range Cruise Wind-Altitude Trade PI.11.7

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Descent PI.11.7
Holding PI.11.7

Advisory Information PI.12.1
Autobrake Landing Distance PI.12.1
Slippery Runway Landing Distance PI.12.1
Non-Normal Configuration Landing Distance PI.12.2
Brake Cooling Schedule PI.12.6

Engine Inoperative PI.13.1
Initial Max Continuous %N1 . . . . . PI.13.1
Max Continuous %N1 . . . . . PI.13.2
Driftdown Speed/Level Off Altitude PI.13.5
Driftdown/LRC Cruise Range Capability PI.13.6
Long Range Cruise Altitude Capability PI.13.7
Long Range Cruise Control PI.13.8
Long Range Cruise Diversion Fuel and Time PI.13.9
Holding PI.13.10

PMC Off PI.14.1
Takeoff %N1 . . . . . PI.14.1
Takeoff %N1 (20K Derate) PI.14.2

Gear Down PI.15.1
220 KIAS Cruise Altitude Capability PI.15.1
220 KIAS Cruise Control PI.15.2
220 KIAS Enroute Fuel and Time PI.15.3
Descent at 220 KIAS PI.15.3
Holding PI.15.4

Text PI.16.1
Introduction PI.16.1
General PI.16.1
All Engines PI.16.6
Advisory Information PI.16.7
Engine Inoperative PI.16.8
PMC Off Power Settings PI.16.10
Gear Down PI.16.10

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737-400 CFM56-3_23.5K KG FAA

General PI.20.1

Takeoff Speeds. PI.20.1
V1(MCG). PI.20.3
Clearway and Stopway V1 Adjustments PI.20.3
Stab Trim Setting PI.20.3
VREF PI.20.4
Flap Maneuver Speeds. PI.20.4
Slush/Standing Water Takeoff PI.20.5
Slippery Runway Takeoff PI.20.7
Takeoff %N1 . . . . . PI.20.9
Assumed Temperature Reduced Thrust. PI.20.10
Takeoff Speeds (22K Derate) PI.20.12
V1(MCG) (22K Derate) PI.20.14
Clearway and Stopway V1 Adjustments (22K Derate) PI.20.14
Stab Trim Setting (22K Derate). PI.20.14
Slush/Standing Water Takeoff (22K Derate) PI.20.15
Slippery Runway Takeoff (22K Derate) PI.20.16
Takeoff %N1 (22K Derate) PI.20.18
Assumed Temperature Reduced Thrust (22K Derate). PI.20.19
Takeoff Speeds (20K Derate) PI.20.21
V1(MCG) (20K Derate) PI.20.21
Clearway and Stopway V1 Adjustments (20K Derate) PI.20.22
Takeoff %N1 (20K Derate) PI.20.23
Assumed Temperature Reduced Thrust (20K Derate). PI.20.24
Max Climb %N1 . . . . . PI.20.26
Go-around %N1. . . . . PI.20.26
Flight With Unreliable Airspeed / Turbulent Air Penetration PI.20.27

All Engine PI.21.1

Long Range Cruise Maximum Operating Altitude PI.21.1
Long Range Cruise Control PI.21.3

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Long Range Cruise Enroute Fuel and Time - Low Altitudes PI.21.4
Long Range Cruise Enroute Fuel and Time - High Altitudes PI.21.5
Long Range Cruise Wind-Altitude Trade PI.21.6
Descent PI.21.6
Holding PI.21.6

Advisory Information PI.22.1
Autobrake Landing Distance PI.22.1
Slippery Runway Landing Distance PI.22.1
Non-Normal Configuration Landing Distance PI.22.2
Brake Cooling Schedule PI.22.6

Engine Inoperative PI.23.1
Initial Max Continuous %N1 . . . . . PI.23.1
Max Continuous %N1 . . . . . PI.23.2
Driftdown Speed/Level Off Altitude PI.23.5
Driftdown/LRC Cruise Range Capability PI.23.6
Long Range Cruise Altitude Capability PI.23.7
Long Range Cruise Control PI.23.8
Long Range Cruise Diversion Fuel and Time PI.23.9
Holding PI.23.10

PMC Off PI.24.1
Takeoff %N1 . . . . . PI.24.1
Takeoff %N1 (22K Derate) PI.24.2
Takeoff %N1 (20K Derate) PI.24.3

Gear Down PI.25.1
220 KIAS Cruise Altitude Capability PI.25.1
220 KIAS Cruise Control PI.25.2
220 KIAS Enroute Fuel and Time PI.25.3
Descent at 220 KIAS PI.25.3
Holding PI.25.4

Text PI.26.1
Introduction PI.26.1
General PI.26.1

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All Engines PI.26.6
Advisory Information PI.26.7
Engine Inoperative. PI.26.8
PMC Off Power Settings PI.26.10
Gear Down. PI.26.10

737-500 CFM56-3_20K KG FAA

General PI.30.1

Takeoff Speeds. PI.30.1
V1(MCG), Minimum VR PI.30.3
Clearway and Stopway V1 Adjustments PI.30.3
Stab Trim Setting PI.30.3
VREF PI.30.4
Flap Maneuver Speeds. PI.30.4
Slush/Standing Water Takeoff PI.30.5
Slippery Runway Takeoff PI.30.6
Takeoff %N1 . . . . . PI.30.8
Assumed Temperature Reduced Thrust. PI.30.9
Takeoff Speeds (18.5K Derate) PI.30.11
V1(MCG), Minimum VR (18.5K Derate). PI.30.12
Clearway and Stopway V1 Adjustments (18.5K Derate) PI.30.13
Stab Trim Setting (18.5K Derate) PI.30.13
Slush/Standing Water Takeoff (18.5K Derate) PI.30.14
Slippery Runway Takeoff (18.5K Derate). PI.30.15
Takeoff %N1 (18.5K Derate). PI.30.17
Assumed Temperature Reduced Thrust (18.5K Derate) PI.30.18
Max Climb %N1 . . . . . PI.30.20
Go-around %N1. . . . . PI.30.20
Flight With Unreliable Airspeed / Turbulent Air Penetration PI.30.21

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All Engine PI.31.1
Long Range Cruise Maximum Operating Altitude PI.31.1
Long Range Cruise Control PI.31.3
Long Range Cruise Enroute Fuel and Time - Low Altitudes PI.31.4
Long Range Cruise Enroute Fuel and Time - High Altitudes PI.31.5
Long Range Cruise Wind-Altitude Trade PI.31.6
Descent PI.31.6
Holding PI.31.6
Advisory Information PI.32.1
Autobrake Landing Distance PI.32.1
Slippery Runway Landing Distance PI.32.1
Non-Normal Configuration Landing Distance PI.32.2
Brake Cooling Schedule PI.32.6
Engine Inoperative PI.33.1
Initial Max Continuous %N1 . . . . . PI.33.1
Max Continuous %N1 . . . . . PI.33.2
Driftdown Speed/Level Off Altitude PI.33.5
Driftdown/LRC Cruise Range Capability PI.33.6
Long Range Cruise Altitude Capability PI.33.7
Long Range Cruise Control PI.33.8
Long Range Cruise Diversion Fuel and Time PI.33.9
Holding PI.33.10
PMC Off PI.34.1
Takeoff %N1 . . . . . PI.34.1
Takeoff %N1 (18.5K Derate) PI.34.2
Gear Down PI.35.1
220 KIAS Cruise Altitude Capability PI.35.1
220 KIAS Cruise Control PI.35.2
220 KIAS Enroute Fuel and Time PI.35.3
Descent at 220 KIAS PI.35.3
Holding PI.35.4

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Text PI.36.1
Introduction PI.36.1
General. PI.36.1
All Engines PI.36.6
Advisory Information PI.36.7
Engine Inoperative. PI.36.9
PMC Off Power Settings PI.36.10
Gear Down. PI.36.11

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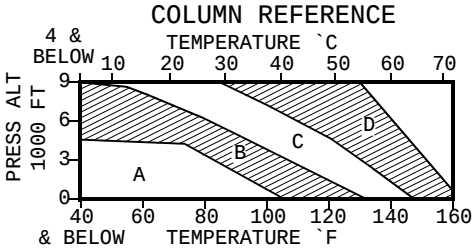
Takeoff Speeds



SLOPE/WIND V_1 ADJUSTMENT*						
WEIGHT 1000 KG	SLOPE % DN		UP	WIND KTS TAIL		HEAD
	-2	0	2	-15	0	40
65	-2	0	3	-3	0	1
55	-2	0	2	-3	0	1
45	-2	0	1	-3	0	1
35	-2	0	1	-3	0	1

*V_1 NOT TO EXCEED V_R

FLAPS	WT 1000 KG	A			B			C			D		
		V_1	V_R	V_2	V_1	V_R	V_2	V_1	V_R	V_2	V_1	V_R	V_2
1	65	158	162	168	160	164	168	161	165	168			
	60	149	153	160	152	155	160	153	156	160			
	55	141	144	153	143	146	152	144	147	152	147	150	152
	50	132	134	145	134	137	145	136	138	144	139	140	144
	45	123	125	138	125	127	137	127	129	137	130	131	136
	40	114	116	131	116	118	130	118	120	129	121	122	128
	35	104	107	124	107	109	122	109	110	122	111	113	121
5	65	151	154	160	152	156	160						
	60	143	146	153	145	148	153	146	149	153			
	55	135	137	146	137	140	146	139	141	146			
	50	127	129	139	129	131	139	131	133	139	132	135	139
	45	118	120	132	120	122	132	123	124	132	125	126	131
	40	109	112	126	112	114	125	114	115	124	117	117	123
	35	100	103	119	103	105	118	105	106	117	107	108	116
15	65	144	146	152	145	148	152						
	60	136	138	145	138	140	145						
	55	129	130	139	131	132	139	133	134	138			
	50	121	122	132	123	124	132	125	126	132			
	45	113	114	126	115	116	125	117	118	125	118	119	125
	40	104	106	120	106	108	119	108	109	118	110	111	118
	35	96	98	113	97	99	112	100	101	112	102	103	111

CHECK V1(MCG).

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V1(MCG)

ACTUAL OAT		PRESS ALT FT					
`C	`F	-1000	0	2000	4000	6000	8000
55	131	106	105	99			
50	122	108	107	103			
40	104	113	111	107	103	99	94
30	86	116	116	111	107	103	99
20	68	117	116	113	111	107	102
10	50	117	116	113	111	108	104
-50	-58	118	118	115	112	109	105

FOR A/C OFF INCREASE V_1(MCG) BY 2 KNOTS.

Clearway and Stopway V1 Adjustments

CLEARWAY MINUS STOPWAY (FT)	NORMAL V1 (KIAS)			
	100	120	140	160
800			-3	-2
600		-5	-3	-2
400	-6	-5	-3	-1
200	-3	-2	-1	0
0	0	0	0	0
-800	0	0	0	0

Maximum Allowable Clearway

FIELD LENGTH (FT)	MAX ALLOWABLE CLEARWAY FOR V1 REDUCTION (FT)
4000	450
6000	550
8000	650
10000	700
12000	750
14000	800

Stab Trim Setting

Max Takeoff Thrust

FLAP POSITION	C.G. % MAC						
	6	10	14	18	22	26	30
	STAB TRIM UNITS (NOSE UP)						
1 & 5	6 1/4	5 3/4	5 1/4	4 3/4	4 1/4	3 3/4	3
15	5 3/4	5	4 1/2	3 3/4	3	2 1/2	1 3/4

For weights at or below 45,360 kg subtract 1/2 unit from above value

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VREF

WEIGHT (1000 KG)	FLAPS		
	40	30	15
65	153	154	165
60	146	147	158
55	138	141	151
50	131	134	144
45	123	127	136
40	115	119	128
35	107	111	119

For approach speed add wind factor of 1/2 headwind component + gust (max 20 knots).

Flap Maneuver Speeds

FLAP POSITION	MANEUVER SPEED (KIAS)		
	WEIGHT		
	AT OR BELOW 53070 KG	ABOVE 53070 KG AND AT OR BELOW 62823 KG	ABOVE 62823 KG
UP	210	220	230
1	190	200	210
5	180	190	200
10	170	180	190
15	150	160	170
25	140	150	160

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ADVISORY INFORMATION

Slush/Standing Water Takeoff

Weight Adjustment (1000 KG)

FIELD/OBSTACLE LIMIT WEIGHT (1000 KG)	SLUSH/STANDING WATER DEPTH								
	3mm (0.12 INCHES)			6mm (0.25 INCHES)			13mm (0.50 INCHES)		
	PRESS ALT (FT)			PRESS ALT (FT)			PRESS ALT (FT)		
	S.L.	4000	8000	S.L.	4000	8000	S.L.	4000	8000
68	-7.1	-7.7	-7.5	-8.3	-9.1	-9.2	-12.5	-12.7	-13.2
64	-6.6	-7.2	-7.8	-7.9	-8.6	-9.4	-11.1	-12.8	-12.8
60	-6.1	-6.8	-7.5	-7.3	-8.2	-9.1	-9.8	-11.9	-12.7
56	-5.6	-6.4	-7.1	-6.6	-7.6	-8.5	-8.9	-10.5	-12.3
52	-5.0	-5.8	-6.5	-5.8	-6.8	-7.8	-7.7	-9.1	-11.3
48	-4.4	-5.0	-5.9	-5.1	-6.0	-6.9	-6.5	-7.8	-9.8
44	-3.7	-4.4	-5.1	-4.2	-5.1	-6.0	-5.3	-6.5	-8.1
40	-3.1	-3.7	-4.3	-3.2	-4.0	-4.8	-4.2	-5.2	-6.0
36	-2.5	-3.0	-3.5	-2.2	-3.0	-3.7	-3.1	-3.9	-3.9

V1 (MCG) Limit Weight (1000 KG)

ADJUSTED FIELD LENGTH (FT)	SLUSH/STANDING WATER DEPTH								
	3mm (0.12 INCHES)			6mm (0.25 INCHES)			13mm (0.50 INCHES)		
	PRESS ALT (FT)			PRESS ALT (FT)			PRESS ALT (FT)		
	S.L.	4000	8000	S.L.	4000	8000	S.L.	4000	8000
4600							30.5		
5000	29.4			31.8			36.6	31.0	
5400	35.1	28.1		37.8	31.0		42.6	35.9	30.0
5800	40.7	33.2		44.0	36.0	29.8	48.4	40.8	34.3
6200	46.7	38.2	32.1	49.9	41.1	34.4	54.4	45.5	38.7
6600	52.9	43.3	36.2	56.3	46.7	38.9	60.7	50.5	42.9
7000	59.6	48.4	40.4	62.6	52.0	43.5	66.9	55.8	47.2
7400	66.9	53.9	44.8	68.8	57.6	48.2	73.2	61.5	51.4
7800	74.1	59.9	49.4	75.1	63.2	52.6		67.2	55.6
8200		66.1	54.0		68.9	57.4		72.9	60.1
8600		72.4	58.9		74.6	62.6			64.5
9000			64.2			67.8			68.9
9400			69.5			73.0			73.3
9800			74.8						

1. **Enter Weight Adjustment table with slush/standing water depth and dry field/obstacle limit weight to obtain slush/standing water weight adjustment.**
2. **Adjust field length available by -108ft/+108ft for every 5°C above/below 4°C**
3. **Find V1(MCG) limit weight for temperature adjusted field length and pressure altitude.**
4. **Max allowable slush/standing water limited weight is lesser of weights from 1 and 3.**

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ADVISORY INFORMATION

Slush/Standing Water Takeoff

V1 Adjustment (KIAS)

WEIGHT (1000 KG)	SLUSH/STANDING WATER DEPTH								
	3mm (0.12 INCHES)			6mm (0.25 INCHES)			13mm (0.50 INCHES)		
	PRESS ALT (FT)			PRESS ALT (FT)			PRESS ALT (FT)		
	S.L.	4000	8000	S.L.	4000	8000	S.L.	4000	8000
68	-12	-11	-9	-6	-6	-8	2	-1	-6
64	-14	-12	-11	-8	-7	-8	2	0	-4
60	-16	-14	-12	-10	-8	-8	2	2	-1
56	-18	-16	-13	-13	-10	-8	0	2	1
52	-20	-18	-15	-16	-12	-9	-4	1	1
48	-22	-19	-17	-19	-15	-11	-8	-1	1
44	-23	-21	-18	-20	-17	-13	-12	-5	1
40	-23	-21	-19	-21	-18	-15	-15	-9	-1
36	-23	-22	-20	-21	-19	-17	-17	-13	-6

1. **Obtain V1, VR and V2 for the actual weight.**
2. **If V1(MCG) limited, set V1 = V1(MCG). If not V1(MCG) limited, enter V1 Adjustment Table with the actual weight to obtain V1 speed adjustment. If adjusted V1 is less than V1(MCG), set V1 = V1(MCG).**

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ADVISORY INFORMATION

Slippery Runway Takeoff
Weight Adjustment (1000 KG)

FIELD/OBSTACLE LIMIT WEIGHT (1000 KG)	REPORTED BRAKING ACTION								
	GOOD			MEDIUM			POOR		
	PRESS ALT (FT)			PRESS ALT (FT)			PRESS ALT (FT)		
	S.L.	4000	8000	S.L.	4000	8000	S.L.	4000	8000
68	0.0	0.0	0.0	-4.0	-4.0	-4.0	-7.3	-7.3	-7.3
64	0.0	0.0	0.0	-3.8	-3.8	-3.8	-7.1	-7.1	-7.1
60	-0.7	-0.7	-0.7	-3.8	-3.8	-3.8	-6.8	-6.8	-6.8
56	-0.9	-0.9	-0.9	-3.8	-3.8	-3.8	-6.6	-6.6	-6.6
52	-1.0	-1.0	-1.0	-3.6	-3.6	-3.6	-6.2	-6.2	-6.2
48	-1.1	-1.1	-1.1	-3.2	-3.2	-3.2	-5.7	-5.7	-5.7
44	-0.8	-0.8	-0.8	-2.9	-2.9	-2.9	-5.1	-5.1	-5.1
40	-0.5	-0.5	-0.5	-2.5	-2.5	-2.5	-4.4	-4.4	-4.4
36	-0.2	-0.2	-0.2	-2.0	-2.0	-2.0	-3.5	-3.5	-3.5

V1 (MCG) Limit Weight (1000 KG)

ADJUSTED FIELD LENGTH (FT)	REPORTED BRAKING ACTION								
	GOOD			MEDIUM			POOR		
	PRESS ALT (FT)			PRESS ALT (FT)			PRESS ALT (FT)		
	S.L.	4000	8000	S.L.	4000	8000	S.L.	4000	8000
3800	31.5								
4200	41.0								
4600	49.9	34.4							
5000	60.3	43.6		32.9					
5400	71.0	52.9	37.2	39.6	28.2				
5800		63.5	46.3	46.6	34.9				
6200		74.2	56.0	54.6	41.7	30.2	32.5		
6600			66.7	63.3	48.7	37.0	37.2		
7000				71.9	57.2	43.8	42.1	29.1	
7400					65.9	51.0	47.2	33.9	
7800					74.5	59.8	52.6	38.7	
8200						68.5	58.5	43.6	30.6
8600							64.6	48.8	35.3
9000							70.6	54.3	40.1
9400								60.3	45.1
9800								66.4	50.4
10200								72.4	56.1
10600									62.1
11000									68.2
11400									74.2

1. **Enter Weight Adjustment table with reported braking action and dry field/obstacle limit weight to obtain slippery runway weight adjustment.**
2. **Adjust "Good" field length available by -90ft /+90ft for every 5°C above/below 4°C**
   Adjust "Medium" field length available by -90ft /+90ft for every 5°C above/below 4°C
   Adjust "Poor" field length available by -117ft /+117ft for every 5°C above/below 4°C
3. **Find V1(MCG) limit weight for temperature adjusted field length and pressure altitude.**
4. **Max allowable slippery runway limited weight is lesser of weights from 1 and 3.**

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ADVISORY INFORMATION

Slippery Runway Takeoff

V1 Adjustment (KIAS)

WEIGHT (1000 KG)	REPORTED BRAKING ACTION								
	GOOD			MEDIUM			POOR		
	PRESS ALT (FT)			PRESS ALT (FT)			PRESS ALT (FT)		
	S.L.	4000	8000	S.L.	4000	8000	S.L.	4000	8000
68	-4	-2	0	-13	-11	-9	-27	-24	-21
64	-6	-4	-2	-15	-13	-11	-30	-27	-24
60	-7	-5	-3	-17	-15	-13	-32	-29	-26
56	-9	-7	-5	-20	-18	-16	-34	-31	-28
52	-10	-8	-6	-22	-20	-18	-36	-33	-30
48	-12	-10	-8	-24	-22	-20	-38	-35	-32
44	-13	-11	-9	-26	-24	-22	-39	-36	-33
40	-14	-12	-10	-27	-25	-23	-40	-37	-34
36	-14	-12	-10	-28	-26	-24	-40	-37	-34

1. **Obtain V1, VR and V2 for the actual weight.**
2. **If V1(MCG) limited, set V1 = V1(MCG). If not V1(MCG) limited, enter V1 Adjustment Table with the actual weight to obtain V1 speed adjustment. If adjusted V1 is less than V1(MCG), set V1 = V1(MCG).**

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Takeoff %N1

Based on engine bleed to packs on (Auto) and anti-ice on or off

AIRPORT OAT		AIRPORT PRESSURE ALTITUDE (FT)									
°C	°F	-1000	0	1000	2000	3000	4000	5000	6000	7000	8000
55	131	93.2	93.8	93.8	93.8						
50	122	93.8	94.3	94.3	94.3	93.9	93.6				
45	113	94.2	94.7	94.7	94.6	94.6	94.6	94.7	94.4	94.2	
40	104	94.6	95.2	95.2	95.1	95.0	95.1	95.1	95.2	95.1	94.9
35	95	95.2	95.6	95.6	95.6	95.5	95.7	95.7	95.7	95.6	95.5
30	86	95.2	96.1	96.1	96.0	96.0	96.3	96.2	96.1	96.0	96.0
25	77	94.4	95.3	95.8	96.2	96.5	96.7	96.6	96.6	96.5	96.4
20	68	93.6	94.5	95.0	95.4	95.9	96.6	97.1	97.1	97.0	96.9
15	59	92.8	93.7	94.2	94.6	95.1	95.8	96.3	96.8	97.2	97.5
10	50	92.0	92.9	93.4	93.8	94.2	95.0	95.4	95.9	96.4	96.8
5	41	91.2	92.1	92.5	92.9	93.4	94.1	94.6	95.1	95.5	96.0
0	32	90.3	91.2	91.7	92.1	92.6	93.3	93.7	94.2	94.7	95.1
-10	14	88.7	89.6	90.0	90.4	90.8	91.5	92.0	92.5	92.9	93.4
-20	-4	87.0	87.8	88.3	88.7	89.1	89.8	90.2	90.7	91.1	91.6
-30	-22	85.2	86.0	86.5	86.9	87.3	88.0	88.4	88.9	89.3	89.7
-40	-40	83.5	84.3	84.7	85.1	85.5	86.2	86.6	87.1	87.4	87.9
-50	-58	81.7	82.5	82.9	83.2	83.7	84.3	84.7	85.2	85.6	86.0

%N1 Adjustments for Engine Bleeds

BLEED CONFIGURATION	AIRPORT PRESSURE ALTITUDE (FT)	
	-2000	9000
ENGINE BLEED TO PACKS OFF	1.0	1.0

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Assumed Temperature Reduced Thrust

Maximum Assumed Temperature (Table 1)

Based on 25% Takeoff Thrust Reduction

OAT (°C)	PRESS ALT (1000 FT)										OAT (°F)	PRESS ALT (1000 FT)									
	-1	0	1	2	3	4	5	6	7	8		-1	0	1	2	3	4	5	6	7	8
55	74	71	71								130	165	159	159							
50	71	69	68	68	69	70					120	160	155	154	154	155	157				
45	69	67	66	66	67	67	67	68	70		110	155	151	149	149	150	151	151	152	155	
40	67	65	64	64	64	64	64	64	66	68	100	151	148	145	145	145	145	147	145	149	151
35	65	63	62	62	62	61	61	62	63	64	90	146	143	141	141	140	140	140	140	143	144
30	63	61	60	60	59	59	59	59	60	61	80	146	142	139	138	136	135	135	135	137	138
25	63	61	59	58	57	56	56	56	57	58	70	146	142	138	136	135	131	129	129	130	132
20	63	61	59	58	57	55	53	54	54	55	60	146	142	138	136	135	131	127	127	126	126
15 & BELOW	63	61	59	58	57	55	53	53	52	52	50 & BELOW	146	142	138	136	135	131	127	127	126	126

Maximum Takeoff %N1 (Table 2)

Based on engine bleed to packs on (Auto) and engine anti-ice on or off

ASSUMED TEMP		AIRPORT PRESSURE ALTITUDE (FT)									
°C	°F	-1000	0	1000	2000	3000	4000	5000	6000	7000	8000
75	167	86.0	85.4	85.4							
70	158	87.8	87.6	87.4	87.4	87.6					
65	149	89.7	89.7	89.4	89.2	89.2	89.2	89.3	89.5	89.9	90.4
60	140	91.6	91.8	91.3	91.0	90.8	90.7	90.7	90.8	91.1	91.4
55	131	93.1	93.8	93.2	92.7	92.4	92.1	92.1	92.0	92.1	92.3
50	122	93.7	94.3	94.3	94.3	93.9	93.6	93.4	93.2	93.2	93.2
45	113	94.0	94.7	94.7	94.6	94.6	94.6	94.7	94.4	94.2	94.0
40	104	94.4	95.2	95.2	95.1	95.0	95.1	95.1	95.2	95.1	94.9
35	95	95.0	95.6	95.6	95.6	95.5	95.7	95.7	95.7	95.6	95.5
30	86	95.2	96.1	96.1	96.0	96.0	96.3	96.2	96.1	96.0	96.0
25	77	94.4		96.6	96.5	96.5	96.7	96.6	96.6	96.5	96.4
20	68	93.6					97.1	97.1	97.1	97.0	96.9
15	59	92.8							97.6	97.5	97.5
MINIMUM ASSUMED TEMP °C (°F)		32(90)	30(86)	28(82)	26(79)	24(75)	22(72)	20(68)	18(64)	16(61)	15(59)

With engine bleed to packs off, increase %N1 by 1.0.

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Assumed Temperature Reduced Thrust

%N1 Adjustment For Temperature Difference (Table 3)

ASSUMED TEMP MINUS OAT		OUTSIDE AIR TEMPERATURE													
°C	°F	°C	-40	-20	0	5	10	15	20	25	30	35	40	45	50	55
		°F	-40	-4	32	41	50	59	68	77	86	95	104	113	122	131
10	18							1.6	1.6	1.6	1.5	1.5	1.5	1.5	1.4	1.4
20	36				3.3	3.2	3.2	3.1	3.0	3.0	2.9	2.8	2.8	2.7	2.5	2.3
30	54				4.8	4.8	4.6	4.5	4.4	4.3	4.0	3.8	3.6	3.6	3.6	3.6
40	72			6.0	6.2	6.1	6.0	5.8	5.7	5.2	5.0	5.0				
50	90			8.2	7.5	7.3	7.2	6.6	6.5							
60	108		10.4	9.5	8.7	8.1	7.9									
70	126		11.8	10.7	9.3											
80	144		13.0	11.8	10.1											
90	162		14.0	12.4												
100	180		15.0	12.8												
110	198		15.4													

1. **Determine Maximum Assumed Temperature allowed from Table 1.**
2. **Find Maximum %N1 from Table 2 using the desired assumed temperature (no greater than temperature from Table 1).**
3. **Use the difference between assumed temperature and OAT to determine the %N1 adjustment from Table 3.**
4. **Subtract %N1 adjustment from Maximum %N1 in Table 2.**

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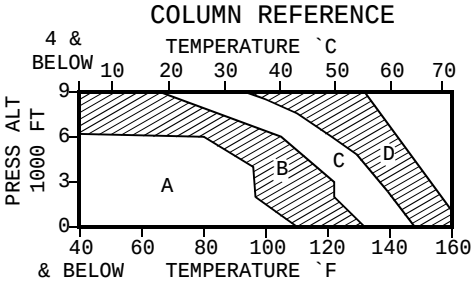
Takeoff Speeds (20K Derate)



SLOPE/WIND V_1 ADJUSTMENT*						
WEIGHT 1000 KG	SLOPE % DN		UP	WIND KTS TAIL		HEAD
	-2	0	2	-15	0	40
65	-2	0	3	-2	0	1
55	-2	0	1	-3	0	1
45	-2	0	0	-3	0	0
35	-2	0	1	-3	0	0

*V_1 NOT TO EXCEED V_R

FLAPS	WT 1000 KG	A			B			C			D		
		V_1	V_R	V_2	V_1	V_R	V_2	V_1	V_R	V_2	V_1	V_R	V_2
1	65	161	163	166	161	164	167						
	60	153	155	160	153	156	160						
	55	144	146	152	145	148	152	147	149	152			
	50	136	137	145	137	139	145	139	140	144			
	45	127	128	137	128	130	137	130	131	137	132	133	136
	40	118	118	130	119	120	129	121	122	129	123	124	128
	35	109	109	122	110	111	122	111	112	121	114	114	120
5	65	153	156	159	154	157	159						
	60	146	148	153	147	149	152	148	151	154			
	55	138	139	146	139	141	145	140	143	146			
	50	130	131	139	131	133	138	133	134	139			
	45	121	122	132	123	124	132	124	126	131	127	127	131
	40	113	113	125	114	115	125	116	117	124	118	118	124
	35	104	104	118	105	106	118	107	108	116	109	109	116
15	65	146	147	152	148	148	151						
	60	139	140	145	141	141	145						
	55	131	132	139	133	134	138	136	136	138			
	50	124	124	132	126	126	132	128	128	132			
	45	115	116	126	118	118	125	119	120	125	121	121	125
	40	107	108	119	109	110	119	110	111	118	112	112	118
	35	98	100	113	100	101	112	101	102	111	103	103	110

CHECK V1(MCG).

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V1(MCG) (20K Derate)

ACTUAL OAT		PRESS ALT FT					
`C	`F	-1000	0	2000	4000	6000	8000
55	131	101	100	97			
50	122	103	102	100	98	93	
40	104	108	107	104	103	99	94
30	86	111	110	108	108	103	97
20	68	111	111	109	108	106	101
10	50	112	111	109	109	106	102
-50	-58	113	112	110	110	107	103

FOR A/C OFF INCREASE V_1(MCG) BY 2 KNOTS.

Clearway and Stopway V1 Adjustments (20K Derate)

CLEARWAY MINUS STOPWAY (FT)	NORMAL V1 (KIAS)			
	100	120	140	160
800			-3	-2
600		-5	-3	-2
400	-6	-5	-3	-1
200	-3	-2	-1	0
0	0	0	0	0
-800	0	0	0	0

Maximum Allowable Clearway (20K Derate)

FIELD LENGTH (FT)	MAX ALLOWABLE CLEARWAY FOR V1 REDUCTION (FT)
4000	450
6000	550
8000	650
10000	700
12000	750
14000	800

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ADVISORY INFORMATION

Slush/Standing Water Takeoff (20K Derate)

Weight Adjustment (1000 KG)

20K DERATE FIELD/OBSTACLE LIMIT WEIGHT (1000 KG)	SLUSH/STANDING WATER DEPTH								
	3mm (0.12 INCHES)			6mm (0.25 INCHES)			13mm (0.50 INCHES)		
	PRESS ALT (FT)			PRESS ALT (FT)			PRESS ALT (FT)		
	S.L.	4000	8000	S.L.	4000	8000	S.L.	4000	8000
68	-7.6	-7.8	-8.2	-9.4	-9.4	-9.4	-12.4	-12.8	-13.5
64	-7.3	-7.6	-8.2	-8.9	-9.4	-9.8	-12.1	-12.9	-12.9
60	-6.6	-7.2	-7.8	-7.9	-8.8	-9.5	-10.9	-12.1	-12.4
56	-5.9	-6.6	-7.3	-6.8	-7.8	-9.0	-9.4	-11.0	-12.0
52	-5.2	-6.0	-6.6	-6.0	-7.0	-8.2	-8.2	-9.6	-11.3
48	-4.6	-5.2	-6.0	-5.2	-6.2	-7.3	-7.0	-8.2	-10.2
44	-3.9	-4.5	-5.2	-4.5	-5.1	-6.4	-5.9	-6.7	-8.3
40	-3.2	-3.5	-4.4	-3.8	-4.3	-5.0	-4.4	-4.7	-6.3
36	-2.6	-2.4	-3.5	-3.2	-3.5	-3.7	-2.9	-2.5	-4.3

V1 (MCG) Limit Weight (1000 KG)

ADJUSTED FIELD LENGTH (FT)	SLUSH/STANDING WATER DEPTH								
	3mm (0.12 INCHES)			6mm (0.25 INCHES)			13mm (0.50 INCHES)		
	PRESS ALT (FT)			PRESS ALT (FT)			PRESS ALT (FT)		
	S.L.	4000	8000	S.L.	4000	8000	S.L.	4000	8000
4600				30.1			34.0		
5000	33.9			36.6			40.5	30.8	
5400	40.6	29.5		43.1	32.4		46.9	36.3	30.2
5800	47.2	35.0	29.2	49.6	37.9	31.5	53.3	41.8	35.3
6200	53.9	40.5	33.9	56.1	43.4	36.3	60.0	47.3	40.3
6600	60.5	46.1	38.6	62.5	48.9	41.1	66.7	52.6	45.5
7000	67.2	51.6	43.3	69.0	54.4	46.0	73.4	58.1	50.5
7400	73.8	57.2	48.0	75.5	59.9	50.8		63.6	55.7
7800		62.9	52.7		65.4	55.7		69.1	60.9
8200		68.5	57.4		70.9	60.6		74.6	66.1
8600		74.1	62.1			65.5			71.3
9000			66.9			70.4			
9400			71.6			75.3			

1. **Enter Weight Adjustment table with slush/standing water depth and 20K derate dry field/obstacle limit weight to obtain slush/standing water weight adjustment.**
2. **Adjust field length available by -108ft/+108ft for every 5°C above/below 4°C.**
3. **Find V1(MCG) limit weight for temperature adjusted field length and pressure altitude.**
4. **Max allowable slush/standing water limited weight is lesser of weights from 1 and 3.**

V1 Adjustment (KIAS)

WEIGHT (1000 KG)	SLUSH/STANDING WATER DEPTH								
	3mm (0.12 INCHES)			6mm (0.25 INCHES)			13mm (0.50 INCHES)		
	PRESS ALT (FT)			PRESS ALT (FT)			PRESS ALT (FT)		
	S.L.	4000	8000	S.L.	4000	8000	S.L.	4000	8000
68	-9	-9	-8	-3	-5	-6	4	0	-4
64	-11	-11	-9	-5	-6	-6	3	1	-2
60	-13	-13	-11	-6	-7	-6	3	2	0
56	-15	-14	-12	-9	-8	-7	2	2	0
52	-17	-16	-14	-12	-11	-7	0	1	1
48	-19	-18	-15	-15	-13	-9	-3	0	1
44	-21	-20	-16	-17	-15	-11	-7	-3	1
40	-21	-21	-18	-19	-17	-13	-11	-8	0
36	-21	-21	-19	-19	-18	-15	-15	-11	-3

1. **Obtain V1, VR and V2 for the actual weight.**
2. **If V1(MCG) limited, set V1 = V1(MCG). If not V1(MCG) limited, enter V1 Adjustment Table with the actual weight to obtain V1 speed adjustment. If adjusted V1 is less than V1(MCG), set V1 = V1(MCG).**

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ADVISORY INFORMATION

Slippery Runway Takeoff (20K Derate)

Weight Adjustment (1000 KG)

20K DERATE FIELD/OBSTACLE LIMIT WEIGHT (1000 KG)	REPORTED BRAKING ACTION								
	GOOD			MEDIUM			POOR		
	PRESS ALT (FT)			PRESS ALT (FT)			PRESS ALT (FT)		
	S.L.	4000	8000	S.L.	4000	8000	S.L.	4000	8000
68	0.0	0.0	0.0	-3.6	-3.6	-3.6	-7.5	-7.5	-7.5
64	-0.2	-0.2	-0.2	-3.9	-3.9	-3.9	-7.3	-7.3	-7.3
60	-0.6	-0.6	-0.6	-4.0	-4.0	-4.0	-6.9	-6.9	-6.9
56	-0.8	-0.8	-0.8	-3.9	-3.9	-3.9	-6.6	-6.6	-6.6
52	-1.1	-1.1	-1.1	-3.8	-3.8	-3.8	-6.3	-6.3	-6.3
48	-1.1	-1.1	-1.1	-3.7	-3.7	-3.7	-5.9	-5.9	-5.9
44	-0.8	-0.8	-0.8	-3.1	-3.1	-3.1	-5.3	-5.3	-5.3
40	-0.3	-0.3	-0.3	-2.6	-2.6	-2.6	-4.5	-4.5	-4.5
36	0.0	0.0	0.0	-2.0	-2.0	-2.0	-3.6	-3.6	-3.6

V1 (MCG) Limit Weight (1000 KG)

ADJUSTED FIELD LENGTH (FT)	REPORTED BRAKING ACTION								
	GOOD			MEDIUM			POOR		
	PRESS ALT (FT)			PRESS ALT (FT)			PRESS ALT (FT)		
	S.L.	4000	8000	S.L.	4000	8000	S.L.	4000	8000
3800	36.3								
4200	46.4	29.2							
4600	56.6	39.3		28.5					
5000	66.1	49.6	32.3	36.7					
5400	75.7	59.4	42.3	44.8	30.9				
5800		69.0	52.7	52.8	39.2		30.5		
6200			62.3	59.1	47.1	33.4	35.6		
6600			71.9	65.0	55.0	41.6	40.8		
7000				70.8	60.9	49.5	46.2	32.0	
7400					66.7	56.8	51.9	37.2	
7800					72.6	62.6	57.8	42.4	28.4
8200						68.5	63.6	47.9	33.6
8600						74.3	69.5	53.7	38.8
9000							75.4	59.6	44.0
9400								65.4	49.6
9800								71.3	55.5
10200									61.3
10600									67.2
11000									73.0

1. **Enter Weight Adjustment table with reported braking action and 20K derate dry field/obstacle limit weight to obtain slippery runway weight adjustment.**
2. **Adjust "Good" field length available by -90ft/+90ft for every 5°C above/below 4°C**
   Adjust "Medium" field length available by -90ft/+90ft for every 5°C above/below 4°C
   Adjust "Poor" field length available by -117ft/+117ft for every 5°C above/below 4°C
3. **Find V1(MCG) limit weight for temperature adjusted field length and pressure altitude.**
4. **Max allowable slippery runway limited weight is lesser of weights from 1 and 3.**

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ADVISORY INFORMATION

Slippery Runway Takeoff (20K Derate)

V1 Adjustment (KIAS)

WEIGHT (1000 KG)	REPORTED BRAKING ACTION								
	GOOD			MEDIUM			POOR		
	PRESS ALT (FT)			PRESS ALT (FT)			PRESS ALT (FT)		
	S.L.	4000	8000	S.L.	4000	8000	S.L.	4000	8000
68	-2	0	2	-9	-7	-5	-23	-20	-17
64	-4	-2	0	-12	-10	-8	-25	-22	-19
60	-6	-4	-2	-15	-13	-11	-28	-25	-22
56	-8	-6	-4	-17	-15	-13	-31	-28	-25
52	-9	-7	-5	-19	-17	-15	-33	-30	-27
48	-11	-9	-7	-21	-19	-17	-35	-32	-29
44	-12	-10	-8	-23	-21	-19	-36	-33	-30
40	-12	-10	-8	-24	-22	-20	-37	-34	-31
36	-13	-11	-9	-25	-23	-21	-38	-35	-32

1. **Obtain V1, VR and V2 for the actual weight.**
2. **If V1(MCG) limited, set V1 = V1(MCG). If not V1(MCG) limited, enter V1 Adjustment Table with the actual weight to obtain V1 speed adjustment. If adjusted V1 is less than V1(MCG), set V1 = V1(MCG).**

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Takeoff %N1 (20K Derate)

Based on engine bleed to packs on (Auto) and anti-ice on or off

AIRPORT OAT		AIRPORT PRESSURE ALTITUDE (FT)									
°C	°F	-1000	0	1000	2000	3000	4000	5000	6000	7000	8000
55	131	89.9	90.5								
50	122	90.6	91.1	91.6	92.3	93.1					
45	113	91.2	91.6	92.1	92.6	93.9	94.9	94.7	94.2		
40	104	91.7	92.1	92.6	93.0	94.3	95.3	95.3	95.3	95.0	94.3
35	95	92.0	92.5	93.0	93.4	94.7	95.9	95.8	95.8	95.1	94.5
30	86	91.8	92.8	93.2	93.6	95.0	96.5	96.4	96.4	95.7	95.0
25	77	91.0	92.1	92.9	93.6	94.5	95.9	96.3	96.7	96.2	95.6
20	68	90.3	91.3	92.1	92.8	93.8	95.1	95.5	95.9	95.9	95.9
15	59	89.5	90.5	91.3	92.0	93.0	94.3	94.7	95.1	95.3	95.3
10	50	88.7	89.7	90.5	91.2	92.1	93.4	93.9	94.3	94.4	94.6
5	41	87.9	88.9	89.7	90.4	91.3	92.6	93.0	93.4	93.6	93.7
0	32	87.1	88.1	88.9	89.6	90.5	91.8	92.2	92.6	92.7	92.9
-10	14	85.5	86.5	87.2	87.9	88.8	90.1	90.5	90.9	91.0	91.2
-20	-4	83.9	84.8	85.6	86.2	87.1	88.3	88.7	89.1	89.2	89.4
-30	-22	82.2	83.1	83.9	84.5	85.4	86.6	87.0	87.4	87.5	87.6
-40	-40	80.5	81.4	82.1	82.7	83.6	84.8	85.1	85.5	85.7	85.8
-50	-58	78.8	79.6	80.3	81.0	81.8	82.9	83.3	83.7	83.8	84.0

%N1 Adjustments for Engine Bleeds

BLEED CONFIGURATION	AIRPORT PRESSURE ALTITUDE (FT)	
	-1000	8000
ENGINE BLEED TO PACKS OFF	1.0	1.0

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Assumed Temperature Reduced Thrust (20K Derate)

Maximum Assumed Temperature (Table 1)

Based on 25% Takeoff Thrust Reduction

OAT (°C)	PRESS ALT (1000 FT)										OAT (°F)	PRESS ALT (1000 FT)									
	-1	0	1	2	3	4	5	6	7	8		-1	0	1	2	3	4	5	6	7	8
55	74	73	73								130	164	162	162							
50	71	71	69	68	66	65					120	159	158	155	153	149	147				
45	69	68	67	66	63	62	62	63	65		110	154	153	151	149	144	142	142	143	145	
40	66	66	65	64	61	60	60	60	60	62	100	150	148	146	145	140	138	138	137	139	141
35	64	63	62	62	59	58	58	57	59	60	90	146	143	142	141	137	134	133	132	136	137
30	63	61	61	60	58	56	55	55	56	58	80	146	141	140	137	135	131	129	128	131	132
25	63	61	60	58	57	55	54	53	54	55	70	146	141	140	136	134	131	129	127	127	126
20 & BELOW	63	61	60	58	57	55	54	53	53	52	60 & BELOW	146	141	140	136	134	131	129	127	127	126

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Assumed Temperature Reduced Thrust (20K Derate)

Maximum Takeoff %N1 (Table 2)

Based on engine bleed to packs on (Auto) and engine anti-ice on or off

ASSUMED TEMP		AIRPORT PRESSURE ALTITUDE (FT)									
°C	°F	-1000	0	1000	2000	3000	4000	5000	6000	7000	8000
75	167	83.0	83.9	83.8	83.5						
70	158	84.7	85.6	85.6	85.4	84.9	84.7				
65	149	86.6	87.3	87.3	87.2	87.0	87.0	87.1	87.2	87.6	87.9
60	140	88.3	88.9	89.0	89.0	89.1	89.1	89.1	89.0	89.1	89.3
55	131	89.7	90.5	90.6	90.7	91.1	91.3	91.0	90.8	90.7	90.6
50	122	90.4	91.1	91.6	92.3	93.1	93.4	92.9	92.5	92.2	91.9
45	113	91.0	91.6	92.1	92.7	93.9	94.9	94.8	94.2	93.7	93.1
40	104	91.6	92.1	92.6	93.0	94.3	95.3	95.3	95.3	95.0	94.3
35	95	91.4	92.5	93.0	93.4	94.8	95.9	95.9	95.8	95.1	94.5
30	86	91.7	92.8	93.2	93.6	95.0	96.5	96.4	96.4	95.7	95.0
25	77	90.9		93.3	93.7	94.5	96.5	96.5	96.7	96.2	95.6
20	68	90.2				94.4				95.9	95.9
15	59	89.4									95.4
MINIMUM ASSUMED TEMP °C (°F)		32(90)	30(86)	28(82)	26(79)	24(75)	29(85)	27(81)	25(77)	20(68)	15(59)

With engine bleed to packs off, increase %N1 by 1.0.

%N1 Adjustment For Temperature Difference (Table 3)

ASSUMED TEMP MINUS OAT		OUTSIDE AIR TEMPERATURE													
°C	°F	°C	-40	-20	0	5	10	15	20	25	30	35	40	45	50	55
		°F	-40	-4	32	41	50	59	68	77	86	95	104	113	122	131
10	18						1.6	1.6	1.6	1.5	1.5	1.5	1.4	1.4	1.4	1.3
20	36				3.3	3.3	3.1	3.1	3.0	3.0	2.9	2.8	2.7	2.6	2.5	2.3
30	54				4.8	4.7	4.6	4.5	4.4	4.3	4.1	3.9	3.7	3.5	3.4	3.3
40	72			6.8	6.1	6.0	5.9	5.7	5.5	5.3	5.1	4.9				
50	90			8.1	7.4	7.2	6.9	6.7	6.3							
60	108		10.4	9.3	8.4	8.1	7.7									
70	126		11.5	10.5	9.0											
80	144		12.7	11.4	9.8											
90	162		13.8	11.9												
100	180		14.5	12.6												
110	198		14.9													

1. **Determine Maximum Assumed Temperature allowed from Table 1.**
2. **Find Maximum %N1 from Table 2 using the desired assumed temperature (no greater than temperature from Table 1).**
3. **Use the difference between assumed temperature and OAT to determine the %N1 adjustment from Table 3.**
4. **Subtract %N1 adjustment from Maximum %N1 in Table 2.**

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Max Climb %N1

Based on engine bleed to packs on (Auto) and anti-ice off

TAT (°C)	PRESSURE ALTITUDE (1000 FT)/SPEED (KIAS OR MACH)								
	0	5	10	15	20	25	30	35	37
	250	250	250	280	280	280	.74	.74	.74
50	90.9	91.1	92.5						
40	92.0	92.2	93.6	93.3	93.6				
30	92.2	93.2	93.5	94.2	94.6	94.7			
20	90.6	92.8	94.3	95.0	95.4	95.6	95.7		
10	89.1	91.2	93.1	95.1	96.1	96.4	96.6	96.6	96.6
0	87.5	89.6	91.5	93.4	95.5	97.2	97.5	97.5	97.5
-10	85.9	87.9	89.8	91.7	93.7	95.9	97.9	98.4	98.4
-20	84.2	86.3	88.1	90.0	91.9	94.0	96.0	99.0	99.3
-30	82.5	84.5	86.3	88.2	90.1	92.1	94.1	97.0	97.6
-40	80.8	82.8	84.5	86.3	88.2	90.2	92.2	95.0	95.6
-50	79.1	81.0	82.7	84.5	86.3	88.3	90.2	92.9	93.5

%N1 Adjustments for Engine Bleeds

BLEED CONFIGURATION	PRESSURE ALTITUDE (1000 FT)								
	0	5	10	15	20	25	30	35	37
ENGINE BLEED TO PACKS OFF	0.5	0.5	0.6	0.7	0.8	0.8	0.9	0.9	0.9
PACKS HIGH	-0.3	-0.3	-0.4	-0.4	-0.4	-0.4	-0.5	-0.6	-0.6
ENGINE ANTI-ICE ON	-0.7	-0.8	-0.9	-1.0	-1.0	-1.0	-1.0	-1.0	-1.0
ENGINE & WING ANTI-ICE ON	-1.9	-2.0	-2.2	-2.4	-2.6	-2.7	-2.8	-3.0	-3.0

Go-around %N1

Based on engine bleed to packs on (Auto), engine anti-ice on or off, and wing anti-ice off

REPORTED OAT		TAT	PRESSURE ALTITUDE (FT)									
°C	°F	°C	-1000	0	1000	2000	3000	4000	5000	6000	7000	8000
55	131	58	93.1	93.9	93.9	93.9						
50	122	53	93.7	94.2	94.2	94.2	94.2	94.2				
45	113	48	94.1	94.7	94.6	94.6	94.6	94.6	94.7	94.8	94.6	
40	104	43	94.5	95.1	95.1	95.1	95.0	95.1	95.1	95.2	95.1	95.0
35	95	38	95.1	95.5	95.6	95.5	95.5	95.7	95.7	95.7	95.6	95.6
30	86	33	95.1	96.1	96.1	96.0	96.0	96.3	96.2	96.2	96.1	96.0
25	77	28	94.3	95.3	95.8	96.2	96.5	96.7	96.7	96.6	96.5	96.5
20	68	23	93.5	94.5	95.0	95.4	95.9	96.6	97.1	97.2	97.0	96.9
15	59	18	92.7	93.7	94.1	94.6	95.1	95.8	96.3	96.8	97.3	97.5
10	50	13	91.9	92.8	93.3	93.7	94.2	95.0	95.4	96.0	96.4	96.9
5	41	8	91.1	92.0	92.5	92.9	93.4	94.1	94.6	95.1	95.6	96.1
0	32	3	90.3	91.2	91.7	92.1	92.6	93.3	93.7	94.3	94.7	95.2
-10	14	-8	88.5	89.5	90.0	90.4	90.8	91.5	92.0	92.5	93.0	93.4
-20	-4	-18	86.8	87.8	88.2	88.6	89.1	89.8	90.3	90.8	91.2	91.6
-30	-22	-28	85.1	86.0	86.5	86.9	87.3	88.0	88.5	89.0	89.4	89.8
-40	-40	-38	83.3	84.2	84.7	85.1	85.5	86.2	86.6	87.1	87.5	87.9
-50	-58	-48	81.5	82.4	82.8	83.2	83.7	84.3	84.7	85.2	85.6	86.0

%N1 Adjustments for Engine Bleeds

BLEED CONFIGURATION	TAT (°C)	
	- 60	60
ENGINE BLEED TO PACKS OFF	0.8	1.0
PACKS HIGH	- 0.3	- 0.3
WING A/I ALL ENGINES	- 1.3	- 1.6
WING A/I 1 ENGINE INOP	- 2.2	- 2.7

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Flight With Unreliable Airspeed / Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

Climb

Flaps Up, Set Max Climb Thrust

PRESSURE ALTITUDE (FT) (SPEED)		WEIGHT (1000 KG)				
		32	34	44	54	64
35000	**PITCH ATT**	**6.5**	**6.5**	**6.0**	**5.5**	**5.5**
(.73M)	V/S (FT/MIN)	3100	2800	1800	1000	300
30000	**PITCH ATT**	**7.0**	**6.5**	**6.0**	**5.5**	**5.5**
(.73M)	V/S (FT/MIN)	3900	3600	2500	1700	1100
25000	**PITCH ATT**	**7.0**	**6.5**	**6.0**	**6.0**	**6.0**
(280 KIAS)	V/S (FT/MIN)	3700	3500	2500	1800	1300
20000	**PITCH ATT**	**8.5**	**8.5**	**7.5**	**7.0**	**7.0**
(280 KIAS)	V/S (FT/MIN)	4600	4300	3100	2300	1800
15000	**PITCH ATT**	**10.5**	**10.0**	**9.0**	**8.0**	**8.0**
(280 KIAS)	V/S (FT/MIN)	5400	5100	3800	2900	2200
10000	**PITCH ATT**	**13.0**	**12.0**	**10.5**	**9.5**	**9.0**
(280 KIAS)	V/S (FT/MIN)	6300	5900	4400	3400	2700
5000	**PITCH ATT**	**15.0**	**14.5**	**12.0**	**11.0**	**10.0**
(280 KIAS)	V/S (FT/MIN)	7100	6700	5000	3900	3100
SEA LEVEL	**PITCH ATT**	**17.5**	**17.0**	**14.0**	**12.5**	**11.5**
(280 KIAS)	V/S (FT/MIN)	7900	7400	5600	4400	3500

Cruise

Flaps Up, %N1 for Level Flight

PRESSURE ALTITUDE (FT) (SPEED)		WEIGHT (1000 KG)					
		34	38	42	46	50	54
30000	**PITCH ATT**	**2.0**	**2.0**	**2.5**	**3.0**	**3.0**	**3.5**
(.73M)	%N1	79	79	80	81	82	83
10000	**PITCH ATT**	**2.0**	**2.0**	**2.5**	**3.0**	**3.0**	**3.5**
(280 KIAS)	%N1	65	65	66	66	67	68

Descent

Flaps Up, Set Idle Thrust

PRESSURE ALTITUDE (FT) (SPEED)		WEIGHT (1000 KG)					
		34	38	42	46	50	54
30000	**PITCH ATT**	**-3.5**	**-3.0**	**-2.5**	**-2.0**	**-1.5**	**-1.0**
(.73M)	V/S (FT/MIN)	-4200	-3800	-3600	-3400	-3300	-3200
20000	**PITCH ATT**	**-2.5**	**-2.0**	**-1.0**	**-0.5**	**0.0**	**0.0**
(280 KIAS)	V/S (FT/MIN)	-2900	-2600	-2500	-2300	-2200	-2100
10000	**PITCH ATT**	**-2.5**	**-2.0**	**-1.5**	**-1.0**	**-0.5**	**0.0**
(280 KIAS)	V/S (FT/MIN)	-2500	-2300	-2200	-2100	-2000	-1900

Holding

Flaps Up, %N1 for Level Flight

PRESSURE ALTITUDE (FT)		WEIGHT (1000 KG)					
		34	38	42	46	50	54
	PITCH ATT	**4.0**	**4.5**	**5.5**	**5.5**	**5.5**	**5.5**
10000	%N1	56	57	59	61	63	65
	KIAS	210	210	210	215	225	230

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Flight With Unreliable Airspeed / Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

Terminal Area

%N1 for Level Flight

FLAP POSITION (SPEED)		WEIGHT (1000 KG)			
		36	41	45	50
FLAPS UP (GEAR UP) (210 KIAS)	**PITCH ATT**	**4.4**	**5.1**	**5.8**	**6.4**
	%N1	54	55	57	59
FLAPS 1 (GEAR UP) (190 KIAS)	**PITCH ATT**	**4.8**	**5.6**	**6.3**	**7.0**
	%N1	56	57	58	61
FLAPS 5 (GEAR UP) (170 KIAS)	**PITCH ATT**	**4.9**	**5.8**	**6.7**	**7.6**
	%N1	54	56	59	62
FLAPS 15 (GEAR DOWN) (150 KIAS)	**PITCH ATT**	**4.9**	**6.0**	**7.0**	**8.1**
	%N1	62	64	66	69

Final Approach

Gear Down, %N1 for 3° Glideslope

FLAP POSITION		WEIGHT (1000 KG)				
		36	41	45	50	54
FLAPS 40	**PITCH ATT**	**0.5**	**0.4**	**0.4**	**0.3**	**0.3**
	%N1	57	60	63	65	67
	KIAS	119	126	133	140	144
FLAPS 30	**PITCH ATT**	**1.9**	**1.9**	**1.9**	**2.0**	**2.0**
	%N1	50	53	56	58	59
	KIAS	123	130	137	144	147
FLAPS 15	**PITCH ATT**	**4.5**	**4.5**	**4.5**	**4.5**	**4.5**
	%N1	46	49	51	53	54
	KIAS	131	139	147	154	158

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Long Range Cruise Maximum Operating Altitude

Max Cruise Thrust

ISA + 10°C and Below

WEIGHT (1000 KG)	OPTIMUM ALT (FT)	TAT (°C)	MARGIN TO INITIAL BUFFET 'G' (BANK ANGLE)				
			1.20 (33°)	1.25 (36°)	1.30 (39°)	1.40 (44°)	1.50 (48°)
64	30800	-10	34100	33200	32300	30700	29200
62	31500	-11	34700	33800	33000	31400	29900
60	32200	-13	35400	34500	33700	32100	30600
58	32900	-14	36100	35300	34400	32800	31400
56	33700	-16	36900	36000	35200	33600	32100
54	34400	-18	37000	36700	35900	34400	32900
52	35200	-19	37000	37000	36700	35200	33700
50	36100	-21	37000	37000	37000	36000	34500
48	36900	-21	37000	37000	37000	36800	35400
46	37000	-21	37000	37000	37000	37000	36300
44	37000	-21	37000	37000	37000	37000	37000

ISA + 15°C

WEIGHT (1000 KG)	OPTIMUM ALT (FT)	TAT (°C)	MARGIN TO INITIAL BUFFET 'G' (BANK ANGLE)				
			1.20 (33°)	1.25 (36°)	1.30 (39°)	1.40 (44°)	1.50 (48°)
64	30800	-4	34100	33200	32300	30700	29200
62	31500	-6	34700	33800	33000	31400	29900
60	32200	-7	35400	34500	33700	32100	30600
58	32900	-9	36100	35300	34400	32800	31400
56	33700	-10	36900	36000	35200	33600	32100
54	34400	-12	37000	36700	35900	34400	32900
52	35200	-14	37000	37000	36700	35200	33700
50	36100	-16	37000	37000	37000	36000	34500
48	36900	-16	37000	37000	37000	36800	35400
46	37000	-16	37000	37000	37000	37000	36300
44	37000	-16	37000	37000	37000	37000	37000

ISA + 20°C

WEIGHT (1000 KG)	OPTIMUM ALT (FT)	TAT (°C)	MARGIN TO INITIAL BUFFET 'G' (BANK ANGLE)				
			1.20 (33°)	1.25 (36°)	1.30 (39°)	1.40 (44°)	1.50 (48°)
64	30800	1	33600*	33200	32300	30700	29200
62	31500	0	34400*	33800	33000	31400	29900
60	32200	-2	35300*	34500	33700	32100	30600
58	32900	-3	36100	35300	34400	32800	31400
56	33700	-5	36800*	36000	35200	33600	32100
54	34400	-7	37000	36700	35900	34400	32900
52	35200	-8	37000	37000	36700	35200	33700
50	36100	-10	37000	37000	37000	36000	34500
48	36900	-10	37000	37000	37000	36800	35400
46	37000	-10	37000	37000	37000	37000	36300
44	37000	-10	37000	37000	37000	37000	37000

***Denotes altitude thrust limited in level flight, 100 fpm residual rate of climb.**

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Long Range Cruise Control

WEIGHT (1000 KG)		PRESSURE ALTITUDE (1000 FT)								
		21	23	25	27	29	31	33	35	37
62	%N1	81.1	82.2	83.3	84.3	85.2	86.4	88.2	91.0	
	MACH	.698	.718	.734	.744	.745	.745	.745	.744	
	KIAS	317	313	308	299	287	275	263	251	
	FF/ENG	1499	1479	1452	1419	1372	1337	1329	1356	
58	%N1	79.7	81.0	82.1	83.1	84.1	85.1	86.3	88.4	92.3
	MACH	.682	.703	.722	.737	.744	.745	.745	.745	.742
	KIAS	309	306	302	296	287	275	263	251	239
	FF/ENG	1413	1396	1376	1348	1313	1271	1244	1244	1293
54	%N1	78.1	79.5	80.7	81.8	82.8	83.8	84.8	86.3	89.0
	MACH	.665	.685	.707	.725	.740	.745	.745	.745	.745
	KIAS	301	298	295	291	285	275	263	251	240
	FF/ENG	1328	1311	1295	1274	1248	1211	1174	1154	1169
50	%N1	76.5	77.8	79.1	80.4	81.5	82.5	83.4	84.5	86.5
	MACH	.646	.667	.688	.709	.728	.741	.745	.745	.745
	KIAS	292	289	287	284	280	274	263	251	240
	FF/ENG	1244	1227	1210	1195	1175	1150	1114	1081	1073
46	%N1	74.9	76.1	77.4	78.7	79.9	81.0	82.1	83.0	84.5
	MACH	.625	.647	.668	.689	.710	.729	.742	.745	.745
	KIAS	282	280	278	276	273	269	262	251	240
	FF/ENG	1160	1144	1128	1112	1098	1079	1055	1021	997
42	%N1	72.9	74.2	75.5	76.8	78.1	79.3	80.5	81.5	82.8
	MACH	.603	.624	.646	.667	.688	.710	.729	.743	.745
	KIAS	271	270	268	266	264	261	257	250	240
	FF/ENG	1077	1062	1046	1029	1015	1003	985	963	936
38	%N1	70.8	72.2	73.5	74.7	76.0	77.4	78.6	79.8	81.2
	MACH	.579	.599	.621	.643	.664	.686	.709	.728	.742
	KIAS	260	259	257	256	254	252	249	245	239
	FF/ENG	994	981	964	948	934	921	910	894	878
34	%N1	68.4	69.8	71.2	72.5	73.8	75.1	76.5	77.7	79.2
	MACH	.553	.573	.593	.615	.637	.659	.682	.705	.725
	KIAS	248	247	246	244	243	241	239	237	233
	FF/ENG	912	898	884	866	852	840	829	818	809

Shaded area approximates optimum altitude.

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Long Range Cruise Enroute Fuel and Time - Low Altitudes

Ground to Air Miles Conversion

AIR DISTANCE (NM) HEADWIND COMPONENT (KTS)					GROUND DISTANCE (NM)	AIR DISTANCE (NM) TAILWIND COMPONENT (KTS)				
100	80	60	40	20		20	40	60	80	100
284	262	243	226	213	200	191	182	174	167	160
428	395	366	341	319	300	286	273	261	250	241
573	528	489	455	426	400	381	364	348	334	321
719	662	612	570	533	500	477	455	435	418	402
865	796	735	684	640	600	572	546	522	501	482
1011	931	859	799	747	700	667	637	610	584	562
1158	1065	983	914	854	800	763	728	697	668	642
1305	1200	1107	1028	961	900	858	819	784	751	722
1453	1335	1231	1143	1068	1000	953	910	871	835	802
1601	1471	1355	1258	1175	1100	1048	1001	957	917	881
1750	1607	1480	1374	1282	1200	1144	1091	1044	1000	961
1900	1743	1605	1489	1389	1300	1239	1183	1131	1084	1041
2050	1880	1730	1605	1497	1400	1334	1274	1218	1167	1121
2201	2018	1856	1721	1604	1500	1429	1364	1304	1250	1201
2352	2156	1982	1836	1712	1600	1524	1455	1391	1332	1280
2504	2293	2108	1952	1819	1700	1620	1545	1477	1415	1359
2656	2432	2234	2068	1926	1800	1715	1636	1564	1498	1438
2809	2571	2360	2185	2034	1900	1810	1727	1650	1580	1517
2963	2710	2487	2301	2142	2000	1905	1817	1737	1663	1597

Reference Fuel and Time Required at Check Point

AIR DIST (NM)	PRESSURE ALTITUDE (1000 FT)									
	10		14		20		24		28	
	FUEL (1000 KG)	TIME (HR:MIN)	FUEL (1000 KG)	TIME (HR:MIN)	FUEL (1000 KG)	TIME (HR:MIN)	FUEL (1000 KG)	TIME (HR:MIN)	FUEL (1000 KG)	TIME (HR:MIN)
200	1.4	0:39	1.3	0:38	1.1	0:37	1.0	0:36	0.9	0:35
300	2.2	0:58	2.0	0:56	1.7	0:53	1.5	0:51	1.4	0:50
400	2.9	1:17	2.7	1:14	2.3	1:09	2.1	1:07	1.9	1:04
500	3.7	1:35	3.3	1:31	2.9	1:26	2.6	1:22	2.4	1:19
600	4.4	1:54	4.0	1:49	3.5	1:42	3.2	1:38	2.9	1:34
700	5.2	2:13	4.7	2:07	4.1	1:59	3.7	1:53	3.4	1:49
800	5.9	2:32	5.4	2:25	4.7	2:15	4.3	2:09	4.0	2:04
900	6.6	2:52	6.0	2:44	5.3	2:32	4.8	2:25	4.5	2:19
1000	7.3	3:11	6.7	3:02	5.8	2:49	5.4	2:41	5.0	2:34
1100	8.1	3:31	7.4	3:21	6.4	3:06	5.9	2:57	5.5	2:49
1200	8.8	3:50	8.0	3:39	7.0	3:23	6.4	3:13	5.9	3:04
1300	9.5	4:10	8.7	3:58	7.6	3:40	7.0	3:29	6.4	3:20
1400	10.2	4:30	9.3	4:17	8.1	3:57	7.5	3:45	6.9	3:35
1500	10.9	4:51	10.0	4:36	8.7	4:15	8.0	4:02	7.4	3:50
1600	11.6	5:11	10.6	4:55	9.3	4:32	8.6	4:18	7.9	4:06
1700	12.3	5:31	11.2	5:14	9.8	4:50	9.1	4:35	8.4	4:21
1800	13.0	5:52	11.9	5:34	10.4	5:08	9.6	4:51	8.9	4:37
1900	13.6	6:13	12.5	5:54	10.9	5:25	10.1	5:08	9.3	4:53
2000	14.3	6:34	13.1	6:13	11.5	5:43	10.6	5:25	9.8	5:09

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Long Range Cruise Enroute Fuel and Time - Low Altitudes

Fuel Required Adjustment (1000 KG)

REFERENCE FUEL REQUIRED (1000 KG)	WEIGHT AT CHECK POINT (1000 KG)						
	30	35	40	45	50	55	60
4	-0.4	-0.3	-0.1	0.0	0.2	0.4	0.6
6	-0.6	-0.4	-0.2	0.0	0.3	0.6	1.0
8	-0.9	-0.6	-0.3	0.0	0.4	0.9	1.3
10	-1.1	-0.7	-0.4	0.0	0.5	1.1	1.6
12	-1.3	-0.9	-0.4	0.0	0.6	1.3	1.9
14	-1.6	-1.0	-0.5	0.0	0.7	1.5	2.3
16	-1.8	-1.2	-0.6	0.0	0.9	1.7	2.6

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Long Range Cruise Enroute Fuel and Time - High Altitudes

Ground to Air Miles Conversion

AIR DISTANCE (NM) HEADWIND COMPONENT (KTS)					GROUND DISTANCE (NM)	AIR DISTANCE (NM) TAILWIND COMPONENT (KTS)				
100	80	60	40	20		20	40	60	80	100
534	503	475	448	423	400	381	363	346	331	317
797	751	708	669	633	600	572	545	521	499	479
1062	999	942	890	843	800	763	728	696	667	642
1328	1248	1176	1111	1053	1000	954	911	872	837	806
1595	1498	1410	1333	1263	1200	1146	1095	1049	1007	970
1862	1748	1646	1555	1474	1400	1337	1279	1226	1177	1133
2130	2000	1881	1777	1685	1600	1529	1463	1402	1346	1296
2399	2251	2118	2000	1896	1800	1720	1646	1578	1515	1458
2668	2504	2355	2224	2107	2000	1911	1828	1752	1683	1620
2939	2757	2592	2447	2318	2200	2102	2011	1927	1850	1781
3211	3011	2830	2671	2529	2400	2293	2194	2102	2019	1943
3484	3266	3068	2895	2741	2600	2484	2377	2277	2187	2104
3757	3521	3307	3119	2952	2800	2675	2559	2452	2355	2266
4032	3777	3546	3344	3164	3000	2866	2742	2627	2522	2427
4309	4034	3786	3569	3376	3200	3057	2924	2801	2689	2588
4586	4292	4026	3794	3588	3400	3247	3106	2975	2856	2748
4864	4551	4268	4020	3800	3600	3438	3288	3149	3023	2909
5144	4811	4509	4246	4012	3800	3629	3470	3323	3190	3069
5425	5071	4751	4472	4225	4000	3819	3652	3498	3357	3229

Reference Fuel and Time Required at Check Point

AIR DIST (NM)	PRESSURE ALTITUDE (1000 FT)									
	29		31		33		35		37	
	FUEL (1000 KG)	TIME (HR:MIN)	FUEL (1000 KG)	TIME (HR:MIN)	FUEL (1000 KG)	TIME (HR:MIN)	FUEL (1000 KG)	TIME (HR:MIN)	FUEL (1000 KG)	TIME (HR:MIN)
400	2.1	0:59	2.0	0:59	1.9	0:59	1.8	1:01	1.8	1:03
600	3.0	1:30	2.9	1:29	2.8	1:28	2.7	1:29	2.6	1:31
800	4.0	2:01	3.8	1:59	3.7	1:57	3.6	1:58	3.5	1:59
1000	4.9	2:31	4.7	2:28	4.6	2:26	4.4	2:26	4.3	2:27
1200	5.9	3:02	5.6	2:58	5.4	2:56	5.3	2:55	5.1	2:55
1400	6.8	3:32	6.6	3:28	6.3	3:25	6.1	3:23	6.0	3:23
1600	7.7	4:03	7.5	3:58	7.2	3:54	7.0	3:52	6.8	3:51
1800	8.7	4:34	8.4	4:28	8.1	4:23	7.8	4:20	7.6	4:19
2000	9.6	5:05	9.3	4:58	9.0	4:52	8.7	4:49	8.4	4:48
2200	10.5	5:36	10.2	5:28	9.8	5:22	9.5	5:18	9.3	5:16
2400	11.5	6:08	11.1	5:59	10.7	5:52	10.3	5:47	10.1	5:44
2600	12.4	6:40	11.9	6:30	11.5	6:22	11.2	6:16	10.8	6:13
2800	13.2	7:12	12.8	7:02	12.4	6:52	12.0	6:46	11.6	6:42
3000	14.1	7:45	13.6	7:33	13.2	7:23	12.8	7:16	12.4	7:11
3200	15.0	8:18	14.5	8:05	14.0	7:54	13.5	7:46	13.2	7:40
3400	15.9	8:52	15.3	8:38	14.8	8:25	14.3	8:16	13.9	8:09
3600	16.7	9:26	16.1	9:10	15.6	8:57	15.1	8:47	14.7	8:39
3800	17.5	10:00	16.9	9:43	16.4	9:29	15.9	9:18	15.4	9:09
4000	18.4	10:35	17.7	10:17	17.1	10:01	16.6	9:49	16.1	9:39

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Long Range Cruise Enroute Fuel and Time - High Altitudes

Fuel Required Adjustment (1000 KG)

REFERENCE FUEL REQUIRED (1000 KG)	WEIGHT AT CHECK POINT (1000 KG)						
	30	35	40	45	50	55	60
4	-0.5	-0.4	-0.2	0.0	0.3	0.7	1.3
6	-0.9	-0.6	-0.3	0.0	0.4	1.0	1.8
8	-1.2	-0.8	-0.4	0.0	0.6	1.3	2.3
10	-1.5	-1.0	-0.5	0.0	0.7	1.6	2.7
12	-1.8	-1.3	-0.6	0.0	0.8	1.9	3.1
14	-2.1	-1.5	-0.7	0.0	1.0	2.1	3.5
16	-2.4	-1.6	-0.9	0.0	1.1	2.4	3.9
18	-2.6	-1.8	-1.0	0.0	1.2	2.6	4.2
20	-2.8	-2.0	-1.1	0.0	1.3	2.8	4.5
22	-3.0	-2.1	-1.2	0.0	1.4	3.0	4.8

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Long Range Cruise Wind-Altitude Trade

PRESSURE ALTITUDE (1000 FT)	CRUISE WEIGHT (1000 KG)													
	62	60	58	56	54	52	50	48	46	44	42	40	38	36
37					14	8	1	0	0	0	0	0	0	0
35		9	5	2	0	0	2	6	8	10	12	13	14	14
33	2	1	0	1	3	7	12	17	20	23	26	27	29	30
31	1	3	4	9	14	19	25	31	35	38	41	43	44	45
29	8	12	17	22	28	33	39	45	50	54	57	59	60	61
27	17	25	30	35	41	48	55	61	66	69	71	74	75	77
25	34	39	44	51	57	63	70	78	82	86	89	92	94	96
23	48	53	59	65	72	79	86	94	99	103	106	109	112	113

The above wind factor tables are for calculation of wind required to maintain present range capability at new pressure altitude, i.e., break-even wind.

Method:

1. **Read wind factors for present and new altitudes from table.**
2. **Determine difference (new altitude wind factor minus present altitude wind factor); This difference may be negative or positive.**
3. **Break-even wind at new altitude is present altitude wind plus difference from step 2.**

Descent

Based on .74/250 descent speed

PRESSURE ALT (1000 FT)	25	27	29	31	33	35	37
DISTANCE (NM)	79	86	92	98	105	111	116
TIME (MINUTES)	18	19	20	21	22	23	23

Holding

Flaps Up

WEIGHT (1000 KG)		PRESSURE ALTITUDE (FT)								
		1500	5000	10000	15000	20000	25000	30000	35000	37000
62	%N1	61.7	64.3	68.3	72.4	76.3	80.4	84.5	90.6	
	KIAS	247	248	250	253	255	257	256	249	
	FF/ENG	1430	1400	1360	1330	1310	1310	1310	1400	
58	%N1	60.1	62.6	66.5	70.7	74.7	78.7	82.8	87.6	92.4
	KIAS	239	240	241	244	246	248	248	243	239
	FF/ENG	1360	1320	1280	1250	1230	1230	1220	1250	1360
54	%N1	58.4	60.8	64.7	68.7	73.0	77.0	81.0	85.4	88.2
	KIAS	231	231	232	235	237	239	241	238	234
	FF/ENG	1280	1240	1200	1170	1150	1150	1140	1150	1180
50	%N1	56.6	59.1	62.7	66.8	71.0	75.0	79.1	83.3	85.6
	KIAS	224	223	223	225	227	230	232	230	228
	FF/ENG	1210	1170	1130	1090	1070	1060	1050	1060	1070
46	%N1	54.8	57.1	60.6	64.6	68.8	73.0	77.1	81.2	83.4
	KIAS	216	214	214	215	218	220	222	222	221
	FF/ENG	1130	1100	1050	1020	990	980	970	970	980
42	%N1	53.1	55.3	58.8	62.6	66.6	70.8	74.9	79.0	81.1
	KIAS	210	210	210	210	210	210	212	213	212
	FF/ENG	1070	1030	990	950	920	900	890	890	890
38	%N1	51.5	53.8	57.2	60.8	64.8	69.0	72.9	76.9	79.0
	KIAS	210	210	210	210	210	210	210	210	210
	FF/ENG	1010	980	940	900	870	840	830	820	820
34	%N1	50.0	52.4	55.7	59.3	63.1	67.2	71.2	75.1	77.0
	KIAS	210	210	210	210	210	210	210	210	210
	FF/ENG	970	940	900	860	820	800	770	760	770

This table includes 5% additional fuel for holding in a racetrack pattern.

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ADVISORY INFORMATION

Autobrake Landing Distance

Reference Landing Distance (FT)

AUTOBRAKE SETTING	APPROACH SPEED (KIAS)						
	100	110	120	130	140	150	160
1	4500	5300	6100	6930	7850	8850	9950
2	3800	4440	5080	5730	6450	7240	8100
3	3200	3680	4150	4650	5220	5840	6500
MAX	2500	2760	3010	3280	3600	4000	4500

Actual (unfactored) distances are shown.
Landing distance required includes 1000 ft of air distance.
The distances are valid with or without reverse thrust and for any flap.
If entering with ground speed, ignore adjustments.

Landing Distance Adjustments (FT)

AUTOBRAKE SETTING	ALTITUDE ADJUSTMENT PER 1000 FT ABOVE SEA LEVEL	TAILWIND ADJUSTMENT PER 10 KTS	TEMP ADJUSTMENT PER 10°C ABOVE ISA
1	250	1100	300
2	200	900	250
3	140	650	200
MAX	80	400	100

Slippery Runway Landing Distance

Reference Landing Distance (FT)

BRAKING CONFIGURATION	REPORTED BRAKING ACTION			
	DRY	GOOD	MEDIUM	POOR
MAX MANUAL BRAKING	2525	3445	4495	5625
AUTOBRAKE SETTING 2	4495	4955	5280	5940
AUTOBRAKE SETTING 3	3165	3840	4575	5625
MAX AUTOBRAKE SETTING	2625	3445	4495	5625

Landing Distance Adjustments (FT)

CONDITIONS		ADJUSTMENTS			
WEIGHT	PER 5000 KG BELOW 48,000 KG	-150	-245	-375	-510
	PER 5000 KG ABOVE 48,000 KG	525	490	525	605
AIRPORT PRESSURE ALTITUDE	PER 1000 FT ABOVE SEA LEVEL	130	150	150	165
WIND	PER 10 KTS HEADWIND	-80	-150	-245	-345
	PER 10 KTS TAILWIND	835	885	1035	1475
APPROACH SPEED	PER 10 KTS ABOVE VREF	510	475	445	425
SLOPE	PER 1% DOWNHILL SLOPE	35	80	195	445
	PER 1% UPHILL SLOPE	-35	-35	-115	-260
REVERSE THRUST	1 REVERSER INOPERATIVE	35	150	410	805
	2 REVERSERS INOPERATIVE	115	560	1770	4890
	MAXIMUM REVERSE THRUST	0	0	-35	-100

Actual (unfactored) distances are shown.
Landing Distance Required Includes 1000 ft of Air Distance.
Based on flaps 40, VREF40 approach speed.
Includes 2 engine detent reverse thrust.

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ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

Dry Runway

LANDING CONFIGURATION	APPROACH SPEED	LANDING DISTANCE AND ADJUSTMENTS (FT)		
		REFERENCE DISTANCE FOR 48000 KG LANDING WEIGHT	WT ADJ PER 2000 KG ABOVE/BELOW 48000 KG	ALTITUDE ADJ PER 1000 FT ABOVE S.L.
ONE ENGINE INOPERATIVE	VREF15	2800	160 / -80	140
ANTI-SKID INOPERATIVE	VREF40	3920	120/-110	80
HYDRAULIC SYSTEM A INOPERATIVE (FLAPS 15)	VREF15	3250	120 / -80	150
HYDRAULIC SYSTEM A INOPERATIVE (FLAPS 30)	VREF30	3280	90/-100	130
HYDRAULIC SYSTEM A INOPERATIVE (FLAPS 40)	VREF40	3180	100/-90	130
HYDRAULIC SYSTEM B INOPERATIVE (FLAPS 15)	VREF15	3050	180 / -80	150
HYDRAULIC SYSTEMS A & B INOPERATIVE (MANUAL REVERSION)	VREF15	4900	170 / -150	230
LEADING EDGE ASYMMETRY OR FAILURE	VREF15+5	2900	170 / -80	150
TRAILING EDGE FAILURE (FLAPS LESS THAN 1)	VREF40+40	3450	250 / -110	190
TRAILING EDGE FAILURE (FLAPS GREATER THAN 1 AND LESS THAN 15)	VREF40+30	3200	210 / -100	180
FLAPS UP	VREF40+55	3950	310 / -140	230
JAMMED STABILIZER	VREF15	2730	150 / -70	130

Actual (unfactored) distances are shown
Includes distances from 50 ft above threshold (1000 ft of air distance)
Assumes maximum manual braking and maximum reverse thrust when available on operating engines.

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ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

Good Reported Braking Action

LANDING CONFIGURATION	APPROACH SPEED	LANDING DISTANCE AND ADJUSTMENTS (FT)		
		REFERENCE DISTANCE FOR 48000 KG LANDING WEIGHT	WT ADJ PER 2000 KG ABOVE/BELOW 48000 KG	ALTITUDE ADJ PER 1000 FT ABOVE S.L.
ONE ENGINE INOPERATIVE	VREF15	3950	130 / -120	170
ANTI-SKID INOPERATIVE	VREF40	4810	170/-150	110
HYDRAULIC SYSTEM A INOPERATIVE (FLAPS 15)	VREF15	4550	150 / -140	210
HYDRAULIC SYSTEM A INOPERATIVE (FLAPS 30)	VREF30	4460	150/-150	180
HYDRAULIC SYSTEM A INOPERATIVE (FLAPS 40)	VREF40	4260	160/-150	180
HYDRAULIC SYSTEM B INOPERATIVE (FLAPS 15)	VREF15	3920	120 / -120	170
HYDRAULIC SYSTEMS A & B INOPERATIVE (MANUAL REVERSION)	VREF15	4950	170 / -150	230
LEADING EDGE ASYMMETRY OR FAILURE	VREF15+5	4050	130 / -120	190
TRAILING EDGE FAILURE (FLAPS LESS THAN 1)	VREF40+40	4540	140 / -130	210
TRAILING EDGE FAILURE (FLAPS GREATER THAN 1 AND LESS THAN 15)	VREF40+30	4200	130 / -120	190
FLAPS UP	VREF40+55	5080	160 / -140	240
JAMMED STABILIZER	VREF15	3750	120 / -110	150

Actual (unfactored) distances are shown
Includes distances from 50 ft above threshold (1000 ft of air distance)
Assumes maximum manual braking and maximum reverse thrust when available on operating engines.

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ADVISORY INFORMATION

Non-Normal Configuration Landing Distance
Medium Reported Braking Action

LANDING CONFIGURATION	APPROACH SPEED	LANDING DISTANCE AND ADJUSTMENTS (FT)		
		REFERENCE DISTANCE FOR 48000 KG LANDING WEIGHT	WT ADJ PER 2000 KG ABOVE/BELOW 48000 KG	ALTITUDE ADJ PER 1000 FT ABOVE S.L.
ONE ENGINE INOPERATIVE	VREF15	5500	200 / -190	240
ANTI-SKID INOPERATIVE	VREF40	5870	220/-200	145
HYDRAULIC SYSTEM A INOPERATIVE (FLAPS 15)	VREF15	5970	230 / -220	280
HYDRAULIC SYSTEM A INOPERATIVE (FLAPS 30)	VREF30	5690	210/-220	240
HYDRAULIC SYSTEM A INOPERATIVE (FLAPS 40)	VREF40	5370	220/-200	230
HYDRAULIC SYSTEM B INOPERATIVE (FLAPS 15)	VREF15	5250	200 / -170	230
HYDRAULIC SYSTEMS A & B INOPERATIVE (MANUAL REVERSION)	VREF15	6450	250 / -230	310
LEADING EDGE ASYMMETRY OR FAILURE	VREF15+5	5420	200 / -190	260
TRAILING EDGE FAILURE (FLAPS LESS THAN 1)	VREF40+40	6050	220 / -190	280
TRAILING EDGE FAILURE (FLAPS GREATER THAN 1 AND LESS THAN 15)	VREF40+30	5620	210 / -180	260
FLAPS UP	VREF40+55	6850	250 / -220	330
JAMMED STABILIZER	VREF15	4970	180 / -170	210

Actual (unfactored) distances are shown
Includes distances from 50 ft above threshold (1000 ft of air distance)
Assumes maximum manual braking and maximum reverse thrust when available on operating engines.

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ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

Poor Reported Braking Action

LANDING CONFIGURATION	APPROACH SPEED	LANDING DISTANCE AND ADJUSTMENTS (FT)		
		REFERENCE DISTANCE FOR 48000 KG LANDING WEIGHT	WT ADJ PER 2000 KG ABOVE/BELOW 48000 KG	ALTITUDE ADJ PER 1000 FT ABOVE S.L.
ONE ENGINE INOPERATIVE	VREF15	7330	310 / -280	330
ANTI-SKID INOPERATIVE	VREF40	7410	290/-280	200
HYDRAULIC SYSTEM A INOPERATIVE (FLAPS 15)	VREF15	7400	310 / -290	360
HYDRAULIC SYSTEM A INOPERATIVE (FLAPS 30)	VREF30	6900	270/-280	290
HYDRAULIC SYSTEM A INOPERATIVE (FLAPS 40)	VREF40	6490	280/-260	280
HYDRAULIC SYSTEM B INOPERATIVE (FLAPS 15)	VREF15	6650	260 / -250	300
HYDRAULIC SYSTEMS A & B INOPERATIVE (MANUAL REVERSION)	VREF15	7930	340 / -300	380
LEADING EDGE ASYMMETRY OR FAILURE	VREF15+5	6800	280 / -250	330
TRAILING EDGE FAILURE (FLAPS LESS THAN 1)	VREF40+40	7600	310 / -280	360
TRAILING EDGE FAILURE (FLAPS GREATER THAN 1 AND LESS THAN 15)	VREF40+30	7100	290 / -250	340
FLAPS UP	VREF40+55	8600	340 / -310	420
JAMMED STABILIZER	VREF15	6250	250 / -240	270

Actual (unfactored) distances are shown
Includes distances from 50 ft above threshold (1000 ft of air distance)
Assumes maximum manual braking and maximum reverse thrust when available on operating engines.

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ADVISORY INFORMATION

Brake Cooling Schedule

Reference Brake Energy Per Brake (Millions of Foot Pounds)

		WIND CORRECTED BRAKES ON SPEED (KIAS)*														
		60			80			100			120			140		
WEIGHT (1000 KG)	OAT (°C)	PRESS ALT			PRESS ALT			PRESS ALT			PRESS ALT			PRESS ALT		
		0	2	4	0	2	4	0	2	4	0	2	4	0	2	4
65	-20	7.0	7.5	8.1	11.3	12.2	13.1	16.2	17.7	19.1	21.9	24.1	25.9			
	0	7.5	8.2	8.9	12.3	13.2	14.3	17.7	19.3	20.8	23.9	26.0	27.9			
	20	8.1	8.8	9.6	13.3	14.3	15.4	19.2	20.8	22.4	25.9	28.3	30.0			
	40	8.7	9.5	10.2	14.3	15.3	16.6	20.7	22.4	24.1	27.9	30.0	32.1			
60	-20	6.6	7.2	7.7	10.6	11.5	12.4	15.3	16.6	18.0	20.5	22.2	24.0	26.5		
	0	7.1	7.8	8.4	11.5	12.5	13.5	16.7	18.1	19.6	22.4	24.3	26.1	28.9		
	20	7.7	8.4	9.1	12.5	13.5	14.5	18.1	19.7	21.2	24.3	26.3	28.3	31.3		
	40	8.2	9.1	9.7	13.4	14.5	15.6	19.5	21.2	22.8	26.2	28.3	30.4	28.9		
55	-20	6.1	6.6	7.1	9.8	10.6	11.5	14.1	15.3	16.5	19.1	20.5	22.1	24.5	26.4	
	0	6.6	7.2	7.7	10.7	11.6	12.5	15.4	16.7	18.0	20.7	22.4	24.1	26.7	28.8	
	20	7.2	7.7	8.3	11.6	12.5	13.5	16.6	18.1	19.5	22.4	24.3	26.1	28.9	31.2	
	40	7.7	8.3	8.9	12.5	13.5	14.5	17.9	19.5	21.0	24.1	26.2	28.2	31.1	33.6	
50	-20	5.7	6.2	6.7	9.1	9.8	10.6	13.2	14.3	15.3	17.6	19.0	20.4	22.6	24.5	26.4
	0	6.2	6.7	7.3	9.9	10.7	11.6	14.4	15.6	16.7	19.2	20.7	22.2	24.7	26.7	28.8
	20	6.7	7.3	7.9	10.6	11.6	12.5	15.5	16.8	18.2	20.7	22.4	24.1	26.7	28.9	31.2
	40	7.2	7.8	8.4	11.5	12.5	13.5	16.7	18.1	19.6	22.3	24.1	26.0	28.8	31.1	33.5
45	-20	5.5	6.0	6.3	8.5	9.2	9.9	12.2	13.1	14.1	16.0	17.4	18.8	20.5	22.2	24.0
	0	5.9	6.4	6.9	9.3	10.0	10.8	13.2	14.3	15.3	17.4	18.9	20.4	22.4	24.3	26.2
	20	6.4	6.9	7.4	10.1	10.9	11.7	14.3	15.4	16.6	18.9	20.5	22.1	24.3	26.3	28.3
	40	6.8	7.4	7.9	10.8	11.7	12.6	15.3	16.5	17.8	20.4	22.0	23.8	26.2	28.3	30.5
40	-20	5.1	5.5	5.9	7.9	8.5	9.1	11.0	12.0	12.9	14.6	15.8	17.2	18.8	20.2	21.7
	0	5.5	5.9	6.4	8.6	9.3	10.0	12.0	13.0	14.0	15.9	17.3	18.7	20.5	22.1	23.8
	20	5.9	6.4	6.9	9.2	10.0	10.8	13.0	14.1	15.2	17.3	18.8	20.2	22.1	24.0	25.8
	40	6.3	6.8	7.4	9.9	10.8	11.6	14.0	15.1	16.3	18.6	20.2	21.7	23.8	25.8	27.8

*** To correct for wind, enter table with the brakes on speed minus one half the headwind or plus 1.5 times the tailwind.**
If ground speed is used for brakes on speed, ignore wind and enter table with sea level, 15°C.

Adjusted Brake Energy Per Brake (Millions of Foot Pounds)
No Reverse Thrust

		REFERENCE BRAKE ENERGY PER BRAKE (MILLIONS OF FOOT POUNDS)						
EVENT		5	10	15	20	25	30	35
RTO MAX MAN		5	10	15	20	25	30	35
LANDING	MAX MAN	3.4	7.4	11.9	16.5	21.1	25.8	30.5
	MAX AUTO	3.2	7.0	11.4	15.5	19.7	24.2	28.5
	AUTOBRAKE 3	3.1	6.7	10.8	14.6	18.2	22.2	25.9
	AUTOBRAKE 2	2.9	6.3	10.3	13.8	17.0	20.1	23.2
	AUTOBRAKE 1	2.9	6.2	10.0	13.4	16.5	19.7	22.6

Two Engine Detent Reverse Thrust

		REFERENCE BRAKE ENERGY PER BRAKE (MILLIONS OF FOOT POUNDS)						
EVENT		5	10	15	20	25	30	35
LANDING	MAX AUTO	2.6	5.6	8.9	12.4	16.0	19.4	22.8
	AUTOBRAKE 3	1.9	3.9	6.4	8.4	10.5	12.6	14.7
	AUTOBRAKE 1 & 2	1.9	3.7	5.8	7.8	9.6	11.3	12.8

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Brake Cooling Schedule

Cooling Time (Minutes) - Category A Brakes

	ADJUSTED BRAKE ENERGY PER BRAKE (MILLIONS OF FOOT POUNDS)							
	8 & BELOW	10	12	14	16	18	18.5 TO 23.7	23.8 & ABOVE
INFLIGHT GEAR DOWN	NO SPECIAL PROCEDURE REQUIRED	2	3	4	5	6	CAUTION	FUSE PLUG MELT ZONE
GROUND		19	32	43	53	61		

Observe maximum quick turnaround limit.

Table shows energy per brake added by a single stop with all brakes operating. Energy is assumed to be equally distributed among the operating brakes. Total energy is the sum of residual energy plus energy added.

Add 1.0 million foot pounds for each taxi mile.

When in caution zone, wheel fuse plugs may melt. Delay takeoff and inspect after 30 minutes. If overheat occurs after takeoff, extend gear soon for at least 7 minutes.

When in fuse plug melt zone, clear runway immediately. Unless required, do not set parking brake. Do not approach gear or attempt to taxi for one hour. Tire, wheel and brake replacement may be required. If overheat occurs after takeoff, extend gear soon for at least 12 minutes.

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ENGINE INOP

Initial Max Continuous %N1

Based on .74M, engine bleed for packs auto and anti-ice off

TAT (°C)	PRESSURE ALTITUDE (1000 FT)					
	27	29	31	33	35	37
20	95.2	95.2	95.2	95.2	95.1	95.1
15	95.8	95.8	95.8	958	95.7	95.7
10	96.1	96.1	96.1	96.1	96.0	96.0
5	96.6	96.6	96.6	96.6	96.5	96.5
0	97.1	97.1	97.1	97.1	97.0	97.0
-5	96.6	97.7	97.7	97.7	97.6	97.6
-10	95.7	96.9	98.1	98.7	98.0	98.0
-15	94.8	96.0	97.1	98.4	98.3	98.3
-20	93.9	95.1	96.2	97.4	98.4	98.7
-25	92.9	94.1	95.3	96.5	97.5	98.0
-30	92.0	93.2	94.3	95.5	96.5	97.0
-40	90.1	91.3	98.3	93.5	94.5	95.0
-50	88.1	89.3	90.3	91.5	92.4	93.0

%N1 Adjustment for Engine Bleeds

BLEED CONFIGURATION	PRESSURE ALTITUDE (1000 FT)						
	25	27	29	31	33	35	37
ENGINE ANTI-ICE ON	-0.9	-0.9	-0.9	-0.9	-0.9	-0.9	-1.0
ENGINE AND WING ANTI-ICE ON	-2.9	-2.9	-2.9	-2.8	-2.8	-2.8	-2.9

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ENGINE INOP

Max Continuous %N1

33000 FT to 23000 FT Pressure Altitudes

Based on engine bleed for packs auto and anti-ice off

33000 FT PRESS ALT.		TAT (°C)										
KIAS	M	-50	-40	-30	-20	-10	0	5	10	15	20	25
200	.58	92.3	94.4	96.4	97.1	96.5	96.0	95.7				
220	.63	92.2	94.3	96.3	97.8	97.1	96.4	96.1	95.7	95.3		
240	.68	92.1	94.2	96.2	98.1	97.8	96.9	96.4	96.0			
260	.74	91.4	93.5	95.5	97.4	98.0	97.0	96.6	96.1	95.6		
31000 FT PRESS ALT.		TAT (°C)										
KIAS	M	-50	-40	-30	-20	-10	0	5	10	15	20	25
200	.55	91.5	93.5	95.5	97.0	96.4	95.8	95.5				
220	.61	91.3	93.3	95.3	97.2	96.8	96.2	95.9	95.6	95.3		
240	.66	91.1	93.1	95.1	97.1	97.4	96.7	96.3	95.9	95.5		
260	.71	90.9	92.9	94.9	96.8	98.0	97.0	96.5	96.1	95.6		
29000 FT PRESS ALT.		TAT (°C)										
KIAS	M	-50	-40	-30	-20	-10	0	5	10	15	20	25
200	.53	90.6	92.6	94.6	96.5	96.2	95.6	95.4	95.1			
220	.58	90.4	92.4	94.4	96.3	96.6	96.1	95.8	95.5	95.1		
240	.63	90.2	92.2	94.1	96.1	97.1	96.5	96.1	95.8	95.4		
260	.68	90.0	92.0	93.9	95.9	97.7	96.9	96.4	96.0	95.6	95.1	
27000 FT PRESS ALT.		TAT (°C)										
KIAS	M	-50	-40	-30	-20	-10	0	5	10	15	20	25
200	.51	89.8	91.8	93.8	95.7	96.0	95.5	95.2	94.9			
220	.56	89.6	91.6	93.5	95.4	96.4	95.9	95.6	95.3	95.0	94.6	
240	.60	89.3	91.3	93.2	95.1	96.9	96.3	96.0	97.7	95.3	95.0	
260	.65	89.1	91.0	93.0	94.9	96.7	96.7	96.3	95.9	95.5	95.1	
25000 FT PRESS ALT.		TAT (°C)										
KIAS	M	-50	-40	-30	-20	-10	0	5	10	15	20	25
200	.49	89.0	91.0	92.9	94.8	95.9	95.4	95.1	94.8	94.5		
220	.53	88.8	90.7	92.7	94.6	96.3	95.7	95.4	95.1	94.8	94.5	
240	.58	88.5	90.5	92.4	94.3	96.1	96.1	95.8	95.5	95.2	94.8	
260	.63	88.2	90.2	92.1	94.0	95.8	96.5	96.1	95.8	95.4	95.0	94.6
23000 FT PRESS ALT.		TAT (°C)										
KIAS	M	-50	-40	-30	-20	-10	0	5	10	15	20	25
200	.47	88.2	90.1	92.0	93.9	95.7	95.3	95.0	94.4	94.0		
220	.51	88.0	89.9	91.8	93.7	95.5	95.6	95.3	95.0	94.7	94.4	
240	.56	87.7	89.7	91.6	93.4	95.3	95.9	95.7	95.4	95.0	94.7	94.3
260	.60	87.5	89.4	91.3	93.2	95.0	96.3	96.0	95.7	95.4	95.0	94.6

Anti-Ice Adjustments

BLEED CONFIGURATION	PRESS ALT (1000 FT)					
	23	25	27	29	31	33
ENGINE ANTI-ICE ON	-0.9	-0.9	-0.9	-0.9	-0.9	-0.9
ENGINE AND WING ANTI-ICE ON	-2.9	-2.9	-2.9	-2.9	-2.8	-2.8

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ENGINE INOP

Max Continuous %N1

21000 FT to 14000 FT Pressure Altitudes

Based on engine bleed for packs auto and anti-ice off

21000 FT PRESS ALT.		TAT (°C)											
KIAS	M	-50	-40	-30	-20	-10	0	5	10	15	20	25	30
200	.45	87.3	89.3	91.2	93.0	94.8	95.1	94.9	94.6	94.3	93.9		
220	.49	87.2	89.1	91.0	92.9	94.7	95.4	95.1	94.9	94.6	94.2	93.9	93.5
240	.54	86.9	88.9	90.8	92.6	94.4	95.8	95.5	95.2	94.9	94.5	94.2	
260	.60	86.7	88.6	90.5	92.3	94.1	95.9	95.8	95.5	95.2	94.8	94.5	94.0
20000 FT PRESS ALT.		TAT (°C)											
KIAS	M	-50	-40	-30	-20	-10	0	5	10	15	20	25	30
200	.44	86.9	88.9	90.7	92.6	94.4	95.1	94.8	94.5	94.2	93.9	93.5	
220	.48	86.8	88.7	90.6	92.4	94.2	95.4	95.1	94.8	94.5	94.2	93.8	93.4
240	.53	86.6	88.5	90.4	92.2	94.0	95.7	95.4	95.1	94.8	94.5	94.1	93.7
260	.57	86.3	88.2	90.1	91.9	93.7	95.5	95.8	95.5	95.1	94.8	94.4	94.0
18000 FT PRESS ALT.		TAT (°C)											
KIAS	M	-50	-40	-30	-20	-10	0	5	10	15	20	25	30
200	.42	86.1	88.0	89.9	91.7	93.5	95.0	94.7	94.4	94.1	93.8	93.4	
220	.46	86.0	87.9	89.7	91.6	93.4	95.1	95.0	94.7	94.4	94.1	93.7	93.3
240	.51	85.8	87.7	89.6	91.4	93.2	94.9	95.3	95.0	94.7	94.3	94.0	93.6
260	.55	85.6	87.4	89.3	91.1	92.9	94.7	95.5	95.3	95.0	94.6	94.3	93.8
17000 FT PRESS ALT.		TAT (°C)											
KIAS	M	-50	-40	-30	-20	-10	0	5	10	15	20	25	30
200	.42	85.7	87.6	89.5	91.3	93.1	94.9	94.7	94.4	94.1	93.7	93.3	92.9
220	.46	85.6	87.5	89.3	91.1	92.9	94.7	94.9	94.6	94.3	94.0	93.6	93.2
240	.50	85.4	87.3	89.2	91.0	92.8	94.5	95.2	94.9	94.6	94.3	93.9	93.5
260	.54	85.2	87.1	88.9	90.7	92.5	94.2	95.1	95.2	94.9	94.6	94.2	93.8
16000 FT PRESS ALT.		TAT (°C)											
KIAS	M	-50	-40	-30	-20	-10	0	5	10	15	20	25	30
200	.41	85.4	87.2	89.1	90.9	92.7	94.4	94.6	94.3	94.0	93.6	93.3	92.8
220	.45	85.2	87.1	88.9	90.7	92.5	94.3	94.9	94.6	94.3	93.9	93.6	93.1
240	.49	85.0	86.9	88.8	90.6	92.3	94.1	94.9	94.8	94.5	94.2	93.8	93.5
260	.53	84.8	86.7	88.5	90.3	92.1	93.8	94.7	95.1	94.8	94.5	94.1	93.7
14000 FT PRESS ALT.		TAT (°C)											
KIAS	M	-50	-40	-30	-20	-10	0	5	10	15	20	25	30
200	.39	84.6	86.5	88.3	90.1	91.9	93.6	94.4	94.3	93.9	93.6	93.2	92.7
220	.43	84.4	86.3	88.1	89.9	91.7	93.4	94.2	94.5	94.2	93.8	93.4	93.0
240	.47	84.3	86.1	87.9	89.7	91.5	93.2	94.1	94.7	94.4	94.1	93.7	93.3
260	.51	84.1	85.9	87.8	89.6	91.3	93.0	93.9	94.7	94.7	94.4	94.0	93.6

Anti-Ice Adjustments

BLEED CONFIGURATION	PRESS ALT (1000 FT)					
	14	16	17	18	20	21
ENGINE ANTI-ICE ON	-0.9	-0.9	-0.9	-0.9	-0.9	-0.9
ENGINE AND WING ANTI-ICE ON	-3.1	-3.1	-3.1	-3.0	-3.0	-3.0

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ENGINE INOP

Max Continuous %N1

12000 FT to 1000 FT Pressure Altitudes

Based on engine bleed for packs auto and anti-ice off

12000 FT PRESS ALT.								TAT (°C)					
KIAS	M	-50	-40	-30	-20	-10	0	5	10	15	20	25	30
200	.38	83.8	85.7	87.5	89.3	91.0	92.7	93.6	94.2	93.9	93.5	93.1	92.7
220	.41	83.6	85.5	87.3	89.1	90.8	92.5	93.4	94.2	94.1	93.7	93.3	92.9
240	.45	83.5	85.3	87.1	88.9	90.7	92.4	93.2	94.0	94.3	94.0	93.6	93.2
260	.49	83.3	85.2	87.0	88.8	90.5	92.2	93.0	93.9	94.6	94.2	93.9	93.5
10000 FT PRESS ALT.								TAT (°C)					
KIAS	M	-50	-40	-30	-20	-10	0	5	10	15	20	25	30
200	.36	83.1	84.9	86.7	88.5	90.2	91.9	92.7	93.6	93.8	93.5	93.1	92.6
220	.40	82.9	84.7	86.5	88.3	90.0	91.7	92.5	93.3	94.0	93.6	93.2	92.8
240	.43	82.7	84.6	86.4	88.1	89.8	91.5	92.4	93.2	94.0	93.9	93.5	93.1
260	.47	82.6	84.4	86.2	88.0	89.7	91.4	92.2	93.0	93.8	94.1	93.8	93.4
7000 FT PRESS ALT.								TAT (°C)					
KIAS	M	-50	-40	-30	-20	-10	0	5	10	15	20	25	30
200	.34	81.9	83.7	85.5	87.2	88.9	90.6	91.4	92.3	93.1	93.4	93.0	92.6
220	.38	81.7	83.5	85.3	87.1	88.8	90.4	91.3	92.1	92.9	93.5	93.2	92.7
240	.41	81.6	83.4	85.2	86.9	88.6	90.3	91.1	91.9	92.7	93.5	93.3	92.9
260	.45	81.4	83.3	85.0	86.7	88.4	90.1	90.9	91.7	92.6	93.4	93.6	93.2
5000 FT PRESS ALT.								TAT (°C)					
KIAS	M	-50	-40	-30	-20	-10	0	5	10	15	20	25	30
200	.33	81.1	82.9	84.7	86.4	88.1	89.8	90.6	91.4	92.2	93.0	93.0	92.5
220	.36	81.0	82.8	84.5	86.2	87.9	89.6	90.4	91.2	92.0	92.8	93.1	92.7
240	.40	80.8	82.6	84.4	86.1	87.8	89.4	90.2	91.0	91.8	92.6	93.3	92.8
260	.43	80.7	82.5	84.2	85.9	87.6	89.3	90.1	90.9	91.7	92.5	93.3	93.1
3000 FT PRESS ALT.								TAT (°C)					
KIAS	M	-50	-40	-30	-20	-10	0	5	10	15	20	25	30
200	.32	80.4	82.2	83.9	85.6	87.3	88.9	89.7	90.5	91.3	92.1	92.9	92.5
220	.35	80.2	82.0	83.7	85.4	87.1	88.8	89.6	90.4	91.2	91.9	92.7	92.6
240	.40	80.1	81.8	83.6	85.3	86.9	88.6	89.4	90.2	91.0	91.8	92.5	92.8
260	.43	79.9	81.7	83.4	85.1	86.8	88.4	89.2	90.0	90.8	91.6	92.4	93.0
1000 FT PRESS ALT.								TAT (°C)					
KIAS	M	-50	-40	-30	-20	-10	0	5	10	15	20	25	30
200	.31	79.6	81.4	83.1	84.8	86.4	88.1	88.9	89.7	90.5	91.2	92.0	92.5
220	.34	79.5	81.2	82.9	84.6	86.3	87.9	88.7	89.5	90.3	91.1	91.8	92.6
240	.37	79.3	81.1	82.8	84.5	86.1	87.7	88.5	89.3	90.1	90.9	91.7	92.4
260	.40	79.2	80.9	82.6	84.3	86.0	87.6	88.4	89.2	90.0	90.7	91.5	92.3

Anti-Ice Adjustments

BLEED CONFIGURATION	PRESS ALT (1000 FT)					
	1	3	5	7	10	12
ENGINE ANTI-ICE ON	-0.6	-0.7	-0.8	-0.8	-0.8	-0.8
ENGINE AND WING ANTI-ICE ON	-2.4	-2.6	-2.8	-2.8	-2.9	-2.9

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ENGINE INOP

MAX CONTINUOUS THRUST

Driftdown Speed/Level Off Altitude
100 ft/min residual rate of climb

WEIGHT (1000 KG)		OPTIMUM DRIFTDOWN SPEED (KIAS)	LEVEL OFF ALTITUDE (FT)		
START DRIFTDOWN	LEVEL OFF		ISA + 10°C & BELOW	ISA + 15°C	ISA + 20 °C
64	61	235	16200	15000	13600
60	57	228	18200	17200	15900
56	53	220	20400	19300	18200
52	49	212	22700	21700	20600
48	46	204	25100	24100	23100
44	42	196	27400	26600	25700
40	38	187	29900	29200	28400

Includes APU fuel burn.

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ENGINE INOP

MAX CONTINUOUS THRUST

Driftdown/LRC Cruise Range Capability

Ground to Air Miles Conversion

AIR DISTANCE (NM)					GROUND DISTANCE (NM)	AIR DISTANCE (NM)				
HEADWIND COMPONENT (KTS)						TAILWIND COMPONENT (KTS)				
100	80	60	40	20		20	40	60	80	100
140	130	121	113	106	100	95	90	85	81	78
288	265	245	228	213	200	188	178	169	161	153
433	398	368	342	320	300	283	267	253	241	229
576	530	490	456	426	400	377	356	338	321	306
718	660	611	569	532	500	471	446	423	402	384
858	790	732	682	638	600	566	536	508	484	461
997	919	852	795	744	700	661	626	594	565	539
1134	1047	972	907	850	800	755	716	680	647	618
1272	1175	1091	1019	956	900	850	806	766	729	696
1408	1302	1211	1131	1062	1000	945	896	852	812	775
1545	1429	1330	1243	1167	1100	1040	986	938	894	854
1682	1557	1449	1355	1273	1200	1135	1077	1024	976	933
1819	1684	1568	1467	1379	1300	1230	1167	1110	1059	1012
1956	1812	1688	1580	1484	1400	1325	1257	1196	1141	1090
2095	1941	1808	1692	1590	1500	1419	1347	1282	1222	1168
2234	2070	1929	1805	1696	1600	1514	1437	1367	1304	1246

Driftdown/Cruise Fuel and Time

AIR DIST (NM)	FUEL REQUIRED (1000 KG)								TIME (HRS:MIN)
	WEIGHT AT START OF DRIFTDOWN (1000 KG)								
	35	40	45	50	55	60	65	70	
100	0.4	0.4	0.4	0.5	0.5	0.5	0.6	0.6	0:17
200	0.8	0.9	1.0	1.1	1.1	1.2	1.3	1.4	0:37
300	1.3	1.4	1.6	1.7	1.9	2.0	2.2	2.3	0:55
400	1.7	1.9	2.1	2.4	2.6	2.8	3.1	3.2	1:13
500	2.1	2.4	2.6	3.0	3.2	3.5	3.8	4.1	1:31
600	2.5	2.8	3.2	3.5	3.9	4.2	4.6	4.9	1:48
700	2.9	3.3	3.7	4.1	4.5	4.9	5.4	5.8	2:05
800	3.3	3.8	4.2	4.7	5.2	5.6	6.1	6.6	2:22
900	3.7	4.2	4.7	5.3	5.8	6.3	6.9	7.4	2:38
1000	4.1	4.7	5.2	5.8	6.4	7.0	7.6	8.2	2:54
1100	4.5	5.1	5.7	6.4	7.0	7.7	8.3	9.0	3:10
1200	4.9	5.5	6.2	6.9	7.6	8.3	9.1	9.8	3:26
1300	5.3	6.0	6.7	7.5	8.2	9.0	9.8	10.6	3:42
1400	5.7	6.4	7.2	8.0	8.8	9.6	10.5	11.4	3:59
1500	6.1	6.9	7.7	8.6	9.4	10.3	11.2	12.2	4:16
1600	6.5	7.3	8.2	9.1	10.0	11.0	11.9	12.9	4:33

Includes APU fuel burn.
Driftdown at optimum driftdown speed and cruise at Long Range Cruise speed.

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ENGINE INOP

MAX CONTINUOUS THRUST

Long Range Cruise Altitude Capability
100 ft/min residual rate of climb

WEIGHT (1000 KG)	PRESSURE ALTITUDE (FT)		
	ISA + 10°C & BELOW	ISA + 15°C	ISA + 20°C
64	9500	7200	4900
60	12300	10200	7900
56	15200	13200	11000
52	18000	16200	14100
48	20900	19200	17400
44	24000	22300	20600
40	27000	25600	24000
36	30000	28800	27500
32	33200	32200	31000

With engine anti-ice on, decrease altitude capability by 1400ft.
With engine and wing anti-ice on, decrease altitude capability by 5300ft.

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ENGINE INOP

MAX CONTINUOUS THRUST

Long Range Cruise Control

WEIGHT (1000 KG)		PRESSURE ALTITUDE (1000 FT)									
		10	12	14	16	18	20	22	24	26	28
64	%N1	90.9	92.4								
	MACH	.561	.577								
	KIAS	311	309								
	FF/ENG	2974	2952								
60	%N1	89.2	90.7	92.2							
	MACH	.547	.564	.580							
	KIAS	303	302	299							
	FF/ENG	2796	2780	2755							
56	%N1	87.5	88.9	90.4	91.9						
	MACH	.531	.549	.566	.582						
	KIAS	294	293	292	289						
	FF/ENG	2613	2603	2584	2553						
52	%N1	85.7	87.0	88.5	90.0	91.6	93.2				
	MACH	.513	.532	.550	.567	.584	.600				
	KIAS	284	284	283	281	278	275				
	FF/ENG	2428	2419	2409	2383	2353	2333				
48	%N1	83.6	85.1	86.5	88.0	89.5	91.1	92.8			
	MACH	.495	.513	.532	.550	.568	.584	.600			
	KIAS	274	274	273	272	271	268	265			
	FF/ENG	2243	2233	2225	2210	2185	2161	2153			
44	%N1	81.5	82.9	84.4	85.8	87.3	88.8	90.4	92.1		
	MACH	.475	.493	.511	.530	.549	.567	.584	.600		
	KIAS	263	263	263	262	261	260	257	254		
	FF/ENG	2059	2049	2040	2030	2014	1994	1981	1970		
40	%N1	79.0	80.5	82.0	83.5	84.9	86.4	88.0	89.6	91.3	93.3
	MACH	.454	.471	.489	.508	.527	.546	.565	.582	.599	.616
	KIAS	251	251	251	251	250	250	248	246	243	239
	FF/ENG	1878	1866	1857	1849	1838	1825	1816	1799	1781	1777
36	%N1	76.4	77.8	79.4	80.9	82.4	83.8	85.3	86.9	88.5	90.2
	MACH	.434	.449	.465	.484	.502	.522	.541	.561	.578	.595
	KIAS	240	239	239	239	238	238	238	236	234	231
	FF/ENG	1707	1689	1677	1668	1660	1650	1646	1635	1613	1594

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ENGINE INOP

MAX CONTINUOUS THRUST

Long Range Cruise Diversion Fuel and Time

Ground to Air Miles Conversion

AIR DISTANCE (NM)					GROUND DISTANCE (NM)	AIR DISTANCE (NM)				
HEADWIND COMPONENT (KTS)						TAILWIND COMPONENT (KTS)				
100	80	60	40	20		20	40	60	80	100
299	272	249	230	214	200	190	181	173	166	159
608	552	504	464	430	400	380	362	345	330	316
920	834	759	697	646	600	570	542	517	494	473
1235	1118	1015	932	862	800	760	722	688	658	630
1553	1403	1273	1167	1078	1000	949	903	860	821	787
1873	1691	1532	1403	1295	1200	1139	1082	1031	984	943
2196	1980	1792	1640	1512	1400	1328	1262	1202	1147	1099
2524	2273	2055	1877	1730	1600	1517	1441	1372	1310	1255

Reference Fuel and Time Required at Check Point

AIR DIST (NM)	PRESSURE ALTITUDE (1000 FT)									
	10		14		18		22		26	
	FUEL (1000 KG)	TIME (HR:MIN)	FUEL (1000 KG)	TIME (HR:MIN)	FUEL (1000 KG)	TIME (HR:MIN)	FUEL (1000 KG)	TIME (HR:MIN)	FUEL (1000 KG)	TIME (HR:MIN)
200	1.4	0:42	1.2	0:40	1.1	0:39	1.0	0:38	0.9	0:37
400	2.8	1:22	2.6	1:18	2.3	1:14	2.1	1:11	2.0	1:09
600	4.2	2:03	3.9	1:56	3.5	1:50	3.3	1:46	3.1	1:42
800	5.6	2:44	5.2	2:35	4.7	2:26	4.4	2:20	4.1	2:15
1000	7.0	3:26	6.4	3:14	5.9	3:03	5.5	2:55	5.2	2:48
1200	8.3	4:08	7.7	3:54	7.1	3:40	6.6	3:30	6.2	3:22
1400	9.6	4:51	8.9	4:34	8.2	4:18	7.7	4:05	7.2	3:56
1600	11.0	5:35	10.1	5:15	9.4	4:57	8.7	4:41	8.2	4:30

Fuel Required Adjustment (1000 KG)

REFERENCE FUEL REQUIRED (1000 KG)	WEIGHT AT CHECK POINT (1000 KG)					
	35	40	45	50	55	60
2	-0.2	-0.1	0.0	0.1	0.3	0.5
4	-0.3	-0.2	0.0	0.3	0.7	1.0
6	-0.5	-0.3	0.0	0.5	1.0	1.5
8	-0.7	-0.4	0.0	0.6	1.3	2.0
10	-0.9	-0.4	0.0	0.7	1.6	2.5
12	-1.0	-0.5	0.0	0.9	1.8	2.9
14	-1.2	-0.6	0.0	1.0	2.1	3.3

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ENGINE INOP

MAX CONTINUOUS THRUST

Holding
Flaps Up

WEIGHT (1000 KG)		PRESSURE ALTITUDE (FT)					
		1500	5000	10000	15000	20000	25000
62	%N1	78.9	81.8	85.5	89.8		
	KIAS	242	243	243	245		
	FF/ENG	2530	2510	2500	2520		
58	%N1	77.0	79.9	83.8	87.8		
	KIAS	234	235	236	236		
	FF/ENG	2360	2340	2330	2330		
54	%N1	75.1	77.8	81.8	85.8	90.5	
	KIAS	225	227	228	228	230	
	FF/ENG	2200	2180	2160	2160	2190	
50	%N1	72.9	75.7	79.8	83.7	88.0	
	KIAS	216	218	219	219	221	
	FF/ENG	2050	2020	2000	1990	2000	
46	%N1	70.8	73.5	77.5	81.5	85.6	90.8
	KIAS	210	210	210	211	211	213
	FF/ENG	1900	1870	1840	1830	1830	1860
42	%N1	68.9	71.6	75.6	79.6	83.5	88.0
	KIAS	210	210	210	210	210	210
	FF/ENG	1790	1760	1720	1700	1690	1700
38	%N1	67.1	69.8	73.8	77.8	81.7	85.8
	KIAS	210	210	210	210	210	210
	FF/ENG	1690	1660	1620	1590	1580	1570
34	%N1	65.4	68.1	72.1	76.0	79.9	83.9
	KIAS	210	210	210	210	210	210
	FF/ENG	1600	1570	1520	1500	1480	1470

This table includes 5% additional fuel for holding in a racetrack pattern.

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Takeoff %N1

Based on engine bleed to packs on (Auto) and anti-ice on or off

AIRPORT OAT		AIRPORT PRESSURE ALTITUDE (FT)									
°C	°F	-1000	-0	1000	2000	3000	4000	5000	6000	7000	8000
55	131	94.7	94.9	94.9	94.9						
50	122	95.1	95.4	95.4	95.4	95.4	95.4				
45	113	95.6	95.9	95.9	95.9	95.9	95.9	95.9	95.9	95.9	
40	104	96.0	96.3	96.3	96.3	96.3	96.3	96.3	96.4	96.4	96.4
35	95	96.3	96.8	96.8	96.8	96.8	96.8	96.8	96.8	96.8	96.8
30	86	95.5	96.6	96.8	97.2	97.2	97.2	97.2	97.2	97.1	97.1
25	77	94.8	95.8	96.0	96.5	97.0	97.4	97.6	97.5	97.5	97.4
20	68	94.0	95.0	95.2	95.7	96.2	96.5	96.9	97.1	97.1	97.1
15	59	93.2	94.1	94.4	94.8	95.4	95.7	96.3	96.3	96.3	96.3
10	50	92.4	93.3	93.5	94.0	94.6	94.9	95.2	95.5	95.6	95.7
5	41	91.5	92.5	92.7	93.2	93.7	94.0	94.4	94.7	94.8	94.9
0	32	90.7	91.7	91.9	92.3	92.9	93.2	93.5	93.8	93.9	94.0
-10	14	89.0	90.0	90.2	90.6	91.2	91.5	91.8	92.1	92.2	92.3
-20	-4	87.3	88.2	88.4	88.9	89.4	89.7	90.0	90.3	90.4	90.5
-30	-22	85.6	86.5	86.7	87.1	87.6	87.9	88.2	88.5	88.6	88.7
-40	-40	83.8	84.7	84.9	85.3	85.8	86.1	86.4	86.7	86.8	86.9
-50	-58	82.0	82.8	83.0	83.5	83.9	84.2	84.5	84.8	84.9	85.0

Takeoff %N1

Based on engine bleed to packs off and anti-ice on or off

AIRPORT OAT		AIRPORT PRESSURE ALTITUDE (FT)									
°C	°F	-1000	0	1000	2000	3000	4000	5000	6000	7000	8000
55	131	95.7	95.9	95.9	95.9						
50	122	96.1	96.4	96.4	96.4	96.4	96.4				
45	113	96.6	96.8	96.8	96.8	96.8	96.8	96.8	96.8	96.9	
40	104	97.0	97.3	97.3	97.3	97.3	97.3	97.3	97.3	97.3	97.3
35	95	97.3	97.7	97.7	97.7	97.7	97.7	97.7	97.7	97.7	97.7
30	86	96.5	97.5	97.7	98.1	98.1	98.1	98.1	98.1	98.1	98.0
25	77	95.7	96.7	96.9	97.4	97.9	97.9	97.9	97.9	97.9	97.9
20	68	94.9	95.9	96.1	96.6	97.1	97.1	97.1	97.1	97.1	97.1
15	59	94.1	95.0	95.3	95.7	96.3	96.3	96.3	96.3	96.3	96.3
10	50	93.3	94.2	94.4	94.9	95.6	95.8	96.1	96.3	96.3	96.3
5	41	92.4	93.4	93.6	94.1	94.6	94.9	95.2	95.4	95.4	95.4
0	32	91.6	92.5	92.7	93.2	93.8	94.1	94.4	94.5	94.5	94.5
-10	14	89.8	90.8	91.0	91.5	92.0	92.3	92.6	92.8	92.8	92.8
-20	-4	88.1	89.1	89.3	89.7	90.3	90.6	90.9	91.0	91.0	91.0
-30	-22	86.4	87.3	87.5	88.0	88.5	88.8	89.1	89.2	89.2	89.2
-40	-40	84.6	85.5	85.7	86.1	86.6	86.9	87.2	87.3	87.4	87.4
-50	-58	82.8	83.6	83.8	84.3	84.7	85.0	85.3	85.5	85.5	85.5

Takeoff Speeds Adjustment

SPEED (KIAS)	PRESSURE ALTITUDE (FT)			
	BELOW 5000		5000 & ABOVE	
	ABOVE 21°C (70°F)	21°C (70°F) & BELOW	ABOVE 21°C (70°F)	21°C (70°F) & BELOW
V1(MCG)	6	4	6	4
V1	0	0	0	0
VR	0	0	0	1

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Takeoff %N1 (20K Derate)

Based on engine bleed to packs on (Auto) and anti-ice on or off

AIRPORT OAT		AIRPORT PRESSURE ALTITUDE (FT)									
°C	°F	-1000	0	1000	2000	3000	4000	5000	6000	7000	8000
55	131	91.5	92.3	92.2	91.7						
50	122	92.0	92.8	93.4	93.7	93.0	93.0				
45	113	92.6	93.1	93.6	93.9	93.9	94.5	94.3	93.7		
40	104	93.1	93.6	94.0	94.2	94.3	95.1	95.0	94.9	94.8	94.5
35	95	93.2	94.1	94.6	94.8	94.8	95.5	95.7	95.6	95.8	95.7
30	86	93.0	94.5	94.3	94.5	95.0	95.4	95.6	95.8	95.8	95.8
25	77	92.2	93.7	93.7	93.7	94.4	95.5	95.3	95.1	95.4	95.8
20	68	91.5	92.9	92.9	92.9	93.6	95.5	95.7	95.8	95.8	95.7
15	59	90.7	92.1	92.1	92.1	92.8	94.9	95.1	95.1	95.1	95.1
10	50	89.9	91.3	91.3	91.3	92.0	94.1	94.3	94.3	94.3	94.3
5	41	89.1	90.5	90.5	90.5	91.2	93.2	93.4	93.4	93.4	93.4
0	32	88.3	89.7	89.7	89.7	90.4	92.4	92.6	92.6	92.6	92.6
-10	14	86.6	88.0	88.0	88.0	88.7	90.7	90.9	90.9	90.9	90.9
-20	-4	85.0	86.3	86.3	86.4	87.0	89.0	89.1	89.1	89.1	89.1
-30	-22	83.3	84.6	84.6	84.6	85.3	87.2	87.4	87.4	87.4	87.4
-40	-40	81.6	82.9	82.9	82.9	83.5	85.4	85.5	85.5	85.6	85.6
-50	-58	79.8	81.1	81.1	81.1	81.7	83.5	83.7	83.7	83.7	83.7

Takeoff %N1 (20K Derate)

Based on engine bleed to packs off and anti-ice on or off

AIRPORT OAT		AIRPORT PRESSURE ALTITUDE (FT)									
°C	°F	-1000	0	1000	2000	3000	4000	5000	6000	7000	8000
55	131	92.4	93.3	93.2	92.7						
50	122	93.0	93.8	94.4	94.6	94.0	94.0				
45	113	93.5	94.0	94.6	94.9	94.8	95.5	95.3	94.7		
40	104	94.0	94.5	94.9	95.1	95.2	96.1	95.9	95.8	95.8	95.4
35	95	94.2	95.0	95.6	95.7	95.7	96.5	96.6	96.6	96.7	96.6
30	86	93.9	95.4	95.2	95.4	96.0	96.3	96.5	96.7	96.7	96.8
25	77	93.1	94.6	94.6	94.6	95.3	96.4	96.2	96.0	96.3	96.7
20	68	92.4	93.8	93.8	93.8	94.6	95.9	95.9	95.9	95.9	95.9
15	59	91.6	93.0	93.0	93.0	93.7	95.1	95.1	95.1	95.1	95.1
10	50	90.8	92.2	92.2	92.2	92.9	94.3	94.3	94.3	94.3	94.3
5	41	90.0	91.4	91.4	91.4	92.1	93.4	93.4	93.4	93.4	93.4
0	32	89.2	90.6	90.6	90.6	91.3	92.6	92.6	92.6	92.6	92.6
-10	14	87.5	88.9	88.9	88.9	89.6	90.9	90.9	90.9	90.9	90.9
-20	-4	85.8	87.2	87.2	87.2	87.9	89.1	89.1	89.1	89.1	89.1
-30	-22	84.1	85.5	85.5	85.5	86.1	87.4	87.4	87.4	87.4	87.4
-40	-40	82.4	83.7	83.7	83.7	84.3	85.5	85.5	85.5	85.6	85.6
-50	-58	80.6	81.9	81.9	81.9	82.5	83.7	83.7	83.7	83.7	83.7

Takeoff Speeds Adjustment

SPEED (KIAS)	PRESSURE ALTITUDE (FT)	
	BELOW 5000	5000 & ABOVE
V1 (MCG)	6	4
V1	0	0
VR	0	0

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GEAR DOWN

220 KIAS Cruise Altitude Capability

Max Cruise Thrust, 100 ft/min residual rate of climb

WEIGHT (1000 KG)	PRESSURE ALTITUDE (FT)		
	ISA + 10°C & BELOW	ISA + 15°C	ISA + 20°C
64	22700	21400	20100
62	24100	23000	21800
60	25000	23800	22600
58	25700	24600	23500
56	26400	25400	24200
54	27100	26100	25000
52	27700	26800	25700
50	28300	27400	26300
48	28900	28000	26900
46	29500	28600	27500
44	30000	29100	28100
42	30500	29600	28700
40	31000	30100	29200
38	31400	30600	29700
36	31800	31000	30100
34	32200	31400	30500
32	32600	31800	30900

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GEAR DOWN

220 KIAS Cruise Control

WEIGHT (1000 KG)		PRESSURE ALTITUDE (1000 FT)									
		13	15	17	19	21	23	25	27	29	31
64	%N1	82.0	83.7	85.3	87.2	89.1	91.5				
	MACH	.441	.458	.476	.495	.515	.536				
	KIAS	230	230	230	230	230	230				
	FF/ENG	1887	1888	1892	1900	1915	1947				
60	%N1	79.9	81.6	83.2	84.9	86.8	88.9	91.4			
	MACH	.422	.438	.456	.474	.493	.513	.534			
	KIAS	220	220	220	220	220	220	220			
	FF/ENG	1739	1739	1741	1746	1753	1771	1807			
56	%N1	78.9	80.5	82.1	83.8	85.5	87.4	89.7	92.3		
	MACH	.422	.438	.456	.474	.493	.513	.534	.557		
	KIAS	220	220	220	220	220	220	220	220		
	FF/ENG	1672	1670	1670	1671	1675	1685	1707	1742		
52	%N1	77.9	79.5	81.1	82.7	84.4	86.2	88.3			
	MACH	.422	.438	.456	.474	.493	.513	.534	.557		
	KIAS	220	220	220	220	220	220	220	220		
	FF/ENG	1613	1609	1607	1606	1608	1614	1629	1651		
48	%N1	76.9	78.6	80.2	81.8	83.5	85.2	87.1	89.4	92.0	
	MACH	.422	.438	.456	.474	.493	.513	.534	.557	.580	
	KIAS	220	220	220	220	220	220	220	220	220	
	FF/ENG	1561	1556	1553	1551	1551	1555	1565	1578	1612	
44	%N1	76.0	77.7	79.3	80.9	82.6	84.3	86.1	88.2	90.6	
	MACH	.422	.438	.456	.474	.493	.513	.534	.557	.580	
	KIAS	220	220	220	220	220	220	220	220	220	
	FF/ENG	1513	1507	1503	1501	1500	1502	1509	1517	1542	
40	%N1	75.3	76.9	78.6	80.1	81.8	83.4	85.2	87.2	89.5	92.4
	MACH	.422	.438	.456	.474	.493	.513	.534	.557	.580	.605
	KIAS	220	220	220	220	220	220	220	220	220	220
	FF/ENG	1472	1466	1461	1458	1456	1456	1461	1466	1483	1522
36	%N1	74.6	76.2	77.9	79.5	81.1	82.7	84.4	86.4	88.6	91.2
	MACH	.422	.438	.456	.474	.493	.513	.534	.557	.580	.605
	KIAS	220	220	220	220	220	220	220	220	220	220
	FF/ENG	1437	1430	1425	1421	1419	1418	1422	1424	1436	1468

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GEAR DOWN

220 KIAS Enroute Fuel and Time

Ground to Air Miles Conversion

AIR DISTANCE (NM) HEADWIND COMPONENT (KTS)					GROUND DISTANCE (NM)	AIR DISTANCE (NM) TAILWIND COMPONENT (KTS)				
100	80	60	40	20		20	40	60	80	100
321	288	259	236	217	200	189	180	171	163	156
650	580	520	473	434	400	378	358	340	324	310
980	874	783	711	652	600	568	538	511	487	465
1309	1167	1045	949	869	800	756	717	680	648	619
1639	1461	1308	1186	1087	1000	945	895	850	810	774
1968	1754	1571	1424	1305	1200	1134	1074	1020	971	928
2298	2048	1833	1662	1522	1400	1323	1253	1189	1132	1082
2627	2341	2096	1900	1740	1600	1512	1432	1360	1295	1237
2956	2635	2358	2138	1958	1800	1701	1611	1529	1456	1391

Reference Fuel and Time Required at Check Point

AIR DIST (NM)	PRESSURE ALTITUDE (1000 FT)									
	10		14		20		24		28	
	FUEL (1000 KG)	TIME (HR:MIN)	FUEL (1000 KG)	TIME (HR:MIN)	FUEL (1000 KG)	TIME (HR:MIN)	FUEL (1000 KG)	TIME (HR:MIN)	FUEL (1000 KG)	TIME (HR:MIN)
200	2.3	0:49	2.1	0:46	1.8	0:43	1.7	0:41	1.5	0:39
400	4.6	1:36	4.3	1:31	3.8	1:24	3.6	1:19	3.3	1:15
600	7.0	2:23	6.4	2:15	5.8	2:04	5.4	1:57	5.1	1:50
800	9.3	3:10	8.6	3:00	7.7	2:44	7.2	2:35	6.8	2:26
1000	11.5	3:57	10.7	3:44	9.7	3:25	9.1	3:13	8.6	3:01
1200	13.8	4:45	12.8	4:28	11.6	4:05	10.9	3:51	10.3	3:37
1400	16.0	5:32	14.9	5:13	13.5	4:46	12.6	4:29	11.9	4:12
1600	18.2	6:19	16.9	5:57	15.3	5:26	14.4	5:06	13.6	4:48
1800	20.4	7:06	19.0	6:41	17.2	6:06	16.1	5:44	15.2	5:23

Fuel Required Adjustments (1000 KG)

REFERENCE FUEL REQUIRED (1000 KG)	WEIGHT AT CHECK POINT (1000 KG)						
	30	35	40	45	50	55	60
5	-0.3	-0.3	-0.1	0.0	0.2	0.5	0.9
10	-0.7	-0.5	-0.3	0.0	0.4	1.0	1.6
15	-0.9	-0.7	-0.4	0.0	0.6	1.3	2.2
20	-1.0	-0.8	-0.4	0.0	0.7	1.5	2.5
25	-1.1	-0.9	-0.5	0.0	0.7	1.6	2.6

Based on 220 KIAS cruise and descent.

Descent at 220 KIAS

PRESSURE ALT (1000 FT)	5	10	15	17	19	21	23	25	27	29	31	33
DISTANCE (NM)	17	26	35	39	43	46	50	54	57	61	65	69
TIME (MINUTES)	6	8	10	11	12	13	13	14	15	16	16	17

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GEAR DOWN

Holding
Flaps Up

WEIGHT (1000 KG)		PRESSURE ALTITUDE (FT)						
		1500	5000	10000	15000	20000	25000	30000
62	%N1	71.2	74.0	78.1	82.2	86.6	92.4	
	KIAS	220	220	220	220	220	220	
	FF/ENG	1930	1890	1870	1870	1880	1960	
58	%N1	70.2	72.9	77.0	81.0	85.2	90.5	
	KIAS	220	220	220	220	220	220	
	FF/ENG	1860	1830	1800	1790	1800	1840	
54	%N1	69.2	71.9	75.9	80.0	84.1	88.9	
	KIAS	220	220	220	220	220	220	
	FF/ENG	1800	1770	1730	1720	1720	1750	
50	%N1	67.0	69.6	73.6	77.7	81.8	86.2	92.5
	KIAS	210	210	210	210	210	210	210
	FF/ENG	1670	1630	1590	1570	1570	1580	1650
46	%N1	65.9	68.5	72.6	76.6	80.7	85.0	90.6
	KIAS	210	210	210	210	210	210	210
	FF/ENG	1620	1580	1540	1510	1510	1510	1550
42	%N1	65.0	67.6	71.6	75.6	79.7	83.9	89.2
	KIAS	210	210	210	210	210	210	210
	FF/ENG	1570	1530	1490	1460	1450	1450	1480
38	%N1	64.2	66.7	70.7	74.7	78.8	82.9	87.9
	KIAS	210	210	210	210	210	210	210
	FF/ENG	1530	1490	1440	1420	1400	1400	1420
34	%N1	63.5	65.9	69.9	73.9	78.0	82.1	86.9
	KIAS	210	210	210	210	210	210	210
	FF/ENG	1490	1450	1410	1380	1360	1360	1370

This table includes 5% additional fuel for holding in a racetrack pattern.

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Introduction

This chapter contains information to supplement performance data from the Flight Management Computer (FMC). In addition, sufficient inflight data is provided to complete a flight with the FMC inoperative. In the event of conflict between data presented in this chapter and that contained in the Approved Flight Manual, the Flight Manual shall always take precedence.

General

Takeoff Speeds

The speeds presented in the Takeoff Speeds table as well as FMC computed takeoff speeds can be used for all performance conditions provided adjustments are made to V1 for clearway, stopway, brake deactivation, improved climb, contaminated runway situations, brake energy limits or obstacle clearance with unbalanced V1. These speeds may be used for weights less than or equal to the performance limited weight.

Normal takeoff speeds, V1, VR, and V2, with anti-skid on, are read from the table by entering with takeoff flap setting, brake release weight, and appropriate column. The appropriate column is obtained by entering the Column Reference chart with the airport pressure altitude and the actual temperature. If an Altitude Adjustment chart is provided, adjust the takeoff speeds appropriately. Slope and wind adjustments to V1 are obtained by entering the Slope and Wind V1 Adjustment table.

If takeoffs are scheduled using these simplified speeds in conjunction with airport analyses that include clearway and/or stopway credits, adjustments to V1 speed are required.

Adjust V1 by the amount shown in the Clearway/Stopway table. The adjusted V1 must not exceed VR.

The maximum allowable clearway limits shown on the takeoff speeds page are provided for guidance when more precise data is unavailable.

V1(MCG)

Regulations prohibit scheduling takeoff with a V1 less than minimum V1 for control on the ground, V1(MCG). Therefore compare the adjusted V1 to the V1(MCG). To find V1(MCG) enter the V1(MCG) table with the airport pressure altitude and actual OAT. If applicable, add the adjustments shown below the table. If the adjusted V1 is less than

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V1(MCG), set V1 equal to V1(MCG). If VR is less than V1(MCG), set VR equal to V1(MCG), and determine a new V2 by adding the difference between the normal VR and V1(MCG) to the normal V2.

If it is necssary to increase V1 to V1(MCG), then provided the actual field length exceeds the minimum field length of 4500ft no takeoff weight adjustment is necessary.

Clearway and Stopway V1 Adjustments

Takeoff speed corrections are to be applied to V1 speed when using takeoff weights based on the use of clearway and stopway.

Adjust V1 speed by the amount shown in the appropriate column. The adjusted V1 speed must not exceed VR. If the adjusted V1 speed is greater than VR, reduce V1 to equal VR.

Maximum allowable clearway limits are provided for guidance when more precise data is not available.

Assumed Temperature Takeoff

For reduced thrust takeoffs based on assumed temperature, determine V1, VR, V2 from the appropriate column for airport pressure altitude and assumed temperature. Adjust V1 for slope, wind, clearway and stopway as required. Compare adjusted V1 with V1(MCG) from table using airport altitude and actual temperature. This is a regulatory requirement to cover the case where the pilot elects to advance thrust levers to maximum takeoff thrust following the engine failure. If adjusted V1 is less than V1(MCG) set V1 equal to V1(MCG).

Stab Trim

To find takeoff stabilizer trim setting, enter the Stab Trim Setting table with takeoff flap setting and center of gravity (C.G. % MAC) and read required stabilizer trim units. For weights that meet the conditions listed, apply the adjustments shown below the table.

VREF

The Reference Speed table contains flaps 40, 30, and 15 landing speeds for a given weight. Apply wind correction shown as required.

Flap Maneuver Speeds

This table provides the flap speed schedule for recommended maneuvering speed. The speed schedule is a function of weight and will provide adequate maneuver margin above stall at all weights.

During flap retraction/extension, movement of the flap to the next position should be initiated when reaching the maneuver speed for the existing flap.

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Slush/Standing Water Takeoff

Experience has shown that aircraft performance may deteriorate significantly on runways covered with snow, slush, standing water, or ice. Therefore a reduction in runway/obstacle limited takeoff weight and revised takeoff speeds is necessary. The following information is intended for guidance in accordance with advisory material and is based on an engine failure at the critical point during takeoff.

The entire runway is assumed to be completely covered by a contaminant of uniform thickness and density. Therefore this information is conservative when operating under typical colder weather conditions where patches of slush exist and some degree of sanding is common.

Takeoffs in slush depths greater than 13mm (0.5 inches) are not recommended because of possible airplane damage as a result of slush impingement on the airplane structure. The use of assumed temperature method for reduced thrust is not allowed on contaminated runways. Interpolation for slush/standing water depths between the values shown is permitted.

Instructions for Using Tables:

Takeoff weight is determined as follows:

1. Determine the dry field/obstacle limit weight for the takeoff flap setting.

2. Enter the Weight Adjustment table with the field/obstacle limit weight to obtain the weight reduction for the slush/standing water depth and airport pressure altitude.

3. Adjust field length available for temperature by amount shown on chart.

4. Enter V1(MCG) Limit Weight table with the adjusted field length and pressure altitude to obtain the slush/standing water limit weight with respect to minimum field length required for V1(MCG) speed.

The maximum allowable takeoff weight in slush/standing water is the lesser of the limit weights found in steps 2 and 4.

Takeoff speeds determination:

1. Determine takeoff speeds V1, VR and V2 for actual brake release weight using the Takeoff Speeds table in this section.

2. If V1(MCG) limited, set V1 = V1(MCG). If not limited by V1(MCG) considerations, enter V1 Adjustment table with actual brake release weight to determine the V1 reduction to apply to V1 speed. If the adjusted V1 is less than V1(MCG), set V1 = V1(MCG).

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Slippery Runway

Airplane braking action is reported as good, medium or poor, depending on existing runway conditions. If braking action is reported as good, conditions should not be expected to be as good as on clean dry runways. The value “good” is comparative and is intended to mean that airplanes should not experience braking or directional control difficulties when stopping. The performance level used to calculate the “good” data is consistent with wet runway testing done on early Boeing jets. The performance level used to calculate the “poor” data reflects a runway covered with wet ice. Performance is based on reversers operating and a 15 ft. screen height at the end of the runway. The tables provided are used in the same manner as the Slush/Standing Water tables.

Anti-skid Inoperative

When operating with anti-skid inoperative, the field limit weight and V1 must be reduced to account for the effect on accelerate-stop performance. A simplified method which conservatively accounts for the effects of anti-skid inoperative is to reduce the normal runway/obstacle limited weight by 7800 kg and the V1 associated with the reduced weight by the amount shown in the table below.

ANTI-SKID INOPERATIVE V1 ADJUSTMENT	
FIELD LENGTH (FT)	V1 ADJUSTMENT (KIAS)
6000	-27
8000	-22
10000	-18
12000	-15
14000	-12

If the resulting V1 is less than minimum V1, takeoff is permitted with V1 set equal to V1(MCG) provided the dry accelerate stop distance adjusted for wind and slope exceeds approximately 7900 ft.

Detailed analysis for the specific case from the Airplane Flight Manual may yield a less restrictive penalty.

Takeoff %N1

To find Max Takeoff %N1 based on normal engine bleed for air conditioning packs on (Auto), enter Takeoff %N1 Table with airport pressure altitude and airport OAT and read %N1. For packs off operation, apply the %N1 adjustment shown below the table. No takeoff %N1 adjustment is required for engine and wing anti-ice.

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Assumed Temperature Reduced Thrust

Regulations permit the use of up to 25% takeoff thrust reduction for operation with assumed temperature reduced thrust. To find the maximum allowable assumed temperature enter the top chart with airport pressure altitude and OAT. Compare this temperature to that at which the airplane is performance limited as determined from available takeoff performance data. Next, enter the center table with airport pressure altitude and the lower of the two temperatures previously determined to obtain a maximum takeoff %N1. Do not use an assumed temperature less than the minimum assumed temperature shown. Enter the %N1 Adjustment table with OAT and the difference between the assumed and actual OAT to obtain a %N1 adjustment. Subtract the %N1 adjustment from the maximum takeoff %N1 found previously to determine the assumed temperature reduced thrust %N1.

Takeoff with assumed temperature reduced thrust is not permitted when: runway is contaminated with ice, slush, snow, or standing water; anti-skid is inoperative; or PMC is off. Use of this procedure is not recommended if potential windshear conditions exist.

Max Climb %N1

This table shows Max Climb %N1 for a 250 KIAS/280 KIAS/.74M climb speed schedule, normal engine bleed for packs on (Auto) and anti-ice off. Enter the table with airport pressure altitude and TAT and read %N1. Apply the appropriate bleed setting %N1 adjustment shown below the table. %N1 adjustments are shown for engine bleed to packs off operation, packs high operation, and wing anti-ice on.

Go-around %N1

To find Max Go-around %N1 based on normal engine bleed for packs on (Auto) and anti-ice off, enter the Go-around %N1 table with airport pressure altitude and reported OAT or TAT and read %N1. Apply the appropriate bleed setting %N1 adjustment shown below the table. %N1 adjustments are shown for engine bleed to packs off operation, packs high operation, and wing anti-ice on.

Flight with Unreliable Airspeed / Turbulent Air Penetration

Pitch attitude and average %N1 information is provided for use in all phases of flight in the event of unreliable airspeed/Mach indications resulting from blocking or freezing of the pitot system. Loss of radome or turbulent air may also cause unreliable airspeed/Mach indications. The cruise table in this section may also be used for turbulent air penetration.

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Pitch attitude is shown in bold type for emphasis since altitude and/or vertical speed indications may also be unreliable.

All Engines

Long Range Cruise Maximum Operating Altitude

These tables provide the maximum operating altitude in the same manner as the FMC. Maximum altitudes are shown for a given cruise weight and maneuver capability. This table considers both thrust and buffet limits, providing the more limiting of the two. Any data that is thrust limited is denoted by an asterisk and represents only a thrust limited condition in level flight with 100 ft/min residual rate of climb. Flying above these altitudes with sustained banks in excess of approximately 15° may cause the airplane to lose speed and/or altitude.

Note that the altitudes shown in the table are limited to the maximum certified altitude of 37000 ft.

Long Range Cruise Control

These tables provide target %N1, Long Range Cruise Mach number, IAS and standard day fuel flow per engine for the airplane weight and pressure altitude. As indicated by the shaded area, at optimum altitude .74M approximates the Long Range Cruise Mach schedule.

Long Range Cruise Enroute Fuel and Time

Long Range Cruise Enroute Fuel and Time tables are provided to determine remaining time and fuel required to destination. The data is based on Long Range Cruise and .74M/250 KIAS descent. Tables are presented for low altitudes and high altitudes.

To determine remaining fuel and time required, first enter the Ground to Air Miles Conversion table to convert ground distance and enroute wind to an equivalent still air distance for use with the Reference Fuel and Time tables. Next, enter the Reference Fuel and Time Table with air distance from the Ground to Air Miles Conversion Table and the desired altitude and read Reference Fuel and Time Required. Lastly, enter the Fuel Required Adjustment Table with the Reference Fuel and the actual weight at checkpoint to obtain fuel required to destination.

Long Range Cruise Wind-Altitude Trade

Wind is a factor which may justify operations considerably below optimum altitude. For example, a favorable wind component may have an effect on ground speed which more than compensates for the loss in air range.

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Using this table, it is possible to determine the break-even wind (advantage necessary or disadvantage that can be tolerated) to maintain the same range at another altitude and long range cruise speed. The tables make no allowance for climb or descent time, fuel or distance, and are based on comparing ground fuel mileage.

Descent

Distance and time for descent are shown for a .74M/250 KIAS descent speed schedule. Enter the table with top of descent pressure altitude and read distance in nautical miles and time in minutes. Data is based on flight idle thrust descent in zero wind. Allowances are included for a straight-in approach with gear down and landing flaps at the outer marker.

Holding

Target %N1, indicated airspeed and fuel flow per engine information is tabulated for holding with flaps up based on the FMC optimum holding speed schedule. This is the higher of the maximum endurance speed and the maneuvering speed. Small variations in airspeed will not appreciably affect the overall endurance time. Enter the table with weight and pressure altitude to read %N1, IAS and fuel flow per engine.

Advisory Information

Autobrake Landing Distance

The Autobrake Landing Distance Table is provided as advisory information to assist in the selection of the most desirable auto brake setting for a given field length. It is not to be used to determine required field length. This data reflects actual landing distances on a dry runway for setting 1 through MAX, from touchdown to full stop, with or without reverse thrust. The table includes typical flare distances from threshold.

To use the Autobrake Landing Distance Table, first determine the available (or desired) landing distance. Enter the chart with the estimated approach speed and determine the actual stopping distance from touchdown for a given autobrake setting. If airspeed is used for approach speed, correct landing distance for pressure altitude, tailwind and temperature effects.

Selection of an autobrake setting results in a constant rate of deceleration. Maximum effort manual braking should achieve shorter landing distance than the MAX setting.

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Slippery Runway Landing Distance

Landing distances are the actual landing distances for the specified runway conditions and do not include the 1.67 regulatory factor. Therefore they cannot be used to determine dispatch required landing field length. When landing on slippery runways or runways contaminated with ice, snow, slush or standing water, the reported braking action must be considered. If the surface is affected by water, snow or ice, and the braking action is reported as “good,” conditions should not be expected to be as good as on clean dry runways. The value “good” is comparative and is intended to mean that airplanes should not experience braking or directional control difficulties when landing. The performance level used to calculate the “good” data is consistent with wet runway testing done on early Boeing jets. The performance level used to calculate the “poor” data reflects runways covered with wet ice. Read landing distance for the reported braking action at the airplane weight, and then apply the adjustments for airport pressure altitude and approach speed as required.

Non-normal Configuration Landing Distance

Advisory information is provided to support non-normal configurations that affect landing performance of the airplane. Landing distances are shown for dry runway and good, medium, and poor reported braking action. Each non-normal configuration is listed with its recommended approach speed. Landing distance can be determined for the reference landing weight and then adjusted for actual weight and pressure altitude.

Brake Cooling Schedule

Advisory information is provided to assist in avoiding problems associated with hot brakes. For normal operation, most landings are at weights below the quick turnaround limit weight. Application of the recommended cooling procedures shown will avoid brake overheat and fuse plug problems that could result from repeated landings at short time intervals or a rejected takeoff.

Engine Inoperative

Initial Max Continuous %N1

The Initial Max Continuous %N1 setting for use following an engine failure is shown. The table is based on the typical all engine cruise speed of .74M to provide a target %N1 setting at the start of driftdown. Once driftdown is established, the Max Continuous %N1 Table should be used to determine %N1 for the given conditions.

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Max Continuous %N1

Power setting is based on one engine operating with one A/C pack operating and all anti-ice bleeds off. Enter the table with pressure altitude and IAS or Mach to read %N1.

It is desirable to maintain engine thrust within the limits of the Max Cruise thrust rating. However, where thrust in excess of Max Cruise rating is required, such as for meeting terrain clearance, ATC altitude assignments, or to attain maximum range capability, it is permissible to use the thrust needed up to the Max Continuous thrust rating. The Max Continuous thrust rating is intended primarily for emergency use at the discretion of the pilot and is the maximum thrust that may be used continuously.

Driftdown Speed/Level Off Altitude

The table shows optimum driftdown speed as a function of cruise weight at start of driftdown. Also shown are the approximate weight and pressure altitude at which the airplane will level off considering 100 ft/min residual rate of climb.

The level off altitude is dependent on air temperature (ISA deviation).

Driftdown/LRC Cruise Range Capability

This table shows the range capability from the start of driftdown. Driftdown is continued to level off altitude. As weight decreases due to fuel burn, the airplane is accelerated to Long Range Cruise speed. Cruise is continued at level off altitude and Long Range Cruise speed.

To determine fuel required, enter the Ground to Air Miles Conversion table with the desired ground distance and correct for anticipated winds to obtain air distance to destination. Then enter the Driftdown/Cruise Fuel and Time table with air distance and weight at start of driftdown to determine fuel and time required. If altitudes other than the level off altitude is used, fuel and time required may be obtained by using the Engine Inoperative Long Range Cruise Enroute Fuel and Time Table.

Long Range Cruise Altitude Capability

The table shows the maximum altitude that can be maintained at a given weight and air temperature (ISA deviation), based on Long Range Cruise speed, Max Continuous thrust, and 100 ft/min residual rate of climb.

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Long Range Cruise Control

The table provides target %N1, engine inoperative Long Range Cruise Mach number, IAS and fuel flow for the airplane weight and pressure altitude. The fuel flow values in this table reflect single engine fuel burn. To conservatively account for APU fuel burn, add 90 kg/hr to fuel flow values.

Long Range Cruise Diversion Fuel and Time

Tables are provided for crews to determine the fuel and time required to proceed to an alternate airfield with one engine inoperative. The data is based on single engine Long Range Cruise speed and .74M/250 KIAS descent. Enter with Air Distance as determined from the Ground to Air Miles Conversion Table and read Fuel and Time required at the cruise pressure altitude. Adjust the fuel obtained for deviation from the reference weight at checkpoint as required by entering the off reference fuel corrections table with the fuel required for the reference weight and the actual weight at checkpoint. Read fuel required and time for the actual weight.

Holding

Single engine holding data is provided in the same format as the all engine holding data and is based on the same assumptions.

PMC Off Power Settings

To find Max Takeoff %N1 with both PMC's Off, enter Max Takeoff %N1 table (PMC Off) for the appropriate bleed configuration with airport pressure altitude and airport OAT and read %N1. No takeoff %N1 correction is required for engine anti-ice operation up to 10°C (50°F) which is the highest temperature recommended for engine anti-ice operation. Do not use reduced thrust with PMC Off.

For Maximum Climb and Go-Around with both PMC's Off, use PMC on %N1.

Gear Down

This section contains performance for airplane operation with the landing gear extended for all phases of flight. The data is based on engine bleeds for normal air conditioning.

Note: The Flight Management Computer System (FMCS) does not contain special provisions for operation with landing gear extended. As a result, the FMCS will generate inaccurate enroute speed schedules, display non-conservative predictions of fuel burn, estimated time of arrival (ETA),

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maximum altitude, and compute overly shallow descent path. To obtain accurate ETA predictions, gear down cruise speed and altitude should be entered on the CLB and CRZ pages. Gear down cruise speed should also be entered on the DES page. Use of VNAV during descent under these circumstances is not recommended.

Tables for gear down performance in this section are identical in format and used in the same manner as tables for the gear up configuration previously described.

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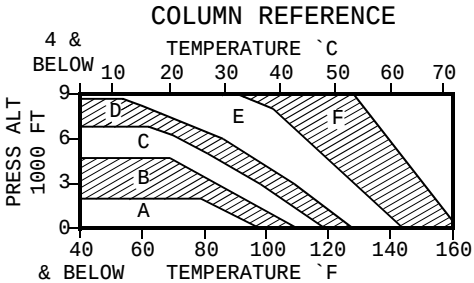
Takeoff Speeds
Flaps 5



SLOPE/WIND V_1 ADJUSTMENT*						
WEIGHT 1000 KG	SLOPE % DN		UP	WIND KTS TAIL		HEAD
	-2	0	2	-15	0	40
70	-3	0	4	-3	0	1
60	-2	0	3	-3	0	1
50	-2	0	2	-3	0	2
40	-2	0	2	-3	0	2

*V_1 NOT TO EXCEED V_R

WT 1000 KG	A			B			C		
	V_1	V_R	V_2	V_1	V_R	V_2	V_1	V_R	V_2
70	157	162	168	157	163	168			
65	150	154	162	151	156	161	152	157	161
60	143	147	155	144	148	155	145	149	155
55	136	139	149	136	140	149	138	141	148
50	128	131	143	129	132	142	130	133	142
45	120	123	136	120	123	136	122	125	135
40	112	115	130	112	115	129	113	116	129

WT 1000 KG	D			E			F		
	V_1	V_R	V_2	V_1	V_R	V_2	V_1	V_R	V_2
70									
65									
60	145	149	155						
55	138	142	148	140	143	148			
50	130	133	142	132	135	141			
45	122	125	135	124	126	134	128	128	134
40	113	116	128	116	118	128	119	120	127

IN BOXED AREA CHECK MINIMUM V_1 (MCG) FOR ACTUAL TEMP.

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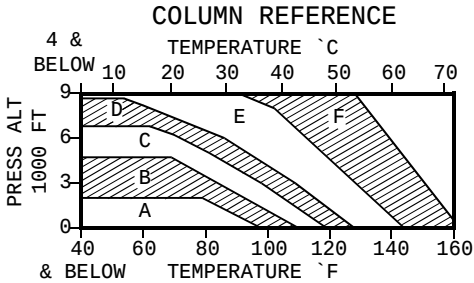
Takeoff Speeds

Flaps 15



SLOPE/WIND V_1 ADJUSTMENT*						
WEIGHT 1000 KG	SLOPE % DN		UP	WIND KTS TAIL		HEAD
	-2	0	2	-15	0	40
70	-3	0	4	-3	0	1
60	-2	0	3	-3	0	1
50	-2	0	2	-3	0	2
40	-2	0	2	-3	0	2

*V_1 NOT TO EXCEED V_R

WT 1000 KG	A			B			C		
	V_1	V_R	V_2	V_1	V_R	V_2	V_1	V_R	V_2
70	148	151	157						
65	142	145	152	143	146	151	145	147	152
60	135	138	146	136	139	146	138	140	146
55	128	131	141	129	132	140	131	133	140
50	121	123	135	122	124	134	123	125	134
45	113	116	129	113	116	128	115	118	128
40	105	107	122	105	108	122	107	110	122

WT 1000 KG	D			E			F		
	V_1	V_R	V_2	V_1	V_R	V_2	V_1	V_R	V_2
70									
65									
60	139	140	146						
55	132	133	140	134	134	139			
50	124	126	134	126	127	133			
45	116	118	128	118	119	127	120	120	127
40	107	110	121	109	111	120	112	113	120

IN BOXED AREA CHECK MINIMUM V_1 (MCG) FOR ACTUAL TEMP.

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V1(MCG)

ACTUAL OAT		PRESS ALT FT					
`C	`F	-1000	0	2000	4000	6000	8000
55	131	107	105				
50	122	111	109	104			
40	104	116	114	110	105	100	96
30	86	120	120	115	110	106	102
20	68	120	120	117	114	110	106
10	50	120	120	117	114	111	108
-50	-58	121	121	118	115	112	109

FOR A/C OFF INCREASE V_1 (MCG) BY 3 KNOTS.

Clearway and Stopway V1 Adjustments

CLEARWAY MINUS STOPWAY (FT)	NORMAL V1 (KIAS)			
	100	120	140	160
800			-3	-2
600		-3	-3	-2
400	-4	-3	-2	-1
200	-2	-2	-1	0
0	0	0	0	0
-400	2	2	2	1
-800	2	2	2	1

Maximum Allowable Clearway

FIELD LENGTH (FT)	MAX ALLOWABLE CLEARWAY FOR V1 REDUCTION (FT)
4000	450
6000	550
8000	600
10000	650
12000	700
14000	750

Stab Trim Setting

Max Takeoff Thrust

Flaps 5

C.G. % MAC	6	10	14	18	22	26	30
STAB TRIM	5 1/2	5	4 1/2	3 3/4	3 1/4	2 3/4	2 1/4

For weights at or below 45,360 kg subtract 1/2 unit from above value.
For weights at or above 61,235 kg add 1/2 unit from above value.

Flaps 15

C.G. % MAC	6	10	14	18	22	26	30
STAB TRIM	5	4 1/4	3 3/4	3	2 1/2	1 3/4	1

For weights at or below 45,360 kg subtract 1/2 unit from above value.
For weights at or above 61,235 kg add 1/2 unit from above value.

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VREF

WEIGHT (1000 KG)	FLAPS		
	40	30	15
70	155	159	177
65	149	154	171
60	143	147	164
55	137	141	156
50	130	134	149
45	124	127	141
40	116	119	132
35	109	111	123

For approach speed add wind factor of 1/2 headwind component + gust (max 20 knots).

Flap Maneuver Speeds

FLAP POSITION	MANEUVER SPEED (KIAS)		
	WEIGHT		
	AT OR BELOW 53070 KG	ABOVE 53070 KG AND AT OR BELOW 62823 KG	ABOVE 62823 KG
UP	210	220	230
1	190	200	210
5	180	190	200
10	170	180	190
15	150	160	170
25	140	150	160

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ADVISORY INFORMATION

Slush/Standing Water Takeoff

Weight Adjustment (1000 KG)

FIELD/OBSTACLE LIMIT WEIGHT (1000 KG)	SLUSH/STANDING WATER DEPTH								
	3mm (0.12 INCHES)			6mm (0.25 INCHES)			13mm (0.50 INCHES)		
	PRESS ALT (FT)			PRESS ALT (FT)			PRESS ALT (FT)		
	S.L.	4000	8000	S.L.	4000	8000	S.L.	4000	8000
68	-7.3	-7.8	-8.1	-9.0	-9.4	-9.3	-12.7	-13.2	-12.3
64	-6.9	-7.5	-7.7	-8.4	-9.1	-9.1	-11.9	-13.0	-12.8
60	-6.5	-7.2	-7.5	-7.8	-8.8	-9.1	-10.8	-12.4	-13.0
56	-5.9	-6.7	-7.5	-7.0	-8.1	-9.0	-9.4	-11.4	-12.7
52	-5.2	-6.1	-7.0	-6.1	-7.2	-8.5	-8.0	-9.9	-11.7
48	-4.5	-5.3	-6.1	-5.2	-6.1	-7.4	-6.7	-8.3	-10.2
44	-3.8	-4.5	-5.2	-4.3	-5.1	-6.2	-5.4	-6.8	-8.5
40	-3.0	-3.6	-4.2	-3.4	-4.1	-4.9	-4.2	-5.0	-6.8
36	-2.0	-2.5	-3.0	-2.2	-3.0	-3.5	-2.9	-3.3	-5.1

V1(MCG) Limit Weight (1000 KG)

ADJUSTED FIELD LENGTH (FT)	SLUSH/STANDING WATER DEPTH								
	3mm (0.12 INCHES)			6mm (0.25 INCHES)			13mm (0.50 INCHES)		
	PRESS ALT (FT)			PRESS ALT (FT)			PRESS ALT (FT)		
	S.L.	4000	8000	S.L.	4000	8000	S.L.	4000	8000
4600							30.5		
5000	28.5			31.6			36.4	29.8	
5400	34.3			37.5	29.9		42.4	35.1	29.9
5800	40.2	32.1		43.3	35.2	28.3	48.4	40.4	34.0
6200	46.1	37.2	30.0	49.4	40.6	32.9	54.4	45.4	38.1
6600	52.4	42.3	34.3	55.9	45.6	37.6	60.6	50.2	42.4
7000	58.9	47.4	38.6	62.1	51.0	42.2	66.8	55.6	46.7
7400	65.4	52.9	43.0	68.5	56.4	46.7	72.9	60.8	50.9
7800	72.4	58.6	47.7	75.0	62.1	51.2		66.3	55.2
8200		64.2	52.3		67.9	55.9		71.8	59.3
8600		70.3	57.1		73.6	60.7			63.7
9000			62.3			65.7			67.8
9400			67.3			70.6			71.9
9800			72.2			75.5			

1. **Enter Weight Adjustment table with slush/standing water depth and dry field/obstacle limit weight to obtain slush/standing water weight adjustment.**
2. **Adjust field length available by -126ft/+126ft for every 5°C above/below 4°C.**
3. **Find V1(MCG) limit weight for temperature adjusted field length and pressure altitude.**
4. **Max allowable slush/standing water limited weight is lesser of weights from 1 and 3.**

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ADVISORY INFORMATION

Slush/Standing Water Takeoff

V1 Adjustment (KIAS)

WEIGHT (1000 KG)	SLUSH/STANDING WATER DEPTH								
	3mm (0.12 INCHES)			6mm (0.25 INCHES)			13mm (0.50 INCHES)		
	PRESS ALT (FT)			PRESS ALT (FT)			PRESS ALT (FT)		
	S.L.	4000	8000	S.L.	4000	8000	S.L.	4000	8000
68	-13	-11	-10	-7	-7	-7	2	1	-2
64	-14	-13	-11	-8	-8	-7	2	1	-2
60	-16	-15	-13	-10	-9	-8	2	1	-1
56	-18	-16	-15	-13	-11	-9	0	1	0
52	-20	-18	-16	-15	-13	-10	-4	0	0
48	-21	-20	-18	-17	-16	-12	-8	-2	0
44	-22	-21	-19	-19	-17	-13	-12	-5	1
40	-22	-22	-20	-20	-19	-15	-14	-9	-1
36	-22	-22	-21	-20	-20	-17	-16	-12	-6

1. **Obtain V1, VR and V2 for the actual weight.**
2. **If V1(MCG) limited, set V1 = V1(MCG). If not V1(MCG) limited, enter V1 Adjustment Table with the actual weight to obtain V1 speed adjustment. If adjusted V1 is less than V1(MCG), set V1 = V1(MCG).**

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ADVISORY INFORMATION

Slippery Runway Takeoff

Weight Adjustment (1000 KG)

FIELD/OBSTACLE LIMIT WEIGHT (1000 KG)	REPORTED BRAKING ACTION								
	GOOD			MEDIUM			POOR		
	PRESS ALT (FT)			PRESS ALT (FT)			PRESS ALT (FT)		
	S.L.	4000	8000	S.L.	4000	8000	S.L.	4000	8000
68	0.0	0.0	0.0	-3.6	-3.6	-3.6	-7.0	-7.0	-7.0
64	-0.3	-0.3	-0.3	-3.9	-3.9	-3.9	-6.9	-6.9	-6.9
60	-0.7	-0.7	-0.7	-3.9	-3.9	-3.9	-6.6	-6.6	-6.6
56	-0.9	-0.9	-0.9	-4.0	-4.0	-4.0	-6.3	-6.3	-6.3
52	-1.1	-1.1	-1.1	-3.8	-3.8	-3.8	-5.8	-5.8	-5.8
48	-1.1	-1.1	-1.1	-3.5	-3.5	-3.5	-5.3	-5.3	-5.3
44	-0.9	-0.9	-0.9	-3.1	-3.1	-3.1	-4.6	-4.6	-4.6
40	-0.6	-0.6	-0.6	-2.5	-2.5	-2.5	-3.7	-3.7	-3.7
36	-0.2	-0.2	-0.2	-1.8	-1.8	-1.8	-2.7	-2.7	-2.7

V1(MCG) Limit Weight (1000 KG)

ADJUSTED FIELD LENGTH (FT)	REPORTED BRAKING ACTION								
	GOOD			MEDIUM			POOR		
	PRESS ALT (FT)			PRESS ALT (FT)			PRESS ALT (FT)		
	S.L.	4000	8000	S.L.	4000	8000	S.L.	4000	8000
3800	29.3								
4200	39.7								
4600	50.0	34.5							
5000	60.6	44.8	29.3	30.7					
5400	70.4	55.2	39.7	38.2					
5800		65.7	50.0	45.6	34.4				
6200		75.2	60.6	53.1	41.9	30.7	31.4		
6600			70.4	61.2	49.4	38.2	36.1		
7000				70.1	57.1	45.6	40.9	29.1	
7400					65.5	53.1	45.9	33.8	
7800					74.6	61.2	51.1	38.5	
8200						70.1	56.6	43.3	31.4
8600							62.3	48.5	36.1
9000							68.2	53.8	40.9
9400							74.3	59.4	45.9
9800								65.3	51.1
10200								71.3	56.6
10600									62.3
11000									68.2
11400									74.3

1. **Enter Weight Adjustment table with reported braking action and dry field/obstacle limit weight to obtain slippery runway weight adjustment.**
2. **Adjust "Good" field length available by -100ft/+100ft for every 5°C above/below 4°C**
   Adjust "Medium" field length available by -100ft/+100ft for every 5°C above/below 4°C
   Adjust "Poor" field length available by -140ft/+130ft for every 5°C above/below 4°C
3. **Find V1(MCG) limit weight for temperature adjusted field length and pressure altitude.**
4. **Max allowable slippery runway limited weight is lesser of weights from 1 and 3.**

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ADVISORY INFORMATION

Slippery Runway Takeoff
V1 Adjustment (KIAS)

WEIGHT (1000 KG)	REPORTED BRAKING ACTION								
	GOOD			MEDIUM			POOR		
	PRESS ALT (FT)			PRESS ALT (FT)			PRESS ALT (FT)		
	S.L.	4000	8000	S.L.	4000	8000	S.L.	4000	8000
68	-4	-2	0	-14	-12	-10	-31	-27	-23
64	-6	-4	-2	-16	-14	-12	-34	-30	-26
60	-7	-5	-3	-18	-16	-14	-36	-32	-28
56	-9	-7	-5	-20	-18	-16	-38	-34	-30
52	-11	-9	-7	-22	-20	-18	-40	-36	-32
48	-12	-10	-8	-24	-22	-20	-41	-37	-33
44	-13	-11	-9	-25	-23	-21	-42	-38	-34
40	-14	-12	-10	-26	-24	-22	-43	-39	-35
36	-15	-13	-11	-27	-25	-23	-43	-39	-35

1. **Obtain V1, VR and V2 for the actual weight.**
2. **If V1(MCG) limited, set V1 = V1(MCG). If not V1(MCG) limited, enter V1 Adjustment Table with the actual weight to obtain V1 speed adjustment. If adjusted V1 is less than V1(MCG), set V1 = V1(MCG).**

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Takeoff %N1

Based on engine bleed to packs on (Auto) and anti-ice on or off

AIRPORT OAT		AIRPORT PRESSURE ALTITUDE (FT)									
°C	°F	-1000	0	1000	2000	3000	4000	5000	6000	7000	8000
55	131	93.6	93.5	93.0	92.5						
50	122	94.8	95.1	94.9	94.8	94.4	93.3				
45	113	95.4	95.9	95.9	96.0	95.8	95.4	95.3	94.7	94.1	
40	104	96.1	96.6	96.6	96.6	96.5	96.4	96.4	96.4	96.4	95.9
35	95	96.6	97.3	97.2	97.2	97.1	97.1	97.2	97.1	97.1	97.1
30	86	96.5	97.8	97.8	97.9	97.8	97.8	97.9	97.9	98.0	97.9
25	77	95.7	97.0	97.6	98.3	98.6	98.6	98.7	98.7	98.8	98.7
20	68	94.9	96.2	96.7	97.5	98.1	98.7	99.3	99.4	99.5	99.4
15	59	94.1	95.4	95.9	96.6	97.2	97.8	98.4	99.1	99.7	99.9
10	50	93.3	94.6	95.1	95.8	96.4	97.0	97.6	98.2	98.8	99.3
5	41	92.5	93.7	94.2	94.9	95.5	96.1	96.7	97.3	97.9	98.4
0	32	91.6	92.9	93.4	94.1	94.7	95.2	95.8	96.5	97.0	97.5
-10	14	89.9	91.1	91.6	92.3	92.9	93.5	94.1	94.7	95.3	95.7
-20	-4	88.2	89.4	89.9	90.6	91.1	91.7	92.3	92.9	93.4	93.9
-30	-22	86.4	87.6	88.1	88.8	89.3	89.9	90.4	91.0	91.6	92.0
-40	-40	84.7	85.8	86.3	86.9	87.4	88.0	88.5	89.1	89.7	90.1
-50	-58	82.8	83.9	84.4	85.0	85.6	86.1	86.6	87.2	87.7	88.1

%N1 Adjustments for Engine Bleeds

BLEED CONFIGURATION	AIRPORT PRESSURE ALTITUDE (FT)	
	-1000	8000
ENGINE BLEED TO PACKS OFF	1.0	1.0

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Assumed Temperature Reduced Thrust

Maximum Assumed Temperature (Table 1)

Based on 25% Takeoff Thrust Reduction

OAT (°C)	PRESS ALT (1000 FT)										OAT (°F)	PRESS ALT (1000 FT)									
	-1	0	1	2	3	4	5	6	7	8		-1	0	1	2	3	4	5	6	7	8
55	70	71	69								130	158	158	156							
50	67	67	67	67	65	63					120	152	152	151	151	148	146				
45	65	65	64	64	62	61	60	59	57		110	149	148	147	145	143	140	139	137	135	
40	64	63	62	62	60	59	58	57	56	55	100	145	144	143	142	139	136	134	133	131	129
35	62	61	61	60	58	57	56	55	54	53	90	142	140	139	138	136	132	130	129	127	126
30	61	59	59	58	57	55	54	53	52	51	80	142	139	136	134	132	129	127	125	123	122
25	61	59	58	56	55	53	52	51	50	49	70	142	139	136	133	130	126	124	122	120	118
20	61	59	58	56	55	52	51	50	49	48	60	142	139	136	133	130	126	123	121	118	116
15 & BELOW	61	59	58	56	55	52	51	49	48	46	50 & BELOW	142	139	136	133	130	126	123	121	118	116

Maximum Takeoff %N1 (Table 2)

Based on engine bleed to packs on (Auto) and engine anti-ice on or off

ASSUMED TEMP		AIRPORT PRESSURE ALTITUDE (FT)									
°C	°F	-1000	0	1000	2000	3000	4000	5000	6000	7000	8000
75	167	83.0	86.2	87.2							
70	158	85.7	86.3	86.5	87.3	87.3					
65	149	88.5	88.8	88.4	88.0	86.8	86.8	87.4	88.2	88.9	89.5
60	140	91.2	91.2	90.8	90.3	89.4	88.2	87.6	87.5	88.2	88.8
55	131	93.5	93.6	93.1	92.6	91.9	90.8	90.2	89.6	89.0	88.3
50	122	94.7	95.1	94.9	94.8	94.4	93.3	92.8	92.2	91.6	90.9
45	113	95.3	95.9	95.9	96.0	95.8	95.4	95.3	94.8	94.1	93.4
40	104	96.0	96.7	96.6	96.6	96.5	96.4	96.4	96.4	96.4	95.9
35	95	96.5	97.3	97.3	97.2	97.2	97.1	97.2	97.2	97.1	97.1
30	86	96.4	97.8	97.8	97.9	97.8	97.8	98.0	97.9	98.0	97.9
25	77	95.6	97.0	97.6	98.3	98.6	98.6	98.7	98.7	98.8	98.7
20	68	94.8	96.2	96.8	97.5	98.1	98.7	99.3	99.4	99.5	99.4
15	59	94.0	95.4	95.9	96.6	97.2	97.8	98.4	99.1	99.7	99.9
MINIMUM ASSUMED TEMP °C (°F)		32(90)	30(86)	28(82)	26(79)	24(75)	22(72)	20(68)	18(64)	16(61)	14(57)

With engine bleed to packs off, increase %N1 by 1.0.

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Assumed Temperature Reduced Thrust

%N1 Adjustment For Temperature Difference (Table 3)

ASSUMED TEMP MINUS OAT		OUTSIDE AIR TEMPERATURE														
°C	°F	°C	-40	-20	0	5	10	15	20	25	30	35	40	45	50	55
		°F	-40	-4	32	41	50	59	68	77	86	95	104	113	122	131
10	18						1.6	1.6	1.6	1.5	1.5	1.5	1.5	1.4	1.4	1.3
20	36				3.3	3.2	3.2	3.1	3.0	3.0	2.9	2.8	2.8	2.7	2.5	2.3
30	54				4.8	4.8	4.6	4.5	4.4	4.3	4.0	3.8	3.6	3.6	3.6	3.6
40	72			6.0	6.2	6.1	6.0	5.8	5.7	5.2	5.0	5.0				
50	90			8.2	7.5	7.3	7.2	6.6								
60	108		10.4	9.5	8.7	8.1	7.9									
70	126		11.8	10.7	9.3											
80	144		13.0	11.8	10.1											
90	162		14.0	12.4												
100	180		15.0	12.8												
110	198		15.4													

1. **Determine Maximum Assumed Temperature allowed from Table 1.**
2. **Find Maximum %N1 from Table 2 using the desired assumed temperature (no greater than temperature from Table 1).**
3. **Use the difference between assumed temperature and OAT to determine the %N1 adjustment from Table 3.**
4. **Subtract %N1 adjustment from Maximum %N1 in Table 2.**

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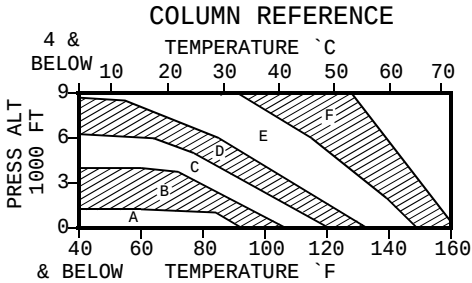
Takeoff Speeds (22K Derate)

Flaps 5



SLOPE/WIND V_1 ADJUSTMENT*						
WEIGHT 1000 KG	SLOPE % DN		UP	WIND KTS TAIL		HEAD
	-2	0	2	-15	0	40
70	-3	0	4	-3	0	1
60	-2	0	2	-3	0	1
50	-2	0	1	-4	0	1
40	-2	0	1	-4	0	1

*V_1 NOT TO EXCEED V_R

WT 1000 KG	A			B			C		
	V_1	V_R	V_2	V_1	V_R	V_2	V_1	V_R	V_2
70	158	163	168	158	164	169			
65	151	155	161	152	156	162	153	157	162
60	144	148	155	145	148	155	146	149	155
55	137	139	149	138	140	149	138	141	148
50	129	131	142	130	132	142	131	133	142
45	121	123	136	122	124	135	122	125	135
40	113	114	130	113	116	129	113	116	128

WT 1000 KG	D			E			F		
	V_1	V_R	V_2	V_1	V_R	V_2	V_1	V_R	V_2
70									
65									
60									
55	140	143	148						
50	132	134	141	133	135	141			
45	124	126	135	125	127	134	128	128	134
40	115	117	128	116	118	127	119	120	126

IN BOXED AREA CHECK MINIMUM V_1 (MCG) FOR ACTUAL TEMP.

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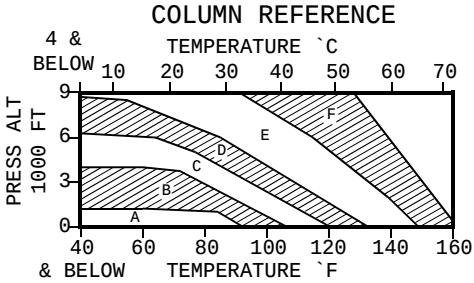
Takeoff Speeds (22K Derate)
Flaps 15



SLOPE/WIND V_1 ADJUSTMENT*						
WEIGHT 1000 KG	SLOPE % DN		UP	WIND KTS TAIL		HEAD
	-2	0	2	-15	0	40
70	-3	0	4	-3	0	1
60	-2	0	2	-3	0	1
50	-2	0	1	-4	0	1
40	-2	0	1	-4	0	1

*V_1 NOT TO EXCEED V_R

WT 1000 KG	A			B			C		
	V_1	V_R	V_2	V_1	V_R	V_2	V_1	V_R	V_2
70	150	152	158						
65	143	145	152	145	146	151			
60	137	139	146	138	140	146	139	140	146
55	130	131	141	131	132	140	131	133	140
50	122	124	135	123	125	134	124	125	134
45	114	116	128	116	117	128	116	118	128
40	106	107	122	108	109	122	108	110	122

WT 1000 KG	D			E			F		
	V_1	V_R	V_2	V_1	V_R	V_2	V_1	V_R	V_2
70									
65									
60									
55	134	134	140						
50	126	126	134	127	128	133			
45	118	119	127	120	120	126	120	121	127
40	109	111	121	111	112	120	112	113	120

IN BOXED AREA CHECK MINIMUM V_1 (MCG) FOR ACTUAL TEMP.

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V1(MCG) (22K Derate)

ACTUAL OAT		PRESS ALT FT					
`C	`F	-1000	0	2000	4000	6000	8000
55	131	106	104				
50	122	109	107	103			
40	104	113	111	107	103	99	94
30	86	117	116	111	107	104	98
20	68	117	116	113	111	107	102
10	50	118	116	113	111	108	104
-50	-58	119	118	115	112	109	105

FOR A/C OFF INCREASE V_1 (MCG) BY 2 KNOTS.

Clearway and Stopway V1 Adjustments (22K Derate)

CLEARWAY MINUS STOPWAY (FT)	NORMAL V1 (KIAS)			
	100	120	140	160
800			-3	-2
600		-3	-2	-1
400	-4	-3	-2	-1
200	-2	-1	-1	0
0	0	0	0	0
-400	1	1	1	1
-800	1	1	1	1

Maximum Allowable Clearway (22K Derate)

FIELD LENGTH (FT)	MAX ALLOWABLE CLEARWAY FOR V1 REDUCTION (FT)
4000	400
6000	500
8000	550
10000	600
12000	700
14000	750

Stab Trim Setting (22K Derate)

Flaps 5

C.G. % MAC	6	10	14	18	22	26	30
STAB TRIM	5 1/2	5	4 1/2	3 3/4	3 1/4	2 3/4	2 1/4

For weights at or below 45,360 kg subtract 1/2 unit from above value.
For weights at or above 61,235 kg add 1/2 unit from above value.

Flaps 15

C.G. % MAC	6	10	14	18	22	26	30
STAB TRIM	5	4 1/4	3 3/4	3	2 1/2	1 3/4	1

For weights at or below 45,360 kg subtract 1/2 unit from above value.
For weights at or above 61,235 kg add 1/2 unit from above value.

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ADVISORY INFORMATION

Slush/Standing Water Takeoff (22K Derate)

Weight Adjustment (1000 KG)

22K DERATE FIELD/OBSTACLE LIMIT WEIGHT (1000 KG)	SLUSH/STANDING WATER DEPTH								
	3mm (0.12 INCHES)			6mm (0.25 INCHES)			13mm (0.50 INCHES)		
	PRESS ALT (FT)			PRESS ALT (FT)			PRESS ALT (FT)		
	S.L.	4000	8000	S.L.	4000	8000	S.L.	4000	8000
68	-7.6	-8.1	-8.6	-9.5	-9.9	-9.7	-13.8	-13.7	-12.3
64	-7.2	-7.8	-8.1	-8.7	-9.5	-9.7	-12.3	-13.2	-13.1
60	-6.6	-7.4	-7.9	-7.9	-8.8	-9.5	-10.9	-12.5	-13.1
56	-6.0	-6.8	-7.5	-7.1	-8.1	-9.0	-9.5	-11.4	-12.7
52	-5.3	-6.2	-7.0	-6.2	-7.3	-8.5	-8.1	-10.1	-11.8
48	-4.6	-5.4	-6.3	-5.3	-6.4	-7.5	-6.8	-8.6	-10.4
44	-3.8	-4.5	-5.2	-4.3	-5.2	-6.3	-5.4	-6.8	-8.6
40	-3.0	-3.6	-4.2	-3.4	-4.1	-4.8	-4.2	-5.0	-6.8
36	-2.0	-2.4	-2.7	-2.2	-2.6	-3.1	-3.0	-3.2	-5.0

V1(MCG) Limit Weight (1000 KG)

ADJUSTED FIELD LENGTH (FT)	SLUSH/STANDING WATER DEPTH								
	3mm (0.12 INCHES)			6mm (0.25 INCHES)			13mm (0.50 INCHES)		
	PRESS ALT (FT)			PRESS ALT (FT)			PRESS ALT (FT)		
	S.L.	4000	8000	S.L.	4000	8000	S.L.	4000	8000
4600							32.1		
5000	30.2			33.0			38.3	32.1	
5400	37.0	29.2		40.0	32.4		44.7	37.2	31.5
5800	43.6	34.8	28.7	46.7	37.9	31.5	50.9	42.3	35.8
6200	49.9	40.4	33.2	52.9	43.3	36.2	56.4	47.4	40.0
6600	56.7	45.8	37.7	59.3	48.7	40.8	62.0	52.3	44.4
7000	63.2	51.4	42.1	66.3	54.3	45.2	67.6	57.3	48.9
7400	70.2	56.9	46.8	73.3	59.9	49.9	73.1	62.1	53.3
7800		62.5	51.8		65.3	54.7		66.9	57.9
8200		68.7	56.6		71.4	59.5		71.8	62.5
8600		75.0	61.6			64.6			67.0
9000			66.7			69.4			71.6
9400			71.7			74.1			

1. **Enter Weight Adjustment table with reported braking action and 22K derate dry field/obstacle limit weight to obtain slippery runway weight adjustment.**
2. **Adjust field length available by -126ft/+126ft for every 5°C above/below 4°C**
3. **Find V1(MCG) limit weight for temperature adjusted field length and pressure altitude.**
4. **Max allowable slippery runway limited weight is lesser of weights from 1 and 3.**

V1 Adjustment (KIAS)

WEIGHT (1000 KG)	SLUSH/STANDING WATER DEPTH								
	3mm (0.12 INCHES)			6mm (0.25 INCHES)			13mm (0.50 INCHES)		
	PRESS ALT (FT)			PRESS ALT (FT)			PRESS ALT (FT)		
	S.L.	4000	8000	S.L.	4000	8000	S.L.	4000	8000
68	-11	-10	-8	-6	-5	-5	2	1	-2
64	-13	-11	-9	-7	-6	-6	2	1	-1
60	-15	-13	-11	-9	-7	-6	2	1	-1
56	-17	-15	-12	-11	-9	-7	2	2	0
52	-18	-16	-14	-14	-11	-8	-1	1	0
48	-20	-18	-15	-16	-13	-9	-5	0	1
44	-21	-20	-17	-18	-15	-11	-10	-3	1
40	-22	-20	-18	-19	-17	-13	-14	-7	0
36	-22	-21	-19	-20	-18	-15	-15	-11	-3

1. **Obtain V1, VR and V2 for the actual weight.**
2. **If V1(MCG) limited, set V1 = V1(MCG). If not V1(MCG) limited, enter V1 Adjustment Table with the actual weight to obtain V1 speed adjustment. If adjusted V1 is less than V1(MCG), set V1 = V1(MCG).**

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ADVISORY INFORMATION

Slippery Runway Takeoff (22K Derate)

Weight Adjustment (1000 KG)

22K DERATE FIELD/OBSTACLE LIMIT WEIGHT (1000 KG)	REPORTED BRAKING ACTION								
	GOOD			MEDIUM			POOR		
	PRESS ALT (FT)			PRESS ALT (FT)			PRESS ALT (FT)		
	S.L.	4000	8000	S.L.	4000	8000	S.L.	4000	8000
68	-0.5	-0.5	-0.5	-4.4	-4.4	-4.4	-7.1	-7.1	-7.1
64	-0.7	-0.7	-0.7	-4.2	-4.2	-4.2	-7.0	-7.0	-7.0
60	-1.0	-1.0	-1.0	-4.1	-4.1	-4.1	-6.7	-6.7	-6.7
56	-1.1	-1.1	-1.1	-4.1	-4.1	-4.1	-6.4	-6.4	-6.4
52	-1.3	-1.3	-1.3	-4.1	-4.1	-4.1	-6.0	-6.0	-6.0
48	-1.3	-1.3	-1.3	-3.9	-3.9	-3.9	-5.4	-5.4	-5.4
44	-1.1	-1.1	-1.1	-3.5	-3.5	-3.5	-4.7	-4.7	-4.7
40	-0.8	-0.8	-0.8	-2.9	-2.9	-2.9	-3.9	-3.9	-3.9
36	-0.5	-0.5	-0.5	-1.9	-1.9	-1.9	-2.8	-2.8	-2.8

V1(MCG) Limit Weight (1000 KG)

ADJUSTED FIELD LENGTH (FT)	REPORTED BRAKING ACTION								
	GOOD			MEDIUM			POOR		
	PRESS ALT (FT)			PRESS ALT (FT)			PRESS ALT (FT)		
	S.L.	4000	8000	S.L.	4000	8000	S.L.	4000	8000
3800	31.6								
4200	42.1								
4600	52.6	36.8							
5000	62.9	47.3	31.6	33.2					
5400	73.2	57.8	42.1	40.9	29.4				
5800		68.0	52.6	48.5	37.1				
6200			62.9	56.3	44.7	33.2	31.4		
6600			73.2	64.4	52.3	40.9	36.1		
7000				72.5	60.3	48.5	40.9	29.1	
7400					68.4	56.3	45.9	33.8	
7800						64.4	51.1	38.5	
8200						72.5	56.6	43.3	31.4
8600							62.3	48.5	36.1
9000							68.2	53.8	40.9
9400							74.3	59.4	45.9
9800								65.3	51.1
10200								71.3	56.6
10600									62.3
11000									68.2
11400									74.3

1. **Enter Weight Adjustment table with reported braking action and 22K derate dry field/obstacle limit weight to obtain slippery runway weight adjustment.**
2. **Adjust "Good" field length available by -99ft/+99ft for every 5°C above/below 4°C**
   Adjust "Medium" field length available by -99ft/+99ft for every 5°C above/below 4°C
   Adjust "Poor" field length available by -135ft/+135ft for every 5°C above/below 4°C
3. **Find V1(MCG) limit weight for temperature adjusted field length and pressure altitude.**
4. **Max allowable slippery runway limited weight is lesser of weights from 1 and 3.**

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ADVISORY INFORMATION

Slippery Runway Takeoff (22K Derate)

V1 Adjustment (KIAS)

WEIGHT (1000 KG)	REPORTED BRAKING ACTION								
	GOOD			MEDIUM			POOR		
	PRESS ALT (FT)			PRESS ALT (FT)			PRESS ALT (FT)		
	S.L.	4000	8000	S.L.	4000	8000	S.L.	4000	8000
68	-4	-2	0	-12	-10	-8	-29	-25	-21
64	-5	-3	-1	-15	-13	-11	-32	-28	-24
60	-7	-5	-3	-17	-15	-13	-34	-30	-26
56	-8	-6	-4	-19	-17	-15	-37	-33	-29
52	-10	-8	-6	-21	-19	-17	-38	-34	-30
48	-11	-9	-7	-23	-21	-19	-40	-36	-32
44	-12	-10	-8	-24	-22	-20	-41	-37	-33
40	-13	-11	-9	-25	-23	-21	-42	-38	-34
36	-14	-12	-10	-27	-25	-23	-42	-38	-34

1. **Obtain V1, VR and V2 for the actual weight.**
2. **If V1(MCG) limited, set V1 = V1(MCG). If not V1(MCG) limited, enter V1 Adjustment Table with the actual weight to obtain V1 speed adjustment. If adjusted V1 is less than V1(MCG), set V1 = V1(MCG).**

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Takeoff %N1 (22K Derate)

Based on engine bleed to packs on (Auto) and anti-ice on or off

AIRPORT OAT		AIRPORT PRESSURE ALTITUDE (FT)									
°C	°F	-1000	0	1000	2000	3000	4000	5000	6000	7000	8000
55	131	93.2	93.8	93.8	93.8						
50	122	93.8	94.3	94.3	94.3	93.9	93.6				
45	113	94.2	94.7	94.7	94.6	94.6	94.6	94.7	94.4	94.2	
40	104	94.6	95.2	95.2	95.1	95.0	95.1	95.1	95.2	95.1	94.9
35	95	95.2	95.6	95.6	95.6	95.5	95.7	95.7	95.7	95.6	95.5
30	86	95.2	96.1	96.1	96.0	96.0	96.3	96.2	96.1	96.0	96.0
25	77	94.4	95.3	95.8	96.2	96.5	96.7	96.6	96.6	96.5	96.4
20	68	93.6	94.5	95.0	95.4	95.9	96.6	97.1	97.1	97.0	96.9
15	59	92.8	93.7	94.2	94.6	95.1	95.8	96.3	96.8	97.2	97.5
10	50	92.0	92.9	93.4	93.8	94.2	95.0	95.4	95.9	96.4	96.8
5	41	91.2	92.1	92.5	92.9	93.4	94.1	94.6	95.1	95.5	96.0
0	32	90.3	91.2	91.7	92.1	92.6	93.3	93.7	94.2	94.7	95.1
-10	14	88.7	89.6	90.0	90.4	90.8	91.5	92.0	92.5	92.9	93.4
-20	-4	87.0	87.8	88.3	88.7	89.1	89.8	90.2	90.7	91.1	91.6
-30	-22	85.2	86.0	86.5	86.9	87.3	88.0	88.4	88.9	89.3	89.7
-40	-40	83.5	84.3	84.7	85.1	85.5	86.2	86.6	87.1	87.4	87.9
-50	-58	81.7	82.5	82.9	83.2	83.7	84.3	84.7	85.2	85.6	86.0

%N1 Adjustments for Engine Bleeds

BLEED CONFIGURATION	AIRPORT PRESSURE ALTITUDE (FT)	
	-2000	9000
ENGINE BLEED TO PACKS OFF	1.0	1.0

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Assumed Temperature Reduced Thrust (22K Derate)

Maximum Assumed Temperature (Table 1)

Based on 25% Takeoff Thrust Reduction

OAT (°C)	PRESS ALT (1000 FT)										OAT (°F)	PRESS ALT (1000 FT)									
	-1	0	1	2	3	4	5	6	7	8		-1	0	1	2	3	4	5	6	7	8
55	74	71	71								130	165	159	159							
50	71	69	68	68	69	70					120	160	155	154	154	155	157				
45	69	67	66	66	67	67	67	68	70		110	155	151	149	149	150	151	151	152	155	
40	67	65	64	64	64	64	64	64	66	68	100	151	148	145	145	145	145	147	145	149	151
35	65	63	62	62	62	61	61	62	63	64	90	146	143	141	141	140	140	140	140	143	144
30	63	61	60	60	59	59	59	59	60	61	80	146	142	139	138	136	135	135	135	137	138
25	63	61	59	58	57	56	56	56	57	58	70	146	142	138	136	135	131	129	129	130	132
20	63	61	59	58	57	55	53	54	54	55	60	146	142	138	136	135	131	127	127	126	126
15 & BELOW	63	61	59	58	57	55	53	53	52	52	50 & BELOW	146	142	138	136	135	131	127	127	126	126

Maximum Takeoff %N1 (Table 2)

Based on engine bleed to packs on (Auto) and engine anti-ice on or off

ASSUMED TEMP		AIRPORT PRESSURE ALTITUDE (FT)									
°C	°F	-1000	0	1000	2000	3000	4000	5000	6000	7000	8000
75	167	86.0	85.4	85.4							
70	158	87.8	87.6	87.4	87.4	87.6					
65	149	89.7	89.7	89.4	89.2	89.2	89.2	89.3	89.5	89.9	90.4
60	140	91.6	91.8	91.3	91.0	90.8	90.7	90.7	90.8	91.1	91.4
55	131	93.1	93.8	93.2	92.7	92.4	92.1	92.1	92.0	92.1	92.3
50	122	93.7	94.3	94.3	94.3	93.9	93.6	93.4	93.2	93.2	93.2
45	113	94.0	94.7	94.7	94.6	94.6	94.6	94.7	94.4	94.2	94.0
40	104	94.4	95.2	95.2	95.1	95.0	95.1	95.1	95.2	95.1	94.9
35	95	95.0	95.6	95.6	95.6	95.5	95.7	95.7	95.7	95.6	95.5
30	86	95.2	96.1	96.1	96.0	96.0	96.3	96.2	96.1	96.0	96.0
25	77	94.4		96.6	96.5	96.5	96.7	96.6	96.6	96.5	96.4
20	68	93.6					97.1	97.1	97.1	97.0	96.9
15	59	92.8							97.6	97.5	97.5
MINIMUM ASSUMED TEMP °C (°F)		32(90)	30(86)	28(82)	26(79)	24(75)	22(72)	20(68)	18(64)	16(61)	15(59)

With engine bleed to packs off, increase %N1 by 1.0.

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Assumed Temperature Reduced Thrust (22K Derate)

%N1 Adjustment For Temperature Difference (Table 3)

ASSUMED TEMP MINUS OAT		OUTSIDE AIR TEMPERATURE													
°C	°F	°C	-40	-20	0	5	10	15	20	25	30	35	40	45	50	55
		°F	-40	-4	32	41	50	59	68	77	86	95	104	113	122	131
10	18							1.6	1.6	1.6	1.5	1.5	1.5	1.5	1.4	1.4
20	36				3.3	3.2	3.2	3.1	3.0	3.0	2.9	2.8	2.8	2.7	2.5	2.3
30	54				4.8	4.8	4.6	4.5	4.4	4.3	4.0	3.8	3.6	3.6	3.6	3.6
40	72			6.0	6.2	6.1	6.0	5.8	5.7	5.2	5.0	5.0				
50	90			8.2	7.5	7.3	7.2	6.6	6.5							
60	108		10.4	9.5	8.7	8.1	7.9									
70	126		11.8	10.7	9.3											
80	144		13.0	11.8	10.1											
90	162		14.0	12.4												
100	180		15.0	12.8												
110	198		15.4													

1. **Determine Maximum Assumed Temperature allowed from Table 1.**
2. **Find Maximum %N1 from Table 2 using the desired assumed temperature (no greater than temperature from Table 1).**
3. **Use the difference between assumed temperature and OAT to determine the %N1 adjustment from Table 3.**
4. **Subtract %N1 adjustment from Maximum %N1 in Table 2.**

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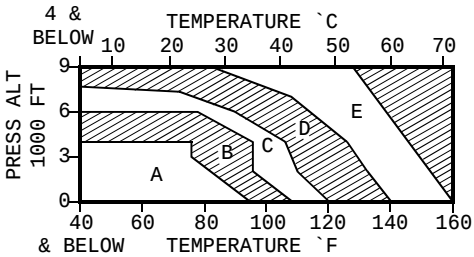
Takeoff Speeds (20K Derate)

COLUMN REFERENCE



SLOPE/WIND V_1 ADJUSTMENT*						
WEIGHT 1000 KG	SLOPE % DN		UP	WIND KTS TAIL		HEAD
	-2	0	2	-15	0	40
70	-2	0	3	-3	0	2
60	-2	0	1	-3	0	1
50	-2	0	1	-4	0	1
40	-2	0	1	-4	0	1

*V_1 NOT TO EXCEED V_R

FLAPS	WT 1000 KG	A			B			C			D			E		
		V_1	V_R	V_2	V_1	V_R	V_2	V_1	V_R	V_2	V_1	V_R	V_2	V_1	V_R	V_2
5	70	161	165	168	161	165	168									
	65	154	157	162	154	158	161	155	159	162						
	60	147	149	155	147	150	155	148	151	155	150	152	155			
	55	139	141	149	140	142	148	140	143	148	142	144	148			
	50	132	133	142	132	134	141	132	135	141	134	136	141	137	137	141
	45	123	125	135	124	125	135	124	126	134	126	127	134	128	128	134
	40	115	116	128	115	117	128	116	117	128	117	119	127	120	120	127
15	70	152	154	159												
	65	146	147	152	148	149	153									
	60	139	140	146	141	141	146	142	143	146						
	55	132	133	140	134	134	140	135	135	140	136	136	140			
	50	124	125	134	126	126	134	127	127	133	128	128	133			
	45	116	118	128	118	118	128	119	119	127	120	120	127	120	121	126
	40	107	110	122	109	110	121	110	111	121	111	112	120	112	113	120

IN BOXED AREA CHECK MINIMUM V_1 (MCG) FOR ACTUAL TEMP.

V1(MCG) (20K Derate)

ACTUAL OAT		PRESS ALT FT					
`C	`F	-1000	0	2000	4000	6000	8000
54	130	101	99				
50	122	103	101	99	97		
40	104	107	106	103	103	98	93
30	86	110	110	107	107	103	96
20	68	111	110	108	108	105	101
10	50	111	110	109	108	105	101
-50	-58	112	111	110	109	106	102

FOR A/C OFF INCREASE V_1 (MCG) BY 3 KNOTS.

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Clearway and Stopway V1 Adjustments (20K Derate)

CLEARWAY MINUS STOPWAY (FT)	NORMAL V1 (KIAS)			
	100	120	140	160
800			-3	-2
600		-3	-2	-1
400	-4	-3	-2	-1
200	-2	-1	-1	0
0	0	0	0	0
-400	1	1	1	1
-800	1	1	1	1

Maximum Allowable Clearway (20K Derate)

FIELD LENGTH (FT)	MAX ALLOWABLE CLEARWAY FOR V1 REDUCTION (FT)
4000	400
6000	500
8000	550
10000	600
12000	700
14000	750

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Takeoff %N1 (20K Derate)

Based on engine bleed to packs on (Auto) and anti-ice on or off

AIRPORT OAT		AIRPORT PRESSURE ALTITUDE (FT)									
°C	°F	-1000	0	1000	2000	3000	4000	5000	6000	7000	8000
55	131	89.9	90.5								
50	122	90.6	91.1	91.6	92.3	93.1					
45	113	91.2	91.6	92.1	92.6	93.9	94.9	94.7	94.2		
40	104	91.7	92.1	92.6	93.0	94.3	95.3	95.3	95.3	95.0	94.3
35	95	92.0	92.5	93.0	93.4	94.7	95.9	95.8	95.8	95.1	94.5
30	86	91.8	92.8	93.2	93.6	95.0	96.5	96.4	96.4	95.7	95.0
25	77	91.0	92.1	92.9	93.6	94.5	95.9	96.3	96.7	96.2	95.6
20	68	90.3	91.3	92.1	92.8	93.8	95.1	95.5	95.9	95.9	95.9
15	59	89.5	90.5	91.3	92.0	93.0	94.3	94.7	95.1	95.3	95.3
10	50	88.7	89.7	90.5	91.2	92.1	93.4	93.9	94.3	94.4	94.6
5	41	87.9	88.9	89.7	90.4	91.3	92.6	93.0	93.4	93.6	93.7
0	32	87.1	88.1	88.9	89.6	90.5	91.8	92.2	92.6	92.7	92.9
-10	14	85.5	86.5	87.2	87.9	88.8	90.1	90.5	90.9	91.0	91.2
-20	-4	83.9	84.8	85.6	86.2	87.1	88.3	88.7	89.1	89.2	89.4
-30	-22	82.2	83.1	83.9	84.5	85.4	86.6	87.0	87.4	87.5	87.6
-40	-40	80.5	81.4	82.1	82.7	83.6	84.8	85.1	85.5	85.7	85.8
-50	-58	78.8	79.6	80.3	81.0	81.8	82.9	83.3	83.7	83.8	84.0

%N1 Adjustments for Engine Bleeds

BLEED CONFIGURATION	AIRPORT PRESSURE ALTITUDE (FT)	
	-1000	8000
ENGINE BLEED TO PACKS OFF	1.0	1.0

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Assumed Temperature Reduced Thrust (20K Derate)

Maximum Assumed Temperature (Table 1)

Based on 25% Takeoff Thrust Reduction

OAT (°C)	PRESS ALT (1000 FT)										OAT (°F)	PRESS ALT (1000 FT)									
	-1	0	1	2	3	4	5	6	7	8		-1	0	1	2	3	4	5	6	7	8
55	74	73	73								130	164	162	162							
50	71	71	69	68	66	65					120	159	158	155	153	149	147				
45	69	68	67	66	63	62	62	63	65		110	154	153	151	149	144	142	142	143	145	
40	66	66	65	64	61	60	60	60	60	62	100	150	148	146	145	140	138	138	137	139	141
35	64	63	62	62	59	58	58	57	59	60	90	146	143	142	141	137	134	133	132	136	137
30	63	61	61	60	58	56	55	55	56	58	80	146	141	140	137	135	131	129	128	131	132
25	63	61	60	58	57	55	54	53	54	55	70	146	141	140	136	134	131	129	127	127	126
20 & BELOW	63	61	60	58	57	55	54	53	53	52	60 & BELOW	146	141	140	136	134	131	129	127	127	126

Maximum Takeoff %N1 (Table 2)

Based on engine bleed to packs on (Auto) and engine anti-ice on or off

ASSUMED TEMP		AIRPORT PRESSURE ALTITUDE (FT)									
°C	°F	-1000	0	1000	2000	3000	4000	5000	6000	7000	8000
75	167	83.0	83.9	83.8	83.5						
70	158	84.7	85.6	85.6	85.4	84.9	84.7				
65	149	86.6	87.3	87.3	87.2	87.0	87.0	87.1	87.2	87.6	87.9
60	140	88.3	88.9	89.0	89.0	89.1	89.1	89.1	89.0	89.1	89.3
55	131	89.7	90.5	90.6	90.7	91.1	91.3	91.0	90.8	90.7	90.6
50	122	90.4	91.1	91.6	92.3	93.1	93.4	92.9	92.5	92.2	91.9
45	113	91.0	91.6	92.1	92.7	93.9	94.9	94.8	94.2	93.7	93.1
40	104	91.6	92.1	92.6	93.0	94.3	95.3	95.3	95.3	95.0	94.3
35	95	91.4	92.5	93.0	93.4	94.8	95.9	95.9	95.8	95.1	94.5
30	86	91.7	92.8	93.2	93.6	95.0	96.5	96.4	96.4	95.7	95.0
25	77	90.9		93.3	93.7	94.5	96.5	96.5	96.7	96.2	95.6
20	68	90.2				94.4				95.9	95.9
15	59	89.4									95.4
MINIMUM ASSUMED TEMP °C (°F)		32(90)	30(86)	28(82)	26(79)	24(75)	29(85)	27(81)	25(77)	20(68)	15(59)

With engine bleed to packs off, increase %N1 by 1.0.

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Assumed Temperature Reduced Thrust (20K Derate)

%N1 Adjustment For Temperature Difference (Table 3)

ASSUMED TEMP MINUS OAT		OUTSIDE AIR TEMPERATURE														
°C	°F	°C	-40	-20	0	5	10	15	20	25	30	35	40	45	50	55
		°F	-40	-4	32	41	50	59	68	77	86	95	104	113	122	131
10	18						1.6	1.6	1.6	1.5	1.5	1.5	1.4	1.4	1.4	1.3
20	36				3.3	3.3	3.1	3.1	3.0	3.0	2.9	2.8	2.7	2.6	2.5	2.3
30	54				4.8	4.7	4.6	4.5	4.4	4.3	4.1	3.9	3.7	3.5	3.4	3.3
40	72			6.8	6.1	6.0	5.9	5.7	5.5	5.3	5.1	4.9				
50	90			8.1	7.4	7.2	6.9	6.7	6.3							
60	108		10.4	9.3	8.4	8.1	7.7									
70	126		11.5	10.5	9.0											
80	144		12.7	11.4	9.8											
90	162		13.8	11.9												
100	180		14.5	12.6												
110	198		14.9													

1. **Determine Maximum Assumed Temperature allowed from Table 1.**
2. **Find Maximum %N1 from Table 2 using the desired assumed temperature (no greater than temperature from Table 1).**
3. **Use the difference between assumed temperature and OAT to determine the %N1 adjustment from Table 3.**
4. **Subtract %N1 adjustment from Maximum %N1 in Table 2.**

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Max Climb %N1

Based on engine bleed to packs on (Auto) and anti-ice off

TAT (°C)	PRESSURE ALTITUDE (1000 FT)/SPEED (KIAS OR MACH)								
	0	5	10	15	20	25	30	35	37
	250	250	250	280	280	280	.74	.74	.74
50	93.7	93.9	94.1						
40	93.9	94.0	94.3	95.0	95.2				
30	93.8	94.8	95.1	95.8.	96.0	96.0			
20	92.2	94.4	96.1	96.8	97.1	97.1	97.1		
10	90.6	92.8	95.2	97.3	98.3	98.5	98.6	98.6	98.6
0	89.0	91.1	93.5	95.5	98.2	100.1	100.3	100.3	100.3
-10	87.4	89.4	91.8	93.8	96.4	98.8	101.4	102.1	102.1
-20	85.7	87.7	90.0	92.0	94.5	96.9	99.4	102.3	102.5
-30	84.0	86.0	88.3	90.1	92.6	95.0	97.4	100.3	100.5
-40	82.3	84.2	86.4	88.3	90.7	93.0	95.4	98.2	98.4
-50	80.5	82.3	84.5	86.4	88.8	91.0	93.4	96.1	96.3

%N1 Adjustments for Engine Bleeds

BLEED CONFIGURATION	PRESSURE ALTITUDE (1000 FT)								
	0	5	10	15	20	25	30	35	37
ENGINE BLEED TO PACKS OFF	1.0	0.9	0.9	1.0	1.0	1.0	1.1	1.1	1.1
PACKS HIGH	-0.2	-0.2	-0.3	-0.3	-0.4	-0.4	-0.4	-0.4	-0.4
ENGINE ANTI-ICE ON	-0.6	-0.7	-0.7	-0.8	-0.9	-0.9	-0.9	-0.9	-0.9
ENGINE & WING ANTI-ICE ON	-1.5	-1.6	-1.7	-2.0	-2.3	-2.6	-2.8	-3.1	-3.1

Go-around %N1

Based on engine bleed for packs on (Auto), engine anti-ice on or off, and wing anti-ice off

REPORTED OAT		TAT	AIRPORT PRESSURE ALTITUDE (FT)									
°C	°F	(°C)	-1000	0	1000	2000	3000	4000	5000	6000	7000	8000
55	131	58	93.6	93.7	93.6	93.4						
50	122	53	94.9	95.1	95.0	94.9	94.9	94.4				
45	113	48	95.5	95.9	95.9	96.0	95.9	95.5	95.5	95.7	95.8	
40	104	43	96.0	96.7	96.6	96.6	96.5	96.5	96.5	96.5	96.5	96.5
35	95	38	96.5	97.3	97.3	97.2	97.2	97.2	97.3	97.3	97.3	97.2
30	86	33	96.5	97.7	97.9	97.9	97.9	97.9	98.0	98.0	98.1	98.0
25	77	28	95.7	96.9	97.4	98.2	98.7	98.7	98.8	98.8	99.0	98.8
20	68	23	94.9	96.1	96.6	97.4	98.0	98.6	99.2	99.5	99.6	99.5
15	59	18	94.1	95.2	95.8	96.5	97.1	97.8	98.4	99.0	99.7	100.1
10	50	13	93.2	94.4	95.0	95.7	96.3	96.9	97.5	98.2	98.8	99.3
5	41	8	92.4	93.6	94.1	94.8	95.5	96.1	96.7	97.3	97.9	98.4
0	32	3	91.6	92.7	93.3	94.0	94.6	95.2	95.8	96.4	97.0	97.5
-10	14	-8	89.8	91.0	91.5	92.3	92.8	93.4	94.0	94.6	95.2	95.7
-20	-4	-18	88.1	89.3	89.8	90.5	91.1	91.6	92.2	92.8	93.4	93.9
-30	-22	-28	86.3	87.5	88.0	88.7	89.2	89.8	90.4	91.0	91.5	92.0
-40	-40	-38	84.5	85.7	86.2	86.8	87.4	87.9	88.5	89.1	89.6	90.1
-50	-58	-48	82.7	83.8	84.3	85.0	85.5	86.0	86.6	87.2	87.7	88.1

%N1 Adjustments for Engine Bleeds

BLEED CONFIGURATION	TAT (°C)	
	- 50	+ 60
ENGINE BLEED TO PACKS OFF	1.0	1.2
PACKS HIGH	- 0.2	- 0.3
WING A/I ALL ENGINES	- 1.3	- 1.7
WING A/I 1 ENGINE INOP	- 2.1	- 2.6

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Flight With Unreliable Airspeed / Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

Climb

Flaps Up, Set Max Climb Thrust

PRESSURE ALTITUDE (FT) (SPEED)		WEIGHT (1000 KG)				
		36	44	52	60	68
35000	**PITCH ATT**	**6.5**	**6.0**	**5.5**	**5.5**	**5.5**
(.73M)	V/S (FT/MIN)	2800	2000	1300	700	100
30000	**PITCH ATT**	**7.0**	**6.0**	**6.0**	**5.5**	**5.5**
(.73M)	V/S (FT/MIN)	3700	2800	2100	1500	1000
25000	**PITCH ATT**	**7.0**	**6.5**	**6.0**	**6.0**	**6.0**
(280 KIAS)	V/S (FT/MIN)	3600	2800	2200	1700	1300
20000	**PITCH ATT**	**9.0**	**8.0**	**7.5**	**7.0**	**7.0**
(280 KIAS)	V/S (FT/MIN)	4400	3500	2800	2200	1800
15000	**PITCH ATT**	**10.5**	**9.5**	**9.0**	**8.5**	**8.0**
(280 KIAS)	V/S (FT/MIN)	5200	4100	3300	2700	2200
10000	**PITCH ATT**	**12.5**	**11.0**	**10.0**	**9.5**	**9.5**
(280 KIAS)	V/S (FT/MIN)	5900	4700	3900	3200	2700
5000	**PITCH ATT**	**14.5**	**13.0**	**11.5**	**11.0**	**10.5**
(280 KIAS)	V/S (FT/MIN)	6700	5400	4400	3700	3100
SEA LEVEL	**PITCH ATT**	**17.0**	**15.0**	**13.5**	**12.5**	**11.5**
(280 KIAS)	V/S (FT/MIN)	7400	6000	4900	4200	3500

Cruise

Flaps Up, %N1 for Level Flight

PRESSURE ALTITUDE (FT) (SPEED)		WEIGHT (1000 KG)				
		36	44	52	60	68
30000	**PITCH ATT**	**2.0**	**2.5**	**3.0**	**3.5**	**4.0**
(.73M)	%N1	79	81	83	85	87
10000	**PITCH ATT**	**2.0**	**2.5**	**3.5**	**4.0**	**4.5**
(280 KIAS)	%N1	66	67	69	70	72

Descent

Flaps Up, Set Idle Thrust

PRESSURE ALTITUDE (FT) (SPEED)		WEIGHT (1000 KG)				
		36	44	52	60	68
30000	**PITCH ATT**	**-3.5**	**-2.0**	**-1.5**	**-0.5**	**0.0**
(.73M)	V/S (FT/MIN)	-4100	-3600	-3300	-3100	-3100
20000	**PITCH ATT**	**-2.0**	**-1.0**	**0.0**	**0.5**	**1.5**
(280 KIAS)	V/S (FT/MIN)	-2800	-2400	-2200	-2100	-2000
10000	**PITCH ATT**	**-2.5**	**-1.0**	**0.0**	**0.5**	**1.5**
(280 KIAS)	V/S (FT/MIN)	-2500	-2100	-2000	-1800	-1800

Holding

Flaps Up, %N1 for Level Flight

PRESSURE ALTITUDE (FT)		WEIGHT (1000 KG)				
		36	44	52	60	68
	PITCH ATT	**4.5**	**5.5**	**5.5**	**5.5**	**5.5**
10000	%N1	57	60	64	68	72
	KIAS	210	210	230	245	265

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Flight With Unreliable Airspeed/Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

Terminal Area

%N1 for Level Flight

FLAP POSITION (SPEED)		WEIGHT (1000 KG)				
		36	45	54	64	68
FLAPS UP (GEAR UP)	**PITCH ATT**	**4.4**	**5.7**	**7.1**	**8.5**	**9.2**
(210 KIAS)	%N1	55	58	62	65	68
FLAPS 1 (GEAR UP)	**PITCH ATT**	**4.6**	**6.2**	**7.8**	**9.3**	**10.1**
(190 KIAS)	%N1	56	60	64	67	70
FLAPS 5 (GEAR UP)	**PITCH ATT**	**4.8**	**6.7**	**8.5**	**10.3**	**11.2**
(170 KIAS)	%N1	55	59	64	69	72
FLAPS 15 (GEAR DOWN)	**PITCH ATT**	**5.1**	**7.0**	**9.4**	**11.7**	**12.9**
(150 KIAS)	%N1	62	66	71	72	75

Final Approach

Gear Down, %N1 for 3° Glideslope

FLAP POSITION		WEIGHT (1000 KG)				
		36	45	54	64	68
FLAPS 40	**PITCH ATT**	**0.3**	**0.4**	**0.5**	**0.6**	**0.7**
	%N1	56	63	68	72	74
	KIAS	120	134	146	158	163
FLAPS 30	**PITCH ATT**	**2.1**	**2.1**	**2.1**	**2.2**	**2.2**
	%N1	50	56	61	65	66
	KIAS	123	137	150	162	167
FLAPS 15	**PITCH ATT**	**3.9**	**3.9**	**3.9**	**3.9**	**3.9**
	%N1	46	51	56	59	60
	KIAS	135	151	166	179	185

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Long Range Cruise Maximum Operating Altitude

Max Cruise Thrust

ISA + 10°C and Below

WEIGHT (1000 KG)	OPTIMUM ALT (FT)	TAT (°C)	MARGIN TO INITIAL BUFFET 'G' (BANK ANGLE)				
			1.20 (33°)	1.25 (36°)	1.30 (39°)	1.40 (44°)	1.50 (48°)
68	29500	-7	32800	31900	31000	29400	27900
66	30100	-8	33400	32500	31700	30000	28500
64	30800	-10	34100	33200	32300	30700	29200
62	31500	-11	34700	33800	33000	31400	29900
60	32200	-13	35400	34500	33700	32100	30600
58	32900	-14	36100	35300	34400	32800	31400
56	33700	-16	36900	36000	35200	33600	32100
54	34400	-18	37000	36700	35900	34400	32900
52	35200	-19	37000	37000	36700	35200	33700
50	36100	-21	37000	37000	37000	36000	34500
48	36900	-21	37000	37000	37000	36800	35400
46	37000	-21	37000	37000	37000	37000	36300
44	37000	-21	37000	37000	37000	37000	37000
42	37000	-21	37000	37000	37000	37000	37000

ISA + 15°C

WEIGHT (1000 KG)	OPTIMUM ALT (FT)	TAT (°C)	MARGIN TO INITIAL BUFFET 'G' (BANK ANGLE)				
			1.20 (33°)	1.25 (36°)	1.30 (39°)	1.40 (44°)	1.50 (48°)
68	29500	-1	32400*	31900	31000	29400	27900
66	30100	-3	33200*	32500	31700	30000	28500
64	30800	-4	34000*	33200	32300	30700	29200
62	31500	-6	34700	33800	33000	31400	29900
60	32200	-7	35400	34500	33700	32100	30600
58	32900	-9	36100	35300	34400	32800	31400
56	33700	-10	36900	36000	35200	33600	32100
54	34400	-12	37000	36700	35900	34400	32900
52	35200	-14	37000	37000	36700	35200	33700
50	36100	-16	37000	37000	37000	36000	34500
48	36900	-16	37000	37000	37000	36800	35400
46	37000	-16	37000	37000	37000	37000	36300
44	37000	-16	37000	37000	37000	37000	37000
42	37000	-16	37000	37000	37000	37000	37000

***Denotes altitude thrust limited in level flight, 100 fpm residual rate of climb.**

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Long Range Cruise Maximum Operating Altitude
Max Cruise Thrust
ISA + 20°C

WEIGHT (1000 KG)	OPTIMUM ALT (FT)	TAT (°C)	MARGIN TO INITIAL BUFFET 'G' (BANK ANGLE)				
			1.20 (33°)	1.25 (36°)	1.30 (39°)	1.40 (44°)	1.50 (48°)
68	29500	4	31500*	31500*	31000	29400	27900
66	30100	3	32400*	32400*	31700	30000	28500
64	30800	1	33300*	33200	32300	30700	29200
62	31500	0	34100*	33800	33000	31400	29900
60	32200	-2	35000*	34500	33700	32100	30600
58	32900	-3	35900*	35300	34400	32800	31400
56	33700	-5	36600*	36000	35200	33600	32100
54	34400	-7	37000	36700	35900	34400	32900
52	35200	-8	37000	37000	36700	35200	33700
50	36100	-10	37000	37000	37000	36000	34500
48	36900	-10	37000	37000	37000	36800	35400
46	37000	-10	37000	37000	37000	37000	36300
44	37000	-10	37000	37000	37000	37000	37000
42	37000	-10	37000	37000	37000	37000	37000

***Denotes altitude thrust limited in level flight, 100 fpm residual rate of climb.**

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Long Range Cruise Control

WEIGHT (1000 KG)		PRESSURE ALTITUDE (1000 FT)								
		21	23	25	27	29	31	33	35	37
66	%N1	82.7	83.8	84.7	85.6	86.6	88.0	90.2		
	MACH	.713	.730	.742	.745	.745	.745	.745		
	KIAS	324	319	311	300	287	275	263		
	FF/ENG	1610	1581	1547	1498	1457	1433	1437		
62	%N1	81.5	82.6	83.6	84.6	85.4	86.5	88.2	90.9	
	MACH	.698	.718	.734	.744	.745	.745	.745	.744	
	KIAS	317	313	308	299	287	275	263	251	
	FF/ENG	1525	1502	1473	1439	1391	1357	1341	1363	
58	%N1	80.2	81.3	82.4	83.4	84.3	85.2	86.4	88.3	92.2
	MACH	.682	.703	.722	.737	.744	.745	.745	.745	.742
	KIAS	309	306	302	296	287	275	263	251	239
	FF/ENG	1440	1420	1397	1369	1331	1289	1261	1253	1301
54	%N1	78.8	79.9	81.1	82.1	83.2	84.0	85.0	86.3	88.8
	MACH	.665	.685	.707	.725	.740	.745	.745	.745	.745
	KIAS	301	298	295	291	285	275	263	251	240
	FF/ENG	1356	1336	1317	1294	1267	1228	1191	1168	1175
50	%N1	77.2	78.4	79.6	80.7	81.8	82.8	83.6	84.6	86.5
	MACH	.646	.667	.688	.709	.728	.741	.745	.745	.745
	KIAS	292	289	287	284	280	274	263	251	240
	FF/ENG	1270	1253	1233	1216	1194	1167	1130	1098	1085
46	%N1	75.5	76.7	77.9	79.1	80.3	81.3	82.3	83.2	84.6
	MACH	.625	.647	.668	.689	.710	.729	.742	.745	.745
	KIAS	282	280	278	276	273	269	262	251	240
	FF/ENG	1184	1168	1151	1132	1117	1096	1071	1035	1012
42	%N1	73.6	74.9	76.1	77.4	78.5	79.7	80.8	81.8	83.0
	MACH	.603	.624	.646	.667	.688	.710	.729	.743	.745
	KIAS	271	270	268	266	264	261	257	250	240
	FF/ENG	1098	1084	1068	1050	1034	1020	1000	978	949
38	%N1	71.6	72.8	74.1	75.4	76.6	77.8	79.0	80.1	81.5
	MACH	.579	.599	.621	.643	.664	.686	.709	.728	.742
	KIAS	260	259	257	256	254	252	249	245	239
	FF/ENG	1012	1000	984	967	953	939	925	908	892

Shaded area approximates optimum altitude.

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Long Range Cruise Enroute Fuel and Time - Low Altitudes

Ground to Air Miles Conversion

AIR DISTANCE (NM)					GROUND DISTANCE (NM)	AIR DISTANCE (NM)				
HEADWIND COMPONENT (KTS)						TAILWIND COMPONENT (KTS)				
100	80	60	40	20		20	40	60	80	100
279	259	241	226	212	200	191	182	174	167	161
421	390	362	339	318	300	286	273	262	251	242
563	521	484	452	425	400	382	365	349	336	323
706	653	606	566	531	500	477	456	437	420	404
849	785	729	680	638	600	573	548	525	504	485
992	917	851	794	744	700	669	639	612	588	566
1136	1050	973	908	851	800	764	731	700	672	646
1280	1183	1096	1022	958	900	859	822	787	755	727
1425	1316	1219	1136	1064	1000	955	913	874	839	807
1570	1449	1342	1251	1171	1100	1050	1004	961	922	888
1716	1583	1465	1365	1278	1200	1145	1095	1048	1006	968
1862	1718	1589	1480	1385	1300	1241	1186	1135	1090	1049
2009	1852	1713	1595	1492	1400	1336	1277	1223	1173	1129
2156	1987	1837	1709	1599	1500	1431	1368	1310	1257	1209
2304	2122	1961	1824	1706	1600	1527	1459	1397	1340	1289
2452	2258	2085	1939	1813	1700	1622	1550	1484	1423	1369
2601	2394	2210	2055	1920	1800	1717	1641	1571	1507	1449
2750	2530	2335	2170	2028	1900	1813	1732	1658	1590	1529
2900	2667	2460	2286	2135	2000	1908	1823	1745	1673	1609

Reference Fuel and Time Required at Check Point

AIR DIST (NM)	PRESSURE ALTITUDE (1000 FT)									
	10		14		20		24		28	
	FUEL (1000 KG)	TIME (HR:MIN)	FUEL (1000 KG)	TIME (HR:MIN)	FUEL (1000 KG)	TIME (HR:MIN)	FUEL (1000 KG)	TIME (HR:MIN)	FUEL (1000 KG)	TIME (HR:MIN)
200	1.5	0:38	1.4	0:37	1.1	0:36	1.0	0:35	0.9	0:35
300	2.3	0:56	2.1	0:54	1.8	0:51	1.6	0:50	1.5	0:49
400	3.1	1:14	2.8	1:11	2.4	1:07	2.2	1:05	2.0	1:03
500	3.9	1:32	3.5	1:28	3.1	1:23	2.8	1:20	2.6	1:17
600	4.6	1:50	4.2	1:45	3.7	1:38	3.4	1:35	3.1	1:31
700	5.4	2:08	5.0	2:02	4.3	1:54	4.0	1:50	3.7	1:46
800	6.2	2:26	5.7	2:19	5.0	2:10	4.6	2:05	4.2	2:00
900	6.9	2:45	6.4	2:37	5.6	2:26	5.2	2:20	4.8	2:15
1000	7.7	3:03	7.1	2:54	6.2	2:42	5.7	2:35	5.3	2:29
1100	8.4	3:22	7.8	3:12	6.8	2:58	6.3	2:51	5.8	2:44
1200	9.2	3:41	8.5	3:30	7.4	3:15	6.9	3:06	6.4	2:58
1300	9.9	4:00	9.1	3:48	8.0	3:31	7.5	3:21	6.9	3:13
1400	10.7	4:19	9.8	4:06	8.7	3:48	8.0	3:37	7.4	3:28
1500	11.4	4:38	10.5	4:24	9.3	4:04	8.6	3:53	7.9	3:42
1600	12.1	4:57	11.2	4:42	9.9	4:21	9.1	4:08	8.5	3:57
1700	12.8	5:17	11.8	5:01	10.5	4:38	9.7	4:24	9.0	4:12
1800	13.6	5:36	12.5	5:19	11.0	4:55	10.2	4:40	9.5	4:27
1900	14.3	5:56	13.2	5:38	11.6	5:12	10.8	4:56	10.0	4:42
2000	15.0	6:16	13.8	5:57	12.2	5:29	11.3	5:12	10.5	4:57

Fuel Required Adjustment (1000 KG)

REFERENCE FUEL REQUIRED (1000 KG)	WEIGHT AT CHECK POINT (1000 KG)				
	35	45	50	55	65
5	-0.6	-0.2	0.0	0.2	0.8
10	-1.1	-0.4	0.0	0.5	1.5
15	-1.7	-0.6	0.0	0.7	2.3

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Long Range Cruise Enroute Fuel and Time - High Altitudes

Ground to Air Miles Conversion

AIR DISTANCE (NM) HEADWIND COMPONENT (KTS)					GROUND DISTANCE (NM)	AIR DISTANCE (NM) TAILWIND COMPONENT (KTS)				
100	80	60	40	20		20	40	60	80	100
269	252	236	223	211	200	191	182	174	167	161
530	498	468	443	421	400	382	365	349	336	323
791	744	702	664	631	600	573	548	525	504	485
1052	991	935	885	841	800	764	731	700	672	647
1315	1238	1168	1106	1051	1000	955	914	875	840	809
1578	1486	1402	1328	1261	1200	1146	1096	1051	1009	971
1842	1734	1635	1549	1471	1400	1337	1279	1226	1177	1133
2107	1983	1870	1771	1682	1600	1529	1463	1402	1347	1296
2372	2232	2105	1993	1892	1800	1720	1646	1578	1515	1458
2638	2482	2341	2215	2103	2000	1911	1829	1753	1683	1620
2905	2732	2575	2437	2313	2200	2102	2012	1928	1852	1782
3172	2983	2811	2660	2524	2400	2293	2195	2103	2020	1944
3441	3235	3047	2882	2735	2600	2485	2377	2279	2188	2106
3710	3487	3285	3106	2946	2800	2676	2560	2454	2356	2268
3981	3740	3522	3329	3157	3000	2867	2743	2629	2525	2430

Reference Fuel and Time Required at Check Point

AIR DIST (NM)	PRESSURE ALTITUDE (1000 FT) 29		31		33		35		37	
	FUEL (1000 KG)	TIME (HR:MIN)	FUEL (1000 KG)	TIME (HR:MIN)	FUEL (1000 KG)	TIME (HR:MIN)	FUEL (1000 KG)	TIME (HR:MIN)	FUEL (1000 KG)	TIME (HR:MIN)
200	0.9	0:35	0.8	0:34	0.8	0:35	0.8	0:35	0.7	0:35
400	2.0	1:03	1.9	1:02	1.8	1:02	1.8	1:03	1.7	1:03
600	3.1	1:31	2.9	1:30	2.8	1:30	2.8	1:31	2.7	1:31
800	4.1	1:59	4.0	1:58	3.9	1:58	3.7	1:59	3.7	1:59
1000	5.2	2:28	5.0	2:26	4.8	2:25	4.7	2:26	4.6	2:27
1200	6.2	2:57	6.0	2:54	5.8	2:53	5.7	2:54	5.6	2:55
1400	7.3	3:26	7.0	3:23	6.8	3:21	6.6	3:22	6.5	3:23
1600	8.3	3:55	8.0	3:51	7.8	3:49	7.5	3:50	7.4	3:51
1800	9.3	4:24	9.0	4:20	8.7	4:18	8.5	4:18	8.3	4:19
2000	10.3	4:54	10.0	4:49	9.6	4:46	9.4	4:46	9.2	4:47
2200	11.3	5:24	10.9	5:18	10.6	5:15	10.3	5:14	10.1	5:15
2400	12.3	5:55	11.9	5:48	11.5	5:43	11.2	5:43	10.9	5:43
2600	13.2	6:25	12.8	6:18	12.4	6:12	12.0	6:11	11.8	6:12
2800	14.2	6:56	13.7	6:48	13.3	6:42	12.9	6:40	12.6	6:40
3000	15.1	7:27	14.6	7:18	14.2	7:11	13.8	7:08	13.5	7:08

Fuel Required Adjustment (1000 KG)

REFERENCE FUEL REQUIRED (1000 KG)	WEIGHT AT CHECK POINT (1000 KG)				
	35	45	50	55	65
4	-0.6	-0.2	0.0	0.5	2.5
8	-1.2	-0.4	0.0	0.9	4.3
12	-1.8	-0.6	0.0	1.2	5.2
16	-2.4	-0.8	0.0	1.4	5.3

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Long Range Cruise Wind-Altitude Trade

PRESSURE ALTITUDE (1000 FT)	CRUISE WEIGHT (1000 KG)																
	68	66	64	62	60	58	56	54	52	50	48	46	44	42	40	38	36
37								14	8	1	0	0	0	0	0	0	0
35					9	5	2	0	0	2	6	8	10	12	13	14	14
33		9	5	2	1	0	1	3	7	12	17	20	23	26	27	29	30
31	2	1	0	1	3	4	9	14	19	25	31	35	38	41	43	44	45
29	1	2	5	8	12	17	22	28	33	39	45	50	54	57	59	60	61
27	8	12	15	17	25	30	35	41	48	55	61	66	69	71	74	75	77
25	19	24	28	34	39	44	51	57	63	70	78	82	86	89	92	94	96
23	32	37	42	48	53	59	65	72	79	86	94	99	103	106	109	112	113

The above wind factor tables are for calculation of wind required to maintain present range capability at new pressure altitude, i.e., break-even wind.

Method:

1. **Read wind factors for present and new altitudes from table.**
2. **Determine difference (new altitude wind factor minus present altitude wind factor); This difference may be negative or positive.**
3. **Break-even wind at new altitude is present altitude wind plus difference from step 2.**

Descent

Based on .74/250 descent speed

PRESSURE ALT (1000 FT)	21	23	25	27	29	31	33	35	37
DISTANCE (NM)	66	72	78	85	91	97	103	109	114
TIME (MINUTES)	15	16	18	18	19	20	21	22	23

Holding

Flaps Up

WEIGHT (1000 KG)		PRESSURE ALTITUDE (FT)								
		1500	5000	10000	15000	20000	25000	30000	35000	37000
66	%N1	63.7	66.7	70.8	74.7	78.5	82.3	86.1		
	KIAS	255	256	259	261	263	265	263		
	FF/ENG	1500	1480	1460	1440	1420	1420	1420		
62	%N1	62.1	64.9	69.2	73.2	77.0	80.8	84.6	90.6	
	KIAS	247	248	250	253	255	257	256	249	
	FF/ENG	1410	1390	1370	1350	1340	1330	1330	1410	
58	%N1	60.5	63.1	67.4	71.4	75.4	79.2	83.0	87.5	92.3
	KIAS	239	240	241	244	246	248	248	243	239
	FF/ENG	1330	1310	1290	1270	1250	1250	1240	1260	1370
54	%N1	58.9	61.3	65.4	69.6	73.6	77.5	81.3	85.3	88.2
	KIAS	231	231	232	235	237	239	241	238	234
	FF/ENG	1260	1230	1210	1190	1170	1160	1150	1160	1200
50	%N1	57.2	59.5	63.3	67.6	71.7	75.6	79.5	83.3	85.5
	KIAS	224	223	223	225	227	230	232	230	228
	FF/ENG	1180	1150	1130	1110	1090	1080	1070	1070	1080
46	%N1	55.5	57.7	61.1	65.4	69.6	73.6	77.5	81.4	83.3
	KIAS	216	214	214	215	218	220	222	222	221
	FF/ENG	1110	1070	1050	1030	1010	1000	980	980	990
42	%N1	53.7	56.0	59.3	63.3	67.5	71.5	75.4	79.3	81.2
	KIAS	210	210	210	210	210	210	212	213	212
	FF/ENG	1040	1010	990	960	930	920	900	900	900
38	%N1	52.4	54.7	57.9	61.5	65.7	69.7	73.5	77.4	79.3
	KIAS	210	210	210	210	210	210	210	210	210
	FF/ENG	990	960	940	900	880	860	840	830	840

This table includes 5% additional fuel for holding in a racetrack pattern.

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ADVISORY INFORMATION

Autobrake Landing Distance

Reference Landing Distance (FT)

AUTOBRAKE SETTING	APPROACH SPEED (KIAS)						
	100	110	120	130	140	150	160
1	4500	5300	6100	6930	7850	8850	9950
2	3800	4440	5080	5730	6450	7240	8100
3	3200	3680	4150	4650	5220	5840	6500
MAX	2500	2760	3010	3280	3600	4000	4500

Actual (unfactored) distances are shown.
Landing distance required includes 1000 ft of air distance.
The distances are valid with or without reverse thrust and for any flap.
If entering with ground speed, ignore adjustments.

Landing Distance Adjustments (FT)

AUTOBRAKE SETTING	ALTITUDE ADJUSTMENT PER 1000 FT ABOVE SEA LEVEL	TAILWIND ADJUSTMENT PER 10 KTS	TEMP ADJUSTMENT PER 10°C ABOVE ISA
1	250	1200	350
2	200	950	300
3	140	700	200
MAX	80	400	150

Slippery Runway Landing Distance

Reference Landing Distance (FT)

BRAKING CONFIGURATION	REPORTED BRAKING ACTION			
	DRY	GOOD	MEDIUM	POOR
MAX MANUAL BRAKING	2660	3660	4825	6055
AUTOBRAKE SETTING 2	4840	5330	5695	6400
AUTOBRAKE SETTING 3	3380	4100	4890	6055
MAX AUTOBRAKE SETTING	2770	3660	4825	6055

Landing Distance Adjustments (FT)

CONDITIONS		ADJUSTMENTS			
WEIGHT	PER 5000 KG BELOW 52,000 KG	-150	-245	-375	-510
	PER 5000 KG ABOVE 52,000 KG	490	445	475	560
AIRPORT PRESSURE ALTITUDE	PER 1000 FT ABOVE SEA LEVEL	150	150	165	195
WIND	PER 10 KTS HEADWIND	-100	-165	-260	-375
	PER 10 KTS TAILWIND	885	935	1085	1540
APPROACH SPEED	PER 10 KTS ABOVE VREF	540	490	460	460
SLOPE	PER 1% DOWNHILL SLOPE	35	80	215	510
	PER 1% UPHILL SLOPE	-35	-35	-130	-295
REVERSE THRUST	1 REVERSER INOPERATIVE	35	165	445	885
	2 REVERSERS INOPERATIVE	130	590	1805	4775
	MAXIMUM REVERSE THRUST	0	0	-35	-115

Actual (unfactored) distances are shown.
Landing distance required includes 1000 ft of air distance.
Based on flaps 40, VREF40 approach speed.
Includes 2 engine detent reverse thrust.

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ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

Dry Runway

LANDING CONFIGURATION	APPROACH SPEED	LANDING DISTANCE AND ADJUSTMENTS (FT)		
		REFERENCE DISTANCE FOR 48000 KG LANDING WEIGHT	WT ADJ PER 2000 KG ABOVE/BELOW 48000 KG	ALTITUDE ADJ PER 1000 FT ABOVE S.L.
ONE ENGINE INOPERATIVE	VREF15	2900	200 / -90	190
ANTI-SKID INOPERATIVE	VREF40	3920	110/-110	90
HYDRAULIC SYSTEM A INOPERATIVE (FLAPS 15)	VREF15	3450	150 / -100	190
HYDRAULIC SYSTEM A INOPERATIVE (FLAPS 30)	VREF30	3280	100/-90	130
HYDRAULIC SYSTEM A INOPERATIVE (FLAPS 40)	VREF40	3190	90/-90	130
HYDRAULIC SYSTEM B INOPERATIVE (FLAPS 15)	VREF15	3200	210 / -100	200
HYDRAULIC SYSTEMS A & B INOPERATIVE (MANUAL REVERSION)	VREF15	5120	210 / -170	260
LEADING EDGE ASYMMETRY OR FAILURE	VREF15+5	3030	200 / -80	200
TRAILING EDGE FAILURE (FLAPS LESS THAN 1)	VREF40+40	3480	250 / -120	250
TRAILING EDGE FAILURE (FLAPS GREATER THAN 1 AND LESS THAN 15)	VREF40+30	3200	210 / -110	210
FLAPS UP	VREF40+55	3950	300 / -160	340
JAMMED STABILIZER	VREF15	2820	180 / -80	160

Actual (unfactored) distances are shown
Includes distances from 50 ft above threshold (1000 ft of air distance)
Assumes maximum manual braking and maximum reverse thrust when available on operating engines.

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ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

Good Reported Braking Action

LANDING CONFIGURATION	APPROACH SPEED	LANDING DISTANCE AND ADJUSTMENTS (FT)		
		REFERENCE DISTANCE FOR 48000 KG LANDING WEIGHT	WT ADJ PER 2000 KG ABOVE/BELOW 48000 KG	ALTITUDE ADJ PER 1000 FT ABOVE S.L.
ONE ENGINE INOPERATIVE	VREF15	4100	140 / -130	210
ANTI-SKID INOPERATIVE	VREF40	4810	160/-150	125
HYDRAULIC SYSTEM A INOPERATIVE (FLAPS 15)	VREF15	4830	170 / -160	250
HYDRAULIC SYSTEM A INOPERATIVE (FLAPS 30)	VREF30	4460	150/-150	180
HYDRAULIC SYSTEM A INOPERATIVE (FLAPS 40)	VREF40	4270	140/-130	180
HYDRAULIC SYSTEM B INOPERATIVE (FLAPS 15)	VREF15	4050	140 / -120	200
HYDRAULIC SYSTEMS A & B INOPERATIVE (MANUAL REVERSION)	VREF15	5200	190 / -170	260
LEADING EDGE ASYMMETRY OR FAILURE	VREF15+5	4200	140 / -130	220
TRAILING EDGE FAILURE (FLAPS LESS THAN 1)	VREF40+40	4550	130 / -120	250
TRAILING EDGE FAILURE (FLAPS GREATER THAN 1 AND LESS THAN 15)	VREF40+30	4200	120 / -120	220
FLAPS UP	VREF40+55	5100	150 / -140	320
JAMMED STABILIZER	VREF15	3880	130 / -120	180

Actual (unfactored) distances are shown
Includes distances from 50 ft above threshold (1000 ft of air distance)
Assumes maximum manual braking and maximum reverse thrust when available on operating engines.

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ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

Medium Reported Braking Action

LANDING CONFIGURATION	APPROACH SPEED	LANDING DISTANCE AND ADJUSTMENTS (FT)		
		REFERENCE DISTANCE FOR 48000 KG LANDING WEIGHT	WT ADJ PER 2000 KG ABOVE/BELOW 48000 KG	ALTITUDE ADJ PER 1000 FT ABOVE S.L.
ONE ENGINE INOPERATIVE	VREF15	5700	220 / -210	300
ANTI-SKID INOPERATIVE	VREF40	5860	210/-200	175
HYDRAULIC SYSTEM A INOPERATIVE (FLAPS 15)	VREF15	6300	260 / -230	330
HYDRAULIC SYSTEM A INOPERATIVE (FLAPS 30)	VREF30	5680	220/-210	240
HYDRAULIC SYSTEM A INOPERATIVE (FLAPS 40)	VREF40	5380	200/-190	230
HYDRAULIC SYSTEM B INOPERATIVE (FLAPS 15)	VREF15	5430	200 / -190	280
HYDRAULIC SYSTEMS A & B INOPERATIVE (MANUAL REVERSION)	VREF15	6730	270 / -250	340
LEADING EDGE ASYMMETRY OR FAILURE	VREF15+5	5600	220 / -200	310
TRAILING EDGE FAILURE (FLAPS LESS THAN 1)	VREF40+40	6050	210 / -190	350
TRAILING EDGE FAILURE (FLAPS GREATER THAN 1 AND LESS THAN 15)	VREF40+30	5700	200 / -180	310
FLAPS UP	VREF40+55	6850	230 / -220	450
JAMMED STABILIZER	VREF15	5100	200 / -180	240

Actual (unfactored) distances are shown
Includes distances from 50 ft above threshold (1000 ft of air distance)
Assumes maximum manual braking and maximum reverse thrust when available on operating engines.

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ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

Poor Reported Braking Action

LANDING CONFIGURATION	APPROACH SPEED	LANDING DISTANCE AND ADJUSTMENTS (FT)		
		REFERENCE DISTANCE FOR 48000 KG LANDING WEIGHT	WT ADJ PER 2000 KG ABOVE/BELOW 48000 KG	ALTITUDE ADJ PER 1000 FT ABOVE S.L.
ONE ENGINE INOPERATIVE	VREF15	7550	320 / -290	410
ANTI-SKID INOPERATIVE	VREF40	7440	300/-280	240
HYDRAULIC SYSTEM A INOPERATIVE (FLAPS 15)	VREF15	7700	340 / -310	420
HYDRAULIC SYSTEM A INOPERATIVE (FLAPS 30)	VREF30	6900	280/-270	290
HYDRAULIC SYSTEM A INOPERATIVE (FLAPS 40)	VREF40	6490	260/-250	280
HYDRAULIC SYSTEM B INOPERATIVE (FLAPS 15)	VREF15	6830	280 / -260	360
HYDRAULIC SYSTEMS A & B INOPERATIVE (MANUAL REVERSION)	VREF15	8200	270 / -320	400
LEADING EDGE ASYMMETRY OR FAILURE	VREF15+5	7000	300 / -270	390
TRAILING EDGE FAILURE (FLAPS LESS THAN 1)	VREF40+40	7650	290 / -270	460
TRAILING EDGE FAILURE (FLAPS GREATER THAN 1 AND LESS THAN 15)	VREF40+30	7100	270 / -250	400
FLAPS UP	VREF40+55	8600	320 / -310	580
JAMMED STABILIZER	VREF15	6420	270 / -240	310

Actual (unfactored) distances are shown
Includes distances from 50 ft above threshold (1000 ft of air distance)
Assumes maximum manual braking and maximum reverse thrust when available on operating engines.

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ADVISORY INFORMATION

Brake Cooling Schedule

Reference Brake Energy Per Brake (Millions of Foot Pounds)

		WIND CORRECTED BRAKES ON SPEED (KIAS)*														
		60			80			100			120			140		
WEIGHT	OAT	PRESS ALT			PRESS ALT			PRESS ALT			PRESS ALT			PRESS ALT		
(1000 KG)	(°C)	0	2	4	0	2	4	0	2	4	0	2	4	0	2	4
65	-20	7.0	7.5	8.1	11.3	12.2	13.1	16.2	17.7	19.1	21.9	24.1	25.9			
	0	7.5	8.2	8.9	12.3	13.2	14.3	17.7	19.3	20.8	23.9	26.0	27.9			
	20	8.1	8.8	9.6	13.3	14.3	15.4	19.2	20.8	22.4	25.9	28.3	30.0			
	40	8.7	9.5	10.2	14.3	15.3	16.6	20.7	22.4	24.1	27.9	30.0	32.1			
60	-20	6.6	7.2	7.7	10.6	11.5	12.4	15.3	16.6	18.0	20.5	22.2	24.0	26.5		
	0	7.1	7.8	8.4	11.5	12.5	13.5	16.7	18.1	19.6	22.4	24.3	26.1	28.9		
	20	7.7	8.4	9.1	12.5	13.5	14.5	18.1	19.7	21.2	24.3	26.3	28.3	31.3		
	40	8.2	9.1	9.7	13.4	14.5	15.6	19.5	21.2	22.8	26.2	28.3	30.4	28.9		
55	-20	6.1	6.6	7.1	9.8	10.6	11.5	14.1	15.3	16.5	19.1	20.5	22.1	24.5	26.4	
	0	6.6	7.2	7.7	10.7	11.6	12.5	15.4	16.7	18.0	20.7	22.4	24.1	26.7	28.8	
	20	7.2	7.7	8.3	11.6	12.5	13.5	16.6	18.1	19.5	22.4	24.3	26.1	28.9	31.2	
	40	7.7	8.3	8.9	12.5	13.5	14.5	17.9	19.5	21.0	24.1	26.2	28.2	31.1	33.6	
50	-20	5.7	6.2	6.7	9.1	9.8	10.6	13.2	14.3	15.3	17.6	19.0	20.4	22.6	24.5	26.4
	0	6.2	6.7	7.3	9.9	10.7	11.6	14.4	15.6	16.7	19.2	20.7	22.2	24.7	26.7	28.8
	20	6.7	7.3	7.9	10.6	11.6	12.5	15.5	16.8	18.2	20.7	22.4	24.1	26.7	28.9	31.2
	40	7.2	7.8	8.4	11.5	12.5	13.5	16.7	18.1	19.6	22.3	24.1	26.0	28.8	31.1	33.5
45	-20	5.5	6.0	6.3	8.5	9.2	9.9	12.2	13.1	14.1	16.0	17.4	18.8	20.5	22.2	24.0
	0	5.9	6.4	6.9	9.3	10.0	10.8	13.2	14.3	15.3	17.4	18.9	20.4	22.4	24.3	26.2
	20	6.4	6.9	7.4	10.1	10.9	11.7	14.3	15.4	16.6	18.9	20.5	22.1	24.3	26.3	28.3
	40	6.8	7.4	7.9	10.8	11.7	12.6	15.3	16.5	17.8	20.4	22.0	23.8	26.2	28.3	30.5
40	-20	5.1	5.5	5.9	7.9	8.5	9.1	11.0	12.0	12.9	14.6	15.8	17.2	18.8	20.2	21.7
	0	5.5	5.9	6.4	8.6	9.3	10.0	12.0	13.0	14.0	15.9	17.3	18.7	20.5	22.1	23.8
	20	5.9	6.4	6.9	9.2	10.0	10.8	13.0	14.1	15.2	17.3	18.8	20.2	22.1	24.0	25.8
	40	6.3	6.8	7.4	9.9	10.8	11.6	14.0	15.1	16.3	18.6	20.2	21.7	23.8	25.8	27.8

*** To correct for wind, enter table with the brakes on speed minus one half the headwind or plus 1.5 times the tailwind.**
If ground speed is used for brakes on speed, ignore wind and enter table with sea level, 15°C.

Adjusted Brake Energy Per Brake (Millions of Foot Pounds)
No Reverse Thrust

		REFERENCE BRAKE ENERGY PER BRAKE (MILLIONS OF FOOT POUNDS)						
EVENT		5	10	15	20	25	30	35
RTO MAX MAN		5	10	15	20	25	30	35
LANDING	MAX MAN	3.4	7.4	11.9	16.5	21.1	25.8	30.5
	MAX AUTO	3.2	7.0	11.4	15.5	19.7	24.2	28.5
	AUTOBRAKE 3	3.1	6.7	10.8	14.6	18.2	22.2	25.9
	AUTOBRAKE 2	2.9	6.3	10.3	13.8	17.0	20.1	23.2
	AUTOBRAKE 1	2.9	6.2	10.0	13.4	16.5	19.7	22.6

Two Engine Detent Reverse Thrust

		REFERENCE BRAKE ENERGY PER BRAKE (MILLIONS OF FOOT POUNDS)						
EVENT		5	10	15	20	25	30	35
LANDING	MAX AUTO	2.6	5.6	8.9	12.4	16.0	19.4	22.8
	AUTOBRAKE 3	1.9	3.9	6.1	8.4	10.5	12.6	14.7
	AUTOBRAKE 1 & 2	1.9	3.7	5.8	7.8	9.6	11.3	12.8

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ADVISORY INFORMATION

Brake Cooling Schedule

Cooling Time (Minutes) - Category A Brakes

	ADJUSTED BRAKE ENERGY PER BRAKE (MILLIONS OF FOOT POUNDS)							
	8 & BELOW	10	12	14	16	18	18.5 TO 23.7	23.8 & ABOVE
INFLIGHT GEAR DOWN	NO SPECIAL PROCEDURE REQUIRED	2	3	4	5	6	CAUTION	FUSE PLUG MELT ZONE
GROUND		19	32	43	53	61		

Observe maximum quick turnaround limit.
Table shows energy per brake added by a single stop with all brakes operating. Energy is assumed to be equally distributed among the operating brakes. Total energy is the sum of residual energy plus energy added.
Add 1.0 million foot pounds for each taxi mile.
When in caution zone, wheel fuse plugs may melt. Delay takeoff and inspect after 30 minutes. If overheat occurs after takeoff, extend gear soon for at least 7 minutes.
When in fuse plug melt zone, clear runway immediately. Unless required, do not set parking brake. Do not approach gear or attempt to taxi for one hour. Tire, wheel and brake replacement may be required. If overheat occurs after takeoff, extend gear soon for at least 12 minutes.

Cooling Time (Minutes) - Category B Brakes

	ADJUSTED BRAKE ENERGY PER BRAKE (MILLIONS OF FOOT POUNDS)								
	8 & BELOW	10	12	14	16	18	20	21.0 TO 28.7	28.8 & ABOVE
INFLIGHT GEAR DOWN	NO SPECIAL PROCEDURE REQUIRED	2	3	4	5	6	7	CAUTION	FUSE PLUG MELT ZONE
GROUND		19	32	43	53	61	67		

Observe maximum quick turnaround limit.
Table shows energy per brake added by a single stop with all brakes operating. Energy is assumed to be equally distributed among the operating brakes. Total energy is the sum of residual energy plus energy added.
Add 1.0 million foot pounds for each taxi mile.
When in caution zone, wheel fuse plugs may melt. Delay takeoff and inspect after 30 minutes. If overheat occurs after takeoff, extend gear soon for at least 7 minutes.
When in fuse plug melt zone, clear runway immediately. Unless required, do not set parking brake. Do not approach gear or attempt to taxi for one hour. Tire, wheel and brake replacement may be required. If overheat occurs after takeoff, extend gear soon for at least 12 minutes.

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ENGINE INOP

Initial Max Continuous %N1

Based on .74M, engine bleed to packs auto and anti-ice off

TAT (°C)	PRESSURE ALTITUDE (1000 FT)					
	27	29	31	33	35	37
20	96.6	96.6	96.6	96.6	96.6	96.6
15	97.3	97.3	97.3	97.3	97.3	97.3
10	98.1	98.1	98.1	98.1	98.1	98.1
5	98.9	98.9	98.9	98.9	98.9	98.9
0	99.8	99.8	99.8	99.8	99.8	99.8
-5	99.5	100.7	100.7	100.7	100.7	100.7
-10	98.5	100.1	101.6	101.6	101.6	101.6
-15	97.6	99.2	100.8	102.5	102.5	102.5
-20	96.7	98.2	99.8	101.5	102.1	102.1
-25	95.7	97.2	98.8	100.4	101.0	101.1
-30	94.7	96.3	97.8	99.4	100.0	100.0
-40	92.8	94.3	95.8	97.4	97.9	98.0
-50	90.7	92.2	93.7	95.3	95.8	95.8

%N1 Adjustment for Engine Bleeds

BLEED CONFIGURATION	PRESSURE ALTITUDE (1000 FT)					
	27	29	31	33	35	37
ENGINE ANTI-ICE ON	-1.1	-1.1	-1.1	-1.1	-1.1	-1.1
ENGINE AND WING ANTI-ICE ON	-3.4	-3.5	-3.5	-3.7	-3.7	-3.8

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ENGINE INOP

Max Continuous %N1

33000 FT to 23000 FT Pressure Altitudes

Based on engine bleed to packs auto and anti-ice off

33000 FT PRESS ALT.		TAT (°C)										
KIAS	M	-50	-40	-30	-20	-10	0	5	10	15	20	25
200	.58	95.2	97.3	99.4	101.1	100.0	98.9	98.2				
220	.63	95.1	97.3	99.3	101.3	100.7	99.3	98.5	97.9	97.2		
240	.68	95.1	97.2	99.2	101.2	101.4	99.6	98.8	98.0			
260	.74	94.5	96.6	98.7	100.7	101.6	99.8	98.9	98.1	97.3		
31000 FT PRESS ALT.		TAT (°C)										
KIAS	M	-50	-40	-30	-20	-10	0	5	10	15	20	25
200	.55	94.7	96.8	98.9	100.8	99.8	98.7	98.1				
220	.61	94.6	96.7	98.7	100.7	100.3	99.1	98.4	97.8	97.1		
240	.66	94.4	96.5	98.5	100.5	101.0	99.4	98.7	97.9	97.2		
260	.71	94.1	96.2	98.2	100.2	101.6	99.8	98.9	98.1	97.3		
29000 FT PRESS ALT.		TAT (°C)										
KIAS	M	-50	-40	-30	-20	-10	0	5	10	15	20	25
200	.53	94.0	96.1	98.1	100.1	99.6	98.5	97.9	97.3			
220	.58	93.8	95.9	97.9	99.9	100.1	98.9	98.3	97.6	97.0		
240	.63	93.6	95.6	97.7	99.6	100.7	99.3	98.6	97.9	97.2		
260	.68	93.3	95.3	97.4	99.3	101.3	99.6	98.8	98.0	97.3	96.6	
27000 FT PRESS ALT.		TAT (°C)										
KIAS	M	-50	-40	-30	-20	-10	0	5	10	15	20	25
200	.51	93.1	95.2	97.2	99.2	99.4	98.3	97.8	97.2			
220	.56	92.9	94.9	96.9	98.9	99.8	98.7	98.1	97.5	96.9	96.3	
240	.60	92.6	94.6	96.7	98.6	100.3	99.1	98.4	97.8	97.2	96.6	
260	.65	92.1	94.2	96.2	98.1	100.1	99.4	98.7	97.9	97.2	96.6	
25000 FT PRESS ALT.		TAT (°C)										
KIAS	M	-50	-40	-30	-20	-10	0	5	10	15	20	25
200	.49	92.3	94.3	96.3	98.3	99.3	98.2	97.6	97.1	96.5		
220	.53	92.0	94.0	96.0	97.9	99.6	98.5	98.0	97.4	96.8	96.2	
240	.58	91.7	93.7	95.7	97.6	99.6	98.9	98.3	97.7	97.0	96.5	
260	.63	91.2	93.2	95.2	97.1	99.0	99.3	98.6	97.9	97.2	96.6	96.0
23000 FT PRESS ALT.		TAT (°C)										
KIAS	M	-50	-40	-30	-20	-10	0	5	10	15	20	25
200	.47	91.3	93.4	95.3	97.3	99.2	98.1	97.6	97.0	96.4	95.8	
220	.51	90.9	93.0	94.9	96.9	98.8	98.4	97.8	97.2	96.7	96.1	
240	.56	90.6	92.6	94.6	96.5	98.4	98.7	98.1	97.5	96.9	96.3	95.8
260	.60	90.3	92.3	94.3	96.2	98.1	99.1	98.4	97.8	97.2	96.6	96.0

Anti-Ice Adjustments

BLEED CONFIGURATION	PRESS ALT (1000 FT)					
	23	25	27	29	31	33
ENGINE ANTI-ICE ON	-1.1	-1.1	-1.1	-1.1	-1.1	-1.2
ENGINE AND WING ANTI-ICE ON	-3.3	-3.4	-3.4	-3.5	-3.5	-3.7

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ENGINE INOP

Max Continuous %N1

21000 FT to 14000 FT Pressure Altitudes

Based on engine bleed to packs auto and anti-ice off

21000 FT PRESS ALT		TAT (°C)											
KIAS	M	-50	-40	-30	-20	-10	0	5	10	15	20	25	30
200	.45	90.4	92.4	94.4	96.3	98.2	98.1	97.5	96.9	96.3	95.7		
220	.49	90.0	92.0	93.9	95.8	97.7	98.3	97.7	97.1	96.5	96.0	95.4	94.9
240	.54	89.6	91.6	93.6	95.5	97.3	98.6	98.0	97.4	96.8	96.2	95.7	
260	.60	89.3	91.3	93.2	95.1	97.0	98.8	98.3	97.7	97.1	96.5	95.9	95.4
20000 FT PRESS ALT.		TAT (°C)											
KIAS	M	-50	-40	-30	-20	-10	0	5	10	15	20	25	30
200	.44	89.9	91.9	93.9	95.8	97.7	98.0	97.5	96.8	96.2	95.6	95.1	
220	.48	89.5	91.5	93.4	95.3	97.2	98.2	97.7	97.1	96.5	95.9	95.4	94.9
240	.53	89.1	91.1	93.0	94.9	96.8	98.5	97.9	97.3	96.8	96.2	95.6	95.1
260	.57	88.8	90.8	92.7	94.6	96.4	98.2	98.2	97.6	97.0	96.4	95.9	95.4
18000 FT PRESS ALT.		TAT (°C)											
KIAS	M	-40	-30	-20	-10	0	5	10	15	20	25	30	35
200	.42	90.9	92.8	94.7	96.6	98.0	97.4	96.8	96.1	95.5	95.0		
220	.46	90.5	92.4	94.3	96.2	98.0	97.6	97.0	96.4	95.8	95.3	94.8	
240	.51	90.1	92.0	93.9	95.8	97.6	97.8	97.2	96.7	96.1	95.5	95.0	
260	.55	89.8	91.7	93.5	95.4	97.2	98.0	97.5	96.9	96.3	95.8	95.3	94.9
17000 FT PRESS ALT.		TAT (°C)											
KIAS	M	-40	-30	-20	-10	0	5	10	15	20	25	30	35
200	.42	90.4	92.3	94.2	96.0	97.8	97.4	96.7	96.1	95.5	94.9	94.4	
220	.46	90.0	91.9	93.8	95.6	97.4	97.6	97.0	96.4	95.8	95.2	94.7	
240	.50	89.6	91.5	93.4	95.2	97.0	97.8	97.2	96.6	96.0	95.5	95.0	94.6
260	.54	89.3	91.2	93.0	94.8	96.6	97.5	97.4	96.9	96.3	95.7	95.2	94.8
16000 FT PRESS ALT.		TAT (°C)											
KIAS	M	-40	-30	-20	-10	0	5	10	15	20	25	30	35
200	.41	89.9	91.8	93.6	95.5	97.3	97.3	96.7	96.1	95.5	94.9	94.4	
220	.45	89.5	91.4	93.3	95.1	96.9	97.5	96.9	96.3	95.7	95.2	94.7	
240	.49	89.1	91.0	92.9	94.7	96.5	97.4	97.1	96.6	96.0	95.4	94.9	94.5
260	.53	88.8	90.7	92.5	94.3	96.1	97.0	97.4	96.8	96.2	95.7	95.2	94.8
14000 FT PRESS ALT.		TAT (°C)											
KIAS	M	-30	-20	-10	0	5	10	15	20	25	30	35	40
200	.39	90.7	92.6	94.4	96.2	97.0	96.7	96.0	95.4	94.8	94.3	93.9	
220	.43	90.4	92.2	94.0	95.8	96.7	96.8	96.2	95.6	95.1	94.6	94.1	
240	.47	90.1	91.9	93.7	95.4	96.3	97.1	96.5	95.9	95.3	94.8	94.4	
260	.51	89.7	91.5	93.3	95.1	96.0	96.8	96.7	96.1	95.6	95.1	94.7	94.3

Anti-Ice Adjustments

BLEED CONFIGURATION	PRESS ALT (1000 FT)					
	14	16	17	18	20	21
ENGINE ANTI-ICE ON	-1.0	-1.0	-1.0	-1.0	-1.0	-1.0
ENGINE AND WING ANTI-ICE ON	-2.9	-3.0	-3.0	-3.1	-3.1	-3.2

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ENGINE INOP

Max Continuous %N1

12000 FT to 1000 FT Pressure Altitudes

Based on engine bleed to packs auto and anti-ice off

12000 FT PRESS ALT								TAT (°C)					
KIAS	M	-20	-10	0	5	10	15	20	25	30	35	40	45
200	.38	91.5	93.3	95.0	95.9	96.6	96.0	95.4	94.8	94.3	93.9		
220	.41	91.2	93.0	94.7	95.6	96.4	96.1	95.5	95.0	94.5	94.0	93.8	
240	.45	90.9	92.7	94.4	95.3	96.1	96.4	95.8	95.2	94.7	94.3	94.0	
260	.49	90.6	92.4	94.1	95.0	95.8	96.6	96.0	95.5	95.0	94.6	94.2	94.0
10000 FT PRESS ALT.								TAT (°C)					
KIAS	M	-20	-10	0	5	10	15	20	25	30	35	40	45
200	.36	90.4	92.2	93.9	94.8	95.6	96.0	95.3	94.8	94.3	93.8	93.6	
220	.40	90.1	91.9	93.6	94.5	95.3	96.1	95.5	94.9	94.4	93.9	93.7	
240	.43	89.9	91.6	93.3	94.2	95.0	95.9	95.7	95.1	94.6	94.2	93.9	93.7
260	.47	89.6	91.3	93.1	93.9	94.8	95.6	95.9	95.4	94.9	94.5	94.1	93.9
7000 FT PRESS ALT.								TAT (°C)					
KIAS	M	-10	0	5	10	15	20	25	30	35	40	45	50
200	.34	90.7	92.4	93.2	94.1	94.9	95.3	94.7	94.2	93.8	93.5	93.3	93.3
220	.38	90.4	92.1	93.0	93.8	94.6	95.4	94.8	94.3	93.9	93.6	93.5	93.4
240	.41	90.2	91.9	92.7	93.6	94.4	95.2	95.0	94.5	94.0	93.8	93.6	93.5
260	.45	90.0	91.7	92.5	93.4	94.2	95.0	95.2	94.7	94.3	94.0	93.8	93.7
5000 FT PRESS ALT.								TAT (°C)					
KIAS	M	-10	0	5	10	15	20	25	30	35	40	45	50
200	.33	89.7	91.4	92.2	93.0	93.8	94.6	94.7	94.2	93.8	93.5	93.3	93.3
220	.36	89.4	91.1	92.0	92.8	93.6	94.4	94.8	94.3	93.9	93.6	93.4	93.4
240	.40	89.2	90.9	91.7	92.5	93.4	94.2	94.9	94.4	94.0	93.7	93.5	93.5
260	.43	89.0	90.7	91.5	92.4	93.2	94.0	94.8	94.6	94.2	93.9	93.7	93.7
3000 FT PRESS ALT.								TAT (°C)					
KIAS	M	-10	0	5	10	15	20	25	30	35	40	45	50
200	.32	88.8	90.5	91.3	92.1	93.0	93.8	94.6	94.2	93.8	83.5	93.3	93.2
220	.35	88.6	90.3	91.1	91.9	92.7	93.5	94.3	94.3	93.9	93.6	93.4	93.3
240	.40	88.4	90.1	90.9	91.7	92.5	93.3	94.1	94.4	93.9	93.7	93.5	93.4
260	.43	88.3	89.9	90.7	91.5	92.3	93.1	93.9	94.5	94.1	93.8	93.6	93.6
1000 FT PRESS ALT.								TAT (°C)					
KIAS	M	-10	0	5	10	15	20	25	30	35	40	45	50
200	.31	88.0	89.6	90.4	91.3	92.1	92.9	93.6	94.1	93.8	93.5	93.3	93.2
220	.34	87.8	89.4	90.3	91.1	91.9	92.7	93.4	94.2	98.8	93.6	93.4	93.3
240	.37	87.6	89.3	90.1	90.9	91.7	92.5	93.3	94.0	93.9	93.6	93.5	93.4
260	.40	87.4	89.1	89.9	90.7	91.5	92.3	93.1	93.8	94.0	93.7	93.6	93.5

Anti-Ice Adjustments

BLEED CONFIGURATION	PRESS ALT (1000 FT)					
	1	3	5	7	10	12
ENGINE ANTI-ICE ON	-0.7	-0.7	-0.7	-0.8	-0.9	-0.9
ENGINE AND WING ANTI-ICE ON	-2.2	-2.3	-2.4	-2.5	-2.7	-2.8

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ENGINE INOP

MAX CONTINUOUS THRUST

Driftdown Speed/Level Off Altitude
100 ft/min residual rate of climb

WEIGHT (1000 KG)		OPTIMUM DRIFTDOWN SPEED (KIAS)	LEVEL OFF ALTITUDE (FT)		
START DRIFTDOWN	LEVEL OFF		ISA + 10°C & BELOW	ISA + 15°C	ISA + 20°C
68	65	242	15900	14500	13000
64	61	235	17900	16700	15300
60	57	228	19900	18900	17600
56	53	220	22000	21100	19900
52	50	213	24100	23300	22300
48	46	205	26300	25600	24700
44	42	196	28500	27900	27200
40	38	187	30700	30300	29700

Includes APU fuel burn.

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ENGINE INOP

MAX CONTINUOUS THRUST

Driftdown/LRC Cruise Range Capability

Ground to Air Miles Conversion

AIR DISTANCE (NM)					GROUND DISTANCE (NM)	AIR DISTANCE (NM)				
HEADWIND COMPONENT (KTS)						TAILWIND COMPONENT (KTS)				
100	80	60	40	20		20	40	60	80	100
140	130	121	113	106	100	95	90	85	81	78
286	263	244	227	213	200	189	179	169	161	154
430	395	366	341	319	300	283	268	254	242	230
571	526	488	454	426	400	377	357	339	323	308
711	656	608	567	532	500	472	447	424	404	386
850	784	728	680	637	600	567	537	510	486	464
987	912	848	792	743	700	662	627	596	568	542
1124	1040	967	904	849	800	756	717	682	650	621
1260	1166	1086	1016	955	900	851	808	768	733	700
1395	1293	1205	1128	1060	1000	946	898	855	815	779
1531	1420	1324	1240	1166	1100	1041	989	941	898	858
1667	1547	1442	1351	1271	1200	1136	1079	1027	980	937
1803	1674	1562	1463	1377	1300	1231	1169	1114	1063	1016
1940	1801	1681	1576	1483	1400	1326	1260	1200	1145	1095
2079	1930	1801	1688	1588	1500	1421	1350	1285	1227	1173
2218	2059	1921	1801	1694	1600	1516	1440	1371	1308	1251

Driftdown/Cruise Fuel and Time

AIR DIST (NM)	FUEL REQUIRED (1000 KG)								TIME (HR:MIN)
	WEIGHT AT START OF DRIFTDOWN (1000 KG)								
	35	40	45	50	55	60	65	70	
100	0.4	0.4	0.4	0.5	0.5	0.6	0.6	0.6	0:17
200	0.8	0.9	1.0	1.1	1.2	1.3	1.4	1.5	0:36
300	1.3	1.4	1.6	1.8	1.9	2.1	2.2	2.4	0:54
400	1.7	1.9	2.2	2.4	2.7	2.9	3.1	3.3	1:12
500	2.1	2.4	2.7	3.0	3.3	3.6	3.9	4.2	1:29
600	2.6	2.9	3.2	3.6	3.9	4.3	4.6	5.0	1:46
700	3.0	3.3	3.7	4.2	4.6	5.0	5.4	5.8	2:02
800	3.4	3.8	4.2	4.7	5.2	5.7	6.1	6.6	2:18
900	3.8	4.3	4.8	5.3	5.8	6.3	6.9	7.4	2:34
1000	4.2	4.7	5.3	5.9	6.4	7.0	7.6	8.2	2:50
1100	4.6	5.2	5.8	6.4	7.1	7.7	8.3	9.0	3:06
1200	5.0	5.6	6.3	7.0	7.7	8.3	9.0	9.8	3:22
1300	5.4	6.0	6.7	7.5	8.3	9.0	9.8	10.6	3:38
1400	5.8	6.5	7.2	8.0	8.9	9.6	10.5	11.4	3:54
1500	6.1	6.9	7.7	8.6	9.4	10.3	11.2	12.1	4:10
1600	6.5	7.3	8.2	9.1	10.0	10.9	11.9	12.9	4:27

Includes APU fuel burn.
Driftdown at optimum driftdown speed and cruise at Long Range Cruise speed.

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ENGINE INOP

MAX CONTINUOUS THRUST

Long Range Cruise Altitude Capability
100 ft/min residual rate of climb

WEIGHT (1000 KG)	PRESSURE ALTITUDE (FT)		
	ISA + 10°C & BELOW	ISA + 15°C	ISA + 20°C
68	9000	6800	4700
66	10400	8100	6000
64	11900	9500	7300
62	13300	11000	8700
60	14800	12500	10100
58	16300	13900	11600
56	17800	15500	13100
54	19200	17000	14600
52	20600	18600	16300
50	21900	20100	17900
48	23300	21600	19500
46	24700	23100	21200
44	26100	24700	22900
42	27400	26200	24700
40	28800	27700	26300
38	30200	29300	27900

With engine anti-ice on, decrease altitude capability by 1400ft.
With engine and wing anti-ice on, decrease altitude capability by 4800ft.

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ENGINE INOP

MAX CONTINUOUS THRUST

Long Range Cruise Control

WEIGHT (1000 KG)		PRESSURE ALTITUDE (1000 FT)								
		9	11	13	15	17	19	21	23	25
66	%N1	91.2	92.5	93.9						
	MACH	.559	.575	.591						
	KIAS	316	314	310						
	FF/ENG	3123	3104	3082						
62	%N1	89.7	91.0	92.3	93.8					
	MACH	.545	.563	.578	.594					
	KIAS	308	306	304	300					
	FF/ENG	2940	2926	2905	2880					
58	%N1	88.0	89.4	90.8	92.1	93.6				
	MACH	.530	.548	.565	.581	.597				
	KIAS	299	298	297	294	290				
	FF/ENG	2753	2744	2728	2703	2667				
54	%N1	86.3	87.7	89.1	90.4	91.8	93.4	95.4		
	MACH	.513	.531	.550	.567	.583	.599	.615		
	KIAS	290	289	288	286	284	280	277		
	FF/ENG	2565	2556	2546	2528	2492	2463	2461		
50	%N1	84.4	85.8	87.2	88.6	89.9	91.4	93.0	95.1	
	MACH	.495	.513	.532	.550	.568	.584	.600	.617	
	KIAS	279	279	279	278	276	273	270	267	
	FF/ENG	2376	2367	2359	2348	2321	2290	2273	2273	
46	%N1	82.3	83.7	85.1	86.6	88.0	89.4	90.8	92.4	94.7
	MACH	.476	.494	.512	.531	.550	.568	.584	.601	.617
	KIAS	268	268	268	268	267	265	262	259	256
	FF/ENG	2187	2178	2170	2163	2144	2122	2100	2086	2084
42	%N1	80.0	81.5	82.9	84.4	85.8	87.2	88.6	90.1	91.7
	MACH	.456	.473	.491	.510	.529	.548	.566	.583	.600
	KIAS	257	257	257	257	256	256	254	251	248
	FF/ENG	2000	1991	1982	1974	1963	1948	1934	1914	1898
38	%N1	77.6	79.0	80.5	81.9	83.4	84.8	86.3	87.7	89.1
	MACH	.437	.452	.469	.487	.505	.525	.544	.563	.580
	KIAS	246	245	245	245	245	244	244	242	240
	FF/ENG	1825	1807	1796	1788	1780	1769	1760	1749	1728

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ENGINE INOP

MAX CONTINUOUS THRUST

Long Range Cruise Diversion Fuel and Time

Ground to Air Miles Conversion

AIR DISTANCE (NM)					GROUND DISTANCE (NM)	AIR DISTANCE (NM)				
HEADWIND COMPONENT (KTS)						TAILWIND COMPONENT (KTS)				
100	80	60	40	20		20	40	60	80	100
286	264	244	227	213	200	190	181	173	166	159
579	533	492	457	427	400	380	362	346	331	318
874	803	740	686	641	600	571	544	519	497	477
1171	1074	989	917	855	800	761	725	692	662	635
1470	1347	1238	1147	1069	1000	951	906	864	827	793
1771	1621	1489	1379	1284	1200	1141	1086	1037	992	951
2075	1898	1742	1611	1500	1400	1331	1267	1209	1156	1109
2382	2176	1995	1844	1715	1600	1521	1448	1381	1321	1267
2691	2456	2250	2077	1931	1800	1711	1628	1553	1485	1424

Reference Fuel and Time Required at Check Point

AIR DIST (NM)	PRESSURE ALTITUDE (1000 FT)									
	10		16		20		24		28	
	FUEL (1000 KG)	TIME (HR:MIN)	FUEL (1000 KG)	TIME (HR:MIN)	FUEL (1000 KG)	TIME (HR:MIN)	FUEL (1000 KG)	TIME (HR:MIN)	FUEL (1000 KG)	TIME (HR:MIN)
200	1.5	0:40	1.2	0:38	1.1	0:37	1.0	0:37	0.9	0:36
400	3.0	1:18	2.6	1:13	2.4	1:11	2.2	1:09	2.1	1:07
600	4.5	1:57	3.9	1:49	3.6	1:44	3.4	1:41	3.3	1:39
800	5.9	2:36	5.3	2:24	4.9	2:18	4.6	2:13	4.5	2:10
1000	7.4	3:16	6.6	3:00	6.1	2:53	5.7	2:46	5.6	2:42
1200	8.8	3:56	7.9	3:37	7.3	3:27	6.9	3:19	6.7	3:14
1400	10.2	4:37	9.1	4:14	8.5	4:02	8.0	3:53	7.8	3:46
1600	11.6	5:19	10.4	4:52	9.6	4:37	9.1	4:26	8.8	4:18
1800	13.0	6:02	11.6	5:30	10.8	5:13	10.2	5:00	9.8	4:51

Fuel Required Adjustments (1000 KG)

REFERENCE FUEL REQUIRED (1000 KG)	WEIGHT AT CHECK POINT (1000 KG)				
	35	45	50	55	65
5	-0.7	-0.2	0.0	0.4	1.3
10	-1.4	-0.5	0.0	0.9	2.6
15	-2.1	-0.7	0.0	1.3	3.9

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ENGINE INOP

MAX CONTINUOUS THRUST

Holding
Flaps Up

WEIGHT (1000 KG)		PRESSURE ALTITUDE (FT)						
		1500	5000	10000	15000	20000	25000	30000
66	%N1	5000	83.8	87.7	92.3			
	KIAS	251	251	251	253			
	FF/ENG	2720	2710	2710	2760			
64	%N1	80.2	83.0	86.8	91.3			
	KIAS	247	247	247	249			
	FF/ENG	2630	2630	2620	2660			
62	%N1	79.3	82.1	85.9	90.3			
	KIAS	242	243	243	245			
	FF/ENG	2550	2540	2530	2560			
60	%N1	78.3	81.2	85.1	89.3	94.7		
	KIAS	238	239	239	241	242		
	FF/ENG	2460	2450	2450	2470	2520		
58	%N1	77.4	80.3	84.1	88.3	93.4		
	KIAS	234	235	236	236	238		
	FF/ENG	2380	2370	2360	2380	2420		
56	%N1	76.4	79.3	83.2	87.2	92.2		
	KIAS	230	231	232	232	234		
	FF/ENG	2290	2290	2280	2280	2320		
54	%N1	75.3	78.2	82.2	86.2	91.0		
	KIAS	225	227	228	228	230		
	FF/ENG	2210	2200	2190	2190	2220		
52	%N1	74.3	77.1	81.2	85.1	89.8	96.5	
	KIAS	221	222	224	223	225	225	
	FF/ENG	2130	2120	2110	2110	2120	2220	
50	%N1	73.2	76.0	80.2	84.1	88.5	94.5	
	KIAS	216	218	219	219	221	221	
	FF/ENG	2050	2030	2030	2020	2030	2100	
48	%N1	72.1	74.9	79.1	83.0	87.3	92.8	
	KIAS	211	213	215	215	216	217	
	FF/ENG	1970	1950	1940	1930	1940	1990	
46	%N1	71.1	73.8	77.9	81.9	86.0	91.2	
	KIAS	210	210	210	211	211	213	
	FF/ENG	1900	1880	1860	1850	1850	1880	
44	%N1	70.1	72.9	76.9	80.9	84.9	89.8	
	KIAS	210	210	210	210	210	210	
	FF/ENG	1840	1820	1800	1780	1780	1790	
42	%N1	69.1	72.0	75.9	80.0	83.9	88.5	95.8
	KIAS	210	210	210	210	210	210	210
	FF/ENG	1780	1760	1740	1720	1710	1720	1810
40	%N1	68.2	71.1	75.0	79.1	83.0	87.3	93.4
	KIAS	210	210	210	210	210	210	210
	FF/ENG	1730	1700	1680	1660	1650	1650	1710
38	%N1	67.3	70.2	74.1	78.2	82.1	86.2	91.7
	KIAS	210	210	210	210	210	210	210
	FF/ENG	1670	1650	1630	1610	1600	1590	1620

This table includes 5% additional fuel for holding in a racetrack pattern.

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Takeoff %N1

Based on engine bleed to packs on (Auto) and anti-ice on or off

AIRPORT OAT		AIRPORT PRESSURE ALTITUDE (FT)									
°C	°F	-1000	-0	1000	2000	3000	4000	5000	6000	7000	8000
55	131	92.8	92.6	92.4	92.4						
50	122	94.1	94.3	94.1	94.0	94.0	94.3				
45	113	94.7	94.9	95.1	95.3	95.5	95.3	94.5	94.8	95.5	
40	104	95.4	95.5	95.6	95.7	96.0	96.0	95.7	95.7	96.0	95.8
35	95	95.6	95.7	96.1	96.2	96.4	96.6	96.6	96.6	96.6	96.8
30	86	95.4	95.7	96.0	96.5	96.8	97.1	97.4	97.5	97.4	97.4
25	77	94.6	94.9	95.5	96.4	97.1	97.5	97.9	98.4	98.2	97.8
20	68	93.8	94.1	94.7	95.6	96.5	97.3	97.9	97.8	97.9	97.9
15	59	93.0	93.3	93.9	94.8	95.7	96.5	97.1	97.1	97.1	97.1
10	50	92.2	92.5	93.1	94.0	94.8	95.7	96.3	96.3	96.3	96.3
5	41	91.4	91.6	92.3	93.1	94.0	94.8	95.4	95.4	95.4	95.4
0	32	90.5	90.8	91.4	92.3	93.1	93.9	94.5	94.5	94.5	94.5
-10	14	88.8	89.1	89.7	90.6	91.4	92.2	92.8	92.8	92.8	92.8
-20	-4	87.2	87.4	88.0	88.9	89.7	90.5	91.0	91.0	91.0	91.0
-30	-22	85.4	85.7	86.3	87.1	87.9	88.6	89.2	89.2	89.2	89.2
-40	-40	83.7	83.9	84.5	85.3	86.0	86.8	87.3	87.3	87.3	87.3
-50	-58	81.8	82.1	82.7	83.4	84.2	84.9	85.5	85.5	85.5	85.5

Takeoff %N1

Based on engine bleed to packs off and anti-ice on or off

AIRPORT OAT		AIRPORT PRESSURE ALTITUDE (FT)									
°C	°F	-1000	0	1000	2000	3000	4000	5000	6000	7000	8000
55	131	93.8	93.6	93.3	93.4						
50	122	95.1	95.2	95.0	94.9	95.0	95.2				
45	113	95.7	95.8	96.0	96.2	96.4	96.2	95.5	95.7	96.4	
40	104	96.4	96.4	96.5	96.7	97.0	97.0	96.6	96.6	97.0	96.7
35	95	96.6	96.7	97.0	97.1	97.3	97.5	97.5	97.5	97.6	97.7
30	86	96.4	96.6	97.0	97.4	97.7	98.0	98.4	98.5	98.4	98.3
25	77	95.5	95.8	96.4	97.3	98.0	98.4	98.8	98.8	98.8	98.7
20	68	94.7	95.0	95.6	96.5	97.4	97.9	97.9	97.9	97.9	97.9
15	59	93.9	94.1	94.8	95.7	96.6	97.1	97.1	97.1	97.1	97.1
10	50	93.1	93.3	94.0	94.9	95.7	96.3	96.3	96.3	96.3	96.3
5	41	92.3	92.5	93.1	94.0	94.9	95.4	95.4	95.4	95.4	95.4
0	32	91.4	91.7	92.3	93.2	94.0	94.5	94.5	94.5	94.5	94.5
-10	14	89.6	90.0	90.6	91.4	92.3	92.8	92.8	92.8	92.8	92.8
-20	-4	88.0	88.3	88.9	89.7	90.5	91.0	91.0	91.0	91.0	91.0
-30	-22	86.2	86.5	87.1	87.9	88.7	89.2	89.2	89.2	89.2	89.2
-40	-40	84.5	84.7	85.3	86.1	86.8	87.3	87.3	87.3	87.3	87.3
-50	-58	82.6	82.9	83.4	84.2	85.0	85.5	85.5	85.5	85.5	85.5

Takeoff Speeds Adjustment

SPEED (KIAS)	PRESSURE ALTITUDE (FT)			
	7000 & BELOW		ABOVE 7000	
	ABOVE 27°C	27°C & BELOW	ABOVE 27°C	27°C & BELOW
V1(MCG)	2	2	2	1
V1	0	0	0	0
VR	0	1	0	1

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Takeoff %N1 (22K Derate)

Based on engine bleed to packs on (Auto) and anti-ice on or off

AIRPORT OAT		AIRPORT PRESSURE ALTITUDE (FT)									
°C	°F	-1000	0	1000	2000	3000	4000	5000	6000	7000	8000
55	131	94.7	94.9	94.9	94.9						
50	122	95.1	95.4	95.4	95.4	95.4	95.4				
45	113	95.6	95.9	95.9	95.9	95.9	95.9	95.9	95.9	95.9	
40	104	96.0	96.3	96.3	96.3	96.3	96.3	96.3	96.4	96.4	96.4
35	95	96.3	96.8	96.8	96.8	96.8	96.8	96.8	96.8	96.8	96.8
30	86	95.5	96.6	96.8	97.2	97.2	97.2	97.2	97.2	97.1	97.1
25	77	94.8	95.8	96.0	96.5	97.0	97.4	97.6	97.5	97.5	97.4
20	68	94.0	95.0	95.2	95.7	96.2	96.5	96.9	97.1	97.1	97.1
15	59	93.2	94.1	94.4	94.8	95.4	95.7	96.3	96.3	96.3	96.3
10	50	92.4	93.3	93.5	94.0	94.6	94.9	95.2	95.5	95.6	95.7
5	41	91.5	92.5	92.7	93.2	93.7	94.0	94.4	94.7	94.8	94.9
0	32	90.7	91.7	91.9	92.3	92.9	93.2	93.5	93.8	93.9	94.0
-10	14	89.0	90.0	90.2	90.6	91.2	91.5	91.8	92.1	92.2	92.3
-20	-4	87.3	88.2	88.4	88.9	89.4	89.7	90.0	90.3	90.4	90.5
-30	-22	85.6	86.5	86.7	87.1	87.6	87.9	88.2	88.5	88.6	88.7
-40	-40	83.8	84.7	84.9	85.3	85.8	86.1	86.4	86.7	86.8	86.9
-50	-58	82.0	82.8	83.0	83.5	83.9	84.2	84.5	84.8	84.9	85.0

Based on engine bleed to packs off and anti-ice on or off

AIRPORT OAT		AIRPORT PRESSURE ALTITUDE (FT)									
°C	°F	-1000	0	1000	2000	3000	4000	5000	6000	7000	8000
55	131	95.7	95.9	95.9	95.9						
50	122	96.1	96.4	96.4	96.4	96.4	96.4				
45	113	96.6	96.8	96.8	96.8	96.8	96.8	96.8	96.8	96.9	
40	104	97.0	97.3	97.3	97.3	97.3	97.3	97.3	97.3	97.3	97.3
35	95	97.3	97.7	97.7	97.7	97.7	97.7	97.7	97.7	97.7	97.7
30	86	96.5	97.5	97.7	98.1	98.1	98.1	98.1	98.1	98.1	98.0
25	77	95.7	96.7	96.9	97.4	97.9	97.9	97.9	97.9	97.9	97.9
20	68	94.9	95.9	96.1	96.6	97.1	97.1	97.1	97.1	97.1	97.1
15	59	94.1	95.0	95.3	95.7	96.3	96.3	96.3	96.3	96.3	96.3
10	50	93.3	94.2	94.4	94.9	95.6	95.8	96.1	96.3	96.3	96.3
5	41	92.4	93.4	93.6	94.1	94.6	94.9	95.2	95.4	95.4	95.4
0	32	91.6	92.5	92.7	93.2	93.8	94.1	94.4	94.5	94.5	94.5
-10	14	89.8	90.8	91.0	91.5	92.0	92.3	92.6	92.8	92.8	92.8
-20	-4	88.1	89.1	89.3	89.7	90.3	90.6	90.9	91.0	91.0	91.0
-30	-22	86.4	87.3	87.5	88.0	88.5	88.8	89.1	89.2	89.2	89.2
-40	-40	84.6	85.5	85.7	86.1	86.6	86.9	87.2	87.3	87.4	87.4
-50	-58	82.8	83.6	83.8	84.3	84.7	85.0	85.3	95.5	85.5	85.5

Takeoff Speeds Adjustment

SPEED (KIAS)	PRESSURE ALTITUDE (FT)			
	BELOW 5000		5000 & ABOVE	
	ABOVE 21°C	21°C & BELOW	ABOVE 21°C	21°C & BELOW
V1(MCG)	6	4	6	4
V1	0	0	0	0
VR	0	0	0	1

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Takeoff %N1 (20K Derate)

Based on engine bleed to packs on (Auto) and anti-ice on or off

AIRPORT OAT		AIRPORT PRESSURE ALTITUDE (FT)									
°C	°F	-1000	0	1000	2000	3000	4000	5000	6000	7000	8000
55	131	91.5	92.3	92.2	91.7						
50	122	92.0	92.8	93.4	93.7	93.0	93.0				
45	113	92.6	93.1	93.6	93.9	93.9	94.5	94.3	93.7		
40	104	93.1	93.6	94.0	94.2	94.3	95.1	95.0	94.9	94.8	94.5
35	95	93.2	94.1	94.6	94.8	94.8	95.5	95.7	95.6	95.8	95.7
30	86	93.0	94.5	94.3	94.5	95.0	95.4	95.6	95.8	95.8	95.8
25	77	92.2	93.7	93.7	93.7	94.4	95.5	95.3	95.1	95.4	95.8
20	68	91.5	92.9	92.9	92.9	93.6	95.5	95.7	95.8	95.8	95.7
15	59	90.7	92.1	92.1	92.1	92.8	94.9	95.1	95.1	95.1	95.1
10	50	89.9	91.3	91.3	91.3	92.0	94.1	94.3	94.3	94.3	94.3
5	41	89.1	90.5	90.5	90.5	91.2	93.2	93.4	93.4	93.4	93.4
0	32	88.3	89.7	89.7	89.7	90.4	92.4	92.6	92.6	92.6	92.6
-10	14	86.6	88.0	88.0	88.0	88.7	90.7	90.9	90.9	90.9	90.9
-20	-4	85.0	86.3	86.3	86.4	87.0	89.0	89.1	89.1	89.1	89.1
-30	-22	83.3	84.6	84.6	84.6	85.3	87.2	87.4	87.4	87.4	87.4
-40	-40	81.6	82.9	82.9	82.9	83.5	85.4	85.5	85.5	85.6	85.6
-50	-58	79.8	81.1	81.1	81.1	81.7	83.5	83.7	83.7	83.7	83.7

Based on engine bleed to packs off and anti-ice on or off

AIRPORT OAT		AIRPORT PRESSURE ALTITUDE (FT)									
°C	°F	-1000	0	1000	2000	3000	4000	5000	6000	7000	8000
55	131	92.4	93.3	93.2	92.7						
50	122	93.0	93.8	94.4	94.6	94.0	94.0				
45	113	93.5	94.0	94.6	94.9	94.8	95.5	95.3	94.7		
40	104	94.0	94.5	94.9	95.1	95.2	96.1	95.9	95.8	95.8	95.4
35	95	94.2	95.0	95.6	95.7	95.7	96.5	96.6	96.6	96.7	96.6
30	86	93.9	95.4	95.2	95.4	96.0	96.3	96.5	96.7	96.7	96.8
25	77	93.1	94.6	94.6	94.6	95.3	96.4	96.2	96.0	96.3	96.7
20	68	92.4	93.8	93.8	93.8	94.6	95.9	95.9	95.9	95.9	95.9
15	59	91.6	93.0	93.0	93.0	93.7	95.1	95.1	95.1	95.1	95.1
10	50	90.8	92.2	92.2	92.2	92.9	94.3	94.3	94.3	94.3	94.3
5	41	90.0	91.4	91.4	91.4	92.1	93.4	93.4	93.4	93.4	93.4
0	32	89.2	90.6	90.6	90.6	91.3	92.6	92.6	92.6	92.6	92.6
-10	14	87.5	88.9	88.9	88.9	89.6	90.9	90.9	90.9	90.9	90.9
-20	-4	85.8	87.2	87.2	87.2	87.9	89.1	89.1	89.1	89.1	89.1
-30	-22	84.1	85.5	85.5	85.5	86.1	87.4	87.4	87.4	87.4	87.4
-40	-40	82.4	83.7	83.7	83.7	84.3	85.5	85.5	85.5	85.6	85.6
-50	-58	80.6	81.9	81.9	81.9	82.5	83.7	83.7	83.7	83.7	83.7

Takeoff Speeds Adjustment

SPEED (KIAS)	PRESSURE ALTITUDE (FT)	
	BELOW 5000	5000 & ABOVE
V1(MCG)	6	4
V1	0	0
VR	1	1

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GEAR DOWN

220 KIAS Cruise Altitude Capability

Max Cruise Thrust, 100 ft/min residual rate of climb

WEIGHT (1000 KG)	PRESSURE ALTITUDE (FT)		
	ISA + 10°C & BELOW	ISA + 15°C	ISA + 20°C
66	21500	20200	18900
64	22200	21000	19700
62	23700	22600	21400
60	24500	23400	22200
58	25300	24200	23000
56	26000	25000	23800
54	26700	25700	24500
52	27400	26400	25200
50	28000	27000	25900
48	28600	27600	26600
46	29200	28200	27200
44	29700	28800	27800
42	30200	29300	28300
40	30700	29800	28800
38	31100	30300	29300

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GEAR DOWN

220 KIAS Cruise Control

WEIGHT (1000 KG)		PRESSURE ALTITUDE (1000 FT)										
		11	13	15	17	19	21	23	25	27	29	31
66	%N1	81.3	82.9	84.6	86.3	88.1	90.2					
	MACH	.424	.441	.458	.476	.495	.515					
	KIAS	230	230	230	230	230	230					
	FF/ENG	1948	1949	1951	1957	1966	1981					
62	%N1	79.3	80.9	82.6	84.2	86.0	87.9	90.1				
	MACH	.406	.422	.438	.456	.474	.493	.513				
	KIAS	220	220	220	220	220	220	220				
	FF/ENG	1800	1799	1801	1804	1809	1818	1835				
58	%N1	78.2	79.8	81.4	83.0	84.7	86.5	88.5	90.8			
	MACH	.406	.422	.438	.456	.474	.493	.513	.534			
	KIAS	220	220	220	220	220	220	220	220			
	FF/ENG	1729	1726	1725	1725	1727	1731	1744	1767			
54	%N1	77.1	78.8	80.4	82.0	83.6	85.3	87.1	89.2	91.7		
	MACH	.406	.422	.438	.456	.474	.493	.513	.534	.557		
	KIAS	220	220	220	220	220	220	220	220	220		
	FF/ENG	1665	1660	1657	1656	1655	1657	1664	1679	1705		
50	%N1	76.1	77.8	79.4	81.0	82.6	84.2	86.0	87.9	90.1		
	MACH	.406	.422	.438	.456	.474	.493	.513	.534	.557		
	KIAS	220	220	220	220	220	220	220	220	220		
	FF/ENG	1608	1602	1598	1595	1593	1592	1597	1608	1621		
46	%N1	75.2	76.9	78.5	80.1	81.7	83.3	85.0	86.8	88.8	91.3	
	MACH	.406	.422	.438	.456	.474	.493	.513	.534	.557	.580	
	KIAS	220	220	220	220	220	220	220	220	220	220	
	FF/ENG	1558	1551	1545	1542	1540	1537	1541	1548	1555	1579	
42	%N1	74.3	76.0	77.7	79.3	80.8	82.4	84.1	85.9	87.8	90.0	93.1
	MACH	.406	.422	.438	.456	.474	.493	.513	.534	.557	.580	.605
	KIAS	220	220	220	220	220	220	220	220	220	220	220
	FF/ENG	1513	1505	1499	1495	1492	1489	1490	1496	1500	1516	1559
38	%N1	73.6	75.3	76.9	78.6	80.1	81.7	83.3	85.1	86.9	88.9	91.7
	MACH	.406	.422	.438	.456	.474	.493	.513	.534	.557	.580	.605
	KIAS	220	220	220	220	220	220	220	220	220	220	220
	FF/ENG	1475	1466	1460	1455	1451	1448	1448	1453	1455	1464	1496

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GEAR DOWN

220 KIAS Enroute Fuel and Time

Ground to Air Miles Conversion

AIR DISTANCE (NM) HEADWIND COMPONENT (KTS)					GROUND DISTANCE (NM)	AIR DISTANCE (NM) TAILWIND COMPONENT (KTS)				
100	80	60	40	20		20	40	60	80	100
321	288	259	236	217	200	189	180	171	163	156
650	580	520	473	434	400	378	358	340	324	310
980	874	783	711	652	600	567	537	510	486	465
1309	1167	1045	949	869	800	756	717	680	648	619
1639	1461	1308	1186	1087	1000	945	895	850	809	773
1968	1754	1571	1424	1305	1200	1134	1074	1020	971	928
2298	2048	1833	1662	1522	1400	1323	1253	1189	1132	1082
2627	2341	2096	1900	1740	1600	1512	1432	1360	1294	1236
2956	2635	2358	2138	1958	1800	1701	1611	1529	1456	1391

Reference Fuel and Time Required at Check Point

AIR DIST (NM)	PRESSURE ALTITUDE (1000 FT)									
	10		14		20		24		28	
	FUEL (1000 KG)	TIME (HR:MIN)	FUEL (1000 KG)	TIME (HR:MIN)	FUEL (1000 KG)	TIME (HR:MIN)	FUEL (1000 KG)	TIME (HR:MIN)	FUEL (1000 KG)	TIME (HR:MIN)
200	2.4	0:49	2.2	0:46	1.9	0:43	1.7	0:41	1.6	0:40
400	4.8	1:36	4.5	1:31	4.0	1:24	3.7	1:19	3.5	1:15
600	7.3	2:23	6.7	2:15	6.0	2:04	5.7	1:57	5.4	1:51
800	9.6	3:10	9.0	3:00	8.1	2:44	7.6	2:35	7.2	2:26
1000	12.0	3:57	11.2	3:44	10.1	3:25	9.5	3:13	9.0	3:02
1200	14.3	4:45	13.3	4:28	12.1	4:05	11.3	3:51	10.7	3:37
1400	16.6	5:32	15.5	5:13	14.0	4:46	13.2	4:29	12.5	4:12
1600	18.9	6:19	17.6	5:57	15.9	5:26	15.0	5:07	14.2	4:48
1800	21.1	7:06	19.7	6:41	17.8	6:06	16.8	5:44	15.9	5:23

Fuel Required Adjustments (1000 KG)

REFERENCE FUEL REQUIRED (1000 KG)	WEIGHT AT CHECK POINT (1000 KG)			
	40	50	60	70
5	-0.3	0.0	0.7	1.7
10	-0.6	0.0	1.2	3.1
15	-0.8	0.0	1.6	4.0
20	-1.0	0.0	1.8	4.5
25	-1.1	0.0	1.9	4.6

Based on 220 KIAS cruise and descent.

Descent at 220 KIAS

PRESSURE ALT (1000 FT)	5	10	15	17	19	21	23	25	27	29	31	33
DISTANCE (NM)	17	26	35	38	42	46	49	53	56	60	64	67
TIME (MINUTES)	6	8	10	11	11	12	13	14	14	15	15	16

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GEAR DOWN

Holding
Flaps Up

WEIGHT (1000 KG)		PRESSURE ALTITUDE (FT)						
		1500	5000	10000	15000	20000	25000	30000
66	%N1	73.7	76.4	80.5	84.6	89.1		
	KIAS	230	230	230	230	230		
	FF/ENG	2070	2060	2050	2050	2070		
62	%N1	71.7	74.3	78.4	82.6	86.9		
	KIAS	220	220	220	220	220		
	FF/ENG	1930	1910	1890	1890	1900		
58	%N1	70.6	73.3	77.3	81.4	85.6	90.8	
	KIAS	220	220	220	220	220	220	
	FF/ENG	1860	1840	1820	1810	1820	1860	
54	%N1	69.6	72.3	76.2	80.4	84.4	89.2	
	KIAS	220	220	220	220	220	220	
	FF/ENG	1800	1770	1750	1740	1740	1760	
50	%N1	67.1	70.0	73.9	78.0	82.1	86.5	92.8
	KIAS	210	210	210	210	210	210	210
	FF/ENG	1660	1630	1610	1590	1590	1590	1660
46	%N1	66.1	68.9	72.8	77.0	81.0	85.2	90.7
	KIAS	210	210	210	210	210	210	210
	FF/ENG	1600	1570	1550	1530	1520	1520	1560
42	%N1	65.1	67.9	71.9	75.9	80.0	84.1	89.2
	KIAS	210	210	210	210	210	210	210
	FF/ENG	1550	1520	1500	1480	1470	1460	1490
38	%N1	64.2	67.0	71.1	75.0	79.1	83.2	87.9
	KIAS	210	210	210	210	210	210	210
	FF/ENG	1510	1480	1460	1430	1420	1410	1430

This table includes 5% additional fuel for holding in a racetrack pattern.

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Introduction

This chapter contains information to supplement performance data from the Flight Management Computer (FMC). In addition, sufficient inflight data is provided to complete a flight with the FMC inoperative. In the event of conflict between data presented in this chapter and that contained in the Approved Flight Manual, the Flight Manual shall always take precedence.

General

Takeoff Speeds

The speeds presented in the Takeoff Speeds table as well as FMC computed takeoff speeds can be used for all performance conditions provided adjustments are made to V1 for clearway, stopway, brake deactivation, improved climb, contaminated runway situations, brake energy limits or obstacle clearance with unbalanced V1. These speeds may be used for weights less than or equal to the performance limited weight.

Normal takeoff speeds, V1, VR, and V2, with anti-skid on, are read from the table by entering with takeoff flap setting, brake release weight, and appropriate column. The appropriate column is obtained by entering the Column Reference chart with the airport pressure altitude and the actual temperature. If an Altitude Adjustment chart is provided, adjust the takeoff speeds appropriately. Slope and wind adjustments to V1 are obtained by entering the Slope and Wind V1 Adjustment table.

If takeoffs are scheduled using these simplified speeds in conjunction with airport analyses that include clearway and/or stopway credits, adjustments to V1 speed are required.

Adjust V1 by the amount shown in the Clearway/Stopway table. The adjusted V1 must not exceed VR.

The maximum allowable clearway limits shown on the takeoff speeds page are provided for guidance when more precise data is unavailable.

V1(MCG)

Regulations prohibit scheduling takeoff with a V1 less than minimum V1 for control on the ground, V1(MCG). Therefore compare the adjusted V1 to the V1(MCG). To find V1(MCG) enter the V1(MCG) table with the airport pressure altitude and actual OAT. If applicable, add the adjustments shown below the table. If the adjusted V1 is less than

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V1(MCG), set V1 equal to V1(MCG). If VR is less than V1(MCG), set VR equal to V1(MCG), and determine a new V2 by adding the difference between the normal VR and V1(MCG) to the normal V2.

If it is necssary to increase V1 to V1(MCG), then provided the actual field length exceeds the minimum field length of 4500ft no takeoff weight adjustment is necessary.

Clearway and Stopway V1 Adjustments

Takeoff speed corrections are to be applied to V1 speed when using takeoff weights based on the use of clearway and stopway.

Adjust V1 speed by the amount shown in the appropriate column. The adjusted V1 speed must not exceed VR. If the adjusted V1 speed is greater than VR, reduce V1 to equal VR.

Maximum allowable clearway limits are provided for guidance when more precise data is not available.

Assumed Temperature Takeoff

For reduced thrust takeoffs based on assumed temperature, determine V1, VR, V2 from the appropriate column for airport pressure altitude and assumed temperature. Adjust V1 for slope, wind, clearway and stopway as required. Compare adjusted V1 with V1(MCG) from table using airport altitude and actual temperature. This is a regulatory requirement to cover the case where the pilot elects to advance thrust levers to maximum takeoff thrust following the engine failure. If adjusted V1 is less than V1(MCG) set V1 equal to V1(MCG).

Stab Trim

To find takeoff stabilizer trim setting, enter the appropriate Stab Trim Setting table with center of gravity (C.G. % MAC) and read required stabilizer trim units. For weights that meet the conditions listed, apply the adjustments shown below the table.

VREF

The Reference Speed table contains flaps 40, 30, and 15 landing speeds for a given weight. Apply wind correction shown as required.

Flap Maneuver Speeds

This table provides the flap speed schedule for recommended maneuvering speed. The speed schedule is a function of weight and will provide adequate maneuver margin above stall at all weights.

During flap retraction/extension, movement of the flap to the next position should be initiated when reaching the maneuver speed for the existing flap.

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Slush/Standing Water Takeoff

Experience has shown that aircraft performance may deteriorate significantly on runways covered with snow, slush, standing water, or ice. Therefore a reduction in runway/obstacle limited takeoff weight and revised takeoff speeds is necessary. The following information is intended for guidance in accordance with advisory material and is based on an engine failure at the critical point during takeoff.

The entire runway is assumed to be completely covered by a contaminant of uniform thickness and density. Therefore this information is conservative when operating under typical colder weather conditions where patches of slush exist and some degree of sanding is common.

Takeoffs in slush depths greater than 13mm (0.5 inches) are not recommended because of possible airplane damage as a result of slush impingement on the airplane structure. The use of assumed temperature method for reduced thrust is not allowed on contaminated runways. Interpolation for slush/standing water depths between the values shown is permitted.

Instructions for Using Tables:

Takeoff weight is determined as follows:

1. Determine the dry field/obstacle limit weight for the takeoff flap setting.

2. Enter the Weight Adjustment table with the field/obstacle limit weight to obtain the weight reduction for the slush/standing water depth and airport pressure altitude.

3. Adjust field length available for temperature by amount shown on chart.

4. Enter V1(MCG) Limit Weight table with the adjusted field length and pressure altitude to obtain the slush/standing water limit weight with respect to minimum field length required for V1(MCG) speed.

The maximum allowable takeoff weight in slush/standing water is the lesser of the limit weights found in steps 2 and 4.

Takeoff speeds determination:

1. Determine takeoff speeds V1, VR and V2 for actual brake release weight using the Takeoff Speeds table in this section.

2. If V1(MCG) limited, set V1 = V1(MCG). If not limited by V1(MCG) considerations, enter V1 Adjustment table with actual brake release weight to determine the V1 reduction to apply to V1 speed. If the adjusted V1 is less than V1(MCG), set V1 = V1(MCG).

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Slippery Runway

Airplane braking action is reported as good, medium or poor, depending on existing runway conditions. If braking action is reported as good, conditions should not be expected to be as good as on clean dry runways. The value “good” is comparative and is intended to mean that airplanes should not experience braking or directional control difficulties when stopping. The performance level used to calculate the “good” data is consistent with wet runway testing done on early Boeing jets. The performance level used to calculate the “poor” data reflects a runway covered with wet ice. Performance is based on reversers operating and a 15 ft. screen height at the end of the runway. The tables provided are used in the same manner as the Slush/Standing Water tables.

Anti-skid Inoperative

When operating with anti-skid inoperative, the field limit weight and V1 must be reduced to account for the effect on accelerate-stop performance. A simplified method which conservatively accounts for the effects of anti-skid inoperative is to reduce the normal runway/obstacle limited weight by 7500 kg and the V1 associated with the reduced weight by the amount shown in the table below.

ANTI-SKID INOPERATIVE V1 ADJUSTMENT	
FIELD LENGTH (FT)	V1 ADJUSTMENT (KIAS)
6000	-28
8000	-22
10000	-18
12000	-14
14000	-13

If the resulting V1 is less than minimum V1, takeoff is permitted with V1 set equal to V1(MCG) provided the dry accelerate stop distance adjusted for wind and slope exceeds approximately 9000 ft.

Detailed analysis for the specific case from the Airplane Flight Manual may yield a less restrictive penalty.

Takeoff %N1

To find Max Takeoff %N1 based on normal engine bleed for air conditioning packs on (Auto), enter Takeoff %N1 Table with airport pressure altitude and airport OAT and read %N1. For packs off operation, apply the %N1 adjustment shown below the table. No takeoff %N1 adjustment is required for engine and wing anti-ice.

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Assumed Temperature Reduced Thrust

Regulations permit the use of up to 25% takeoff thrust reduction for operation with assumed temperature reduced thrust. To find the maximum allowable assumed temperature enter the top chart with airport pressure altitude and OAT. Compare this temperature to that at which the airplane is performance limited as determined from available takeoff performance data. Next, enter the center table with airport pressure altitude and the lower of the two temperatures previously determined to obtain a maximum takeoff %N1. Do not use an assumed temperature less than the minimum assumed temperature shown. Enter the %N1 Adjustment table with OAT and the difference between the assumed and actual OAT to obtain a %N1 adjustment. Subtract the %N1 adjustment from the maximum takeoff %N1 found previously to determine the assumed temperature reduced thrust %N1.

Takeoff with assumed temperature reduced thrust is not permitted when: runway is contaminated with ice, slush, snow, or standing water; anti-skid is inoperative; or PMC is off. Use of this procedure is not recommended if potential windshear conditions exist.

Max Climb %N1

This table shows Max Climb %N1 for a 250 KIAS/280 KIAS/.74M climb speed schedule, normal engine bleed for packs on (Auto) and anti-ice off. Enter the table with airport pressure altitude and TAT and read %N1. Apply the appropriate bleed setting %N1 adjustment shown below the table. %N1 adjustments are shown for engine bleed to packs off operation, packs high operation, and wing anti-ice on.

Go-around %N1

To find Max Go-around %N1 based on normal engine bleed for packs on (Auto) and anti-ice off, enter the Go-around %N1 table with airport pressure altitude and reported OAT or TAT and read %N1. Apply the appropriate bleed setting %N1 adjustment shown below the table. %N1 adjustments are shown for engine bleed to packs off operation, packs high operation, and wing anti-ice on.

Flight with Unreliable Airspeed / Turbulent Air Penetration

Pitch attitude and average %N1 information is provided for use in all phases of flight in the event of unreliable airspeed/Mach indications resulting from blocking or freezing of the pitot system. Loss of radome or turbulent air may also cause unreliable airspeed/Mach indications. The cruise table in this section may also be used for turbulent air penetration.

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Pitch attitude is shown in bold type for emphasis since altitude and/or vertical speed indications may also be unreliable.

All Engines

Long Range Cruise Maximum Operating Altitude

These tables provide the maximum operating altitude in the same manner as the FMC. Maximum altitudes are shown for a given cruise weight and maneuver capability. This table considers both thrust and buffet limits, providing the more limiting of the two. Any data that is thrust limited is denoted by an asterisk and represents only a thrust limited condition in level flight with 100 ft/min residual rate of climb. Flying above these altitudes with sustained banks in excess of approximately 15° may cause the airplane to lose speed and/or altitude.

Note that the altitudes shown in the table are limited to the maximum certified altitude of 37000 ft.

Long Range Cruise Control

These tables provide target %N1, Long Range Cruise Mach number, IAS and standard day fuel flow per engine for the airplane weight and pressure altitude. As indicated by the shaded area, at optimum altitude .74M approximates the Long Range Cruise Mach schedule.

Long Range Cruise Enroute Fuel and Time

Long Range Cruise Enroute Fuel and Time tables are provided to determine remaining time and fuel required to destination. The data is based on Long Range Cruise and .74M/250 KIAS descent. Tables are presented for low altitudes and high altitudes.

To determine remaining fuel and time required, first enter the Ground to Air Miles Conversion table to convert ground distance and enroute wind to an equivalent still air distance for use with the Reference Fuel and Time tables. Next, enter the Reference Fuel and Time Table with air distance from the Ground to Air Miles Conversion Table and the desired altitude and read Reference Fuel and Time Required. Lastly, enter the Fuel Required Adjustment Table with the Reference Fuel and the actual weight at checkpoint to obtain fuel required to destination.

Long Range Cruise Wind-Altitude Trade

Wind is a factor which may justify operations considerably below optimum altitude. For example, a favorable wind component may have an effect on ground speed which more than compensates for the loss in air range.

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Using this table, it is possible to determine the break-even wind (advantage necessary or disadvantage that can be tolerated) to maintain the same range at another altitude and long range cruise speed. The tables make no allowance for climb or descent time, fuel or distance, and are based on comparing ground fuel mileage.

Descent

Distance and time for descent are shown for a .74M/250 KIAS descent speed schedule. Enter the table with top of descent pressure altitude and read distance in nautical miles and time in minutes. Data is based on flight idle thrust descent in zero wind. Allowances are included for a straight-in approach with gear down and landing flaps at the outer marker.

Holding

Target %N1, indicated airspeed and fuel flow per engine information is tabulated for holding with flaps up based on the FMC optimum holding speed schedule. This is the higher of the maximum endurance speed and the maneuvering speed. Small variations in airspeed will not appreciably affect the overall endurance time. Enter the table with weight and pressure altitude to read %N1, IAS and fuel flow per engine.

Advisory Information

Autobrake Landing Distance

The Autobrake Landing Distance Table is provided as advisory information to assist in the selection of the most desirable auto brake setting for a given field length. It is not to be used to determine required field length. This data reflects actual landing distances on a dry runway for setting 1 through MAX, from touchdown to full stop, with or without reverse thrust. The table includes typical flare distances from threshold.

To use the Autobrake Landing Distance Table, first determine the available (or desired) landing distance. Enter the chart with the estimated approach speed and determine the actual stopping distance from touchdown for a given autobrake setting. If airspeed is used for approach speed, correct landing distance for pressure altitude, tailwind and temperature effects.

Selection of an autobrake setting results in a constant rate of deceleration. Maximum effort manual braking should achieve shorter landing distance than the MAX setting.

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Slippery Runway Landing Distance

Landing distances are the actual landing distances for the specified runway conditions and do not include the 1.67 regulatory factor. Therefore they cannot be used to determine dispatch required landing field length. When landing on slippery runways or runways contaminated with ice, snow, slush or standing water, the reported braking action must be considered. If the surface is affected by water, snow or ice, and the braking action is reported as “good,” conditions should not be expected to be as good as on clean dry runways. The value “good” is comparative and is intended to mean that airplanes should not experience braking or directional control difficulties when landing. The performance level used to calculate the “good” data is consistent with wet runway testing done on early Boeing jets. The performance level used to calculate the “poor” data reflects runways covered with wet ice. Read landing distance for the reported braking action at the airplane weight, and then apply the adjustments for airport pressure altitude and approach speed as required.

Non-normal Configuration Landing Distance

Advisory information is provided to support non-normal configurations that affect landing performance of the airplane. Landing distances are shown for dry runway and good, medium, and poor reported braking action. Each non-normal configuration is listed with its recommended approach speed. Landing distance can be determined for the reference landing weight and then adjusted for actual weight and pressure altitude.

Brake Cooling Schedule

Advisory information is provided to assist in avoiding problems associated with hot brakes. For normal operation, most landings are at weights below the quick turnaround limit weight. Application of the recommended cooling procedures shown will avoid brake overheat and fuse plug problems that could result from repeated landings at short time intervals or a rejected takeoff.

Engine Inoperative

Initial Max Continuous %N1

The Initial Max Continuous %N1 setting for use following an engine failure is shown. The table is based on the typical all engine cruise speed of .74M to provide a target %N1 setting at the start of driftdown. Once driftdown is established, the Max Continuous %N1 Table should be used to determine %N1 for the given conditions.

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Max Continuous %N1

Power setting is based on one engine operating with one A/C pack operating and all anti-ice bleeds off. Enter the table with pressure altitude and IAS or Mach to read %N1.

It is desirable to maintain engine thrust within the limits of the Max Cruise thrust rating. However, where thrust in excess of Max Cruise rating is required, such as for meeting terrain clearance, ATC altitude assignments, or to attain maximum range capability, it is permissible to use the thrust needed up to the Max Continuous thrust rating. The Max Continuous thrust rating is intended primarily for emergency use at the discretion of the pilot and is the maximum thrust that may be used continuously.

Driftdown Speed/Level Off Altitude

The table shows optimum driftdown speed as a function of cruise weight at start of driftdown. Also shown are the approximate weight and pressure altitude at which the airplane will level off considering 100 ft/min residual rate of climb.

The level off altitude is dependent on air temperature (ISA deviation).

Driftdown/LRC Cruise Range Capability

This table shows the range capability from the start of driftdown. Driftdown is continued to level off altitude. As weight decreases due to fuel burn, the airplane is accelerated to Long Range Cruise speed. Cruise is continued at level off altitude and Long Range Cruise speed.

To determine fuel required, enter the Ground to Air Miles Conversion table with the desired ground distance and correct for anticipated winds to obtain air distance to destination. Then enter the Driftdown/Cruise Fuel and Time table with air distance and weight at start of driftdown to determine fuel and time required. If altitudes other than the level off altitude is used, fuel and time required may be obtained by using the Engine Inoperative Long Range Cruise Enroute Fuel and Time Table.

Long Range Cruise Altitude Capability

The table shows the maximum altitude that can be maintained at a given weight and air temperature (ISA deviation), based on Long Range Cruise speed, Max Continuous thrust, and 100 ft/min residual rate of climb.

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Long Range Cruise Control

The table provides target %N1, engine inoperative Long Range Cruise Mach number, IAS and fuel flow for the airplane weight and pressure altitude. The fuel flow values in this table reflect single engine fuel burn. To conservatively account for APU fuel burn, add 90 kg/hr to fuel flow values.

Long Range Cruise Diversion Fuel and Time

Tables are provided for crews to determine the fuel and time required to proceed to an alternate airfield with one engine inoperative. The data is based on single engine Long Range Cruise speed and .74M/250 KIAS descent. Enter with Air Distance as determined from the Ground to Air Miles Conversion Table and read Fuel and Time required at the cruise pressure altitude. Adjust the fuel obtained for deviation from the reference weight at checkpoint as required by entering the off reference fuel corrections table with the fuel required for the reference weight and the actual weight at checkpoint. Read fuel required and time for the actual weight.

Holding

Single engine holding data is provided in the same format as the all engine holding data and is based on the same assumptions.

PMC Off Power Settings

To find Max Takeoff %N1 with both PMC's Off, enter Max Takeoff %N1 table (PMC Off) for the appropriate bleed configuration with airport pressure altitude and airport OAT and read %N1. No takeoff %N1 correction is required for engine anti-ice operation up to 10°C (50°F) which is the highest temperature recommended for engine anti-ice operation. Do not use reduced thrust with PMC Off.

For Maximum Climb and Go-Around with both PMC's Off, use PMC on %N1.

Gear Down

This section contains performance for airplane operation with the landing gear extended for all phases of flight. The data is based on engine bleeds for normal air conditioning.

Note: The Flight Management Computer System (FMCS) does not contain special provisions for operation with landing gear extended. As a result, the FMCS will generate inaccurate enroute speed schedules, display non-conservative predictions of fuel burn, estimated time of arrival (ETA),

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maximum altitude, and compute overly shallow descent path. To obtain accurate ETA predictions, gear down cruise speed and altitude should be entered on the CLB and CRZ pages. Gear down cruise speed should also be entered on the DES page. Use of VNAV during descent under these circumstances is not recommended.

Tables for gear down performance in this section are identical in format and used in the same manner as tables for the gear up configuration previously described.

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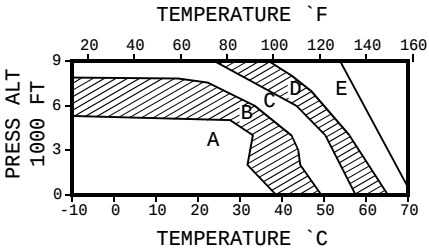
Takeoff Speeds
Flaps 5

COLUMN REFERENCE



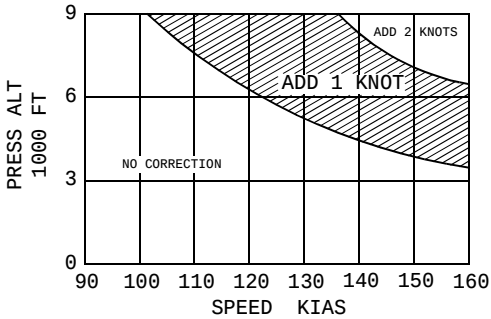
SLOPE/WIND V_1 ADJUSTMENT*						
WEIGHT 1000 KG	SLOPE % DN		UP	WIND KTS TAIL		HEAD
	-2	0	2	-15	0	40
65	-2	0	2	-2	0	2
55	-2	0	1	-3	0	1
45	-2	0	1	-3	0	1
35	-2	0	2	-3	0	2

*V_1 NOT TO EXCEED V_R

WT 1000 KG	A			B			C			D			E		
	V_1	V_R	V_2	V_1	V_R	V_2	V_1	V_R	V_2	V_1	V_R	V_2	V_1	V_R	V_2
65	150	154	158	152	155	158	153	156	158	155	157	158	159	160	163
60	144	147	152	146	148	152	147	149	152	149	151	152	152	153	155
55	137	139	146	139	141	146	140	142	146	143	143	146	145	145	146
50	129	131	139	131	132	139	132	134	139	135	135	139	137	137	139
45	121	122	132	122	124	132	123	125	132	126	127	132	129	129	132
40	111	114	125	113	115	125	114	116	124	117	117	124	119	119	124
35	101	104	118	103	106	118	104	107	117	106	108	117	109	110	116
30	89	95	110	91	97	110	93	97	109	95	99	109	98	100	108

CHECK V1(MCG) AND MINIMUM VR.

ALTITUDE V_1, V_R, V_2 ADJUSTMENT



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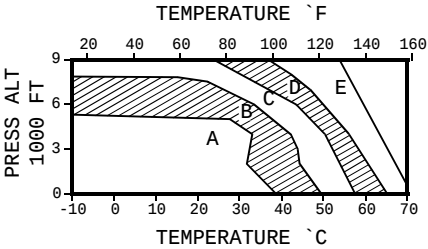
Takeoff Speeds
Flaps 15

COLUMN REFERENCE



SLOPE/WIND V_1 ADJUSTMENT*						
WEIGHT 1000	SLOPE %			WIND KTS		
	DN		UP	TAIL		HEAD
KG	-2	0	2	-15	0	40
65	-2	0	2	-2	0	2
55	-2	0	1	-3	0	1
45	-2	0	1	-3	0	1
35	-2	0	2	-3	0	2

*V_1 NOT TO EXCEED V_R

WT 1000 KG	A			B			C			D			E		
	V_1	V_R	V_2	V_1	V_R	V_2	V_1	V_R	V_2	V_1	V_R	V_2	V_1	V_R	V_2
60	137	138	143	139	140	144	141	141	144	142	143	145	145	145	146
55	131	131	138	132	132	138	134	134	138	135	135	138	137	137	139
50	123	124	132	125	125	131	126	126	131	127	127	131	129	129	132
45	115	116	126	117	117	125	118	118	125	119	119	124	121	121	125
40	106	108	119	107	109	119	109	110	118	111	111	118	113	113	117
35	96	99	112	97	101	112	99	102	112	102	103	111	104	104	110
30	84	90	106	86	92	106	88	93	105	90	94	103	92	95	102

CHECK V1(MCG) AND MINIMUM VR.

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V1(MCG), Minimum VR

ACTUAL OAT		PRESSURE ALTITUDE FT											
		-1000		0		2000		4000		6000		8000	
`C	`F	V1(MCG)	MIN VR	V1(MCG)	MIN VR	V1(MCG)	MIN VR	V1(MCG)	MIN VR	V1(MCG)	MIN VR	V1(MCG)	MIN VR
55	131	99	102	98	101								
50	122	102	104	100	103	98	101	97	100	93	97	89	93
40	104	106	108	105	107	103	105	102	104	97	100	92	96
30	86	109	111	109	110	106	109	106	109	101	104	96	99
20	68	110	111	109	111	107	110	106	109	104	106	100	102
10	50	110	111	109	111	108	110	107	109	104	106	100	103
-50	-58	111	112	110	111	109	110	108	109	105	106	101	103

FOR A/C OFF INCREASE V1(MCG) BY 2 KNOTS
FOR PMC OFF INCREASE V1(MCG) BY 6 KNOTS
FOR PMC OFF INCREASE MIN VR BY 5 KNOTS

Clearway and Stopway V1 Adjustments

CLEARWAY MINUS STOPWAY (FT)	NORMAL V1 (KIAS)			
	100	120	140	160
800			-2	-1
600		-3	-2	-1
400	-4	-3	-2	0
200	-2	-1	-1	0
0	0	0	0	0
-400	1	1	1	1
-800	1	1	1	1

Maximum Allowable Clearway

FIELD LENGTH (FT)	MAX ALLOWABLE CLEARWAY FOR V1 REDUCTION (FT)
4000	450
6000	500
8000	550
10000	600
12000	650
14000	700

Stab Trim Setting

Max Takeoff Thrust

C.G. % MAC	6	10	14	18	22	26	30
STAB TRIM	6 1/4	5 1/2	5	4 1/4	3 3/4	3	2 1/2

FOR WEIGHTS AT OR BELOW 45,360 KG SUBTRACT 1/2 UNIT.
FOR WEIGHTS AT OR ABOVE 54,430 KG ADD 1/2 UNIT.

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VREF

WEIGHT (1000 KG)	FLAPS		
	40	30	15
64	145	149	159
60	140	144	154
56	135	140	149
52	130	135	143
48	125	129	138
44	120	123	132
40	114	117	125
36	108	111	118
32	101	105	111

Increase VREF by 1 knot per 4000 ft above sea level.
For approach speed add wind factor of 1/2 headwind component + gust (max 20 knots).

Flap Maneuver Speeds

FLAP POSITION	MANEUVER SPEED (KIAS)		
	WEIGHT		
	AT OR BELOW 53070 KG	ABOVE 53070 KG AND AT OR BELOW 62823 KG	ABOVE 62823 KG
UP	210	220	230
1	190	200	210
5	180	190	200
10	170	180	190
15	150	160	170
25	140	150	160

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ADVISORY INFORMATION

Slush/Standing Water Takeoff

Weight Adjustment (1000 KG)

FIELD/OBSTACLE LIMIT WEIGHT (1000 KG)	SLUSH/STANDING WATER DEPTH								
	3mm (0.12 INCHES)			6mm (0.25 INCHES)			13mm (0.50 INCHES)		
	PRESS ALT (FT)			PRESS ALT (FT)			PRESS ALT (FT)		
	S.L.	4000	8000	S.L.	4000	8000	S.L.	4000	8000
68	-7.6	-7.8	-8.5	-9.4	-10.1	-10.8	-13.1	-13.7	-13.6
64	-7.2	-7.4	-8.3	-8.8	-9.6	-10.4	-12.3	-13.5	-13.4
60	-6.6	-6.8	-7.9	-7.9	-8.8	-9.8	-11.2	-12.5	-12.9
56	-5.9	-6.2	-7.3	-7.1	-8.0	-9.0	-9.8	-11.1	-12.3
52	-5.2	-5.4	-6.7	-6.2	-7.0	-8.1	-8.4	-9.7	-11.4
48	-4.6	-4.8	-5.9	-5.5	-6.1	-7.2	-7.1	-8.4	-10.3
44	-4.2	-4.3	-5.1	-4.8	-5.3	-6.1	-6.0	-7.3	-8.9
40	-3.8	-3.9	-4.4	-4.5	-4.9	-5.3	-5.5	-6.9	-7.2
36	-3.7	-3.9	-3.9	-4.3	-4.8	-4.6	-5.6	-7.5	-5.2

V1(MCG) Limit Weight (1000 KG)

ADJUSTED FIELD LENGTH (FT)	SLUSH/STANDING WATER DEPTH								
	3mm (0.12 INCHES)			6mm (0.25 INCHES)			13mm (0.50 INCHES)		
	PRESS ALT (FT)			PRESS ALT (FT)			PRESS ALT (FT)		
	S.L.	4000	8000	S.L.	4000	8000	S.L.	4000	8000
4200							28.9		
4600				30.7			35.5		
5000	34.6			37.6	28.1		42.1	33.1	28.4
5400	41.5	30.6		44.5	33.7	28.1	48.6	38.6	32.9
5800	48.7	36.3	30.0	51.8	39.3	33.1	55.2	44.0	37.3
6200	56.3	42.0	35.0	59.1	44.9	38.1	61.7	49.3	41.8
6600	63.8	47.9	40.0	66.5	51.0	43.1	68.2	54.5	46.2
7000	71.3	54.1	45.0	73.8	57.1	48.3	74.8	59.7	50.5
7400		60.4	50.4		63.2	53.5		65.0	54.8
7800		66.6	55.8		69.3	58.8		70.2	59.0
8200		72.8	61.3		75.4	64.1		75.4	63.3
8600			66.8			69.3			67.6
9000			72.2			74.6			71.8

1. **Enter Weight Adjustment table with slush/standing water depth and dry field/obstacle limit weight to obtain slush/standing water weight adjustment.**
2. **Adjust field length available by -99ft/+99ft for every 5°C above/below 4°C.**
3. **Find V1(MCG) limit weight for temperature adjusted field length and pressure altitude.**
4. **Max allowable slush/standing water limited weight is lesser of weights from 1 and 3.**

V1 Adjustment (KIAS)

WEIGHT (1000 KG)	SLUSH/STANDING WATER DEPTH								
	3mm (0.12 INCHES)			6mm (0.25 INCHES)			13mm (0.50 INCHES)		
	PRESS ALT (FT)			PRESS ALT (FT)			PRESS ALT (FT)		
	S.L.	4000	8000	S.L.	4000	8000	S.L.	4000	8000
68	-10	-10	-7	-3	-4	-4	-2	2	2
64	-12	-11	-9	-5	-5	-4	0	2	1
60	-14	-13	-10	-7	-6	-5	2	2	0
56	-16	-15	-12	-10	-8	-6	1	2	0
52	-18	-17	-14	-13	-11	-7	-1	1	1
48	-20	-19	-15	-16	-13	-8	-5	0	1
44	-21	-20	-17	-18	-16	-11	-8	-4	1
40	-22	-21	-18	-19	-17	-13	-11	-8	-1
36	-22	-21	-19	-19	-18	-15	-13	-10	-3

1. **Obtain V1, VR and V2 for the actual weight.**
2. **If V1(MCG) limited, set V1 = V1(MCG). If not V1(MCG) limited, enter V1 Adjustment Table with the actual weight to obtain V1 speed adjustment. If adjusted V1 is less than V1(MCG), set V1 = V1(MCG).**

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ADVISORY INFORMATION

Slippery Runway Takeoff
Weight Adjustment (1000 KG)

FIELD/OBSTACLE LIMIT WEIGHT (1000 KG)	REPORTED BRAKING ACTION								
	GOOD			MEDIUM			POOR		
	PRESS ALT (FT)			PRESS ALT (FT)			PRESS ALT (FT)		
	S.L.	4000	8000	S.L.	4000	8000	S.L.	4000	8000
68	-0.1	-0.1	-0.1	-4.1	-4.1	-4.1	-6.6	-6.6	-6.6
64	-0.5	-0.5	-0.5	-4.2	-4.2	-4.2	-6.5	-6.5	-6.5
60	-0.8	-0.8	-0.8	-4.2	-4.2	-4.2	-6.5	-6.5	-6.5
56	-0.8	-0.8	-0.8	-4.1	-4.1	-4.1	-6.3	-6.3	-6.3
52	-0.9	-0.9	-0.9	-3.8	-3.8	-3.8	-6.1	-6.1	-6.1
48	-0.8	-0.8	-0.8	-3.6	-3.6	-3.6	-5.9	-5.9	-5.9
44	-0.7	-0.7	-0.7	-3.6	-3.6	-3.6	-5.7	-5.7	-5.7
40	-0.7	-0.7	-0.7	-3.7	-3.7	-3.7	-5.6	-5.6	-5.6
36	-0.5	-0.5	-0.5	-4.1	-4.1	-4.1	-5.4	-5.4	-5.4

V1(MCG) Limit Weight (1000 KG)

ADJUSTED FIELD LENGTH (FT)	REPORTED BRAKING ACTION								
	GOOD			MEDIUM			POOR		
	PRESS ALT (FT)			PRESS ALT (FT)			PRESS ALT (FT)		
	S.L.	4000	8000	S.L.	4000	8000	S.L.	4000	8000
3800	37.8								
4200	49.7	34.2							
4600	61.1	46.2	30.5	29.3					
5000	72.6	57.7	42.6	37.5					
5400		69.1	54.2	45.8	35.1				
5800			65.7	54.9	43.3	32.6			
6200				64.0	52.2	40.8	32.9		
6600				73.0	61.2	49.4	38.6		
7000					70.3	58.5	44.2	32.9	
7400						67.6	50.6	38.6	
7800							57.1	44.2	32.9
8200							63.6	50.6	38.6
8600							70.2	57.1	44.2
9000								63.6	50.6
9400								70.2	57.1
9800									63.6
10200									70.2

1. **Enter Weight Adjustment table with reported braking action and dry field/obstacle limit weight to obtain slippery runway weight adjustment.**
2. **Adjust "Good" field length available by -81ft/+81ft for every 5°C above/below 4°C**
   Adjust "Medium" field length available by -81ft/+81ft for every 5°C above/below 4°C
   Adjust "Poor" field length available by -117ft/+117ft for every 5°C above/below 4°C
3. **Find V1(MCG) limit weight for temperature adjusted field length and pressure altitude.**
4. **Max allowable slippery runway limited weight is lesser of weights from 1 and 3.**

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ADVISORY INFORMATION

Slippery Runway Takeoff

V1 Adjustment (KIAS)

WEIGHT (1000 KG)	REPORTED BRAKING ACTION								
	GOOD			MEDIUM			POOR		
	PRESS ALT (FT)			PRESS ALT (FT)			PRESS ALT (FT)		
	S.L.	4000	8000	S.L.	4000	8000	S.L.	4000	8000
68	-4	-2	0	-10	-8	-6	-24	-21	-18
64	-5	-3	-1	-13	-11	-9	-26	-23	-20
60	-6	-4	-2	-15	-13	-11	-29	-26	-23
56	-8	-6	-4	-17	-15	-13	-32	-29	-26
52	-9	-7	-5	-19	-17	-15	-34	-31	-28
48	-10	-8	-6	-21	-19	-17	-36	-33	-30
44	-11	-9	-7	-22	-20	-18	-37	-34	-31
40	-11	-9	-7	-23	-21	-19	-37	-34	-31
36	-12	-10	-8	-25	-23	-21	-38	-35	-32

1. **Obtain V1, VR and V2 for the actual weight.**
2. **If V1(MCG) limited, set V1 = V1(MCG). If not V1(MCG) limited, enter V1 Adjustment Table with the actual weight to obtain V1 speed adjustment. If adjusted V1 is less than V1(MCG), set V1 = V1(MCG).**

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Takeoff %N1

Based on engine bleed to packs on (Auto) and anti-ice on or off

AIRPORT OAT		AIRPORT PRESSURE ALTITUDE (FT)									
°C	°F	-1000	0	1000	2000	3000	4000	5000	6000	7000	8000
55	131	89.9	90.5								
50	122	90.6	91.1	91.6	92.3	93.1					
45	113	91.2	91.6	92.1	92.6	93.9	94.9	94.7	94.2		
40	104	91.7	92.1	92.6	93.0	94.3	95.3	95.3	95.3	95.0	94.3
35	95	92.0	92.5	93.0	93.4	94.7	95.9	95.8	95.8	95.1	94.5
30	86	91.8	92.8	93.2	93.6	95.0	96.5	96.4	96.4	95.7	95.0
25	77	91.0	92.1	92.9	93.6	94.5	95.9	96.3	96.7	96.2	95.6
20	68	90.3	91.3	92.1	92.8	93.8	95.1	95.5	95.9	95.9	95.9
15	59	89.5	90.5	91.3	92.0	93.0	94.3	94.7	95.1	95.3	95.3
10	50	88.7	89.7	90.5	91.2	92.1	93.4	93.9	94.3	94.4	94.6
5	41	87.9	88.9	89.7	90.4	91.3	92.6	93.0	93.4	93.6	93.7
0	32	87.1	88.1	88.9	89.6	90.5	91.8	92.2	92.6	92.7	92.9
-10	14	85.5	86.5	87.2	87.9	88.8	90.1	90.5	90.9	91.0	91.2
-20	-4	83.9	84.8	85.6	86.2	87.1	88.3	88.7	89.1	89.2	89.4
-30	-22	82.2	83.1	83.9	84.5	85.4	86.6	87.0	87.4	87.5	87.6
-40	-40	80.5	81.4	82.1	82.7	83.6	84.8	85.1	85.5	85.7	85.8
-50	-58	78.8	79.6	80.3	81.0	81.8	82.9	83.3	83.7	83.8	84.0

%N1 Adjustments for Engine Bleeds

BLEED CONFIGURATION	AIRPORT PRESSURE ALTITUDE (FT)	
	-1000	8000
ENGINE BLEED TO PACKS OFF	1.0	1.0

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Assumed Temperature Reduced Thrust

Maximum Assumed Temperature (Table 1)

Based on 25% Takeoff Thrust Reduction

OAT (°C)	PRESS ALT (1000 FT)										OAT (°F)	PRESS ALT (1000 FT)									
	-1	0	1	2	3	4	5	6	7	8		-1	0	1	2	3	4	5	6	7	8
55	74	73	73								130	164	162	162							
50	71	71	69	68	66	65					120	159	158	155	153	149	147				
45	69	68	67	66	63	62	62	63	65		110	154	153	151	149	144	142	142	143	145	
40	66	66	65	64	61	60	60	60	60	62	100	150	148	146	145	140	138	138	137	139	141
35	64	63	62	62	59	58	58	57	59	60	90	146	143	142	141	137	134	133	132	136	137
30	63	61	61	60	58	56	55	55	56	58	80	146	141	140	137	135	131	129	128	131	132
25	63	61	60	58	57	55	54	53	54	55	70	146	141	140	136	134	131	129	127	127	126
20 & BELOW	63	61	60	58	57	55	54	53	53	52	60 & BELOW	146	141	140	136	134	131	129	127	127	126

Maximum Takeoff %N1 (Table 2)

Based on engine bleed to packs on (Auto) and engine anti-ice on or off

ASSUMED TEMP		AIRPORT PRESSURE ALTITUDE (FT)									
°C	°F	-1000	0	1000	2000	3000	4000	5000	6000	7000	8000
75	167	83.0	83.9	83.8	83.5						
70	158	84.7	85.6	85.6	85.4	84.9	84.7				
65	149	86.6	87.3	87.3	87.2	87.0	87.0	87.1	87.2	87.6	87.9
60	140	88.3	88.9	89.0	89.0	89.1	89.1	89.1	89.0	89.1	89.3
55	131	89.7	90.5	90.6	90.7	91.1	91.3	91.0	90.8	90.7	90.6
50	122	90.4	91.1	91.6	92.3	93.1	93.4	92.9	92.5	92.2	91.9
45	113	91.0	91.6	92.1	92.7	93.9	94.9	94.8	94.2	93.7	93.1
40	104	91.6	92.1	92.6	93.0	94.3	95.3	95.3	95.3	95.0	94.3
35	95	91.4	92.5	93.0	93.4	94.8	95.9	95.9	95.8	95.1	94.5
30	86	91.7	92.8	93.2	93.6	95.0	96.5	96.4	96.4	95.7	95.0
25	77	90.9		93.3	93.7	94.5	96.5	96.5	96.7	96.2	95.6
20	68	90.2				94.4				95.9	95.9
15	59	89.4									95.4
MINIMUM ASSUMED TEMP °C (°F)		32(90)	30(86)	28(82)	26(79)	24(75)	29(85)	27(81)	25(77)	20(68)	15(59)

With engine bleed to packs off, increase %N1 by 1.0.

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Assumed Temperature Reduced Thrust

%N1 Adjustment For Temperature Difference (Table 3)

ASSUMED TEMP MINUS OAT		OUTSIDE AIR TEMPERATURE													
°C	°F	°C	-40	-20	0	5	10	15	20	25	30	35	40	45	50	55
		°F	-40	-4	32	41	50	59	68	77	86	95	104	113	122	131
10	18						1.6	1.6	1.6	1.5	1.5	1.5	1.4	1.4	1.4	1.3
20	36				3.3	3.3	3.1	3.1	3.0	3.0	2.9	2.8	2.7	2.6	2.5	2.3
30	54				4.8	4.7	4.6	4.5	4.4	4.3	4.1	3.9	3.7	3.5	3.4	3.3
40	72			6.8	6.1	6.0	5.9	5.7	5.5	5.3	5.1	4.9				
50	90			8.1	7.4	7.2	6.9	6.7	6.3							
60	108		10.4	9.3	8.4	8.1	7.7									
70	126		11.5	10.5	9.0											
80	144		12.7	11.4	9.8											
90	162		13.8	11.9												
100	180		14.5	12.6												
110	198		14.9													

1. **Determine Maximum Assumed Temperature allowed from Table 1.**
2. **Find Maximum %N1 from Table 2 using the desired assumed temperature (no greater than temperature from Table 1).**
3. **Use the difference between assumed temperature and OAT to determine the %N1 adjustment from Table 3.**
4. **Subtract %N1 adjustment from Maximum %N1 in Table 2.**

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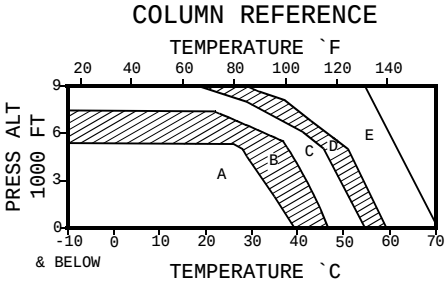
Takeoff Speeds (18.5K Derate)

Flaps 5



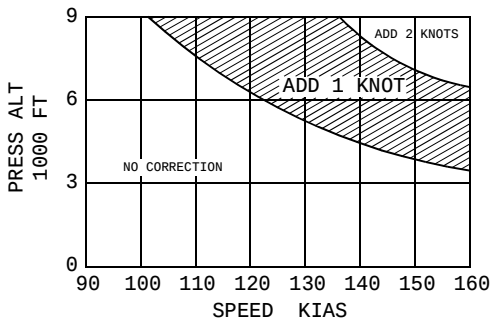
SLOPE/WIND V_1 ADJUSTMENT*						
WEIGHT 1000 KG	SLOPE % DN		UP	WIND KTS TAIL		HEAD
	-2	0	2	-15	0	40
65	-2	0	2	-1	0	2
55	-2	0	1	-2	0	1
45	-2	0	1	-3	0	1
35	-2	0	2	-3	0	1

*V_1 NOT TO EXCEED V_R

WT 1000 KG	A			B			C			D			E		
	V_1	V_R	V_2	V_1	V_R	V_2	V_1	V_R	V_2	V_1	V_R	V_2	V_1	V_R	V_2
65	152	155	158	153	156	158	154	157	158	156	157	158	157	158	158
60	146	148	152	147	149	152	149	150	152	150	151	152	151	152	152
55	139	140	146	140	142	146	142	143	146	143	144	146	145	145	146
50	131	132	139	133	134	139	134	135	139	135	136	139	137	137	139
45	122	123	132	124	125	132	125	126	132	127	127	132	128	128	131
40	113	115	125	115	116	124	116	117	124	117	118	124	119	119	123
35	103	106	118	105	107	117	106	108	117	107	109	116	109	110	116
30	92	96	110	93	97	109	95	98	109	97	100	110	98	100	108

CHECK V1(MCG) AND MINIMUM VR.

ALTITUDE V_1, V_R, V_2 ADJUSTMENT



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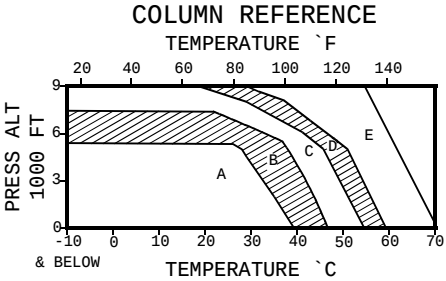
Takeoff Speeds (18.5K Derate)
Flaps 15



SLOPE/WIND V_1 ADJUSTMENT*						
WEIGHT 1000 KG	SLOPE % DN		UP	WIND KTS TAIL		HEAD
	-2	0	2	-15	0	40
65	-2	0	2	-1	0	2
55	-2	0	1	-2	0	1
45	-2	0	1	-3	0	1
35	-2	0	2	-3	0	1

*V_1 NOT TO EXCEED V_R

WT 1000 KG	A			B			C			D			E		
	V_1	V_R	V_2	V_1	V_R	V_2	V_1	V_R	V_2	V_1	V_R	V_2	V_1	V_R	V_2
55	132	132	137	132	133	137	134	134	138	136	136	138	137	137	139
50	124	125	131	126	126	131	127	127	131	128	128	131	129	129	132
45	117	117	125	118	118	125	119	119	124	120	120	125	121	121	125
40	107	109	119	109	110	118	111	111	118	112	112	118	113	113	118
35	97	100	112	99	101	112	101	102	111	103	103	111	104	105	110
30	87	91	104	88	92	105	89	94	105	92	95	104	94	96	104

CHECK V1(MCG) AND MINIMUM VR.

V1(MCG), Minimum VR (18.5K Derate)

ACTUAL OAT		PRESSURE ALTITUDE FT											
		-1000		0		2000		4000		6000		8000	
`C	`F	V1(MCG)	MIN VR	V1(MCG)	MIN VR	V1(MCG)	MIN VR	V1(MCG)	MIN VR	V1(MCG)	MIN VR	V1(MCG)	MIN VR
55	131	95	98	94	97								
50	122	98	100	97	99	95	97	92	95	88	91	81	85
40	104	102	104	102	104	100	103	98	101	94	97	88	92
30	86	105	107	104	107	103	106	101	104	99	102	93	96
20	68	105	107	105	107	104	106	102	105	100	103	96	99
10	50	106	107	106	107	104	106	102	105	101	103	97	100
-50	-58	107	108	107	108	105	107	104	106	102	104	98	100

FOR A/C OFF INCREASE V1(MCG) BY 2 KNOTS
FOR PMC OFF INCREASE V1(MCG) BY 6 KNOTS
FOR PMC OFF INCREASE MIN VR BY 5 KNOTS

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Clearway and Stopway V1 Adjustments (18.5K Derate)

CLEARWAY MINUS STOPWAY (FT)	NORMAL V1 (KIAS)			
	100	120	140	160
800			-2	-1
600		-3	-2	-1
400	-4	-3	-2	0
200	-2	-1	-1	0
0	0	0	0	0
-400	1	1	1	1
-800	1	1	1	1

Maximum Allowable Clearway (18.5K Derate)

FIELD LENGTH (FT)	MAX ALLOWABLE CLEARWAY FOR V1 REDUCTION (FT)
4000	450
6000	500
8000	550
10000	600
12000	650
14000	700

Stab Trim Setting (18.5K Derate)

C.G. % MAC	6	10	14	18	22	26	30
STAB TRIM	6 1/4	5 1/2	5	4 1/4	3 3/4	3	2 1/2

FOR WEIGHTS AT OR BELOW 45,360 KG SUBTRACT 1/2 UNIT.
FOR WEIGHTS AT OR ABOVE 54,430 KG ADD 1/2 UNIT.

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ADVISORY INFORMATION

Slush/Standing Water Takeoff (18.5K Derate)

Weight Adjustment (1000 KG)

18.5K DERATE FIELD/OBSTACLE LIMIT WEIGHT (1000 KG)	SLUSH/STANDING WATER DEPTH								
	3mm (0.12 INCHES)			6mm (0.25 INCHES)			13mm (0.50 INCHES)		
	PRESS ALT (FT)			PRESS ALT (FT)			PRESS ALT (FT)		
	S.L.	4000	8000	S.L.	4000	8000	S.L.	4000	8000
68	-7.7	-8.0	-8.4	-9.6	-10.0	-10.4	-13.1	-10.5	-12.3
64	-7.4	-7.9	-8.4	-9.1	-9.9	-10.5	-12.6	-12.2	-12.7
60	-6.8	-7.5	-8.1	-8.3	-9.3	-10.1	-11.6	-12.4	-12.6
56	-6.1	-6.8	-7.6	-7.3	-8.4	-9.4	-10.2	-11.6	-12.1
52	-5.4	-6.1	-6.9	-6.4	-7.4	-8.5	-8.8	-10.2	-11.2
48	-4.8	-5.3	-6.0	-5.5	-6.3	-7.5	-7.5	-8.7	-10.1
44	-4.2	-4.6	-5.2	-4.8	-5.4	-6.3	-6.5	-7.4	-8.8
40	-3.8	-3.9	-4.4	-4.4	-4.7	-5.3	-6.3	-6.8	-7.5
36	-3.6	-3.6	-3.7	-4.2	-4.3	-4.3	-6.8	-7.3	-6.1

V1(MCG) Limit Weight (1000 KG)

ADJUSTED FIELD LENGTH (FT)	SLUSH/STANDING WATER DEPTH								
	3mm (0.12 INCHES)			6mm (0.25 INCHES)			13mm (0.50 INCHES)		
	PRESS ALT (FT)			PRESS ALT (FT)			PRESS ALT (FT)		
	S.L.	4000	8000	S.L.	4000	8000	S.L.	4000	8000
4200							31.2		
4600	31.2			33.5			38.0	30.4	
5000	38.3	28.9		40.6	31.8		44.7	35.9	30.2
5400	45.4	34.9		47.8	37.7	30.8	51.2	41.5	35.1
5800	52.8	40.8	33.3	55.4	43.6	36.0	57.6	47.1	40.1
6200	60.6	46.8	38.6	63.2	49.5	41.3	64.1	52.9	45.0
6600	68.5	53.0	43.9	70.9	55.8	46.6	70.6	58.8	49.9
7000		59.6	49.2		62.1	51.9		64.6	54.7
7400		66.1	54.7		68.4	57.2		70.4	59.6
7800		72.7	60.3		74.7	62.5			64.5
8200			65.9			67.8			69.4
8600			71.6			73.1			74.2

1. **Enter Weight Adjustment table with slush/standing water depth and 18.5K derate dry field/obstacle limit weight to obtain slush/standing water weight adjustment.**
2. **Adjust field length available by -99ft/+99ft for every 5°C above/below 4°C**
3. **Find V1(MCG) limit weight for temperature adjusted field length and pressure altitude.**
4. **Max allowable slush/standing water limited weight is lesser of weights from 1 and 3.**

V1 Adjustment (KIAS)

WEIGHT (1000 KG)	SLUSH/STANDING WATER DEPTH								
	3mm (0.12 INCHES)			6mm (0.25 INCHES)			13mm (0.50 INCHES)		
	PRESS ALT (FT)			PRESS ALT (FT)			PRESS ALT (FT)		
	S.L.	4000	8000	S.L.	4000	8000	S.L.	4000	8000
68	-7	-6	-4	0	-1	-2	-4	-2	3
64	-9	-8	-6	-2	-2	-3	-2	-1	2
60	-11	-10	-8	-5	-3	-3	1	0	0
56	-14	-12	-9	-7	-5	-3	1	1	-1
52	-16	-14	-11	-10	-8	-5	0	0	0
48	-18	-16	-13	-13	-10	-6	-3	-1	0
44	-19	-18	-15	-16	-13	-8	-6	-2	0
40	-20	-19	-16	-18	-15	-10	-9	-4	0
36	-21	-20	-18	-18	-17	-13	-11	-7	-2

1. **Obtain V1, VR and V2 for the actual weight.**
2. **If V1(MCG) limited, set V1 = V1(MCG). If not V1(MCG) limited, enter V1 Adjustment Table with the actual weight to obtain V1 speed adjustment. If adjusted V1 is less than V1(MCG), set V1 = V1(MCG).**

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ADVISORY INFORMATION

Slippery Runway Takeoff (18.5K Derate)

Weight Adjustment (1000 KG)

18.5K DERATE FIELD/OBSTACLE LIMIT WEIGHT (1000 KG)	REPORTED BRAKING ACTION								
	GOOD			MEDIUM			POOR		
	PRESS ALT (FT)			PRESS ALT (FT)			PRESS ALT (FT)		
	S.L.	4000	8000	S.L.	4000	8000	S.L.	4000	8000
68	0.0	0.0	0.0	-3.5	-3.5	-3.5	-6.8	-6.8	-6.8
64	-0.4	-0.4	-0.4	-4.0	-4.0	-4.0	-7.1	-7.1	-7.1
60	-0.8	-0.8	-0.8	-4.2	-4.2	-4.2	-7.1	-7.1	-7.1
56	-1.0	-1.0	-1.0	-4.2	-4.2	-4.2	-6.9	-6.9	-6.9
52	-1.0	-1.0	-1.0	-4.0	-4.0	-4.0	-6.4	-6.4	-6.4
48	-0.9	-0.9	-0.9	-3.7	-3.7	-3.7	-5.9	-5.9	-5.9
44	-0.8	-0.8	-0.8	-3.4	-3.4	-3.4	-5.4	-5.4	-5.4
40	-0.8	-0.8	-0.8	-3.2	-3.2	-3.2	-5.0	-5.0	-5.0
36	-1.0	-1.0	-1.0	-3.2	-3.2	-3.2	-4.7	-4.7	-4.7

V1(MCG) Limit Weight (1000 KG)

ADJUSTED FIELD LENGTH (FT)	REPORTED BRAKING ACTION								
	GOOD			MEDIUM			POOR		
	PRESS ALT (FT)			PRESS ALT (FT)			PRESS ALT (FT)		
	S.L.	4000	8000	S.L.	4000	8000	S.L.	4000	8000
3400	29.3								
3800	41.6								
4200	52.9	39.2							
4600	63.8	50.8	36.7	32.6					
5000	74.6	61.6	48.6	41.1	30.9				
5400		72.4	59.4	49.9	39.4	29.2			
5800			70.3	59.0	48.1	37.7	30.1		
6200				68.0	57.2	46.3	36.0		
6600					66.2	55.3	41.8	29.0	
7000					75.3	64.4	47.7	34.8	
7400						73.5	54.0	40.7	
7800							60.6	46.5	33.7
8200							67.1	52.7	39.5
8600							73.7	59.2	45.4
9000								65.8	51.4
9400								72.4	57.9
9800									64.5
10200									71.0

1. **Enter Weight Adjustment table with reported braking action and 18.5K derate dry field/obstacle limit weight to obtain slippery runway weight adjustment.**
2. **Adjust "Good" field length available by -81ft/+81ft for every 5°C above/below 4°C**
   Adjust "Medium" field length available by -81ft/+81ft for every 5°C above/below 4°C
   Adjust "Poor" field length available by -117ft/+117ft for every 5°C above/below 4°C
3. **Find V1(MCG) limit weight for temperature adjusted field length and pressure altitude.**
4. **Max allowable slippery runway limited weight is lesser of weights from 1 and 3.**

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ADVISORY INFORMATION

Slippery Runway Takeoff (18.5K Derate)

V1 Adjustment (KIAS)

WEIGHT (1000 KG)	REPORTED BRAKING ACTION								
	GOOD			MEDIUM			POOR		
	PRESS ALT (FT)			PRESS ALT (FT)			PRESS ALT (FT)		
	S.L.	4000	8000	S.L.	4000	8000	S.L.	4000	8000
68	-2	0	2	-8	-6	-4	-18	-15	-12
64	-4	-2	0	-10	-8	-6	-20	-17	-14
60	-5	-3	-1	-13	-11	-9	-23	-20	-17
56	-7	-5	-3	-15	-13	-11	-25	-22	-19
52	-9	-7	-5	-17	-15	-13	-28	-25	-22
48	-10	-8	-6	-19	-17	-15	-30	-27	-24
44	-10	-8	-6	-21	-19	-17	-32	-29	-26
40	-11	-9	-7	-22	-20	-18	-34	-31	-28
36	-12	-10	-8	-24	-22	-20	-37	-34	-31

1. **Obtain V1, VR and V2 for the actual weight.**
2. **If V1(MCG) limited, set V1 = V1(MCG). If not V1(MCG) limited, enter V1 Adjustment Table with the actual weight to obtain V1 speed adjustment. If adjusted V1 is less than V1(MCG), set V1 = V1(MCG).**

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Takeoff %N1 (18.5K Derate)

Based on engine bleed to packs on (Auto) and anti-ice on or off

AIRPORT OAT		AIRPORT PRESSURE ALTITUDE (FT)									
°C	°F	-1000	0	1000	2000	3000	4000	5000	6000	7000	8000
55	131	87.3	87.9	87.7	87.6						
50	122	87.9	88.7	89.4	90.1	90.2					
45	113	88.5	89.3	90.0	90.6	91.3	92.0	92.6			
40	104	89.2	90.0	90.6	91.2	91.8	92.7	93.3	93.1	92.6	
35	95	89.5	90.4	91.2	91.8	92.4	93.2	93.8	93.6	93.1	92.6
30	86	89.3	90.6	91.2	91.9	92.7	93.7	94.3	94.1	93.6	93.1
25	77	88.6	89.9	90.7	91.5	92.3	93.1	94.2	94.5	94.1	93.6
20	68	87.8	89.1	89.9	90.7	91.5	92.3	93.4	93.8	93.7	93.8
15	59	87.1	88.3	89.2	90.0	90.8	91.5	92.6	92.9	92.9	93.0
10	50	86.3	87.6	88.4	89.2	90.0	90.7	91.8	92.1	92.1	92.1
5	41	85.6	86.8	87.6	88.4	89.2	89.9	91.0	91.3	91.3	91.3
0	32	84.8	86.0	86.8	87.6	88.4	89.1	90.2	90.5	90.5	90.5
-10	14	83.2	84.4	85.2	86.0	86.8	87.5	88.5	88.8	88.8	88.8
-20	-4	81.6	82.8	83.6	84.3	85.1	85.8	86.8	87.1	87.1	87.1
-30	-22	80.0	81.2	81.9	82.6	83.4	84.1	85.1	85.4	85.4	85.4
-40	-40	78.4	79.5	80.2	80.9	81.7	82.3	83.3	83.6	83.6	83.6
-50	-58	76.7	77.7	78.5	79.2	79.9	80.5	81.5	81.8	81.8	81.8

%N1 Adjustments for Engine Bleeds

BLEED CONFIGURATION	AIRPORT PRESSURE ALTITUDE (FT)	
	-1000	8000
ENGINE BLEED TO PACKS OFF	0.9	0.9

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Assumed Temperature Reduced Thrust (18.5K Derate)

Maximum Assumed Temperature (Table 1)

Based on 25% Takeoff Thrust Reduction

OAT (°C)	PRESS ALT (1000 FT)										OAT (°F)	PRESS ALT (1000 FT)									
	-1	0	1	2	3	4	5	6	7	8		-1	0	1	2	3	4	5	6	7	8
55	69	69	69	69							130	155	155	155	155						
50	67	67	66	65	66	66					120	152	151	149	148	149	149				
45	65	65	64	63	63	62	62	62	62		110	148	148	145	144	144	142	142	141	141	
40	63	63	62	61	61	60	60	59	58	57	100	145	144	142	140	140	138	138	136	134	132
35	62	61	60	59	59	58	58	57	56	54	90	142	141	139	137	136	134	134	132	130	128
30	61	60	59	58	57	56	56	55	54	52	80	142	140	137	135	135	132	131	128	126	124
25	61	60	58	57	57	56	55	53	52	50	70	142	140	137	135	134	132	131	127	124	120
20 & BELOW	61	60	58	57	57	56	55	53	51	49	60 & BELOW	142	140	137	135	134	132	131	127	124	120

Maximum Takeoff %N1 (Table 2)

Based on engine bleed to packs on (Auto) and engine anti-ice on or off

ASSUMED TEMP		AIRPORT PRESSURE ALTITUDE (FT)									
°C	°F	-1000	0	1000	2000	3000	4000	5000	6000	7000	8000
75	167	76.8	76.6	76.5	76.6	78.3					
70	158	79.6	79.5	79.4	79.4	80.8	80.9				
65	149	82.3	82.4	82.2	82.2	83.2	83.3	84.1	83.3	82.2	81.0
60	140	84.9	85.2	85.0	84.9	85.6	85.6	86.3	85.5	84.4	83.3
55	131	87.2	87.9	87.7	87.6	87.9	87.9	88.5	87.7	86.6	85.5
50	122	87.8	88.7	89.4	90.1	90.2	90.2	90.6	89.8	88.7	87.6
45	113	88.8	89.3	90.0	90.6	91.3	92.0	92.6	91.9	90.8	89.7
40	104	89.1	90.0	90.6	91.2	91.8	92.7	93.3	93.1	92.6	91.7
35	95	89.5	90.4	91.2	91.8	92.4	93.2	93.8	93.6	93.1	92.6
30	86	89.2	90.6	91.2	91.9	92.7	93.7	94.3	94.1	93.6	93.1
25	77	88.5	89.9	90.7	91.5	92.3	93.1	94.2	94.5	94.1	93.6
20	68	87.8	89.1	89.9	90.7	91.5	92.3	93.4	93.8	93.7	93.8
15	59	87.0	88.3	89.2	90.0	90.8	91.5	92.6	92.9	92.9	93.0
MINIMUM ASSUMED TEMP °C (°F)		32(90)	30(86)	28(82)	26(79)	24(75)	29(85)	27(81)	25(77)	23(73)	21(70)

With engine bleed to packs off, increase %N1 by 1.0.

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Assumed Temperature Reduced Thrust (18.5K Derate)

%N1 Adjustment For Temperature Difference (Table 3)

ASSUMED TEMP MINUS OAT		OUTSIDE AIR TEMPERATURE														
°C	°F	°C	-40	-20	0	5	10	15	20	25	30	35	40	45	50	55
		°F	-40	-4	32	41	50	59	68	77	86	95	104	113	122	131
10	18						1.5	1.5	1.5	1.5	1.4	1.4	1.4	1.4	1.3	1.2
20	36				3.1	3.1	3.1	3.0	2.9	2.9	2.8	2.7	2.6	2.5	2.3	2.2
30	54				4.7	4.6	4.5	4.4	4.3	4.1	3.9	3.7	3.6	3.4	3.2	3.0
40	72			6.5	6.1	5.9	5.8	5.5	5.3	5.1	4.9	4.6				
50	90			8.1	7.3	7.0	6.7	6.4	5.3	5.1						
60	108		10.1	9.4	8.2	7.8	7.5	6.4								
70	126		11.6	10.3	8.7	8.4										
80	144		12.7	10.8	9.1	8.4										
90	162		13.2	11.1	9.2											
100	180		13.6	11.4	9.3											
110	198		13.8	11.4												

1. **Determine Maximum Assumed Temperature allowed from Table 1.**
2. **Find Maximum %N1 from Table 2 using the desired assumed temperature (no greater than temperature from Table 1).**
3. **Use the difference between assumed temperature and OAT to determine the %N1 adjustment from Table 3.**
4. **Subtract %N1 adjustment from Maximum %N1 in Table 2.**

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Max Climb %N1

Based on engine bleed to packs on (Auto) and anti-ice off

TAT (°C)	PRESSURE ALTITUDE (1000 FT)/SPEED (KIAS OR MACH)								
	0	5	10	15	20	25	30	35	37
	250	250	250	280	280	280	.74	.74	.74
50	88.9	89.0	89.2						
40	89.8	90.0	90.2	90.7					
30	89.9	90.9	91.1	91.6	91.9	92.1			
20	88.4	90.5	91.8	92.5	92.8	93.0	93.2		
10	86.8	88.9	91.0	92.7	93.5	93.8	94.0	94.0	94.0
0	85.3	87.4	89.4	91.1	93.1	94.4	94.6	94.6	94.6
-10	83.7	85.7	87.7	89.4	91.3	93.1	94.6	95.2	95.2
-20	82.1	84.1	86.0	87.7	89.6	91.3	92.8	95.7	96.0
-30	80.5	82.4	84.3	85.9	87.8	89.5	90.9	93.8	94.5
-40	78.8	80.7	82.6	84.1	86.0	87.6	89.0	91.9	92.5
-50	77.1	79.0	80.8	82.3	84.1	85.7	87.1	89.9	90.5

%N1 Adjustments for Engine Bleeds

BLEED CONFIGURATION	PRESSURE ALTITUDE (1000 FT)								
	0	5	10	15	20	25	30	35	37
ENGINE BLEED TO PACKS OFF	0.5	0.5	0.6	0.7	0.8	0.9	0.9	0.8	0.7
PACKS HIGH	-0.2	-0.3	-0.3	-0.3	-0.4	-0.4	-0.4	-0.5	-0.6
ENGINE ANTI-ICE ON	-0.7	-0.8	-0.9	-1.0	-1.0	-1.0	-1.0	-1.0	-1.0
ENGINE & WING ANTI-ICE ON	-1.9	-2.0	-2.2	-2.4	-2.6	-2.9	-2.8	-2.9	-2.9

Go-around %N1

Based on engine bleed to packs on (Auto), engine anti-ice on or off, and wing anti-ice off

REPORTED OAT		TAT	PRESSURE ALTITUDE (FT)									
°C	°F	°C	-1000	0	1000	2000	3000	4000	5000	6000	7000	8000
55	131	58	89.8	90.5	91.1	91.8	93.3	94.2	93.9	93.6	94.6	95.6
50	122	53	90.5	91.0	91.6	92.2	93.6	94.6	94.4	94.2	94.8	95.4
45	113	48	91.1	91.5	92.1	92.6	93.9	94.9	94.8	94.8	94.9	95.1
40	104	43	91.6	92.1	92.4	93.0	94.3	95.3	95.3	95.3	95.1	94.8
35	95	38	92.0	92.5	93.0	93.4	94.8	95.9	95.9	95.9	95.1	94.5
30	86	33	92.0	92.8	93.2	93.6	95.0	96.5	96.4	96.4	95.7	95.0
25	77	28	91.2	92.0	92.8	93.5	94.5	95.9	96.3	96.8	96.3	95.7
20	68	23	90.4	91.2	92.0	92.8	93.8	95.1	95.5	96.0	96.0	96.0
15	59	18	89.7	90.4	91.2	92.0	93.0	94.3	94.7	95.2	95.3	95.4
10	50	13	88.9	89.6	90.4	91.2	92.1	93.4	93.9	94.3	94.4	94.6
5	41	8	88.1	88.8	89.6	90.4	91.3	92.6	93.0	93.5	93.6	93.8
0	32	3	87.4	88.0	88.8	89.5	90.5	91.8	92.2	92.6	92.7	92.9
-10	14	-8	85.6	86.4	87.2	87.9	88.8	90.1	90.5	90.9	91.0	91.2
-20	-4	-18	84.0	84.8	85.5	86.2	87.1	88.3	88.8	89.2	89.3	89.5
-30	-22	-28	82.4	83.1	83.8	84.5	85.4	86.6	87.0	87.4	87.5	87.7
-40	-40	-38	80.7	81.3	82.1	82.7	83.6	84.8	85.2	85.6	85.7	85.9
-50	-58	-48	78.9	79.6	80.3	80.9	81.8	82.9	83.3	83.7	83.8	84.0

%N1 Adjustments for Engine Bleeds

BLEED CONFIGURATION	TAT (°C)	
	- 60	+ 60
ENGINE BLEED TO PACKS OFF	0.8	1.0
PACKS HIGH	- 0.4	- 0.4
WING A/I ALL ENGINES	- 1.3	- 1.6
WING A/I 1 ENGINE INOP	- 2.1	- 2.6

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Flight With Unreliable Airspeed / Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

Climb

Flaps Up, Set Max Climb Thrust

PRESSURE ALTITUDE (FT) (SPEED)		WEIGHT (1000 KG)				
		32	34	44	54	64
35000	**PITCH ATT**	**6.0**	**6.0**	**5.5**	**5.0**	
(.73M)	V/S (FT/MIN)	2700	2500	1500	800	
30000	**PITCH ATT**	**6.0**	**6.0**	**5.5**	**5.0**	**5.0**
(.73M)	V/S (FT/MIN)	3400	3100	2100	1400	800
25000	**PITCH ATT**	**6.5**	**6.0**	**5.5**	**5.5**	**5.5**
(280 KIAS)	V/S (FT/MIN)	3200	3000	2100	1500	1100
20000	**PITCH ATT**	**8.0**	**7.5**	**7.0**	**6.5**	**6.5**
(280 KIAS)	V/S (FT/MIN)	4000	3800	2700	2000	1500
15000	**PITCH ATT**	**9.5**	**9.5**	**8.0**	**7.5**	**7.5**
(280 KIAS)	V/S (FT/MIN)	4800	4500	3400	2500	1900
10000	**PITCH ATT**	**11.5**	**11.0**	**9.5**	**9.0**	**8.5**
(280 KIAS)	V/S (FT/MIN)	5600	5200	3900	3000	2400
5000	**PITCH ATT**	**13.5**	**13.0**	**11.0**	**10.0**	**9.5**
(280 KIAS)	V/S (FT/MIN)	6300	5900	4400	3500	2800
SEA LEVEL	**PITCH ATT**	**16.0**	**15.5**	**12.5**	**11.5**	**10.5**
(280 KIAS)	V/S (FT/MIN)	7100	6600	5000	3900	3100

Cruise

Flaps Up, %N1 for Level Flight

PRESSURE ALTITUDE (FT) (SPEED)		WEIGHT (1000 KG)					
		34	38	42	46	50	54
30000	**PITCH ATT**	**2.0**	**2.0**	**2.5**	**2.5**	**3.0**	**3.0**
(.73M)	%N1	79	79	80	81	82	83
10000	**PITCH ATT**	**2.0**	**2.0**	**2.5**	**3.0**	**3.0**	**3.5**
(280 KIAS)	%N1	65	66	66	67	68	68

Descent

Flaps Up, Set Idle Thrust

PRESSURE ALTITUDE (FT) (SPEED)		WEIGHT (1000 KG)					
		34	38	42	46	50	54
30000	**PITCH ATT**	**-3.5**	**-3**	**-2.5**	**-2**	**-1.5**	**-1**
(.73M)	V/S (FT/MIN)	-4100	-3800	-3600	-3400	-3300	-3200
20000	**PITCH ATT**	**-2.5**	**-1.5**	**-1**	**-0.5**	**0**	**0.5**
(280 KIAS)	V/S (FT/MIN)	-2800	-2600	-2400	-2300	-2200	-2100
10000	**PITCH ATT**	**-2.5**	**-2**	**-1.5**	**-0.5**	**0**	**0.5**
(280 KIAS)	V/S (FT/MIN)	-2500	-2300	-2100	-2000	-1900	-1900

Holding

Flaps Up, %N1 for Level Flight

PRESSURE ALTITUDE (FT)		WEIGHT (1000 KG)					
		34	38	42	46	50	54
	PITCH ATT	**4.0**	**5.0**	**5.5**	**6.0**	**5.5**	**5.5**
10000	%N1	56	57	59	61	63	65
	KIAS	210	210	210	215	225	230

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Flight With Unreliable Airspeed/Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

Terminal Area

%N1 for Level Flight

FLAP POSITION (SPEED)		WEIGHT (1000 KG)				
		32	36	41	45	50
FLAPS UP (GEAR UP)	**PITCH ATT**	**3.7**	**4.4**	**5.0**	**5.7**	**6.4**
(210 KIAS)	%N1	53	54	56	58	60
FLAPS 1 (GEAR UP)	**PITCH ATT**	**3.9**	**4.7**	**5.4**	**6.2**	**7.0**
(190 KIAS)	%N1	55	57	58	60	62
FLAPS 5 (GEAR UP)	**PITCH ATT**	**3.9**	**4.8**	**5.8**	**6.7**	**7.6**
(170 KIAS)	%N1	53	56	58	61	63
FLAPS 15 (GEAR DOWN)	**PITCH ATT**	**3.8**	**4.9**	**5.9**	**7.0**	**8.1**
(150 KIAS)	%N1	55	58	61	64	67

Final Approach

Gear Down, %N1 for 3° Glideslope

FLAP POSITION		WEIGHT (1000 KG)					
		32	36	41	45	50	54
FLAPS 40	**PITCH ATT**	**0.7**	**0.8**	**0.9**	**0.9**	**1.0**	**1.1**
	%N1	49	53	56	58	61	63
	KIAS	111	115	125	132	138	144
FLAPS 30	**PITCH ATT**	**2.4**	**2.4**	**2.5**	**2.5**	**2.5**	**2.6**
	%N1	43	46	49	52	54	57
	KIAS	114	121	129	135	142	148
FLAPS 15	**PITCH ATT**	**4.9**	**4.9**	**5.0**	**5.0**	**5.0**	**5.0**
	%N1	39	42	44	46	49	51
	KIAS	120	128	136	143	150	157

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Long Range Cruise Maximum Operating Altitude

Max Cruise Thrust

ISA + 10°C and Below

WEIGHT (1000 KG)	OPTIMUM ALT (FT)	TAT (°C)	MARGIN TO INITIAL BUFFET 'G' (BANK ANGLE)				
			1.20 (33°)	1.25 (36°)	1.30 (39°)	1.40 (44°)	1.50 (48°)
64	30800	-10	33400*	33200	32300	30700	29200
62	31500	-11	34300*	33800	33000	31400	29900
60	32200	-13	35100*	34500	33700	32100	30600
58	32900	-14	36000*	35300	34400	32800	31400
56	33700	-16	36600*	36000	35200	33600	32100
54	34400	-18	37000	36700	35900	34400	32900
52	35200	-19	37000	37000	36700	35200	33700
50	36100	-21	37000	37000	37000	36000	34500
48	36900	-21	37000	37000	37000	36800	35400
46	37000	-21	37000	37000	37000	37000	36300
44	37000	-21	37000	37000	37000	37000	37000
42	37000	-21	37000	37000	37000	37000	37000
40	37000	-21	37000	37000	37000	37000	37000
38	37000	-21	37000	37000	37000	37000	37000
36	37000	-21	37000	37000	37000	37000	37000

ISA + 15°C

WEIGHT (1000 KG)	OPTIMUM ALT (FT)	TAT (°C)	MARGIN TO INITIAL BUFFET 'G' (BANK ANGLE)				
			1.20 (33°)	1.25 (36°)	1.30 (39°)	1.40 (44°)	1.50 (48°)
64	30800	-4	32500*	32500*	32300	30700	29200
62	31500	-6	33400*	33400*	33000	31400	29900
60	32200	-7	34300*	34300*	33700	32100	30600
58	32900	-9	35200*	35200*	34400	32800	31400
56	33700	-10	36100*	36000	35200	33600	32100
54	34400	-12	36700*	36700	35900	34400	32900
52	35200	-14	37000	37000	36700	35200	33700
50	36100	-16	37000	37000	37000	36000	34500
48	36900	-16	37000	37000	37000	36800	35400
46	37000	-16	37000	37000	37000	37000	36300
44	37000	-16	37000	37000	37000	37000	37000
42	37000	-16	37000	37000	37000	37000	37000
40	37000	-16	37000	37000	37000	37000	37000
38	37000	-16	37000	37000	37000	37000	37000
36	37000	-16	37000	37000	37000	37000	37000

***Denotes altitude thrust limited in level flight, 100 fpm residual rate of climb.**

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Long Range Cruise Maximum Operating Altitude
Max Cruise Thrust
ISA + 20°C

WEIGHT (1000 KG)	OPTIMUM ALT (FT)	TAT (°C)	MARGIN TO INITIAL BUFFET 'G' (BANK ANGLE)				
			1.20 (33°)	1.25 (36°)	1.30 (39°)	1.40 (44°)	1.50 (48°)
64	30800	1	31400*	31400*	31400*	30700	29200
62	31500	0	32300*	32300*	32300*	31400	29900
60	32200	-2	33300*	33300*	33300*	32100	30600
58	32900	-3	34200*	34200*	34200*	32800	31400
56	33700	-5	35200*	35200*	35200	33600	32100
54	34400	-7	36100*	36100*	35900	34400	32900
52	35200	-8	36800*	36800*	36700	35200	33700
50	36100	-10	37000	37000	37000	36000	34500
48	36900	-10	37000	37000	37000	36800	35400
46	37000	-10	37000	37000	37000	37000	36300
44	37000	-10	37000	37000	37000	37000	37000
42	37000	-10	37000	37000	37000	37000	37000
40	37000	-10	37000	37000	37000	37000	37000
38	37000	-10	37000	37000	37000	37000	37000
36	37000	-10	37000	37000	37000	37000	37000

***Denotes altitude thrust limited in level flight, 100 fpm residual rate of climb.**

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Long Range Cruise Control

WEIGHT (1000 KG)		PRESSURE ALTITUDE (1000 FT)								
		21	23	25	27	29	31	33	35	37
64	%N1	82.0	83.0	84.0	84.8	85.7	87.0	89.2		
	MACH	.706	.724	.739	.744	.745	.745	.745		
	KIAS	320	316	310	300	287	275	263		
	FF/ENG	1537	1513	1484	1443	1392	1368	1378		
60	%N1	80.6	81.8	82.8	83.8	84.6	85.5	87.1	89.7	
	MACH	.690	.711	.728	.741	.745	.745	.745	.745	
	KIAS	313	310	305	298	287	275	263	251	
	FF/ENG	1451	1433	1409	1381	1336	1292	1280	1298	
56	%N1	79.2	80.4	81.6	82.6	83.6	84.3	85.4	87.2	90.7
	MACH	.674	.694	.715	.732	.743	.745	.745	.745	.744
	KIAS	305	302	299	294	287	275	263	252	240
	FF/ENG	1367	1348	1331	1308	1280	1234	1198	1195	1227
52	%N1	77.7	78.9	80.1	81.3	82.3	83.2	84.0	85.2	87.6
	MACH	.656	.676	.697	.718	.734	.744	.745	.745	.745
	KIAS	296	294	291	288	283	275	263	251	240
	FF/ENG	1283	1266	1248	1231	1209	1180	1136	1109	1118
48	%N1	76.0	77.3	78.5	79.7	80.9	81.9	82.8	83.6	85.3
	MACH	.636	.657	.678	.699	.720	.736	.744	.745	.745
	KIAS	287	285	283	280	277	271	263	251	240
	FF/ENG	1199	1183	1166	1150	1134	1112	1082	1043	1028
44	%N1	74.1	75.5	76.8	78.0	79.2	80.4	81.4	82.3	83.4
	MACH	.614	.636	.657	.678	.700	.720	.737	.744	.745
	KIAS	277	275	273	271	269	265	260	251	240
	FF/ENG	1115	1100	1084	1067	1053	1038	1018	990	958
40	%N1	72.1	73.5	74.8	76.1	77.4	78.6	79.7	80.7	82.0
	MACH	.591	.612	.633	.655	.677	.699	.720	.737	.744
	KIAS	266	264	263	261	259	257	253	248	240
	FF/ENG	1031	1018	1003	986	971	959	945	927	905
36	%N1	69.8	71.2	72.6	73.9	75.3	76.5	77.8	78.9	80.3
	MACH	.566	.586	.607	.629	.651	.673	.696	.718	.735
	KIAS	254	253	252	250	249	247	244	241	237
	FF/ENG	948	936	922	905	891	879	867	855	842

Shaded area approximates optimum altitude.

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Long Range Cruise Enroute Fuel and Time - Low Altitudes

Ground to Air Miles Conversion

AIR DISTANCE (NM)					GROUND DISTANCE (NM)	AIR DISTANCE (NM)				
HEADWIND COMPONENT (KTS)						TAILWIND COMPONENT (KTS)				
100	80	60	40	20		20	40	60	80	100
284	262	243	226	213	200	191	182	174	167	160
428	395	366	341	319	300	286	273	261	250	241
573	528	489	455	426	400	381	364	348	334	321
718	662	612	570	533	500	477	455	435	418	402
864	796	735	684	640	600	572	546	522	501	482
1011	931	859	799	747	700	667	637	610	584	562
1158	1065	983	914	854	800	763	728	697	668	642
1305	1200	1107	1028	961	900	858	819	784	751	722
1453	1335	1231	1143	1068	1000	953	910	871	835	802
1601	1471	1355	1258	1175	1100	1048	1001	957	917	881
1750	1607	1480	1374	1282	1200	1144	1091	1044	1000	961
1899	1743	1605	1489	1389	1300	1239	1183	1131	1084	1041
2049	1880	1730	1605	1497	1400	1334	1274	1218	1167	1121
2200	2017	1856	1721	1604	1500	1429	1364	1304	1250	1201
2351	2155	1982	1836	1712	1600	1524	1455	1391	1332	1280
2503	2293	2108	1952	1819	1700	1620	1545	1477	1415	1359
2655	2431	2234	2068	1926	1800	1715	1636	1564	1498	1438
2808	2570	2360	2185	2034	1900	1810	1727	1650	1580	1517
2961	2709	2487	2301	2142	2000	1905	1817	1737	1663	1597

Reference Fuel and Time Required at Check Point

AIR DIST (NM)	PRESSURE ALTITUDE (1000 FT)									
	10		14		20		24		28	
	FUEL (1000 KG)	TIME (HR:MIN)	FUEL (1000 KG)	TIME (HR:MIN)	FUEL (1000 KG)	TIME (HR:MIN)	FUEL (1000 KG)	TIME (HR:MIN)	FUEL (1000 KG)	TIME (HR:MIN)
200	1.4	0:39	1.3	0:38	1.1	0:37	0.9	0:36	0.9	0:35
300	2.2	0:58	1.9	0:56	1.7	0:53	1.5	0:51	1.4	0:50
400	2.9	1:17	2.6	1:14	2.3	1:09	2.1	1:07	1.9	1:05
500	3.6	1:35	3.3	1:31	2.9	1:26	2.6	1:22	2.4	1:19
600	4.4	1:54	4.0	1:49	3.5	1:42	3.2	1:38	2.9	1:34
700	5.1	2:13	4.6	2:07	4.0	1:59	3.7	1:53	3.4	1:49
800	5.8	2:32	5.3	2:25	4.6	2:15	4.3	2:09	3.9	2:04
900	6.5	2:52	6.0	2:44	5.2	2:32	4.8	2:25	4.4	2:19
1000	7.2	3:11	6.6	3:02	5.8	2:49	5.3	2:41	4.9	2:34
1100	7.9	3:31	7.3	3:20	6.4	3:06	5.9	2:57	5.4	2:49
1200	8.6	3:50	7.9	3:39	6.9	3:23	6.4	3:13	5.9	3:04
1300	9.3	4:10	8.6	3:58	7.5	3:40	6.9	3:29	6.4	3:20
1400	10.0	4:30	9.2	4:17	8.1	3:57	7.5	3:45	6.9	3:35
1500	10.7	4:50	9.8	4:36	8.7	4:15	8.0	4:02	7.4	3:50
1600	11.4	5:11	10.5	4:55	9.2	4:32	8.5	4:18	7.9	4:06
1700	12.1	5:31	11.1	5:14	9.8	4:50	9.0	4:35	8.3	4:21
1800	12.8	5:52	11.7	5:34	10.3	5:07	9.6	4:51	8.8	4:37
1900	13.4	6:13	12.4	5:53	10.9	5:25	10.1	5:08	9.3	4:53
2000	14.1	6:34	13.0	6:13	11.4	5:43	10.6	5:25	9.8	5:08

Fuel Required Adjustment (1000 KG)

REFERENCE FUEL REQUIRED (1000 KG)	WEIGHT AT CHECK POINT (1000 KG)			
	35	45	55	65
5	-0.4	0.0	0.5	1.1
10	-0.8	0.0	1.1	2.2
15	-1.1	0.0	1.6	3.3

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Long Range Cruise Enroute Fuel and Time - High Altitudes

Ground to Air Miles Conversion

AIR DISTANCE (NM) HEADWIND COMPONENT (KTS)					GROUND DISTANCE (NM)	AIR DISTANCE (NM) TAILWIND COMPONENT (KTS)				
100	80	60	40	20		20	40	60	80	100
269	252	236	223	211	200	191	182	174	167	161
533	500	471	445	421	400	382	365	349	336	323
797	748	704	666	631	600	573	548	525	504	485
1062	997	939	888	842	800	764	731	700	672	647
1328	1247	1174	1110	1052	1000	955	914	875	840	809
1595	1498	1410	1332	1263	1200	1146	1096	1051	1009	971
1862	1748	1646	1555	1474	1400	1337	1279	1226	1177	1133
2130	2000	1882	1778	1685	1600	1529	1463	1402	1347	1296
2399	2251	2118	2000	1896	1800	1720	1646	1578	1515	1458
2669	2504	2355	2224	2107	2000	1911	1829	1753	1683	1620
2939	2757	2592	2447	2318	2200	2102	2012	1928	1852	1782
3211	3011	2829	2671	2529	2400	2293	2195	2103	2020	1944
3484	3266	3068	2895	2741	2600	2485	2377	2279	2188	2106
3757	3520	3307	3119	2952	2800	2676	2560	2454	2356	2268
4032	3777	3546	3344	3164	3000	2867	2743	2629	2525	2430

Reference Fuel and Time Required at Check Point

AIR DIST (NM)	PRESSURE ALTITUDE (1000 FT) 29		31		33		35		37	
	FUEL (1000 KG)	TIME (HR:MIN)	FUEL (1000 KG)	TIME (HR:MIN)	FUEL (1000 KG)	TIME (HR:MIN)	FUEL (1000 KG)	TIME (HR:MIN)	FUEL (1000 KG)	TIME (HR:MIN)
200	0.8	0:35	0.8	0:35	0.7	0:35	0.7	0:35	0.7	0:35
400	1.8	1:04	1.8	1:03	1.7	1:03	1.6	1:03	1.6	1:03
600	2.9	1:33	2.7	1:32	2.6	1:31	2.5	1:31	2.5	1:31
800	3.8	2:03	3.7	2:00	3.6	1:59	3.5	1:59	3.3	1:59
1000	4.8	2:32	4.7	2:29	4.5	2:27	4.3	2:27	4.2	2:27
1200	5.8	3:02	5.6	2:59	5.4	2:56	5.2	2:55	5.1	2:55
1400	6.8	3:32	6.5	3:28	6.3	3:25	6.1	3:23	5.9	3:23
1600	7.7	4:03	7.4	3:58	7.2	3:54	7.0	3:51	6.8	3:51
1800	8.7	4:34	8.3	4:28	8.1	4:23	7.8	4:20	7.6	4:19
2000	9.6	5:05	9.2	4:58	8.9	4:52	8.6	4:49	8.4	4:48
2200	10.5	5:36	10.1	5:28	9.8	5:22	9.5	5:18	9.2	5:16
2400	11.4	6:08	11.0	5:59	10.6	5:52	10.3	5:47	10.0	5:44
2600	12.3	6:40	11.9	6:30	11.5	6:22	11.1	6:16	10.8	6:13
2800	13.2	7:12	12.7	7:02	12.3	6:52	11.9	6:46	11.6	6:42
3000	14.1	7:45	13.6	7:33	13.1	7:23	12.7	7:16	12.3	7:11

Fuel Required Adjustment (1000 KG)

REFERENCE FUEL REQUIRED (1000 KG)	WEIGHT AT CHECK POINT (1000 KG)			
	35	45	55	65
4	-0.4	0.0	1.0	3.2
8	-0.8	0.0	1.8	5.4
12	-1.3	0.0	2.3	6.4
16	-1.7	0.0	2.6	6.4

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Long Range Cruise Wind-Altitude Trade

PRESSURE ALTITUDE (1000 FT)	CRUISE WEIGHT (1000 KG)														
	64	62	60	58	56	54	52	50	48	46	44	42	40	38	36
37					17	10	6	2	0	0	0	0	0	0	0
35			11	6	2	0	0	1	4	9	12	13	12	13	15
33	7	3	1	0	0	3	7	11	17	22	25	26	27	29	30
31	0	0	1	4	9	15	20	24	30	36	40	42	42	43	46
29	4	8	12	17	23	28	33	38	45	52	55	57	57	59	62
27	17	22	26	31	36	42	48	54	61	67	71	73	74	75	78
25	29	34	40	45	51	58	63	69	76	83	88	90	92	93	96
23	43	49	55	60	66	73	80	85	92	101	104	107	109	111	113

The above wind factor tables are for calculation of wind required to maintain present range capability at new pressure altitude, i.e., break-even wind.

Method:

1. **Read wind factors for present and new altitudes from table.**
2. **Determine difference (new altitude wind factor minus present altitude wind factor); This difference may be negative or positive.**
3. **Break-even wind at new altitude is present altitude wind plus difference from step 2.**

Descent

Based on .74/250 descent speed

PRESSURE ALT (1000 FT)	21	23	25	27	29	31	33	35	37
DISTANCE (NM)	67	73	79	85	91	97	104	110	115
TIME (MINUTES)	15	17	18	19	20	21	21	22	23

Holding

Flaps Up

WEIGHT (1000 KG)		PRESSURE ALTITUDE (FT)								
		1500	5000	10000	15000	20000	25000	30000	35000	37000
64	%N1	62.6	65.4	69.4	73.4	77.6	81.3	85.0		
	KIAS	251	252	254	257	259	261	259		
	FF/ENG	1440	1410	1390	1370	1350	1340	1340		
60	%N1	61.0	63.6	67.8	71.8	76.0	79.8	83.4	88.9	
	KIAS	243	244	246	248	251	253	252	246	
	FF/ENG	1360	1330	1310	1280	1270	1260	1250	1320	
56	%N1	59.3	61.9	65.9	70.0	74.1	78.2	81.9	86.1	90.2
	KIAS	235	235	237	239	242	244	245	241	237
	FF/ENG	1280	1250	1230	1200	1180	1180	1170	1190	1260
52	%N1	57.6	60.0	63.9	68.1	72.2	76.4	80.2	84.0	86.6
	KIAS	228	227	228	230	232	235	236	234	231
	FF/ENG	1210	1180	1150	1120	1100	1100	1080	1090	1120
48	%N1	55.8	58.1	61.9	66.1	70.2	74.4	78.3	82.0	84.1
	KIAS	220	219	219	220	223	225	227	226	225
	FF/ENG	1140	1100	1080	1050	1030	1020	1000	1000	1010
44	%N1	54.0	56.2	59.8	63.7	67.9	72.1	76.4	80.1	81.9
	KIAS	211	210	210	210	212	215	217	218	217
	FF/ENG	1060	1030	1000	970	950	940	920	920	920
40	%N1	52.5	54.7	58.1	61.9	66.0	70.1	74.3	78.1	79.9
	KIAS	210	210	210	210	210	210	210	210	210
	FF/ENG	1010	980	950	920	890	870	850	840	850
36	%N1	51.2	53.3	56.6	60.3	64.3	68.3	72.4	76.4	78.2
	KIAS	210	210	210	210	210	210	210	210	210
	FF/ENG	960	930	900	870	840	820	800	790	790

This table includes 5% additional fuel for holding in a racetrack pattern.

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Autobrake Landing Distance

Reference Landing Distance (FT)

AUTOBRAKE SETTING	APPROACH SPEED (KIAS)						
	100	110	120	130	140	150	160
1	4500	5300	6100	6930	7850	8850	9950
2	3800	4440	5080	5730	6450	7240	8100
3	3200	3680	4150	4650	5220	5840	6500
MAX	2500	2760	3010	3280	3600	4000	4500

Actual (unfactored) distances are shown.
Landing distance required includes 1000 ft of air distance.
The distances are valid with or without reverse thrust and for any flap.
If entering with ground speed, ignore adjustments.

Landing Distance Adjustments (FT)

AUTOBRAKE SETTING	ALTITUDE ADJUSTMENT PER 1000 FT ABOVE SEA LEVEL	TAILWIND ADJUSTMENT PER 10 KTS	TEMP ADJUSTMENT PER 10°C ABOVE ISA
1	250	1100	300
2	200	900	250
3	140	650	200
MAX	80	400	100

Slippery Runway Landing Distance

Reference Landing Distance (FT)

BRAKING CONFIGURATION	REPORTED BRAKING ACTION			
	DRY	GOOD	MEDIUM	POOR
MAX MANUAL BRAKING	2475	3315	4280	5315
AUTOBRAKE SETTING 2	4215	4660	4985	5595
AUTOBRAKE SETTING 3	3020	3660	4330	5315
MAX AUTOBRAKE SETTING	2560	3315	4280	5315

Landing Distance Adjustments (FT)

CONDITIONS		ADJUSTMENTS			
WEIGHT	PER 5000 KG BELOW 45,000 KG	-150	-245	-360	-490
	PER 5000 KG ABOVE 45,000 KG	445	425	445	525
AIRPORT PRESSURE ALTITUDE	PER 1000 FT ABOVE SEA LEVEL	130	130	150	165
WIND	PER 10 KTS HEADWIND	-80	-150	-230	-345
	PER 10 KTS TAILWIND	805	855	1015	1425
APPROACH SPEED	PER 10 KTS ABOVE VREF	510	475	445	425
SLOPE	PER 1% DOWNHILL SLOPE	35	80	180	425
	PER 1% UPHILL SLOPE	-35	-35	-100	-245
REVERSE THRUST	1 REVERSER INOPERATIVE	35	150	375	755
	2 REVERSERS INOPERATIVE	50	525	1690	4775
	MAXIMUM REVERSE THRUST	0	0	-35	-80

Actual (unfactored) distances are shown.
Landing distance required includes 1000 ft of air distance.
Based on flaps 40, VREF40 approach speed.
Includes 2 engine detent reverse thrust.

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ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

Dry Runway

LANDING CONFIGURATION	APPROACH SPEED	LANDING DISTANCE AND ADJUSTMENTS (FT)		
		REFERENCE DISTANCE FOR 48000 KG LANDING WEIGHT	WT ADJ PER 2000 KG ABOVE/BELOW 48000 KG	ALTITUDE ADJ PER 1000 FT ABOVE S.L.
ONE ENGINE INOPERATIVE	VREF15	2800	150 / -80	150
ANTI-SKID INOPERATIVE	VREF40	3840	110/-110	90
HYDRAULIC SYSTEM A INOPERATIVE (FLAPS 15)	VREF15	3270	90 / -90	160
HYDRAULIC SYSTEM A INOPERATIVE (FLAPS 30)	VREF30	3220	90/-90	130
HYDRAULIC SYSTEM A INOPERATIVE (FLAPS 40)	VREF40	3150	80/-90	130
HYDRAULIC SYSTEM B INOPERATIVE (FLAPS 15)	VREF15	3100	150 / -90	160
HYDRAULIC SYSTEMS A & B INOPERATIVE (MANUAL REVERSION)	VREF15	4800	130 / -150	210
LEADING EDGE ASYMMETRY OR FAILURE	VREF15+5	2950	150 / -70	160
TRAILING EDGE FAILURE (FLAPS LESS THAN 1)	VREF40+40	3520	210 / -110	200
TRAILING EDGE FAILURE (FLAPS GREATER THAN 1 AND LESS THAN 15)	VREF40+30	3260	190 / -90	170
FLAPS UP	VREF40+55	4000	260 / -140	260
JAMMED STABILIZER	VREF15	2760	140 / -70	140

Actual (unfactored) distances are shown
Includes distances from 50 ft above threshold (1000 ft of air distance)
Assumes maximum manual braking and maximum reverse thrust when available on operating engines.

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ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

Good Reported Braking Action

LANDING CONFIGURATION	APPROACH SPEED	LANDING DISTANCE AND ADJUSTMENTS (FT)		
		REFERENCE DISTANCE FOR 48000 KG LANDING WEIGHT	WT ADJ PER 2000 KG ABOVE/BELOW 48000 KG	ALTITUDE ADJ PER 1000 FT ABOVE S.L.
ONE ENGINE INOPERATIVE	VREF15	3950	130 / -120	190
ANTI-SKID INOPERATIVE	VREF40	4720	150/-150	125
HYDRAULIC SYSTEM A INOPERATIVE (FLAPS 15)	VREF15	4570	150 / -140	230
HYDRAULIC SYSTEM A INOPERATIVE (FLAPS 30)	VREF30	4380	140/-140	180
HYDRAULIC SYSTEM A INOPERATIVE (FLAPS 40)	VREF40	4210	130/-140	180
HYDRAULIC SYSTEM B INOPERATIVE (FLAPS 15)	VREF15	3920	120 / -110	180
HYDRAULIC SYSTEMS A & B INOPERATIVE (MANUAL REVERSION)	VREF15	4870	130 / -150	220
LEADING EDGE ASYMMETRY OR FAILURE	VREF15+5	4070	130 / -120	200
TRAILING EDGE FAILURE (FLAPS LESS THAN 1)	VREF40+40	4580	130 / -120	220
TRAILING EDGE FAILURE (FLAPS GREATER THAN 1 AND LESS THAN 15)	VREF40+30	4260	120 / -120	190
FLAPS UP	VREF40+55	5160	140 / -140	270
JAMMED STABILIZER	VREF15	3760	120 / -110	170

Actual (unfactored) distances are shown
Includes distances from 50 ft above threshold (1000 ft of air distance)
Assumes maximum manual braking and maximum reverse thrust when available on operating engines.

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ADVISORY INFORMATION

Non-Normal Configuration Landing Distance
Medium Reported Braking Action

LANDING CONFIGURATION	APPROACH SPEED	LANDING DISTANCE AND ADJUSTMENTS (FT)		
		REFERENCE DISTANCE FOR 48000 KG LANDING WEIGHT	WT ADJ PER 2000 KG ABOVE/BELOW 48000 KG	ALTITUDE ADJ PER 1000 FT ABOVE S.L.
ONE ENGINE INOPERATIVE	VREF15	5470	200 / -190	270
ANTI-SKID INOPERATIVE	VREF40	5770	200/-190	175
HYDRAULIC SYSTEM A INOPERATIVE (FLAPS 15)	VREF15	5960	230 / -210	310
HYDRAULIC SYSTEM A INOPERATIVE (FLAPS 30)	VREF30	5610	200/-200	240
HYDRAULIC SYSTEM A INOPERATIVE (FLAPS 40)	VREF40	5310	180/-190	230
HYDRAULIC SYSTEM B INOPERATIVE (FLAPS 15)	VREF15	5210	180 / -170	240
HYDRAULIC SYSTEM A INOPERATIVE (FLAPS 15)	VREF15	6350	200 / -220	290
LEADING EDGE ASYMMETRY OR FAILURE	VREF15+5	5370	190 / -180	270
TRAILING EDGE FAILURE (FLAPS LESS THAN 1)	VREF40+40	6100	200 / -190	300
TRAILING EDGE FAILURE (FLAPS GREATER THAN 1 AND LESS THAN 15)	VREF40+30	5670	190 / -180	260
FLAPS UP	VREF40+55	6900	230 / -220	370
JAMMED STABILIZER	VREF15	4950	180 / -170	220

Actual (unfactored) distances are shown
Includes distances from 50 ft above threshold (1000 ft of air distance)
Assumes maximum manual braking and maximum reverse thrust when available on operating engines.

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ADVISORY INFORMATION

Non-Normal Configuration Landing Distance

Poor Reported Braking Action

LANDING CONFIGURATION	APPROACH SPEED	LANDING DISTANCE AND ADJUSTMENTS (FT)		
		REFERENCE DISTANCE FOR 48000 KG LANDING WEIGHT	WT ADJ PER 2000 KG ABOVE/BELOW 48000 KG	ALTITUDE ADJ PER 1000 FT ABOVE S.L.
ONE ENGINE INOPERATIVE	VREF15	7250	290 / -270	370
ANTI-SKID INOPERATIVE	VREF40	7340	290/-270	240
HYDRAULIC SYSTEM A INOPERATIVE (FLAPS 15)	VREF15	7360	300 / -280	380
HYDRAULIC SYSTEM A INOPERATIVE (FLAPS 30)	VREF30	6800	270/-260	290
HYDRAULIC SYSTEM A INOPERATIVE (FLAPS 40)	VREF40	6410	240/-240	280
HYDRAULIC SYSTEM B INOPERATIVE (FLAPS 15)	VREF15	6570	260 / -240	310
HYDRAULIC SYSTEMS A & B INOPERATIVE (MANUAL REVERSION)	VREF15	7800	270 / -300	360
LEADING EDGE ASYMMETRY OR FAILURE	VREF15+5	6750	270 / -250	340
TRAILING EDGE FAILURE (FLAPS LESS THAN 1)	VREF40+40	7650	280 / -270	390
TRAILING EDGE FAILURE (FLAPS GREATER THAN 1 AND LESS THAN 15)	VREF40+30	7120	260 / -250	330
FLAPS UP	VREF40+55	8640	310 / -300	480
JAMMED STABILIZER	VREF15	6200	240 / -230	290

Actual (unfactored) distances are shown
Includes distances from 50 ft above threshold (1000 ft of air distance)
Assumes maximum manual braking and maximum reverse thrust when available on operating engines.

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ADVISORY INFORMATION

Brake Cooling Schedule

Reference Brake Energy Per Brake (Millions of Foot Pounds)

		WIND CORRECTED BRAKES ON SPEED (KIAS)*														
		60			80			100			120			140		
WEIGHT	OAT	PRESS ALT			PRESS ALT			PRESS ALT			PRESS ALT			PRESS ALT		
(1000 KG)	(°C)	0	2	4	0	2	4	0	2	4	0	2	4	0	2	4
65	-20	7.0	7.5	8.1	11.3	12.2	13.1	16.2	17.7	19.1	21.9	24.1	25.9			
	0	7.5	8.2	8.9	12.3	13.2	14.3	17.7	19.3	20.8	23.9	26.0	27.9			
	20	8.1	8.8	9.6	13.3	14.3	15.4	19.2	20.8	22.4	25.9	28.3	30.0			
	40	8.7	9.5	10.2	14.3	15.3	16.6	20.7	22.4	24.1	27.9	30.7	32.1			
60	-20	6.6	7.2	7.7	10.6	11.5	12.4	15.3	16.6	18.0	20.5	22.2	24.0	26.5		
	0	7.1	7.8	8.4	11.5	12.5	13.5	16.7	18.1	19.6	22.4	24.3	26.1	28.9		
	20	7.7	8.4	9.1	12.5	13.5	14.5	18.1	19.7	21.2	24.3	26.3	28.3	31.3		
	40	8.2	9.1	9.7	13.4	14.5	15.6	19.5	21.2	22.8	26.2	28.3	30.4	33.4		
55	-20	6.1	6.6	7.1	9.8	10.6	11.5	14.1	15.3	16.5	19.1	20.5	22.1	24.5	26.4	
	0	6.6	7.2	7.7	10.7	11.6	12.5	15.4	16.7	18.0	20.7	22.4	24.1	26.7	28.8	
	20	7.2	7.7	8.3	11.6	12.5	13.5	16.6	18.1	19.5	22.4	24.3	26.1	28.9	31.2	
	40	7.7	8.3	8.9	12.5	13.5	14.5	17.9	19.5	21.0	24.1	26.2	28.2	31.1	33.6	
50	-20	5.7	6.2	6.7	9.1	9.8	10.6	13.2	14.3	15.3	17.6	19.0	20.4	22.6	24.5	26.4
	0	6.2	6.7	7.3	9.9	10.7	11.6	14.4	15.6	16.7	19.2	20.7	22.2	24.7	26.7	28.8
	20	6.7	7.3	7.9	10.6	11.6	12.5	15.5	16.8	18.2	20.7	22.4	24.1	26.7	28.9	31.2
	40	7.2	7.8	8.4	11.5	12.5	13.5	16.7	18.1	19.6	22.3	24.1	26.0	28.8	31.1	33.5
45	-20	5.5	6.0	6.3	8.5	9.2	9.9	12.2	13.1	14.1	16.0	17.4	18.8	20.5	22.2	24.0
	0	5.9	6.4	6.9	9.3	10.0	10.8	13.2	14.3	15.3	17.4	18.9	20.4	22.4	24.3	26.2
	20	6.4	6.9	7.4	10.1	10.9	11.7	14.3	15.4	16.6	18.9	20.5	22.1	24.3	26.3	28.3
	40	6.8	7.4	7.9	10.8	11.7	12.6	15.3	16.5	17.8	20.4	22.0	23.8	26.2	28.3	30.5
40	-20	5.1	5.5	5.9	7.9	8.5	9.1	11.0	12.0	12.9	14.6	15.8	17.2	18.8	20.2	21.7
	0	5.5	5.9	6.4	8.6	9.3	10.0	12.0	13.0	14.0	15.9	17.3	18.7	20.5	22.1	23.8
	20	5.9	6.4	6.9	9.2	10.0	10.8	13.0	14.1	15.2	17.3	18.8	20.2	22.1	24.0	25.8
	40	6.3	6.8	7.4	9.9	10.8	11.6	14.0	15.1	16.3	18.6	20.2	21.7	23.8	25.8	27.8

*** To correct for wind, enter table with the brakes on speed minus one half the headwind or plus 1.5 times the tailwind.**
If ground speed is used for brakes on speed, ignore wind and enter table with sea level, 15°C.

Adjusted Brake Energy Per Brake (Millions of Foot Pounds)

No Reverse Thrust

		REFERENCE BRAKE ENERGY PER BRAKE (MILLIONS OF FOOT POUNDS)						
EVENT		5	10	15	20	25	30	35
RTO MAX MAN		5	10	15	20	25	30	35
LANDING	MAX MAN	3.4	7.4	11.9	16.5	21.1	25.8	30.5
	MAX AUTO	3.2	7.0	11.4	15.5	19.7	24.2	28.5
	AUTOBRAKE 3	3.1	6.7	10.8	14.6	18.2	22.2	25.9
	AUTOBRAKE 2	2.9	6.3	10.3	13.8	17.0	20.1	23.2
	AUTOBRAKE 1	2.9	6.2	10.0	13.4	16.5	19.7	22.6

Adjusted Brake Energy Per Brake (Millions of Foot Pounds)

Two Engine Detent Reverse Thrust

		REFERENCE BRAKE ENERGY PER BRAKE (MILLIONS OF FOOT POUNDS)						
EVENT		5	10	15	20	25	30	35
LANDING	MAX AUTO	2.6	5.6	8.9	12.4	16.0	19.4	22.8
	AUTOBRAKE 3	1.9	3.9	6.1	8.4	10.5	12.6	14.7
	AUTOBRAKE 1 & 2	1.9	3.7	5.8	7.8	9.6	11.3	12.8

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Brake Cooling Schedule

Cooling Time (Minutes) - Category A Brakes

	ADJUSTED BRAKE ENERGY PER BRAKE (MILLIONS OF FOOT POUNDS)							
	8 & BELOW	10	12	14	16	18	18.5 TO 23.7	23.8 & ABOVE
INFLIGHT GEAR DOWN	NO SPECIAL PROCEDURE REQUIRED	2	3	4	5	6	CAUTION	FUSE PLUG MELT ZONE
GROUND		19	32	43	53	61		

Observe maximum quick turnaround limit.

Table shows energy per brake added by a single stop with all brakes operating. Energy is assumed to be equally distributed among the operating brakes. Total energy is the sum of residual energy plus energy added.

Add 1.0 million foot pounds for each taxi mile.

When in caution zone, wheel fuse plugs may melt. Delay takeoff and inspect after 30 minutes. If overheat occurs after takeoff, extend gear soon for at least 7 minutes.

When in fuse plug melt zone, clear runway immediately. Unless required, do not set parking brake. Do not approach gear or attempt to taxi for one hour. Tire, wheel and brake replacement may be required. If overheat occurs after takeoff, extend gear soon for at least 12 minutes.

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ENGINE INOP

Initial Max Continuous %N1

Based on .74M, engine bleed for packs auto and anti-ice off

TAT (°C)	PRESSURE ALTITUDE (1000 FT)					
	27	29	31	33	35	37
20	92.9	92.9	92.8	92.8	92.8	92.7
15	93.3	93.3	93.2	93.2	93.2	93.1
10	93.7	93.7	93.6	93.6	93.6	93.5
5	94.0	94.0	93.9	93.9	93.9	93.8
0	94.3	94.3	94.2	94.2	94.2	94.1
-5	93.7	94.6	94.5	94.5	94.5	94.4
-10	92.8	93.9	94.7	94.7	94.7	94.6
-15	91.9	93.0	93.9	95.0	95.0	94.9
-20	91.0	92.0	93.0	94.1	95.3	95.4
-25	90.1	91.1	92.0	93.2	94.4	94.9
-30	89.2	90.2	91.1	92.2	93.4	94.0
-40	87.4	88.3	89.2	90.3	91.5	92.0
-50	85.5	86.4	87.3	88.3	89.5	90.0

%N1 Adjustment for Engine Bleeds

BLEED CONFIGURATION	PRESSURE ALTITUDE (1000 FT)					
	27	29	31	33	35	37
ENGINE ANTI-ICE ON	-1.0	-1.0	-1.0	-1.0	-1.0	-1.0
ENGINE AND WING ANTI-ICE ON	-3.1	-3.0	-3.0	-3.0	-3.0	-3.0

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ENGINE INOP

Max Continuous %N1

33000 FT to 23000 FT Pressure Altitudes

Based on engine bleed for packs auto and anti-ice off

33000 FT PRESS ALT.		TAT (°C)										
KIAS	M	-50	-40	-30	-20	-10	0	5	10	15	20	25
200	.58	89.6	91.6	93.5	94.4	94.0	93.5	93.2				
220	.63	89.4	91.4	93.3	94.8	94.3	93.8	93.5	93.1			
240	.68	89.1	91.0	93.0	94.8	94.6	94.0	93.7	93.4			
260	.74	88.3	90.2	92.2	94.0	94.7	94.2	93.9	93.5	93.2		
31000 FT PRESS ALT.		TAT (°C)										
KIAS	M	-50	-40	-30	-20	-10	0	5	10	15	20	25
200	.55	88.8	90.8	92.7	94.2	93.9	93.4	93.1				
220	.61	88.7	90.6	92.4	94.4	94.2	93.7	93.4	93.1			
240	.66	88.3	90.2	92.1	94.0	94.5	93.9	93.6	93.3	92.9		
260	.71	87.8	89.7	91.7	93.5	94.6	94.1	93.8	93.5	93.1		
29000 FT PRESS ALT.		TAT (°C)										
KIAS	M	-50	-40	-30	-20	-10	0	5	10	15	20	25
200	.53	88.1	90.0	91.9	93.8	93.7	93.2	92.9	92.6			
220	.58	87.9	89.8	91.8	93.6	94.1	93.6	93.3	92.9	92.6		
240	.63	87.6	89.5	91.4	93.3	94.4	93.8	93.5	93.2	92.8		
260	.68	87.1	89.1	91.0	92.8	94.5	94.0	93.7	93.4	93.0	92.6	
27000 FT PRESS ALT.		TAT (°C)										
KIAS	M	-50	-40	-30	-20	-10	0	5	10	15	20	25
200	.51	87.3	89.3	91.2	93.0	93.5	93.0	92.8	92.4			
220	.56	87.1	89.1	91.0	92.8	93.9	93.4	93.1	92.8	92.4		
240	.60	86.9	88.9	90.7	92.6	94.3	93.8	93.5	93.1	92.7	92.3	
260	.65	86.5	88.4	90.3	92.1	93.9	93.9	93.7	93.3	92.9	92.5	
25000 FT PRESS ALT.		TAT (°C)										
KIAS	M	-50	-40	-30	-20	-10	0	5	10	15	20	25
200	.49	86.6	88.5	90.4	92.2	93.4	93.0	92.7	92.4	92.0		
220	.53	86.4	88.3	90.2	92.0	93.7	93.3	93.0	92.7	92.3	91.9	
240	.58	86.2	88.1	90.0	91.8	93.6	93.6	93.3	93.0	92.6	92.2	
260	.63	85.8	87.7	89.6	91.4	93.2	93.8	93.6	93.2	92.9	92.4	92.0
23000 FT PRESS ALT.		TAT (°C)										
KIAS	M	-50	-40	-30	-20	-10	0	5	10	15	20	25
200	.47	85.8	87.7	89.6	91.4	93.2	92.8	92.6	92.3	91.9	91.6	
220	.51	85.6	87.5	89.4	91.2	93.0	93.1	92.9	92.5	92.2	91.8	
240	.56	85.4	87.3	89.2	91.0	92.8	93.4	93.2	92.8	92.5	92.1	91.7
260	.60	85.2	87.1	88.9	90.8	92.5	93.8	93.5	93.1	92.8	92.4	92.0

Anti-Ice Adjustments

BLEED CONFIGURATION	PRESS ALT (1000 FT)					
	23	25	27	29	31	33
ENGINE ANTI-ICE ON	-1.0	-1.0	-1.0	-1.0	-1.0	-1.0
ENGINE AND WING ANTI-ICE ON	-3.1	-3.1	-3.1	-3.0	-3.0	-3.0

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ENGINE INOP

Max Continuous %N1

21000 FT to 14000 FT Pressure Altitudes

Based on engine bleed for packs auto and anti-ice off

21000 FT PRESS ALT.		TAT (°C)											
KIAS	M	-50	-40	-30	-20	-10	0	5	10	15	20	25	30
200	.45	85.0	86.9	88.8	90.6	92.4	92.7	92.5	92.2	91.9	91.5		
220	.49	84.9	86.8	88.6	90.4	92.2	93.0	92.7	92.4	92.1	91.7	91.3	
240	.54	84.7	86.6	88.4	90.2	92.0	93.3	93.0	92.7	92.4	92.0	91.6	
260	.60	84.5	86.3	88.2	90.0	91.7	93.4	93.3	93.0	92.6	92.3	91.8	91.4
20000 FT PRESS ALT.		TAT (°C)											
KIAS	M	-50	-40	-30	-20	-10	0	5	10	15	20	25	30
200	.44	84.7	86.5	88.4	90.2	91.9	92.7	92.4	92.1	91.8	91.5	91.1	
220	.48	84.5	86.4	88.2	90.0	91.8	92.9	92.7	92.4	92.0	91.7	91.3	
240	.53	84.3	86.2	88.0	89.8	91.6	93.2	92.9	92.6	92.3	91.9	91.5	91.1
260	.57	84.1	86.0	87.8	89.6	91.3	93.0	93.2	92.9	92.6	92.2	91.8	91.3
18000 FT PRESS ALT.		TAT (°C)											
KIAS	M	-40	-30	-20	-10	0	5	10	15	20	25	30	35
200	.42	85.8	87.6	89.4	91.1	92.6	92.3	92.0	91.7	91.4	91.0		
220	.46	85.6	87.4	89.2	91.0	92.7	92.6	92.3	92.0	91.6	91.2	90.8	
240	.51	86.6	83.4	90.2	92.0	93.7	94.1	93.8	93.5	93.1	92.7	92.3	
260	.55	86.4	88.2	90.0	91.8	93.5	94.3	94.1	93.7	93.4	93.0	92.5	92.1
17000 FT PRESS ALT.		TAT (°C)											
KIAS	M	-40	-30	-20	-10	0	5	10	15	20	25	30	35
200	.42	87.1	89.0	90.8	92.6	94.3	94.2	93.9	93.6	93.3	92.9	92.5	
220	.46	87.0	88.8	90.6	92.4	94.1	94.4	94.1	93.8	93.5	93.1	92.7	
240	.50	86.8	88.6	90.4	92.2	93.9	94.7	94.4	94.1	93.7	93.3	92.9	92.5
260	.54	86.6	88.4	90.2	92.0	93.7	94.5	94.6	94.3	93.9	93.6	93.2	92.7
16000 FT PRESS ALT.		TAT (°C)											
KIAS	M	-40	-30	-20	-10	0	5	10	15	20	25	30	35
200	.41	86.8	88.6	90.4	92.2	93.9	94.1	93.9	93.6	93.3	92.9	92.5	
220	.45	86.6	88.4	90.2	92.0	93.7	94.3	94.1	93.8	93.4	93.1	92.7	
240	.49	86.4	88.2	90.0	91.8	93.5	94.4	94.3	94.0	93.7	93.3	92.9	92.5
260	.53	86.2	88.0	89.8	91.6	93.3	94.1	94.6	94.2	93.9	93.5	93.1	92.7
14000 FT PRESS ALT.		TAT (°C)											
KIAS	M	-30	-20	-10	0	5	10	15	20	25	30	35	40
200	.39	87.8	89.6	91.3	93.1	93.9	93.8	93.5	93.2	92.8	92.4	92.0	
220	.43	87.6	89.4	91.2	92.9	93.7	94.0	93.7	93.4	93.0	92.6	92.2	
240	.47	87.5	89.2	91.0	92.7	93.5	94.2	93.9	93.6	93.2	92.8	92.4	
260	.51	87.3	89.0	90.8	92.5	93.3	94.2	94.1	93.8	93.4	93.0	92.6	92.1

Anti-Ice Adjustments

BLEED CONFIGURATION	PRESS ALT (1000 FT)					
	14	16	17	18	20	21
ENGINE ANTI-ICE ON	-1.0	-1.0	-1.0	-1.0	-1.0	-1.0
ENGINE AND WING ANTI-ICE ON	-3.2	-3.2	-3.2	-3.2	-3.2	-3.2

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ENGINE INOP

Max Continuous %N1

12000 FT to 1000 FT Pressure Altitudes

Based on engine bleed for packs auto and anti-ice off

12000 FT PRESS ALT.								TAT (°C)					
KIAS	M	-20	-10	0	5	10	15	20	25	30	35	40	45
200	.38	88.8	90.5	92.2	93.1	93.7	93.4	93.1	92.7	92.4	91.9		
220	.41	88.6	90.4	92.1	92.9	93.7	93.6	93.3	92.9	92.5	92.1	91.7	
240	.45	88.4	90.2	91.9	92.7	93.5	93.8	93.5	93.1	92.7	92.3	91.9	
260	.49	88.3	90.0	91.7	92.5	93.3	94.0	93.7	93.3	92.9	92.5	92.0	91.6
10000 FT PRESS ALT.								TAT (°C)					
KIAS	M	-20	-10	0	5	10	15	20	25	30	35	40	45
200	.36	88.0	89.7	91.4	92.2	93.1	93.3	93.0	92.7	92.3	91.9	91.4	
220	.40	87.9	89.6	91.3	92.1	92.9	93.5	93.2	92.9	92.5	92.1	91.6	
240	.43	87.7	89.4	91.1	91.9	92.7	93.5	93.4	93.0	92.6	92.2	91.8	91.3
260	.47	87.5	89.2	90.9	91.7	92.5	93.3	93.6	93.2	92.8	92.4	92.0	91.5
7000 FT PRESS ALT.								TAT (°C)					
KIAS	M	-10	0	5	10	15	20	25	30	35	40	45	50
200	.34	86.6	88.3	89.1	89.9	90.7	91.0	91.1	91.9	92.0	91.8	91.5	91.1
220	.38	86.5	88.1	89.0	89.7	90.5	91.2	91.0	91.8	92.1	92.0	91.6	91.3
240	.41	86.4	88.0	88.8	89.6	90.4	91.2	91.0	91.6	92.1	92.1	91.8	91.4
260	.45	86.2	87.8	88.6	89.4	90.2	90.0	91.1	91.3	92.2	92.1	91.9	91.5
5000 FT PRESS ALT.								TAT (°C)					
KIAS	M	-10	0	5	10	15	20	25	30	35	40	45	50
200	.33	85.8	87.5	88.3	89.0	89.8	90.6	90.6	90.9	91.3	91.1	90.7	90.3
220	.36	85.7	87.3	88.1	88.9	89.7	90.5	90.7	90.8	91.4	91.2	90.9	90.4
240	.40	85.6	87.2	88.0	88.8	89.6	90.3	90.9	90.7	91.3	91.3	91.0	90.6
260	.43	85.4	87.0	87.8	88.6	89.3	90.1	90.9	90.8	91.2	91.4	91.1	90.7
3000 FT PRESS ALT.								TAT (°C)					
KIAS	M	-10	0	5	10	15	20	25	30	35	40	45	50
200	.32	85.0	86.6	87.4	88.2	89.0	89.8	90.5	90.2	89.8	89.4	89.0	88.5
220	.35	84.9	86.5	87.3	88.1	88.8	89.6	90.4	90.3	90.0	89.6	89.1	88.6
240	.40	84.8	86.4	87.2	87.9	88.7	89.5	90.2	90.5	90.1	89.7	89.3	88.8
260	.43	84.6	86.2	87.0	87.8	88.5	89.3	90.0	90.6	90.3	89.9	89.4	89.0
1000 FT PRESS ALT.								TAT (°C)					
KIAS	M	-10	0	5	10	15	20	25	30	35	40	45	50
200	.31	84.2	85.8	86.6	87.4	88.2	88.9	89.7	90.1	89.7	89.2	88.8	88.3
220	.34	84.1	85.7	86.5	87.2	88.0	88.8	89.5	90.2	89.8	89.4	88.9	88.4
240	.37	84.0	85.5	86.3	87.1	87.9	88.6	89.4	90.1	90.0	89.5	89.1	88.6
260	.40	83.8	85.4	86.2	87.0	87.7	88.5	89.2	90.0	90.1	89.7	89.2	88.7

Anti-Ice Adjustments

BLEED CONFIGURATION	PRESS ALT (1000 FT)					
	1	3	5	7	10	12
ENGINE ANTI-ICE ON	-0.7	-0.7	-0.8	-0.8	-0.9	-0.9
ENGINE AND WING ANTI-ICE ON	-2.6	-2.6	-2.8	-2.9	-3.1	-3.1

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ENGINE INOP

MAX CONTINUOUS THRUST

Driftdown Speed/Level Off Altitude
100 ft/min residual rate of climb

WEIGHT (1000 KG)		OPTIMUM DRIFTDOWN SPEED (KIAS)	LEVEL OFF ALTITUDE (FT)		
START DRIFTDOWN	LEVEL OFF		ISA + 10°C & BELOW	ISA + 15°C	ISA + 20°C
64	61	235	15500	14200	12800
60	57	228	17300	16400	15100
56	53	220	18800	18000	17200
52	49	212	20600	19600	18800
48	46	204	23100	22100	20900
44	42	196	25800	24800	23700
40	38	187	28500	27600	26600
36	34	177	31100	30500	29600
32	31	167	33700	33400	32600

Includes APU fuel burn.

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ENGINE INOP

MAX CONTINUOUS THRUST

Driftdown/LRC Cruise Range Capability

Ground to Air Miles Conversion

AIR DISTANCE (NM)					GROUND DISTANCE (NM)	AIR DISTANCE (NM)				
HEADWIND COMPONENT (KTS)						TAILWIND COMPONENT (KTS)				
100	80	60	40	20		20	40	60	80	100
141	131	121	113	106	100	94	90	85	81	77
290	266	246	228	213	200	188	178	169	160	153
436	399	369	343	320	300	282	267	253	240	229
580	532	491	457	426	400	377	356	337	321	305
722	663	613	570	533	500	471	445	422	401	382
863	793	734	683	639	600	566	535	507	482	460
1002	923	855	796	745	700	660	625	593	564	538
1141	1052	975	909	851	800	755	715	678	646	616
1280	1180	1095	1021	957	900	850	805	764	727	694
1417	1308	1215	1134	1063	1000	944	895	850	809	773
1555	1436	1334	1246	1168	1100	1039	985	936	891	851
1693	1564	1454	1358	1274	1200	1134	1075	1022	973	929
1831	1693	1574	1471	1380	1300	1229	1165	1107	1055	1008
1970	1822	1694	1583	1486	1400	1323	1255	1193	1137	1086
2110	1951	1815	1696	1592	1500	1418	1345	1278	1218	1164
2250	2081	1936	1809	1698	1600	1513	1434	1364	1300	1241

Driftdown/Cruise Fuel and Time

AIR DIST (NM)	FUEL REQUIRED (1000 KG)							TIME (HRS:MIN)
	WEIGHT AT START OF DRIFTDOWN (1000 KG)							
	35	40	45	50	55	60	65	
100	0.4	0.4	0.4	0.4	0.5	0.5	0.5	0:18
200	0.8	0.9	0.9	1.0	1.1	1.2	1.3	0:37
300	1.2	1.4	1.5	1.7	1.8	2.0	2.2	0:56
400	1.7	1.8	2.1	2.3	2.5	2.8	3.0	1:14
500	2.1	2.3	2.6	2.9	3.2	3.5	3.8	1:32
600	2.5	2.8	3.1	3.5	3.9	4.2	4.6	1:50
700	2.9	3.3	3.7	4.1	4.5	4.9	5.3	2:07
800	3.3	3.7	4.2	4.7	5.1	5.6	6.1	2:24
900	3.7	4.2	4.7	5.2	5.8	6.3	6.9	2:40
1000	4.1	4.6	5.2	5.8	6.4	6.9	7.6	2:57
1100	4.5	5.1	5.7	6.4	7.0	7.6	8.3	3:13
1200	4.9	5.5	6.2	7.0	7.7	8.3	9.1	3:30
1300	5.3	6.0	6.7	7.5	8.3	9.0	9.8	3:46
1400	5.7	6.4	7.2	8.1	8.9	9.6	10.5	4:03
1500	6.1	6.8	7.7	8.6	9.5	10.3	11.2	4:20
1600	6.5	7.3	8.2	9.2	10.1	10.9	12.0	4:37

Includes APU fuel burn.
Driftdown at optimum speed and cruise at Long Range Cruise speed.

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ENGINE INOP

MAX CONTINUOUS THRUST

Long Range Cruise Altitude Capability
100 ft/min residual rate of climb

WEIGHT (1000 KG)	PRESSURE ALTITUDE (FT)		
	ISA + 10°C & BELOW	ISA + 15°C	ISA + 20°C
64	5200	3000	800
62	6700	4700	3400
60	11500	6100	5600
58	12900	10700	7600
56	14400	12200	10000
54	15900	13800	11600
52	17200	15300	13200
50	17900	16900	14800
48	18700	17800	16500
46	19600	18600	17600
44	20800	19500	18500
42	22500	20700	19400
40	24200	22500	20700
38	25900	24300	22500
36	27700	26100	24500

With engine anti-ice on, decrease altitude capability by 3700ft.
With engine and wing anti-ice on, decrease altitude capability by 8700ft.

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ENGINE INOP

MAX CONTINUOUS THRUST

Long Range Cruise Control

WEIGHT (1000 KG)		PRESSURE ALTITUDE (1000 FT)									
		10	12	14	16	18	20	22	24	26	28
64	%N1	90.4									
	MACH	.561									
	KIAS	311									
	FF/ENG	2950									
60	%N1	88.8	90.2								
	MACH	.547	.564								
	KIAS	303	302								
	FF/ENG	2772	2758								
56	%N1	87.2	88.5	89.9	91.3						
	MACH	.531	.549	.566	.582						
	KIAS	294	293	292	289						
	FF/ENG	2589	2580	2563	2533						
52	%N1	85.4	86.8	88.1	89.5	90.9					
	MACH	.513	.532	.550	.567	.584					
	KIAS	284	284	283	281	278					
	FF/ENG	2406	2396	2387	2363	2333					
48	%N1	83.5	84.8	86.2	87.6	89.0					
	MACH	.495	.513	.532	.550	.568					
	KIAS	274	274	273	272	271					
	FF/ENG	2223	2211	2204	2189	2165					
44	%N1	81.3	82.8	84.1	85.5	86.9	88.3	89.7			
	MACH	.475	.493	.511	.530	.549	.567	.584			
	KIAS	263	263	263	262	261	260	257			
	FF/ENG	2042	2029	2019	2010	1994	1974	1958			
40	%N1	78.9	80.4	81.9	83.3	84.7	86.0	87.5	88.9	90.6	
	MACH	.454	.471	.489	.508	.527	.546	.565	.582	.599	
	KIAS	251	251	251	251	250	250	248	246	243	
	FF/ENG	1864	1851	1839	1830	1819	1806	1795	1777	1754	
36	%N1	76.3	77.8	79.3	80.8	82.2	83.6	85.0	86.4	87.9	89.6
	MACH	.434	.449	.465	.484	.502	.522	.541	.561	.578	.595
	KIAS	240	239	239	239	238	238	238	236	234	231
	FF/ENG	1697	1677	1663	1653	1643	1632	1627	1615	1592	1570

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ENGINE INOP

MAX CONTINUOUS THRUST

Long Range Cruise Diversion Fuel and Time

Ground to Air Miles Conversion

AIR DISTANCE (NM)					GROUND DISTANCE (NM)	AIR DISTANCE (NM)				
HEADWIND COMPONENT (KTS)						TAILWIND COMPONENT (KTS)				
100	80	60	40	20		20	40	60	80	100
299	272	249	230	214	200	190	181	173	166	159
608	552	504	464	430	400	380	362	345	330	316
920	834	759	697	646	600	570	542	517	494	473
1235	1118	1015	932	862	800	760	722	688	658	630
1552	1403	1273	1167	1078	1000	949	903	860	821	787
1872	1690	1532	1403	1295	1200	1139	1082	1031	984	943
2196	1980	1792	1640	1512	1400	1328	1262	1202	1147	1099
2524	2273	2055	1877	1730	1600	1517	1441	1372	1310	1255
2854	2566	2317	2115	1948	1800	1707	1621	1543	1473	1410

Reference Fuel and Time Required at Check Point

AIR DIST (NM)	PRESSURE ALTITUDE (1000 FT)									
	10		14		18		22		26	
	FUEL (1000 KG)	TIME (HR:MIN)	FUEL (1000 KG)	TIME (HR:MIN)	FUEL (1000 KG)	TIME (HR:MIN)	FUEL (1000 KG)	TIME (HR:MIN)	FUEL (1000 KG)	TIME (HR:MIN)
200	1.4	0:42	1.2	0:40	1.1	0:39	1.0	0:38	0.9	0:38
400	2.8	1:22	2.6	1:18	2.3	1:14	2.1	1:11	2.0	1:11
600	4.2	2:03	3.9	1:56	3.5	1:50	3.3	1:46	3.1	1:46
800	5.6	2:44	5.2	2:35	4.7	2:26	4.4	2:20	4.1	2:20
1000	7.0	3:26	6.4	3:14	5.9	3:03	5.5	2:55	5.2	2:55
1200	8.3	4:08	7.7	3:54	7.1	3:40	6.6	3:30	6.2	3:30
1400	9.6	4:51	8.9	4:34	8.2	4:18	7.7	4:05	7.2	4:05
1600	10.9	5:35	10.1	5:15	9.3	4:57	8.7	4:41	8.2	4:41
1800	12.2	6:20	11.3	5:57	10.5	5:36	9.8	5:18	9.1	5:18

Fuel Required Adjustments (1000 KG)

REFERENCE FUEL REQUIRED (1000 KG)	WEIGHT AT CHECK POINT (1000 KG)			
	35	45	55	65
5	-0.4	0.0	0.8	1.8
10	-0.9	0.0	1.6	3.5
15	-1.3	0.0	2.2	5.0

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ENGINE INOP

MAX CONTINUOUS THRUST

Holding
Flaps Up

WEIGHT (1000 KG)		PRESSURE ALTITUDE (FT)						
		1500	5000	10000	15000	20000	25000	30000
64	%N1	79.9	82.5	86.2	90.5			
	KIAS	247	247	247	249			
	FF/ENG	2600	2580	2570	2590			
60	%N1	78.1	80.8	84.5	88.5			
	KIAS	238	239	239	241			
	FF/ENG	2440	2420	2400	2410			
56	%N1	76.1	78.9	82.7	86.6			
	KIAS	230	231	232	232			
	FF/ENG	2270	2260	2240	2230			
52	%N1	74.1	76.9	80.8	84.6	88.9		
	KIAS	221	222	224	223	225		
	FF/ENG	2110	2100	2080	2060	2070		
48	%N1	71.8	74.7	78.7	82.5	86.5		
	KIAS	211	213	215	215	216		
	FF/ENG	1950	1940	1920	1900	1890		
44	%N1	69.8	72.6	76.6	80.4	84.3	88.9	
	KIAS	210	210	210	210	210	210	
	FF/ENG	1820	1800	1780	1750	1740	1740	
40	%N1	67.9	70.7	74.6	78.6	82.4	86.6	
	KIAS	210	210	210	210	210	210	
	FF/ENG	1710	1690	1660	1630	1620	1610	
36	%N1	66.2	68.9	72.8	76.7	80.6	84.5	89.5
	KIAS	210	210	210	210	210	210	210
	FF/ENG	1610	1590	1560	1530	1510	1500	1510

This table includes 5% additional fuel for holding in a racetrack pattern.

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Takeoff %N1

Based on engine bleed to packs on (Auto) and anti-ice on or off

AIRPORT OAT		AIRPORT PRESSURE ALTITUDE (FT)									
°C	°F	-1000	-0	1000	2000	3000	4000	5000	6000	7000	8000
55	131	91.5	92.3	92.2	91.7						
50	122	92.0	92.8	93.4	93.7	93.0	93.0				
45	113	92.6	93.1	93.6	93.9	93.9	94.5	94.3	93.7		
40	104	93.1	93.6	94.0	94.2	94.3	95.1	95.0	94.9	94.8	94.5
35	95	93.2	94.1	94.6	94.8	94.8	95.5	95.7	95.6	95.8	95.7
30	86	93.0	94.5	94.3	94.5	95.0	95.4	95.6	95.8	95.8	95.8
25	77	92.2	93.7	93.7	93.7	94.4	95.5	95.3	95.1	95.4	95.8
20	68	91.5	92.9	92.9	92.9	93.6	95.5	95.7	95.8	95.8	95.7
15	59	90.7	92.1	92.1	92.1	92.8	94.9	95.1	95.1	95.1	95.1
10	50	89.9	91.3	91.3	91.3	92.0	94.1	94.3	94.3	94.3	94.3
5	41	89.1	90.5	90.5	90.5	91.2	93.2	93.4	93.4	93.4	93.4
0	32	88.3	89.7	89.7	89.7	90.4	92.4	92.6	92.6	92.6	92.6
-10	14	86.6	88.0	88.0	88.0	88.7	90.7	90.9	90.9	90.9	90.9
-20	-4	85.0	86.3	86.3	86.4	87.0	89.0	89.1	89.1	89.1	89.1
-30	-22	83.3	84.6	84.6	84.6	85.3	87.2	87.4	87.4	87.4	87.4
-40	-40	81.6	82.9	82.9	82.9	83.5	85.4	85.5	85.5	85.6	85.6
-50	-58	79.8	81.1	81.1	81.1	81.7	83.5	83.7	83.7	83.7	83.7

Takeoff %N1

Based on engine bleed to packs off and anti-ice on or off

AIRPORT OAT		AIRPORT PRESSURE ALTITUDE (FT)									
°C	°F	-1000	0	1000	2000	3000	4000	5000	6000	7000	8000
55	131	92.4	93.3	93.2	92.7						
50	122	93.0	93.8	94.4	94.6	94.0	94.0				
45	113	93.5	94.0	94.6	94.9	94.8	95.5	95.3	94.7		
40	104	94.0	94.5	94.9	95.1	95.2	96.1	95.9	95.8	95.8	95.4
35	95	94.2	95.0	95.6	95.7	95.7	96.5	96.6	96.6	96.7	96.6
30	86	93.9	95.4	95.2	95.4	96.0	96.3	96.5	96.7	96.7	96.8
25	77	93.1	94.6	94.6	94.6	95.3	96.4	96.2	96.0	96.3	96.7
20	68	92.4	93.8	93.8	93.8	94.6	95.9	95.9	95.9	95.9	95.9
15	59	91.6	93.0	93.0	93.0	93.7	95.1	95.1	95.1	95.1	95.1
10	50	90.8	92.2	92.2	92.2	92.9	94.3	94.3	94.3	94.3	94.3
5	41	90.0	91.4	91.4	91.4	92.1	93.4	93.4	93.4	93.4	93.4
0	32	89.2	90.6	90.6	90.6	91.3	92.6	92.6	92.6	92.6	92.6
-10	14	87.5	88.9	88.9	88.9	89.6	90.9	90.9	90.9	90.9	90.9
-20	-4	85.8	87.2	87.2	87.2	87.9	89.1	89.1	89.1	89.1	89.1
-30	-22	84.1	85.5	85.5	85.5	86.1	87.4	87.4	87.4	87.4	87.4
-40	-40	82.4	83.7	83.7	83.7	84.3	85.5	85.5	85.5	85.6	85.6
-50	-58	80.6	81.9	81.9	81.9	82.5	83.7	83.7	83.7	83.7	83.7

Takeoff Speeds Adjustment

SPEED (KIAS)	PRESSURE ALTITUDE (FT)			
	3000 & BELOW		ABOVE 3000	
	ABOVE 27°C	27°C & BELOW	ABOVE 27°C	27°C & BELOW
V1(MCG)	6	6	4	3
V1	0	0	0	0
VR	0	0	0	0

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Takeoff %N1 (18.5K Derate)

Based on engine bleed to packs on (Auto) and anti-ice on or off

AIRPORT OAT		AIRPORT PRESSURE ALTITUDE (FT)									
°C	°F	-1000	0	1000	2000	3000	4000	5000	6000	7000	8000
55	131	88.5	88.9	89.1	89.1						
50	122	89.0	89.5	90.0	90.6	91.3	91.5				
45	113	89.5	89.9	90.3	90.8	92.0	92.8	92.7	92.4		
40	104	90.1	90.3	90.7	91.2	92.4	93.1	93.2	93.3	92.9	92.3
35	95	90.3	90.4	91.1	91.5	92.8	93.4	93.4	93.7	93.2	92.8
30	86	90.1	90.4	90.8	91.3	92.9	93.7	93.7	94.0	93.5	93.1
25	77	89.3	89.6	90.3	90.8	92.1	93.1	93.5	94.2	93.8	93.4
20	68	88.6	88.8	89.5	90.0	91.3	92.3	92.7	93.4	93.4	93.4
15	59	87.8	88.1	88.7	89.2	90.5	91.5	91.9	92.6	92.6	92.6
10	50	87.1	87.3	88.0	88.5	89.8	90.7	91.1	91.8	91.8	91.8
5	41	86.3	86.6	87.2	87.7	89.0	89.9	90.3	91.0	91.0	91.0
0	32	85.5	85.8	86.4	86.9	88.2	89.1	89.5	90.2	90.2	90.2
-10	14	83.9	84.2	84.8	85.3	86.5	87.4	87.8	88.5	88.5	88.5
-20	-4	82.3	82.6	83.2	83.6	84.9	85.7	86.2	86.8	86.8	86.8
-30	-22	80.7	80.9	81.5	82.0	83.2	84.0	84.4	85.1	85.1	85.1
-40	-40	79.0	79.2	79.8	80.3	81.4	82.3	82.7	83.3	83.3	83.3
-50	-58	77.3	77.5	78.1	78.5	79.7	80.5	80.9	81.5	81.5	81.5

Takeoff %N1 (18.5K Derate)

Based on engine bleed to packs off and anti-ice on or off

AIRPORT OAT		AIRPORT PRESSURE ALTITUDE (FT)									
°C	°F	-1000	0	1000	2000	3000	4000	5000	6000	7000	8000
55	131	89.4	89.9	90.1	90.1						
50	122	89.9	90.4	91.0	91.5	92.2	92.4				
45	113	90.5	90.8	91.3	91.8	92.9	93.8	93.7	93.3		
40	104	91.0	91.2	91.7	92.1	93.3	94.1	94.1	94.3	93.8	93.2
35	95	91.2	91.4	92.1	92.5	93.7	94.3	94.4	94.6	94.2	93.8
30	86	91.0	91.3	91.8	92.2	93.8	94.6	94.7	94.9	94.5	94.1
25	77	90.2	90.5	91.2	91.7	93.0	94.0	94.4	95.1	94.8	94.3
20	68	89.5	89.7	90.4	90.9	92.2	93.2	93.6	94.3	94.3	94.3
15	59	88.7	89.0	89.6	90.1	91.4	92.4	92.8	93.5	93.5	93.5
10	50	88.0	88.2	88.9	89.3	90.6	91.6	92.0	92.7	92.7	92.7
5	41	87.2	87.4	88.1	88.6	89.9	90.8	91.2	91.9	91.9	91.9
0	32	86.4	86.6	87.3	87.8	89.0	89.9	90.4	91.0	91.0	91.0
-10	14	84.8	85.0	85.7	86.1	87.4	88.3	88.7	89.4	89.4	89.4
-20	-4	83.2	83.4	84.0	84.5	85.7	86.6	87.0	87.6	87.6	87.6
-30	-22	81.5	81.7	82.3	82.8	84.0	84.8	85.3	85.9	85.9	85.9
-40	-40	79.8	80.0	80.6	81.1	82.3	83.1	83.5	84.1	84.1	84.1
-50	-58	78.1	78.3	78.9	79.3	80.5	81.3	81.7	82.3	82.3	82.3

Takeoff Speeds Adjustment

SPEED (KIAS)	PRESSURE ALTITUDE (FT)			
	3000 & BELOW		ABOVE 3000	
	ABOVE 27°C	27°C & BELOW	ABOVE 27°C	27°C & BELOW
V1(MCG)	6	6	4	3
V1	0	0	0	0
VR	0	0	0	0

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220 KIAS Cruise Altitude Capability

Max Cruise Thrust, 100 ft/min residual rate of climb

WEIGHT (1000 KG)	PRESSURE ALTITUDE (FT)		
	ISA + 10°C & BELOW	ISA + 15°C	ISA + 20°C
64	20500	19200	17700
62	22200	20900	19600
60	23000	21700	20400
58	23800	22600	21200
56	24600	23400	22000
54	25400	24100	22800
52	26100	24900	23600
50	26700	25500	24300
48	27300	26200	24900
46	27900	26700	25500
44	28400	27300	26100
42	29000	27900	26700
40	29500	28400	27200
38	30000	28900	27700
36	30400	29300	28100

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220 KIAS Cruise Control

WEIGHT (1000 KG)		PRESSURE ALTITUDE (1000 FT)										
		10	12	14	16	18	20	22	24	26	28	30
64	%N1	79.7	81.3	82.8	84.4	86.1	87.9					
	MACH	.417	.432	.449	.467	.485	.505					
	KIAS	230	230	230	230	230	230					
	FF/ENG	1890	1886	1884	1884	1887	1893					
60	%N1	77.6	79.2	80.8	82.4	84.0	85.8	87.6	89.7			
	MACH	.399	.414	.430	.447	.465	.483	.503	.524			
	KIAS	220	220	220	220	220	220	220	220			
	FF/ENG	1749	1742	1739	1737	1737	1740	1746	1760			
56	%N1	76.5	78.1	79.7	81.3	82.9	84.5	86.3	88.2			
	MACH	.399	.414	.430	.447	.465	.483	.503	.524			
	KIAS	220	220	220	220	220	220	220	220			
	FF/ENG	1683	1675	1669	1666	1664	1664	1666	1676			
52	%N1	75.5	77.1	78.7	80.3	81.9	83.5	85.1	86.9	89.0		
	MACH	.399	.414	.430	.447	.465	.483	.503	.524	.545		
	KIAS	220	220	220	220	220	220	220	220	220		
	FF/ENG	1625	1615	1608	1603	1600	1598	1598	1604	1613		
48	%N1	74.6	76.2	77.8	79.4	80.9	82.5	84.2	85.9	87.8	90.1	
	MACH	.399	.414	.430	.447	.465	.483	.503	.524	.545	.568	
	KIAS	220	220	220	220	220	220	220	220	220	220	
	FF/ENG	1573	1563	1555	1549	1545	1542	1541	1545	1551	1564	
44	%N1	73.7	75.3	76.9	78.5	80.1	81.7	83.3	84.9	86.8	88.9	
	MACH	.399	.414	.430	.447	.465	.483	.503	.524	.545	.568	
	KIAS	220	220	220	220	220	220	220	220	220	220	
	FF/ENG	1526	1515	1507	1501	1497	1494	1491	1494	1498	1507	
40	%N1	72.8	74.5	76.1	77.7	79.4	80.9	82.5	84.1	85.9	87.9	90.4
	MACH	.399	.414	.430	.447	.465	.483	.503	.524	.545	.568	.592
	KIAS	220	220	220	220	220	220	220	220	220	220	220
	FF/ENG	1485	1474	1465	1458	1454	1450	1448	1450	1452	1457	1481
36	%N1	72.1	73.8	75.4	77.0	78.7	80.3	81.8	83.4	85.1	87.0	89.3
	MACH	.399	.414	.430	.447	.465	.483	.503	.524	.545	.568	.592
	KIAS	220	220	220	220	220	220	220	220	220	220	220
	FF/ENG	1450	1439	1430	1423	1418	1414	1411	1411	1412	1417	1432

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GEAR DOWN

220 KIAS Enroute Fuel and Time

Ground to Air Miles Conversion

AIR DISTANCE (NM) HEADWIND COMPONENT (KTS)					GROUND DISTANCE (NM)	AIR DISTANCE (NM) TAILWIND COMPONENT (KTS)				
100	80	60	40	20		20	40	60	80	100
333	295	263	238	218	200	189	180	171	162	155
676	598	531	479	437	400	378	357	339	323	308
1019	900	798	719	655	600	567	536	509	484	462
1362	1202	1066	960	874	800	755	714	677	644	615
1705	1505	1334	1201	1093	1000	943	892	845	804	768
2048	1807	1601	1441	1312	1200	1132	1071	1015	965	921
2391	2109	1869	1682	1531	1400	1321	1248	1183	1125	1074
2734	2412	2137	1923	1750	1600	1510	1427	1353	1286	1228
3077	2714	2404	2163	1969	1800	1698	1605	1521	1447	1381

Reference Fuel and Time Required at Check Point

AIR DIST (NM)	PRESSURE ALTITUDE (1000 FT)									
	10		14		20		24		28	
	FUEL (1000 KG)	TIME (HR:MIN)	FUEL (1000 KG)	TIME (HR:MIN)	FUEL (1000 KG)	TIME (HR:MIN)	FUEL (1000 KG)	TIME (HR:MIN)	FUEL (1000 KG)	TIME (HR:MIN)
200	2.2	0:49	2.0	0:46	1.9	0:44	1.7	0:42	1.6	0:40
400	4.6	1:36	4.2	1:31	3.9	1:26	3.6	1:21	3.4	1:17
600	6.9	2:23	6.4	2:15	6.0	2:08	5.5	2:00	5.2	1:54
800	9.2	3:10	8.6	3:00	8.0	2:49	7.4	2:40	7.0	2:30
1000	11.5	3:57	10.7	3:44	9.9	3:31	9.3	3:19	8.7	3:07
1200	13.8	4:45	12.8	4:28	11.9	4:13	11.1	3:58	10.4	3:44
1400	16.0	5:32	14.8	5:13	13.8	4:54	12.9	4:37	12.2	4:20
1600	18.2	6:19	16.9	5:57	15.8	5:36	14.7	5:16	13.8	4:57
1800	20.4	7:06	18.9	6:41	17.7	6:18	16.5	5:55	15.5	5:34

Fuel Required Adjustments (1000 KG)

REFERENCE FUEL REQUIRED (1000 KG)	WEIGHT AT CHECK POINT (1000 KG)				
	30	40	45	50	60
5	-0.4	-0.1	0.0	0.2	0.7
10	-0.6	-0.3	0.0	0.4	1.4
15	-0.9	-0.4	0.0	0.5	1.8
20	-1.0	-0.4	0.0	0.6	2.2
25	-1.0	-0.5	0.0	0.7	2.4

Based on 220 KIAS cruise and descent.

Descent at 220 KIAS

PRESSURE ALT (1000 FT)	5	10	15	17	19	21	23	25	27	29	31	33
DISTANCE (NM)	17	26	35	39	42	46	50	53	57	61	64	68
TIME (MINUTES)	6	8	10	11	12	13	13	14	15	15	16	17

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GEAR DOWN

Holding
Flaps Up

WEIGHT (1000 KG)		PRESSURE ALTITUDE (FT)						
		1500	5000	10000	15000	20000	25000	30000
64	%N1	72.9	75.7	79.7	83.6	87.9		
	KIAS	230	230	230	230	230		
	FF/ENG	2020	2000	1980	1980	1990		
62	%N1	71.3	74.1	78.2	82.2	86.4		
	KIAS	220	220	220	220	220		
	FF/ENG	1910	1890	1870	1860	1870		
60	%N1	70.8	73.6	77.6	81.6	85.8		
	KIAS	220	220	220	220	220		
	FF/ENG	1880	1860	1840	1820	1830		
58	%N1	70.3	73.0	77.1	81.0	85.1		
	KIAS	220	220	220	220	220		
	FF/ENG	1850	1820	1800	1790	1790		
56	%N1	69.8	72.5	76.5	80.5	84.5	89.3	
	KIAS	220	220	220	220	220	220	
	FF/ENG	1810	1790	1770	1750	1750	1770	
54	%N1	69.3	72.0	76.0	80.0	84.0	88.6	
	KIAS	220	220	220	220	220	220	
	FF/ENG	1780	1760	1740	1720	1710	1730	
52	%N1	67.5	70.2	74.2	78.3	82.3	86.7	
	KIAS	210	210	210	210	210	210	
	FF/ENG	1680	1650	1630	1610	1600	1600	
50	%N1	67.0	69.7	73.7	77.7	81.7	86.0	
	KIAS	210	210	210	210	210	210	
	FF/ENG	1650	1620	1600	1570	1560	1560	
48	%N1	66.5	69.1	73.1	77.2	81.1	85.3	
	KIAS	210	210	210	210	210	210	
	FF/ENG	1620	1590	1570	1540	1530	1530	
46	%N1	66.0	68.6	72.6	76.6	80.6	84.7	90.1
	KIAS	210	210	210	210	210	210	210
	FF/ENG	1590	1560	1540	1510	1500	1500	1530
44	%N1	65.5	68.1	72.0	76.1	80.1	84.2	89.3
	KIAS	210	210	210	210	210	210	210
	FF/ENG	1560	1540	1510	1490	1470	1470	1490
42	%N1	65.0	67.6	71.5	75.6	79.7	83.7	88.7
	KIAS	210	210	210	210	210	210	210
	FF/ENG	1540	1510	1490	1460	1450	1440	1460
40	%N1	64.6	67.2	71.1	75.1	79.2	83.2	88.1
	KIAS	210	210	210	210	210	210	210
	FF/ENG	1520	1490	1470	1440	1420	1420	1430
38	%N1	64.2	66.7	70.6	74.7	78.8	82.8	87.5
	KIAS	210	210	210	210	210	210	210
	FF/ENG	1500	1470	1440	1410	1400	1390	1400
36	%N1	63.8	66.3	70.2	74.3	78.4	82.4	87.0
	KIAS	210	210	210	210	210	210	210
	FF/ENG	1480	1450	1420	1390	1380	1370	1380

This table includes 5% additional fuel for holding in a racetrack pattern.

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Introduction

This chapter contains information to supplement performance data from the Flight Management Computer (FMC). In addition, sufficient inflight data is provided to complete a flight with the FMC inoperative. In the event of conflict between data presented in this chapter and that contained in the Approved Flight Manual, the Flight Manual shall always take precedence.

General

Takeoff Speeds

The speeds presented in the Takeoff Speeds table as well as FMC computed takeoff speeds can be used for all performance conditions provided adjustments are made to V1 for clearway, stopway, brake deactivation, improved climb, contaminated runway situations, brake energy limits or obstacle clearance with unbalanced V1. These speeds may be used for weights less than or equal to the performance limited weight.

Normal takeoff speeds, V1, VR, and V2, with anti-skid on, are read from the table by entering with takeoff flap setting, brake release weight, and appropriate column. The appropriate column is obtained by entering the Column Reference chart with the airport pressure altitude and the actual temperature. If an Altitude Adjustment chart is provided, adjust the takeoff speeds appropriately. Slope and wind adjustments to V1 are obtained by entering the Slope and Wind V1 Adjustment table.

If takeoffs are scheduled using these simplified speeds in conjunction with airport analyses that include clearway and/or stopway credits, adjustments to V1 speed are required.

Adjust V1 by the amount shown in the Clearway/Stopway table. The adjusted V1 must not exceed VR.

The maximum allowable clearway limits shown on the takeoff speeds page are provided for guidance when more precise data is unavailable.

V1(MCG), Minimum VR

Regulations prohibit scheduling takeoff with a V1 less than minimum V1 for control on the ground, V1(MCG) and VR less than minimum VR, (1.05) VMCA. Therefore compare the adjusted V1 and VR to the V1(MCG) and Minimum VR respectively. To find V1(MCG) and Minimum VR, enter the V1(MCG), Minimum VR table with the airport pressure altitude and actual

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OAT. If applicable, add the adjustments shown below the table. If the adjusted V1 is less than V1(MCG), set V1 equal to V1(MCG). If the adjusted VR is less than Minimum VR, set VR equal to Minimum VR. If VR is less than V1(MCG), set VR equal to V1(MCG). If VR is limited by either V1(MCG) or Minimum VR, determine a new V2 by adding the difference between the normal VR and the higher of V1(MCG) and Min VR to the normal V2.

If it is necssary to increase V1 to V1(MCG), then provided the actual field length exceeds the minimum field length of 3900ft no takeoff weight adjustment is necessary.

Clearway and Stopway V1 Adjustments

Takeoff speed corrections are to be applied to V1 speed when using takeoff weights based on the use of clearway and stopway.

Adjust V1 speed by the amount shown in the appropriate column. The adjusted V1 speed must not exceed VR. If the adjusted V1 speed is greater than VR, reduce V1 to equal VR.

Maximum allowable clearway limits are provided for guidance when more precise data is not available.

Assumed Temperature Takeoff

For reduced thrust takeoffs based on assumed temperature, determine V1, VR, V2 from the appropriate column for airport pressure altitude and assumed temperature. Adjust V1 for slope, wind, clearway and stopway as required. Compare adjusted V1 with V1(MCG) from table using airport altitude and actual temperature. This is a regulatory requirement to cover the case where the pilot elects to advance thrust levers to maximum takeoff thrust following the engine failure. If adjusted V1 is less than V1(MCG) set V1 equal to V1(MCG).

Stab Trim

To find takeoff stabilizer trim setting, enter the Stab Trim Setting table with takeoff flap setting and center of gravity (C.G. % MAC) and read required stabilizer trim units. For weights that meet the conditions listed, apply the adjustments shown below the table.

VREF

The Reference Speed table contains flaps 40, 30, and 15 landing speeds for a given weight. Apply wind correction shown as required.

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Flap Maneuver Speeds

This table provides the flap speed schedule for recommended maneuvering speed. The speed schedule is a function of weight and will provide adequate maneuver margin above stall at all weights.

During flap retraction/extension, movement of the flap to the next position should be initiated when reaching the maneuver speed for the existing flap.

Slush/Standing Water Takeoff

Experience has shown that aircraft performance may deteriorate significantly on runways covered with snow, slush, standing water, or ice. Therefore a reduction in runway/obstacle limited takeoff weight and revised takeoff speeds is necessary. The following information is intended for guidance in accordance with advisory material and is based on an engine failure at the critical point during takeoff.

The entire runway is assumed to be completely covered by a contaminant of uniform thickness and density. Therefore this information is conservative when operating under typical colder weather conditions where patches of slush exist and some degree of sanding is common.

Takeoffs in slush depths greater than 13mm (0.5 inches) are not recommended because of possible airplane damage as a result of slush impingement on the airplane structure. The use of assumed temperature method for reduced thrust is not allowed on contaminated runways. Interpolation for slush/standing water depths between the values shown is permitted.

Instructions for Using Tables:

Takeoff weight is determined as follows:

1. Determine the dry field/obstacle limit weight for the takeoff flap setting.

2. Enter the Weight Adjustment table with the field/obstacle limit weight to obtain the weight reduction for the slush/standing water depth and airport pressure altitude.

3. Adjust field length available for temperature by amount shown on chart.

4. Enter V1(MCG) Limit Weight table with the adjusted field length and pressure altitude to obtain the slush/standing water limit weight with respect to minimum field length required for V1(MCG) speed.

The maximum allowable takeoff weight in slush/standing water is the lesser of the limit weights found in steps 2 and 4.

Takeoff speeds determination:

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1. Determine takeoff speeds V1, VR and V2 for actual brake release weight using the Takeoff Speeds table in this section.

2. If V1(MCG) limited, set V1 = V1(MCG). If not limited by V1(MCG) considerations, enter V1 Adjustment table with actual brake release weight to determine the V1 reduction to apply to V1 speed. If the adjusted V1 is less than V1(MCG), set V1 = V1(MCG).

Slippery Runway

Airplane braking action is reported as good, medium or poor, depending on existing runway conditions. If braking action is reported as good, conditions should not be expected to be as good as on clean dry runways. The value “good” is comparative and is intended to mean that airplanes should not experience braking or directional control difficulties when stopping. The performance level used to calculate the “good” data is consistent with wet runway testing done on early Boeing jets. The performance level used to calculate the “poor” data reflects a runway covered with wet ice. Performance is based on reversers operating and a 15 ft. screen height at the end of the runway. The tables provided are used in the same manner as the Slush/Standing Water tables.

Anti-skid Inoperative

When operating with anti-skid inoperative, the field limit weight and V1 must be reduced to account for the effect on accelerate-stop performance. A simplified method which conservatively accounts for the effects of anti-skid inoperative is to reduce the normal runway/obstacle limited weight by 7500 kg and the V1 associated with the reduced weight by the amount shown in the table below.

ANTI-SKID INOPERATIVE V1 ADJUSTMENT	
FIELD LENGTH (FT)	V1 ADJUSTMENT (KIAS)
6000	-31
8000	-24
10000	-19
12000	-15
14000	-12

If the resulting V1 is less than minimum V1, takeoff is permitted with V1 set equal to V1(MCG) provided the dry accelerate stop distance adjusted for wind and slope exceeds approximately 7900 ft.

Detailed analysis for the specific case from the Airplane Flight Manual may yield a less restrictive penalty.

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Takeoff %N1

To find Max Takeoff %N1 based on normal engine bleed for air conditioning packs on (Auto), enter Takeoff %N1 Table with airport pressure altitude and airport OAT and read %N1. For packs off operation, apply the %N1 adjustment shown below the table. No takeoff %N1 adjustment is required for engine and wing anti-ice.

Assumed Temperature Reduced Thrust

Regulations permit the use of up to 25% takeoff thrust reduction for operation with assumed temperature reduced thrust. To find the maximum allowable assumed temperature enter the top chart with airport pressure altitude and OAT. Compare this temperature to that at which the airplane is performance limited as determined from available takeoff performance data. Next, enter the center table with airport pressure altitude and the lower of the two temperatures previously determined to obtain a maximum takeoff %N1. Do not use an assumed temperature less than the minimum assumed temperature shown. Enter the %N1 Adjustment table with OAT and the difference between the assumed and actual OAT to obtain a %N1 adjustment. Subtract the %N1 adjustment from the maximum takeoff %N1 found previously to determine the assumed temperature reduced thrust %N1.

Takeoff with assumed temperature reduced thrust is not permitted when: runway is contaminated with ice, slush, snow, or standing water; anti-skid is inoperative; or PMC is off. Use of this procedure is not recommended if potential windshear conditions exist.

Max Climb %N1

This table shows Max Climb %N1 for a 250 KIAS/280 KIAS/.74M climb speed schedule, normal engine bleed for packs on (Auto) and anti-ice off. Enter the table with airport pressure altitude and TAT and read %N1. Apply the appropriate bleed setting %N1 adjustment shown below the table. %N1 adjustments are shown for engine bleed to packs off operation, packs high operation, and wing anti-ice on.

Go-around %N1

To find Max Go-around %N1 based on normal engine bleed for packs on (Auto) and anti-ice off, enter the Go-around %N1 table with airport pressure altitude and reported OAT or TAT and read %N1. Apply the appropriate bleed setting %N1 adjustment shown below the table. %N1 adjustments are shown for engine bleed to packs off operation, packs high operation, and wing anti-ice on.

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Flight with Unreliable Airspeed / Turbulent Air Penetration

Pitch attitude and average %N1 information is provided for use in all phases of flight in the event of unreliable airspeed/Mach indications resulting from blocking or freezing of the pitot system. Loss of radome or turbulent air may also cause unreliable airspeed/Mach indications. The cruise table in this section may also be used for turbulent air penetration.

Pitch attitude is shown in bold type for emphasis since altitude and/or vertical speed indications may also be unreliable.

All Engines

Long Range Cruise Maximum Operating Altitude

These tables provide the maximum operating altitude in the same manner as the FMC. Maximum altitudes are shown for a given cruise weight and maneuver capability. This table considers both thrust and buffet limits, providing the more limiting of the two. Any data that is thrust limited is denoted by an asterisk and represents only a thrust limited condition in level flight with 100 ft/min residual rate of climb. Flying above these altitudes with sustained banks in excess of approximately 15° may cause the airplane to lose speed and/or altitude.

Note that the altitudes shown in the table are limited to the maximum certified altitude of 37000 ft.

Long Range Cruise Control

These tables provide target %N1, Long Range Cruise Mach number, IAS and standard day fuel flow per engine for the airplane weight and pressure altitude. As indicated by the shaded area, at optimum altitude .74M approximates the Long Range Cruise Mach schedule.

Long Range Cruise Enroute Fuel and Time

Long Range Cruise Enroute Fuel and Time tables are provided to determine remaining time and fuel required to destination. The data is based on Long Range Cruise and .74M/250 KIAS descent. Tables are presented for low altitudes and high altitudes.

To determine remaining fuel and time required, first enter the Ground to Air Miles Conversion table to convert ground distance and enroute wind to an equivalent still air distance for use with the Reference Fuel and Time tables. Next, enter the Reference Fuel and Time Table with air distance from the Ground to Air Miles Conversion Table and the desired altitude

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and read Reference Fuel and Time Required. Lastly, enter the Fuel Required Adjustment Table with the Reference Fuel and the actual weight at checkpoint to obtain fuel required to destination.

Long Range Cruise Wind-Altitude Trade

Wind is a factor which may justify operations considerably below optimum altitude. For example, a favorable wind component may have an effect on ground speed which more than compensates for the loss in air range.

Using this table, it is possible to determine the break-even wind (advantage necessary or disadvantage that can be tolerated) to maintain the same range at another altitude and long range cruise speed. The tables make no allowance for climb or descent time, fuel or distance, and are based on comparing ground fuel mileage.

Descent

Distance and time for descent are shown for a .74M/250 KIAS descent speed schedule. Enter the table with top of descent pressure altitude and read distance in nautical miles and time in minutes. Data is based on flight idle thrust descent in zero wind. Allowances are included for a straight-in approach with gear down and landing flaps at the outer marker.

Holding

Target %N1, indicated airspeed and fuel flow per engine information is tabulated for holding with flaps up based on the FMC optimum holding speed schedule. This is the higher of the maximum endurance speed and the maneuvering speed. Small variations in airspeed will not appreciably affect the overall endurance time. Enter the table with weight and pressure altitude to read %N1, IAS and fuel flow per engine.

Advisory Information

Autobrake Landing Distance

The Autobrake Landing Distance Table is provided as advisory information to assist in the selection of the most desirable auto brake setting for a given field length. It is not to be used to determine required field length. This data reflects actual landing distances on a dry runway for setting 1 through MAX, from touchdown to full stop, with or without reverse thrust. The table includes typical flare distances from threshold.

To use the Autobrake Landing Distance Table, first determine the available (or desired) landing distance. Enter the chart with the estimated approach speed and determine the actual stopping distance from

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touchdown for a given autobrake setting. If airspeed is used for approach speed, correct landing distance for pressure altitude, tailwind and temperature effects.

Selection of an autobrake setting results in a constant rate of deceleration. Maximum effort manual braking should achieve shorter landing distance than the MAX setting.

Slippery Runway Landing Distance

Landing distances are the actual landing distances for the specified runway conditions and do not include the 1.67 regulatory factor. Therefore they cannot be used to determine dispatch required landing field length. When landing on slippery runways or runways contaminated with ice, snow, slush or standing water, the reported braking action must be considered. If the surface is affected by water, snow or ice, and the braking action is reported as “good,” conditions should not be expected to be as good as on clean dry runways. The value “good” is comparative and is intended to mean that airplanes should not experience braking or directional control difficulties when landing. The performance level used to calculate the “good” data is consistent with wet runway testing done on early Boeing jets. The performance level used to calculate the “poor” data reflects runways covered with wet ice. Read landing distance for the reported braking action at the airplane weight, and then apply the adjustments for airport pressure altitude and approach speed as required.

Non-normal Configuration Landing Distance

Advisory information is provided to support non-normal configurations that affect landing performance of the airplane. Landing distances are shown for dry runway and good, medium, and poor reported braking action. Each non-normal configuration is listed with its recommended approach speed. Landing distance can be determined for the reference landing weight and then adjusted for actual weight and pressure altitude.

Brake Cooling Schedule

Advisory information is provided to assist in avoiding problems associated with hot brakes. For normal operation, most landings are at weights below the quick turnaround limit weight. Application of the recommended cooling procedures shown will avoid brake overheat and fuse plug problems that could result from repeated landings at short time intervals or a rejected takeoff.

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Engine Inoperative

Initial Max Continuous %N1

The Initial Max Continuous %N1 setting for use following an engine failure is shown. The table is based on the typical all engine cruise speed of .74M to provide a target %N1 setting at the start of driftdown. Once driftdown is established, the Max Continuous %N1 Table should be used to determine %N1 for the given conditions.

Max Continuous %N1

Power setting is based on one engine operating with one A/C pack operating and all anti-ice bleeds off. Enter the table with pressure altitude and IAS or Mach to read %N1.

It is desirable to maintain engine thrust within the limits of the Max Cruise thrust rating. However, where thrust in excess of Max Cruise rating is required, such as for meeting terrain clearance, ATC altitude assignments, or to attain maximum range capability, it is permissible to use the thrust needed up to the Max Continuous thrust rating. The Max Continuous thrust rating is intended primarily for emergency use at the discretion of the pilot and is the maximum thrust that may be used continuously.

Driftdown Speed/Level Off Altitude

The table shows optimum driftdown speed as a function of cruise weight at start of driftdown. Also shown are the approximate weight and pressure altitude at which the airplane will level off considering 100 ft/min residual rate of climb.

The level off altitude is dependent on air temperature (ISA deviation).

Driftdown/LRC Cruise Range Capability

This table shows the range capability from the start of driftdown. Driftdown is continued to level off altitude. As weight decreases due to fuel burn, the airplane is accelerated to Long Range Cruise speed. Cruise is continued at level off altitude and Long Range Cruise speed.

To determine fuel required, enter the Ground to Air Miles Conversion table with the desired ground distance and correct for anticipated winds to obtain air distance to destination. Then enter the Driftdown/Cruise Fuel and Time table with air distance and weight at start of driftdown to determine fuel and time required. If altitudes other than the level off altitude is used, fuel and time required may be obtained by using the Engine Inoperative Long Range Cruise Enroute Fuel and Time Table.

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Long Range Cruise Altitude Capability

The table shows the maximum altitude that can be maintained at a given weight and air temperature (ISA deviation), based on Long Range Cruise speed, Max Continuous thrust, and 100 ft/min residual rate of climb.

Long Range Cruise Control

The table provides target %N1, engine inoperative Long Range Cruise Mach number, IAS and fuel flow for the airplane weight and pressure altitude. The fuel flow values in this table reflect single engine fuel burn. To conservatively account for APU fuel burn, add 90 kg/hr to fuel flow values.

Long Range Cruise Diversion Fuel and Time

Tables are provided for crews to determine the fuel and time required to proceed to an alternate airfield with one engine inoperative. The data is based on single engine Long Range Cruise speed and .74M/250 KIAS descent. Enter with Air Distance as determined from the Ground to Air Miles Conversion Table and read Fuel and Time required at the cruise pressure altitude. Adjust the fuel obtained for deviation from the reference weight at checkpoint as required by entering the off reference fuel corrections table with the fuel required for the reference weight and the actual weight at checkpoint. Read fuel required and time for the actual weight.

Holding

Single engine holding data is provided in the same format as the all engine holding data and is based on the same assumptions.

PMC Off Power Settings

To find Max Takeoff %N1 with both PMC's Off, enter Max Takeoff %N1 table (PMC Off) for the appropriate bleed configuration with airport pressure altitude and airport OAT and read %N1. No takeoff %N1 correction is required for engine anti-ice operation up to 10°C (50°F) which is the highest temperature recommended for engine anti-ice operation. Do not use reduced thrust with PMC Off.

For Maximum Climb and Go-Around with both PMC's Off, use PMC on %N1.

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Gear Down

This section contains performance for airplane operation with the landing gear extended for all phases of flight. The data is based on engine bleeds for normal air conditioning.

Note: The Flight Management Computer System (FMCS) does not contain special provisions for operation with landing gear extended. As a result, the FMCS will generate inaccurate enroute speed schedules, display non-conservative predictions of fuel burn, estimated time of arrival (ETA), maximum altitude, and compute overly shallow descent path. To obtain accurate ETA predictions, gear down cruise speed and altitude should be entered on the CLB and CRZ pages. Gear down cruise speed should also be entered on the DES page. Use of VNAV during descent under these circumstances is not recommended.

Tables for gear down performance in this section are identical in format and used in the same manner as tables for the gear up configuration previously described.

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ABORTED ENGINE START

Condition: **One or more of the following conditions:**

- No N1 rotation before the engine start lever is raised to IDLE
- No oil pressure indication by the time the engine is stabilized at idle
- No increase in EGT, within 10 seconds on the ground or 30 seconds in flight, after the engine start lever is raised to IDLE
- No increase in, or a very slow increase in N1 or N2 after EGT indication
- EGT rapidly approaching or exceeding the start limit.

Before engine start lever raised to IDLE:

ENGINE START SWITCH . . . . . OFF

After engine start lever raised to IDLE:

Before starter cutout:

ENGINE START LEVER . . . . . CUTOFF

Continue to motor the engine for 60 seconds.

[Clears fuel and cools engine components.]

ENGINE START SWITCH . . . . . OFF

After starter cutout:

ENGINE START LEVER . . . . . CUTOFF

After N2 decreases to below 20%:

ENGINE START SWITCH . . . . . GRD

Motor the engine for 60 seconds.

[Clears fuel and cools engine components.]

ENGINE START SWITCH . . . . . OFF

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AIR CONDITIONING SMOKE/FUMES

Condition: **A concentration of air conditioning smoke/fumes is identified.**

OXYGEN MASKS AND SMOKE GOGGLES (if required) . . . . . ON

CREW COMMUNICATIONS (if required) . . . . . ESTABLISH

RECIRCULATION FAN SWITCH(ES) . . . . . OFF

[Eliminates possible source of smoke/fumes.]

If smoke/fumes stop:

Continue flight with the recirculation fan switch(es) OFF.

If smoke/fumes continue:

ISOLATION VALVE SWITCH . . . . . CLOSE

R PACK SWITCH . . . . . OFF

If smoke/fumes stop:

Continue flight with the R PACK switch OFF and ISOLATION VALVE switch CLOSE.

If smoke/fumes continue:

R PACK SWITCH . . . . . AUTO

L PACK SWITCH . . . . . OFF

If smoke/fumes stop:

Continue flight with the L PACK switch OFF and ISOLATION VALVE switch CLOSE.

If smoke/fumes continue:

L PACK SWITCH . . . . . AUTO

Plan to land at nearest suitable airport.

Accomplish the SMOKE/FUMES REMOVAL checklist, if required.

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AIRSPEED UNRELIABLE

Condition: **Pitch attitude not consistent with existing phase of flight, altitude, thrust, and weight, or noise and/or low frequency buffeting.**

Crosscheck ground speed and winds provided by the IRS and FMC to determine airspeed accuracy if indicated airspeed is questionable.

Note: Erroneous or unreliable airspeed indications may be caused by blocked or frozen pitot-static system(s), or a severely damaged or missing radome.

AIRPLANE ATTITUDE/THRUST . . . . . ADJUST

Maintain airplane control. Attitude and thrust information is provided in the Performance-Inflight section.

PITOT STATIC HEAT . . . . . CHECK ON

MACH/AIRSPEED INDICATORS . . . . . CROSS CHECK

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AUTO FAIL/UNSCHEDULED PRESSURIZATION CHANGE

Condition: **Automatic pressurization mode has failed, cabin altitude warning horn sounds, or the cabin altitude is not under control.**

Increasing thrust may ensure adequate air supply to control cabin altitude.

ENGINE BLEED AIR SWITCHES . . . . . .ON (ONE AT A TIME)

PACK SWITCHES . . . . . AUTO (ONE AT A TIME)

Allow cabin rate to stabilize before placing second switch on.

If the AUTO FAIL light is illuminated or pressurization is not under control:

PRESSURIZATION MODE SELECTOR. . . . . .STBY

Verify AUTO FAIL light extinguishes. Set cabin altitude using the cabin/flight altitude placard. Before descent, set landing field elevation minus 200 feet.

If the AUTO FAIL light remains illuminated or the STBY mode cannot maintain cabin pressurization:

PRESSURIZATION MODE SELECTOR . . . . . MAN

OUTFLOW VALVE SWITCH . . . . . AS REQUIRED

Operate the outflow valve to maintain proper cabin altitude and cabin rate of change.

At traffic pattern altitude, position the outflow valve to full open.

CONFIGURATION WARNING

Condition: **An intermittent warning horn sounds when advancing thrust levers to takeoff, or a steady warning horn sounds inflight.**

Assure proper airplane configuration.

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DITCHING

Condition: **Airplane ditching and evacuation are required.**

Send distress signals. Determine position, course, speed, altitude, situation, intention, time and position of intended touchdown and transmit mayday. Report type of aircraft and request intercept.

Alert cabin crew to prepare for ditching and seat passengers as far forward as possible.

Burn off fuel to reduce touchdown speed and increase buoyancy.

Plan to touch down on the windward side and parallel to waves and swells.

Plan a flaps 40 landing unless other configuration is required.

---------- DEFERRED ITEMS ----------

==> BELOW 5000 FEET

AURAL WARN C/B (P6) . . . . . PULL

[Prevents warning horn with gear retracted and landing flaps selected.]

PASSENGER SIGNS . . . . . ON

ENGINE BLEED AIR SWITCHES . . . . . OFF

[Permits depressurizing the airplane with outflow valve closed.]

PRESSURIZATION MODE SELECTOR . . . . . MAN DC

OUTFLOW VALVE SWITCH . . . . . CLOSE

[Prevents water from entering the airplane.]

APU SWITCH . . . . . OFF

GROUND PROXIMITY GEAR INHIBIT SWITCH . . . . . GEAR INHIBIT

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LIFE VESTS, SHOULDER HARNESSES AND SEAT BELTS . . . . . ON

PASSENGER CABIN PREPARATION . . . . . COMPLETE

Confirm that passenger cabin preparations are complete.

CAUTION: (737-500) Do not open aft entry or service doors as they may be partially submerged.

FINAL POSITION . . . . . TRANSMIT

Transmit all pertinent information regarding final ditching position.

Review After Impact Procedure:

- **Engine start levers to CUTOFF. Closes fuel shutoff valves to prevent discharge of fuel from ruptured fuel lines.**
- **(737-500) Open flight deck windows to ensure no cabin differential pressure prevents the opening of the doors or emergency exits.**
- **Initiate evacuation.**
- **Proceed to assigned ditching stations, launch rafts and evacuate the airplane as soon as practicable. The airplane may remain afloat indefinitely if fuel load is minimal and no serious damage was sustained during landing.**

---------- DEFERRED ITEMS ----------

==> DITCHING FINAL

Omit normal LANDING checklist.

LANDING GEAR . . . . . UP & OFF

FLAPS . . . . . _____, GREEN LIGHT

Advise cabin crew, at 500 feet, ditching is imminent; at 50 feet, brace for impact.

Maintain airspeed at VREF. Flare airplane to achieve minimum rate of descent at touchdown. Maintain 200-300 fpm rate of descent until start of flare. At flare, rotate smoothly to touchdown attitude of 10-12 degrees, maintaining desired airspeed and rate of descent with thrust. At touchdown, reduce thrust to idle.

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ELECTRICAL SMOKE/FUMES OR FIRE

Condition: **Electrical smoke/fumes/fire is identified.**

OXYGEN MASKS AND SMOKE GOGGLES (if required) . . . . . ON

CREW COMMUNICATIONS (if required) . . . . . ESTABLISH

RECIRCULATION FAN SWITCH(ES) . . . . . OFF

[Removes fan as a possible source of smoke/fumes. Stops recirculation of smoke/fumes and increases fresh air flow.]

If smoke/fumes/fire source is known:

ELECTRICAL POWER (affected equipment) . . . . . REMOVE

If practical, remove power from affected equipment by switch or circuit breaker in flight deck or cabin.

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If smoke/fumes/fire persists or source is unknown:

BUS TRANSFER SWITCH . . . . . OFF

[Prevents unwanted transfer of power.]

GALLEY POWER SWITCH . . . . . OFF

EQUIPMENT COOLING SUPPLY/EXHAUST SWITCHES . . . . . ALTERNATE

[Removes normal fans as a possible source of smoke/fumes.]

CABIN READING & GALLEY ATTENDANT WORK LIGHTS . . . . . ON

Instruct flight attendants to:
- **turn on cabin reading lights**
- **turn on galley attendants work lights.**

[Prepares cabin lighting prior to depowering cabin fluorescent lights.]

CABIN EQUIPMENT . . . . . OFF

Instruct flight attendants to:
- **turn off galley power switches**
- **turn off cabin fluorescent light switches.**

Plan to land at the nearest suitable airport.

Accomplish the SMOKE/FUMES REMOVAL checklist, if required.

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EMERGENCY DESCENT

Condition: **Unable to control cabin pressure with airplane above 14,000 feet MSL or conditions require a rapid descent.**

EMERGENCY DESCENT . . . . . ANNOUNCE

The captain will advise the cabin crew, on the PA system, of impending rapid descent. First officer will advise ATC and obtain the area altimeter setting.

ENGINE START SWITCHES . . . . . ON

THRUST LEVERS . . . . . CLOSE

Reduce thrust to minimum or as required for anti-ice.

SPEED BRAKE . . . . . FLIGHT DETENT

DESCENT . . . . . INITIATE

TARGET SPEED . . . . . Mmo/Vmo

If structural integrity is in doubt, limit speed as much as possible and avoid high maneuvering loads.

LEVEL-OFF ALTITUDE . . . . . LOWEST SAFE ALTITUDE OR 10,000 FT, whichever is higher

SPEED BRAKE . . . . . DOWN DETENT

Smoothly lower the SPEED BRAKE lever and level off. Add thrust and stabilize on altitude at desired airspeed.

CREW OXYGEN REGULATORS . . . . . NORMAL

Flight crew must use oxygen when cabin altitude is above 10,000 feet. To conserve oxygen, position the NORMAL/100% selector to NORMAL.

ENGINE START SWITCHES . . . . . AS REQUIRED

The new course of action is based on weather, oxygen, fuel remaining and available airports. Use of long range cruise may be appropriate.

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INFLIGHT ENGINE START

Condition: **Engine start is desired after a shutdown with no fire or apparent damage.**

Complete the ENGINE FAILURE AND SHUTDOWN checklist before attempting an inflight engine start.

INFLIGHT START ENVELOPE . . . . . CHECK

CAUTION: Starter assist should be used if N2 is below 15%.

THRUST LEVER . . . . . CLOSE

ENGINE START LEVER . . . . . CUTOFF

If starter assist is required:

PACK SWITCH (affected side) . . . . . OFF

DUCT PRESSURE . . . . . MINIMUM 30 PSI

If required, advance the thrust lever to increase duct pressure.

IGNITION SELECT SWITCH . . . . . BOTH

ENGINE START SWITCH . . . . . GRD/FLT

ENGINE START LEVER . . . . . IDLE DETENT

Position engine start lever to IDLE detent at a minimum of 15% N2.

If no increase in EGT is observed within 30 seconds:

ENGINE START LEVER . . . . . CUTOFF

ENGINE START SWITCH . . . . . OFF

After engine start:

ELECTRICAL . . . . . GENERATOR ON

PACK SWITCH . . . . . AUTO

ENGINE START SWITCH . . . . . AS REQUIRED

APU . . . . . AS REQUIRED

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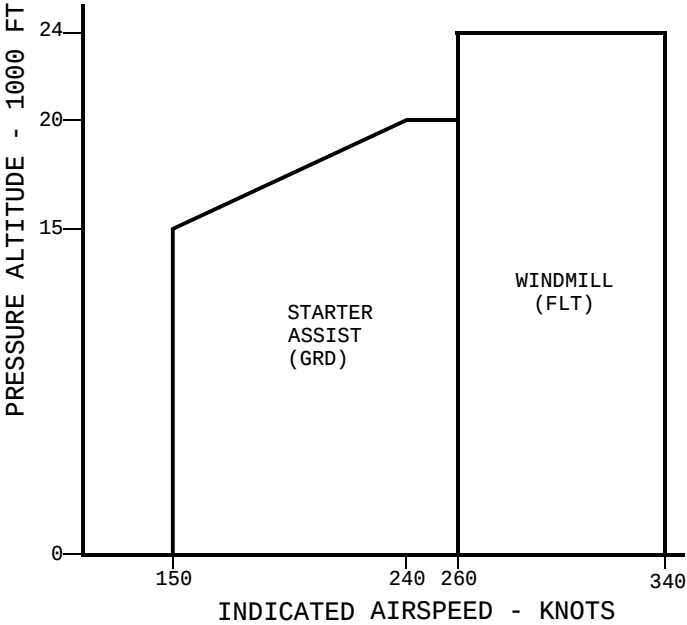
After engine start: (continued)

TRANSPONDER MODE SELECTOR
(TCAS equipped airplanes) . . . . . TA/RA

INFLIGHT START ENVELOPE



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JAMMED OR RESTRICTED ELEVATOR OR AILERON

Condition: **Restricted movement of the elevator or aileron/spoiler is experienced.**

AUTOPILOT (if engaged) . . . . . .DISENGAGE

AUTOTHROTTLE (if engaged) . . . . . .DISENGAGE

Verify thrust is symmetrical.

JAMMED OR RESTRICTED SYSTEM . . . . . .OVERPOWER

Use maximum force, including a combined effort of both pilots, if required. A maximum two-pilot effort on the controls will not cause a cable or system failure.

Do not turn off any flight control switches.

If controls are normal:

Accomplish the normal DESCENT-APPROACH and LANDING checklists.

If controls are not normal:

If electric trim is desired, the Stabilizer Trim Override switch must be positioned to OVERRIDE.

Do not make abrupt thrust changes. Extend or retract speedbrake slowly and smoothly.

Limit bank angle to 15°.

Plan to land at the nearest suitable airport.

Plan a flaps 15 landing.

Set VREF 15 + 5.

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If controls are not normal: (continued)

---------- DEFERRED ITEMS ----------

==> **DESCENT - APPROACH**

ANTI-ICE . . . . . AS REQUIRED

AIR CONDITIONING & PRESSURIZATION . . . . . SET

ALTIMETERS & INSTRUMENTS . . . . . SET & X-CHECKED

N1 & IAS BUGS . . . . . CHECKED & SET, VREF 15 + 5 KNOTS

GROUND PROXIMITY FLAP INHIBIT SWITCH . . . . . FLAP INHIBIT

GO-AROUND PROCEDURE . . . . . REVIEW

Accomplish normal go-around procedure. Advance thrust to go-around smoothly and slowly to avoid excessive pitch-up.

---------- DEFERRED ITEMS ----------

==> **LANDING**

ENGINE START SWITCHES . . . . . ON

RECALL . . . . . CHECKED

SPEED BRAKE . . . . . ARMED, GREEN LIGHT

LANDING GEAR . . . . . DOWN, 3 GREEN

FLAPS . . . . . 15, GREEN LIGHT

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JAMMED STABILIZER

Condition: **The stabilizer fails to respond to electric trim inputs.**

MANUAL TRIM . . . . . .APPLY

Use force as needed to cause the disconnect clutch to disengage. Steady pressure on the manual trim handles is required until the desired trim is attained. A maximum two-pilot effort on the trim wheels will not cause a cable or system failure.

Note: Handle should be folded inside stabilizer trim wheel when not in use.

If the trim wheel moves when released:

STABILIZER TRIM AUTOPILOT CUTOUT SWITCH . . . . . CUTOUT

[Autopilot is not available.]

Maintain in-trim airspeed until start of approach.

Plan a flaps 15 landing.

Set VREF 15.

Establish landing configuration early.

Note: Anticipate higher than normal elevator forces during approach and landing.

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---------- DEFERRED ITEMS ----------

==> DESCENT - APPROACH

ANTI-ICE . . . . . AS REQUIRED

AIR CONDITIONING &
PRESSURIZATION . . . . . SET

ALTIMETERS &
INSTRUMENTS . . . . . SET & X-CHECKED

N1 & IAS BUGS . . . . . CHECKED & SET, VREF 15

GROUND PROXIMITY FLAP
INHIBIT SWITCH . . . . . FLAP INHIBIT

---------- DEFERRED ITEMS ----------

==> LANDING

ENGINE START SWITCHES . . . . . ON

RECALL . . . . . CHECKED

SPEED BRAKE . . . . . ARMED, GREEN LIGHT

LANDING GEAR . . . . . DOWN, 3 GREEN

FLAPS . . . . . 15, GREEN LIGHT

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LOSS OF THRUST ON BOTH ENGINES

Condition: **Loss of all thrust on both engines or loss of thrust lever response from both engines.**

ENGINE START SWITCHES . . . . . FLT

ENGINE START LEVERS . . . . . CUTOFF

EGT decreasing:

ENGINE START LEVERS . . . . . .IDLE DETENT

If EGT reaches 930°C, repeat the above steps.

Note: In moderate to heavy rain it may take up to 3 minutes to accelerate to idle.

APU (if available) . . . . . START & ON BUS

Do not wait for successful engine start(s) prior to starting APU.

[The APU has demonstrated the capability to provide electrical and pneumatic power up to 20,000 feet.]

If neither restart is successful and N2 is below 15%:

WING ANTI–ICE SWITCH . . . . . .OFF

PACK SWITCHES . . . . . .OFF

APU BLEED AIR SWITCH . . . . . .ON

IGNITION SELECT SWITCH . . . . . BOTH

EITHER ENGINE START SWITCH . . . . . GRD

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When engine parameters have stabilized:

APU BLEED AIR SWITCH . . . . . OFF

ENGINE START SWITCH . . . . . FLT

THRUST LEVER . . . . . ADVANCE

ENGINE GENERATOR SWITCH . . . . . ON

PACK SWITCH . . . . . AUTO

Accomplish the INFLIGHT ENGINE START checklist to start the other engine.

If neither IRS attitude display recovers after a generator bus is restored:

IRS MODE SELECTOR SWITCHES . . . . . ATT

Maintain straight and level, constant airspeed flight until attitude displays recover (approximately 30 seconds).

MAGNETIC HEADING . . . . . ENTER

APU . . . . . AS REQUIRED

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PASSENGER EVACUATION

Condition: **Evacuation of passengers and crew is required.**

On the captain's command, simultaneously accomplish individual checklist items.

CAPTAIN

PARKING BRAKE . . . . . SET

SPEED BRAKE LEVER . . . . . DOWN DETENT

[Prevents possible interference or injury to passengers evacuating through the overwing escape hatches.]

ENGINE START LEVERS (both) . . . . . CUTOFF

If time permits, verify flaps are full down before placing the engine start levers to CUTOFF.

[Shuts down the engines to reduce the possibility of slide damage or injury.]

EVACUATION . . . . . INITIATE

Notify the flight attendants.

ENGINE AND APU FIRE WARNING SWITCHES . . . . . OVERRIDE, PULL & ROTATE

Rotate engine fire warning switches in opposite directions. Rotate all switches to the stop and hold for one second.

[Reduces the risk of fire and injury.]

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FIRST OFFICER

FLAP LEVER 40
[Aids in evacuating passengers over the wing.]

STANDBY POWER SWITCH . . . . . .BAT

PRESSURIZATION MODE SELECTOR. . . . . MAN DC

OUTFLOW VALVE SWITCH. . . . . OPEN
[Ensures the airplane is depressurized for opening exits.]

TOWER . . . . . .NOTIFY

RAPID DEPRESSURIZATION

Condition: **A rapid loss of cabin pressure with airplane altitude above 14,000 feet MSL.**

OXYGEN MASKS . . . . . .ON

CREW COMMUNICATIONS . . . . . ESTABLISH

PRESSURIZATION MODE SELECTOR . . . . . MAN

OUTFLOW VALVE SWITCH . . . . . CLOSE
If pressurization is restored, continue manual operation to maintain proper cabin altitude.

PASSENGER SIGNS. . . . . .ON

PASSENGER OXYGEN SWITCH (if required) . . . . . .ON
Activate passenger oxygen if cabin altitude exceeds or is expected to exceed 14,000 feet.

EMERGENCY DESCENT (if required). . . . . INITIATE
Accomplish the EMERGENCY DESCENT checklist if the airplane is above 14,000 feet MSL and control of cabin pressure is not possible, or cabin pressure is lost.

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SMOKE/FUMES REMOVAL

Condition: **Smoke/fumes removal is required.**

OXYGEN MASKS AND SMOKE GOGGLES (if required) . . . . . ON

CABIN DOOR . . . . . CLOSE

[Prevents smoke/fumes contamination of/from other compartments.]

CREW COMMUNICATIONS (if required) . . . . . ESTABLISH

If pack(s) are on and smoke/fumes source is confirmed to be on the flight deck or main cabin:

L AND R PACK SWITCHES . . . . . HIGH

RECIRCULATION FAN SWITCH(ES) . . . . . OFF

PRESSURIZATION MODE SELECTOR . . . . . STBY

CABIN ALTITUDE INDICATOR (maximum 10,000 feet) . . . . . INCREASE

[Increases the ventilation rate.]

CABIN RATE SELECTOR . . . . . MAXIMUM INCR

ENGINE No. 1 and No. 2 BLEED AIR SWITCHES . . . . . VERIFY ON

ENGINE THRUST . . . . . MAXIMUM PRACTICAL N1 (minimum 45%)

[Provides maximum cabin ventilation.]

FLIGHT DECK AIR CONDITIONING AND GASPER OUTLETS . . . . . OPEN

CAUTION: Do not open any flight deck window. Keep the cabin door closed.

If smoke/fumes are uncontrollable:

AIRPLANE ALTITUDE . . . . . LOWEST SAFE ALTITUDE or 10,000 FT, whichever is higher

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If smoke/fumes are uncontrollable: (continued)

At 14,000 feet or below:

PRESSURIZATION MODE SELECTOR . . . . . MAN

OUTFLOW VALVE SWITCH . . . . . OPEN

Position outflow valve to full open. This causes the cabin airflow to carry smoke/fumes aft.

If packs are off and smoke/fumes source is confirmed to be on the flight deck:

CAUTION: Window should not be opened unless the source is confirmed to be originating on the flight deck.

NORMAL HOLDING AIRSPEED . . . . . ESTABLISH

[High airspeed may prevent opening the window.]

FIRST OFFICER'S SLIDING WINDOW . . . . . OPEN

TAILSTRIKE ON TAKEOFF

Condition: **Airplane tail has contacted the ground during takeoff.**

CAUTION: Do not pressurize airplane due to possible structural damage.

PRESSURIZATION MODE SELECTOR . . . . . MAN

OUTFLOW VALVE SWITCH . . . . . OPEN

Hold outflow valve switch in the OPEN position until outflow VALVE position indicator shows valve full open.

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UNCOMMANDED RUDDER

Condition: **Uncommanded rudder pedal displacement or pedal kicks.**

AUTOPILOT (if engaged) . . . . . DISENGAGE

Maintain control of the airplane with all available flight controls. If roll is uncontrollable, immediately reduce pitch/angle of attack and increase airspeed. Do not attempt to maintain altitude until control is recovered.

AUTOTHROTTLE (if engaged) . . . . . DISENGAGE

Verify thrust is symmetrical.

YAW DAMPER SWITCH . . . . . OFF

RUDDER TRIM . . . . . CENTER

RUDDER PEDALS . . . . . FREE & CENTER

Use maximum force including a combined effort of both pilots, if required, to free and center the rudder pedals.

If rudder pedal position or movement is not normal and the condition is the not the result of rudder trim:

SYSTEM B FLIGHT CONTROL SWITCH . . . . . STBY RUD

A slight rudder deflection may remain, but continued rudder pedal pressure may help maintain an in-trim conditon.

Sufficient directional control is available on landing using differential braking and nose wheel steering.

Crosswind capability may be reduced.

Do not use auto brakes.

Consider checking rudder freedom of movement at a safe altitude using slow rudder inputs while in the landing configuration and at approach airspeed.

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If condition was the result of rudder trim or environmental factors:

YAW DAMPER SWITCH . . . . . ON

Accomplish the normal DESCENT - APPROACH and LANDING checklists.

UNCOMMANDED YAW OR ROLL

Condition: **Uncommanded yaw or roll occurs in flight.**

AUTOPILOT (if engaged) . . . . . DISENGAGE

Maintain control of the airplane with all available flight controls. If roll is uncontrollable, immediately reduce pitch/angle of attack and increase airspeed. Do not attempt to maintain altitude until control is recovered.

AUTOTHROTTLE (if engaged) . . . . . DISENGAGE

Verify thrust is symmetrical.

YAW DAMPER SWITCH . . . . . OFF

If it is confirmed that the autopilot or autothrottle is not the cause of the uncommanded yaw or roll, the autopilot and autothrottle may be re-engaged at the pilot's discretion.

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VOLCANIC ASH

Condition: **Static discharge around the windshield, bright glow in the engine inlets, smoke or dust on the flight deck or an acrid odor indicates the airplane is in volcanic ash.**

Exit volcanic ash as quickly as possible. Consider a 180 degree turn.

OXYGEN MASKS AND SMOKE GOGGLES (if required) . . . . . ON

CREW COMMUNICATIONS (if required) . . . . . ESTABLISH

AUTOTHROTTLE (if engaged) . . . . . DISENGAGE

[Allows thrust lever to remain where manually positioned.]

THRUST LEVERS . . . . . CLOSE

Conditions permitting, operate at idle thrust.

[Reduces possible engine damage and/or flameout by decreasing EGT.]

ENGINE START SWITCHES . . . . . FLT

PACK SWITCHES . . . . . HIGH

WING ANTI–ICE . . . . . ON

[Increases bleed air extraction to improve engine stall margin.]

ENGINE ANTI–ICE . . . . . ON

[Increases bleed air extraction to improve engine stall margin.]

APU (if available) . . . . . START

[Provides backup electrical and pneumatic source, if required.]

Note: Encountering volcanic ash can lead to abnormal systems reactions such as:

- **Engine malfunctions, increasing EGT, engine stall or flameout.**
- **Decrease or loss of airspeed indications.**
- **Equipment cooling OFF light.**

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If engines have flamed out or stalled, or EGT rapidly approaches or exceeds limit:

Accomplish the LOSS OF THRUST ON BOTH ENGINES checklist.

Note: Engines are very slow to accelerate to idle at high altitudes, which may be interpreted as a hung start or an engine malfunction.

Plan to land at the nearest suitable airport.

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WINDOW DAMAGE

Condition: **Arcing, substantial delamination, shattering or cracking of any flight deck window.**

WINDOW HEAT SWITCH (affected window) . . . . . OFF
- **Limit maximum airspeed to 250 knots below 10,000 feet.**
- **Use crew and passenger oxygen, if required.**

CABIN ALTITUDE INDICATOR . . . . . 10,000 FT

PRESSURIZATION MODE SELECTOR . . . . . .STBY

Reduce pressure differential by limiting flight altitude as indicated in the following tables.

Note: For MEA between 15,000 and 20,000 feet select a higher CAB ALT to maintain 2 psi differential. Above 20,000 feet, select MAN and maintain 2 psi differential.

Window 1, 2, 3 or 5:

CRACKED PANE	MAX DIFF PRESSURE	APPROX FLT ALT
Outer	— No Restriction —	
Inner	5 PSI	26,000 FT
Both	2 PSI	15,000 FT

Window 4: A failed middle pane usually appears shattered and transparency is virtually lost.

CRACKED PANE	MAX DIFF PRESSURE	APPROX FLT ALT
Outer	— No Restriction —	
Middle	5 PSI	26,000 FT
Both	2 PSI	15,000 FT

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AUTOMATIC UNLOCK

Condition: **The flight deck door AUTO UNLK light illuminated indicates the correct access code has been entered and the flight deck door is programmed to automatically unlock after a time delay.**

FLIGHT DECK DOOR SELECTOR . . . . . DENY

Rotate and hold for 1 second.

[Prohibits unauthorized access to the flight deck.]

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DOOR ANNUNCIATOR

Condition: **A door annunciator light illuminated indicates an exterior door is not properly latched.**

ENTRY/SERVICE

HANDLE (affected door) . . . . . CHECK CLOSED

Check door handle and position to close if necessary.

If the door handle is in the closed position and cabin pressurization is normal:

Proceed normally.

If the door handle will not close and/or cabin pressurization is not normal:

Plan to land at nearest suitable airport.

EQUIP/CARGO/AIRSTAIR

If EQUIP, FWD/AFT CARGO or AIRSTAIR light is illuminated and pressurization is normal:

Proceed normally.

If EQUIP, FWD/AFT CARGO or AIRSTAIR light is illuminated and pressurization is not normal:

PASSENGER SIGNS . . . . . ON

DESCENT . . . . . INITIATE

Descend to 13,000 feet or MEA, whichever is higher.

OXYGEN MASKS (if required) . . . . . ON

Use oxygen masks at cabin altitudes in excess of 10,000 feet.

CREW COMMUNICATIONS . . . . . ESTABLISH

PRESSURIZATION MODE SELECTOR . . . . . STBY

CABIN ALTITUDE INDICATOR 13,000

CABIN RATE SELECTOR . . . . . AS DESIRED

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If EQUIP, FWD/AFT CARGO or AIRSTAIR light is illuminated and pressurization is not normal: (continued)

If MEA is above 13,000 feet:

PRESSURIZATION MODE SELECTOR . . . . . MAN

Adjust outflow valve to increase cabin altitude to MEA.

PASSENGER OXYGEN (if required) . . . . . .ON

Activate passenger oxygen if cabin altitude exceeds 14,000 feet.

Plan to land at nearest suitable airport.

EMERGENCY EXIT LIGHTS NOT ARMED

Condition: **The NOT ARMED light illuminated indicates the EMER EXIT LIGHTS switch is not in the ARMED position.**

If the EMER EXIT LIGHTS switch is ON, individual emergency exit light batteries will provide a minimum of 10 minutes of lighting.

If the EMER EXIT LIGHTS switch is OFF, emergency lighting is not available.

LOCK FAIL

Condition: **The flight deck door LOCK FAIL light illuminated indicates the flight deck door lock has failed or the Flight Deck Access System switch is OFF.**

PASSENGER OXYGEN ON

Condition: **The PASS OXY ON light illuminated indicates the passenger oxygen system is activated.**

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TAILSTRIKE ON TAKEOFF

Condition: **Airplane tail has contacted the ground during takeoff.**

CAUTION: Do not pressurize airplane due to possible structural damage.

PRESSURIZATION MODE SELECTOR . . . . . MAN

OUTFLOW VALVE SWITCH . . . . . OPEN

Hold outflow valve switch in the OPEN position until outflow VALVE position indicator shows valve full open.

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WINDOW DAMAGE

Condition: **Arcing, substantial delamination, shattering or cracking of any flight deck window.**

WINDOW HEAT SWITCH (affected window) . . . . . OFF
- **Limit maximum airspeed to 250 knots below 10,000 feet.**
- **Use crew and passenger oxygen, if required.**

CABIN ALTITUDE INDICATOR. . . . . 10,000 FT

PRESSURIZATION MODE SELECTOR . . . . . .STBY

Reduce pressure differential by limiting flight altitude as indicated in the following tables.

Note: For MEA between 15,000 and 20,000 feet select a higher CAB ALT to maintain 2 psi differential. Above 20,000 feet, select MAN and maintain 2 psi differential.

Window 1, 2, 3 or 5:

CRACKED PANE	MAX DIFF PRESSURE	APPROX FLT ALT
Outer	— No Restriction —	
Inner	5 PSI	26,000 FT
Both	2 PSI	15,000 FT

Window 4: A failed middle pane usually appears shattered and transparency is virtually lost.

CRACKED PANE	MAX DIFF PRESSURE	APPROX FLT ALT
Outer	— No Restriction —	
Middle	5 PSI	26,000 FT
Both	2 PSI	15,000 FT

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AIR CONDITIONING SMOKE/FUMES

Condition: **A concentration of air conditioning smoke/fumes is identified.**

OXYGEN MASKS AND SMOKE GOGGLES (if required) . . . . . ON

CREW COMMUNICATIONS (if required) . . . . . ESTABLISH

RECIRCULATION FAN SWITCH(ES) . . . . . OFF

[Eliminates possible source of smoke/fumes.]

If smoke/fumes stop:

Continue flight with the recirculation fan switch(es) OFF.

If smoke/fumes continue:

ISOLATION VALVE SWITCH . . . . . CLOSE

R PACK SWITCH . . . . . OFF

If smoke/fumes stop:

Continue flight with the R PACK switch OFF and ISOLATION VALVE switch CLOSE.

If smoke/fumes continue:

R PACK SWITCH . . . . . AUTO

L PACK SWITCH . . . . . OFF

If smoke/fumes stop:

Continue flight with the L PACK switch OFF and ISOLATION VALVE switch CLOSE.

If smoke/fumes continue:

L PACK SWITCH . . . . . AUTO

Plan to land at nearest suitable airport.

Accomplish the SMOKE/FUMES REMOVAL checklist, if required.

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AUTO FAIL/UNSCHEDULED PRESSURIZATION CHANGE

Condition: **Automatic pressurization mode has failed, cabin altitude warning horn sounds, or the cabin altitude is not under control.**

Increasing thrust may ensure adequate air supply to control cabin altitude.

ENGINE BLEED AIR SWITCHES . . . . . ON (ONE AT A TIME)

PACK SWITCHES . . . . . AUTO (ONE AT A TIME)

Allow cabin rate to stabilize before placing second switch on.

If the AUTO FAIL light is illuminated or pressurization is not under control:

PRESSURIZATION MODE SELECTOR . . . . . .STBY

Verify AUTO FAIL light extinguishes. Set cabin altitude using the cabin/flight altitude placard. Before descent, set landing field elevation minus 200 feet.

If the AUTO FAIL light remains illuminated or the STBY mode cannot maintain cabin pressurization:

PRESSURIZATION MODE SELECTOR . . . . . MAN

OUTFLOW VALVE SWITCH . . . . . AS REQUIRED

Operate the outflow valve to maintain proper cabin altitude and cabin rate of change.

At traffic pattern altitude, position the outflow valve to full open.

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BLEED TRIP OFF

Condition: **The BLEED TRIP OFF light illuminated indicates the related engine bleed air temperature or pressure is excessive.**

WING ANTI-ICE SWITCH . . . . . OFF

TRIP RESET SWITCH . . . . . PUSH

[The BLEED TRIP OFF light extinguishes if bleed air temperature has cooled below limits.]

If the BLEED TRIP OFF light remains illuminated:

PACK SWITCH (affected side) . . . . . OFF

[Causes operating pack to regulate to high flow in flight with flaps up.]

Avoid icing conditions.

If the BLEED TRIP OFF light extinguishes:

WING ANTI-ICE . . . . . AS REQUIRED

CAUTION: Use of wing anti-ice above approximately FL350 may cause bleed trip off and possible loss of cabin pressure.

DUAL BLEED

Condition: **The DUAL BLEED light illuminated indicates the APU bleed valve open and No. 1 engine BLEED air switch ON, or No. 2 engine BLEED air switch ON with APU bleed air valve and isolation valve open.**

Limit engine thrust to idle while the light is illuminated.

After engine start:

APU BLEED AIR SWITCH . . . . . OFF

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EMERGENCY DESCENT

Condition: **Unable to control cabin pressure with airplane above 14,000 feet MSL or conditions require a rapid descent.**

EMERGENCY DESCENT . . . . . ANNOUNCE

The captain will advise the cabin crew, on the PA system, of impending rapid descent. First officer will advise ATC and obtain the area altimeter setting.

ENGINE START SWITCHES. . . . . .ON

THRUST LEVERS . . . . . CLOSE

Reduce thrust to minimum or as required for anti-ice.

SPEED BRAKE . . . . . FLIGHT DETENT

DESCENT . . . . . INITIATE

TARGET SPEED . . . . . .Mmo/Vmo

If structural integrity is in doubt, limit speed as much as possible and avoid high maneuvering loads.

LEVEL-OFF ALTITUDE. . . . . LOWEST SAFE ALTITUDE OR 10,000 FT, whichever is higher

SPEED BRAKE . . . . . DOWN DETENT

Smoothly lower the SPEED BRAKE lever and level off. Add thrust and stabilize on altitude at desired airspeed.

CREW OXYGEN REGULATORS . . . . . .NORMAL

Flight crew must use oxygen when cabin altitude is above 10,000 feet. To conserve oxygen, position the NORMAL/100% selector to NORMAL.

ENGINE START SWITCHES. . . . . AS REQUIRED

The new course of action is based on weather, oxygen, fuel remaining and available airports. Use of long range cruise may be appropriate.

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EQUIPMENT COOLING OFF

Condition: **The Equipment Cooling Supply or Exhaust OFF light illuminated indicates a loss of airflow from the selected cooling fan.**

EQUIPMENT COOLING SUPPLY/EXHAUST SWITCH (as required) . . . . . ALTERNATE

[The light should extinguish within approximately 5 seconds.]

No further action is necessary in flight if the equipment cooling OFF light does not extinguish.

OFF SCHEDULE DESCENT

Condition: **The OFF SCHED DESCENT light illuminated indicates the airplane descended before reaching the planned cruise altitude set in the FLT ALT indicator.**

No action is necessary if the airplane is returned to the airport of departure for land‘ing.

If not landing at airport of departure:

FLIGHT ALTITUDE INDICATOR . . . . . RESET

Reset to actual airplane altitude.

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PACK

Condition: **A PACK light illuminated indicates both primary and standby pack controls have failed or the related pack valve closed due to temperature exceeded limits.**

ALL TEMPERATURE SELECTORS . . . . . WARMER TEMPERATURE

[Reduces the work load on the affected air conditioning pack.]

TRIP RESET SWITCH . . . . . PUSH

[If the PACK light illuminated as a result of the pack temperature exceeding limits, the light extinguishes if the pack temperature has cooled below limits.]

If one PACK light remains illuminated:

ISOLATION VALVE SWITCH . . . . . CLOSE

PACK SWITCH . . . . . OFF

If both PACK lights remain illuminated:

Note: Both pack valves may have closed resulting in a gradual loss of cabin pressure and an eventual CABIN ALTITUDE warning.

Monitor cabin altitude during the remainder of the flight.

If cabin altitude increases:

DESCENT . . . . . ACCOMPLISH

Monitor cabin altitude and rate. Descend to lowest safe altitude or 10,000 feet, whichever is higher.

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If cabin altitude increases: (continued)

At level off:

AIRSPEED . . . . . 290 KNOTS MINIMUM

[Flight deck and cabin temperatures may increase rapidly below 290 knots.]

PRESSURIZATION MODE SELECTOR . . . . . MAN DC

OUTFLOW VALVE SWITCH . . . . . FULL OPEN

[Increases airplane ventilation.]

RIGHT RECIRCULATION FAN SWITCH . . . . . AUTO

LEFT RECIRCULARION FAN SWITCH . . . . . OFF

If flight deck and cabin temperatures are excessively warm:

FLIGHT DECK DOOR . . . . . OPEN

[Improves flight deck ventilation.]

CABIN LIGHTING . . . . . DIM

IN-FLIGHT ENTERTAINMENT SYSTEMS . . . . . OFF

FLIGHT DECK AND CABIN WINDOW SHADES . . . . . CLOSED

GALLEY POWER . . . . . OFF

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PACK TRIP OFF

Condition: **A PACK TRIP OFF light illuminated indicates the related pack valve closed due to temperature exceeded limits.**

TEMPERATURE SELECTOR . . . . . WARMER TEMPERATURE

[Reduces the work load on the affected air conditioning pack.]

TRIP RESET SWITCH . . . . . PUSH

[The PACK TRIP OFF light extinguishes if the pack temperature has cooled below limits.]

If both PACK TRIP OFF lights remain illuminated:

DESCENT . . . . . ACCOMPLISH

Monitor cabin altitude and rate. Descend to lowest safe altitude or 10,000 feet, whichever is higher.

At level off:

AIRSPEED . . . . . 290 KNOTS MINIMUM

[Flight deck and cabin temperatures may increase rapidly below 290 knots.]

PRESSURIZATION MODE SELECTOR . . . . . MAN DC

OUTFLOW VALVE SWITCH . . . . . FULL OPEN

[Increases airplane ventilation.]

If flight deck and cabin temperatures are excessively warm:

FLIGHT DECK DOOR . . . . . OPEN

[Improves flight deck ventilation.]

CABIN LIGHTING . . . . . DIM

IN-FLIGHT ENTERTAINMENT SYSTEMS . . . . . OFF

FLIGHT DECK AND CABIN WINDOW SHADES . . . . . CLOSED

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If flight deck and cabin temperatures are excessively warm: (continued)

GALLEY POWER . . . . . OFF

RAPID DEPRESSURIZATION

Condition: **A rapid loss of cabin pressure with airplane altitude above 14,000 feet MSL.**

OXYGEN MASKS . . . . . ON

CREW COMMUNICATIONS . . . . . ESTABLISH

PRESSURIZATION MODE SELECTOR . . . . . MAN

OUTFLOW VALVE SWITCH . . . . . CLOSE

If pressurization is restored, continue manual operation to maintain proper cabin altitude.

PASSENGER SIGNS . . . . . ON

PASSENGER OXYGEN SWITCH (if required) . . . . . ON

Activate passenger oxygen if cabin altitude exceeds or is expected to exceed 14,000 feet.

EMERGENCY DESCENT (if required) . . . . . INITIATE

Accomplish the EMERGENCY DESCENT checklist if the airplane is above 14,000 feet MSL and control of cabin pressure is not possible, or cabin pressure is lost.

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WING-BODY OVERHEAT

Condition: **A WING-BODY OVERHEAT light illuminated indicates a bleed air duct leak.**

ISOLATION VALVE SWITCH . . . . . CLOSE

PACK SWITCH (affected side). . . . . OFF

[Causes operating pack to regulate to high flow in flight with flaps up.]

ENGINE BLEED AIR SWITCH (affected side) . . . . . OFF

WING ANTI-ICE SWITCH . . . . . OFF

[Prevents possible asymmetrical ice buildup on the wings.]

Avoid icing conditions.

If the left WING-BODY OVERHEAT light remains illuminated:

APU BLEED AIR SWITCH (if APU running). . . . . OFF

[Stops the flow of bleed air from the APU to the left side pneumatic ducting.]

If the light remains illuminated:

APU SWITCH . . . . . OFF

Do not operate the APU.

If the light extinguishes:

ISOLATION VALVE SWITCH . . . . . AUTO

ENGINE No. 1 BLEED AIR SWITCH . . . . . ON

LEFT PACK SWITCH . . . . . AUTO

WING ANTI-ICE SWITCH . . . . . AS REQUIRED

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ZONE TEMP/DUCT OVERHEAT

Condition: **The (737-400) ZONE TEMP/(737-300/500) DUCT OVERHEAT light illuminated indicates air temperature in the related duct has exceeded limits or also (737-400) that control of the flight deck trim air modulating valve has been lost.**

TEMPERATURE SELECTOR . . . . . .COOLER TEMPERATURE

[Prevents the air mix valves from returning to an overheat condition.]

TRIP RESET SWITCH . . . . . PUSH

[The DUCT OVERHEAT light extinguishes if the duct temperature has cooled below limits.]

(737-400) If supply duct temperature increases rapidly:

TRIM AIR SWITCH . . . . . .OFF

(737-300/500) If duct temperature increases rapidly or the air mix valve indicator moves toward full hot:

TEMPERATURE SELECTOR . . . . . .MANUAL

Adjust the air mix valve position as required.

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ENGINE COWL ANTI-ICE

Condition: **An engine COWL ANTI-ICE light illuminated indicates an overtemperature or overpressure condition in the cowl anti-ice duct.**

Flight conditions permitting:

THRUST LEVER . . . . . RETARD

Reduce thrust until the COWL ANTI-ICE light extinguishes.

AUTOTHROTTLE (If engaged) . . . . . .DISENGAGE

[Prevents undesired autothrottle activity.]

ENGINE COWL VALVE OPEN

Condition: **An engine COWL VALVE OPEN light remaining illuminated bright blue in flight indicates the cowl anti-ice valve position disagrees with the engine anti-ice switch position.**

Valve Open:

If total air temperature is above 10°C, limit thrust on the affected engine to 80% N1 if possible.

Valve Closed:

Avoid icing conditions.

PITOT STATIC HEAT MALFUNCTIONS

Condition: **Pitot static light(s) illuminated indicate related pitot static malfunction(s).**

Avoid icing conditions.

Note: Flight in icing conditions may result in erroneous flight instrument indications.

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WINDOW OVERHEAT

Condition: **A window OVERHEAT light illuminated indicates an overheat condition has been detected.**

WINDOW HEAT SWITCH (affected window) . . . . . OFF

[Extinguishes the OVERHEAT light and resets the system.]

After 2-5 minutes:

WINDOW HEAT SWITCH . . . . . ON

If the window OVERHEAT light re-illuminates:

WINDOW HEAT SWITCH . . . . . OFF

Limit airspeed to 250 knots maximum below 10,000 feet.

WINDSHIELD AIR CONTROLS . . . . . PULL

[Vents conditioned air to the inside of the windshield for defogging.]

WING ANTI-ICE VALVE OPEN

Condition: **A WING ANTI-ICE L VALVE OPEN and/or R VALVE OPEN light remaining illuminated bright blue in flight indicates the related wing anti-ice valve position disagrees with the wing anti-ice switch position.**

Valve Open:

If total air temperature is above 10°C or no visible moisture:

ISOLATION VALVE SWITCH . . . . . CLOSE

PACK SWITCH (affected side) . . . . . OFF

[Causes operating pack to regulate to high flow in flight with flaps up.]

ENGINE BLEED AIR SWITCH (affected side) . . . . . OFF

Valve Closed:

Avoid icing conditions.

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AUTOTHROTTLE DISENGAGE NNC.4.1

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AUTOPILOT DISENGAGE

Condition: **The flashing red A/P light illuminated and the aural tone sounding indicates the autopilot has disengaged.**

Fly the airplane manually or re-engage an autopilot.

AUTOTHROTTLE DISENGAGE

Condition: **The flashing red A/T light illuminated indicates the autothrottle has disengaged.**

Control thrust manually or re-engage the autothrottle.

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ACARS MU FAIL OR CU FAIL NNC.5.1

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ACARS ELECTRICAL POWER LOSS

Condition: **Airplane AC power is lost.**

Note: The ACARS automatically reverts to VOX MODE (the DATA MODE is inoperative).

ACARS MU FAIL OR CU FAIL

Condition: **The ACARS system is inoperative.**

Use normal ARINC voice procedures for reporting.

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BUS OFF

Condition: **A BUS OFF light illuminated indicates the related generator bus is not powered.**

ENGINE GENERATOR SWITCH . . . . . ON

If BUS OFF light remains illuminated:

APU (if available) . . . . . START & ON BUS

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ELECTRICAL SMOKE/FUMES OR FIRE

Condition: **Electrical smoke/fumes/fire is identified.**

OXYGEN MASKS AND SMOKE GOGGLES (if required) . . . . . ON

CREW COMMUNICATIONS (if required) . . . . . ESTABLISH

RECIRCULATION FAN SWITCH(ES) . . . . . OFF

[Removes fan as a possible source of smoke/fumes. Stops recirculation of smoke/fumes and increases fresh air flow.]

If smoke/fumes/fire source is known:

ELECTRICAL POWER (affected equipment) . . . . . REMOVE

If practical, remove power from affected equipment by switch or circuit breaker in flight deck or cabin.

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If smoke/fumes/fire persists or source is unknown:

BUS TRANSFER SWITCH . . . . . OFF

[Prevents unwanted transfer of power.]

GALLEY POWER SWITCH . . . . . OFF

EQUIPMENT COOLING SUPPLY/EXHAUST SWITCHES . . . . . ALTERNATE

[Removes normal fans as a possible source of smoke/fumes.]

CABIN READING & GALLEY ATTENDANT WORK LIGHTS . . . . . ON

Instruct flight attendants to:
- **turn on cabin reading lights**
- **turn on galley attendants work lights.**

[Prepares cabin lighting prior to depowering cabin fluorescent lights.]

CABIN EQUIPMENT . . . . . OFF

Instruct flight attendants to:
- **turn off galley power switches**
- **turn off cabin fluorescent light switches.**

Plan to land at the nearest suitable airport.

Accomplish the SMOKE/FUMES REMOVAL checklist, if required.

GENERATOR DRIVE LOW OIL PRESSURE OR HIGH OIL TEMPERATURE

Condition: **Either light illuminated indicates a malfunction in the related generator drive.**

GENERATOR DRIVE DISCONNECT SWITCH . . . . . DISCONNECT

Hold in the DISCONNECT position momentarily.

[Prevents generator drive damage.]

APU (if available) . . . . . START & ON BUS

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LOSS OF BOTH ENGINE DRIVEN GENERATORS

Condition: **All TRANSFER BUS OFF, BUS OFF and GEN OFF BUS lights illuminated indicate the loss of both engine driven generators.**

Note: At high altitude, thrust deterioration or engine flameout may occur.

BUS TRANSFER SWITCH . . . . . OFF

[Prevents high electrical loads during attempts to restore power.]

ELECTRIC HYDRAULIC PUMP SWITCHES . . . . . OFF

[Prevents high electrical loads during attempts to restore power.]

ENGINE No. 2 GENERATOR SWITCH. . . . . ON

ENGINE No. 1 GENERATOR SWITCH. . . . . ON

If either or both BUS OFF lights remain illuminated:

Note: APU start attempts are not recommended above 25,000 feet.

APU (if available) . . . . . START & ON BUS

[With both buses off, only one start attempt is recommended. Multiple start attempts reduce standby power capacity.]

Note: If the APU is the only operating generator, connect it to the No. 2 bus, as it will power TR No. 3. If the APU cannot be connected to No. 2 bus, connect it to No. 1 bus.

If either or both BUS OFF lights extinguish:

BUS TRANSFER SWITCH . . . . . AUTO

[Restores power to the remaining transfer bus if one BUS OFF light remains illuminated.]

ELECTRIC HYDRAULIC PUMP SWITCHES . . . . . ON (ONE AT A TIME)

[Prevents high peak electrical loads.]

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If either or both BUS OFF lights extinguish: (continued)

If both primary attitude displays are inoperative:

IRS MODE SELECTOR SWITCHES . . . . . ATT

Maintain straight and level, constant airspeed flight until attitude displays recover (approximately 30 seconds).

MAGNETIC HEADING . . . . . ENTER

[Heading may be entered on the POS INIT page or on the overhead IRS display unit by selecting HDG/STS.]

If both BUS OFF lights remain illuminated:

ALL EXTERIOR LIGHTS . . . . . OFF

[Conserves battery power.]

Avoid icing conditions.

Note: Flight in icing conditions may result in erroneous flight instrument indications.

Plan to land at the nearest suitable airport. A fully charged battery will provide a minimum of 30 minutes of standby power.

STANDBY POWER OFF

Condition: **The STANDBY PWR OFF light illuminated indicates the AC standby bus is not powered.**

STANDBY POWER SWITCH . . . . . BAT

[A fully charged battery provides a minimum of 30 minutes of standby power.]

If either generator bus is re-established:

STANDBY POWER SWITCH . . . . . AUTO

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TRANSFER BUS OFF

Condition: **A TRANSFER BUS OFF light illuminated indicates the related transfer bus is not powered.**

BUS TRANSFER SWITCH . . . . . OFF THEN AUTO

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ABORTED ENGINE START

Condition: **One or more of the following conditions:**

- No N1 rotation before the engine start lever is raised to IDLE
- No oil pressure indication by the time the engine is stabilized at idle
- No increase in EGT, within 10 seconds on the ground or 30 seconds in flight, after the engine start lever is raised to IDLE
- No increase in, or a very slow increase in N1 or N2 after EGT indication
- EGT rapidly approaching or exceeding the start limit.

Before engine start lever raised to IDLE:

ENGINE START SWITCH . . . . . OFF

After engine start lever raised to IDLE:

Before starter cutout:

ENGINE START LEVER . . . . . CUTOFF

Continue to motor the engine for 60 seconds.

[Clears fuel and cools engine components.]

ENGINE START SWITCH . . . . . OFF

After starter cutout:

ENGINE START LEVER . . . . . CUTOFF

After N2 decreases to below 20%:

ENGINE START SWITCH . . . . . GRD

Motor the engine for 60 seconds.

[Clears fuel and cools engine components.]

ENGINE START SWITCH . . . . . OFF

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APU DET INOP

Condition: **The APU DET INOP light illuminated indicates the APU fire detection loop has failed.**

APU SWITCH . . . . . OFF

CAUTION: Do not operate the APU. An APU fire would not be detected and the APU would continue to run.

APU FAULT/LOW OIL PRESSURE

Condition: **The APU FAULT/LOW OIL PRESSURE light illuminated indicates an APU malfunction exists.**

Note: The APU shuts down automatically.

APU SWITCH . . . . . OFF

APU FIRE

Condition: **Fire is detected in the APU.**

APU FIRE WARNING SWITCH . . . . . PULL & ROTATE

Rotate to the stop and hold for one second. To manually unlock the APU fire warning switch, press the override and pull.

APU SWITCH . . . . . OFF

If the fire warning switch remains illuminated:

In flight:

Plan to land at the nearest suitable airport.

On the ground:

STANDBY POWER SWITCH . . . . . BAT

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APU OVERSPEED

Condition: **The APU OVERSPEED light illuminated indicates one of the following:**

- APU RPM limit has been exceeded resulting in an automatic shutdown
- the overspeed shutdown protection feature has failed a self-test during a normal APU shutdown
- an APU start has been manually aborted prior to the APU reaching normal operating speed.

APU SWITCH . . . . . OFF

Note: If the OVERSPEED light illuminated due to an RPM limit being exceeded, the APU OVERSPEED RESET switch in the E/E compartment must be reset before another APU start can be attempted.

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ENGINE FAILURE/SHUTDOWN

Condition: **Loss of thrust on an engine as indicated by the engine indicators, or abnormal engine indications.**

Accomplish an engine shutdown only when flight conditions permit.

AUTOTHROTTLE (if engaged) . . . . . .DISENGAGE

[Allows thrust lever to remain where manually positioned.]

THRUST LEVER . . . . . CLOSE

[Assists in recognition of affected engine.]

Conditions permitting, operate for three minutes at idle thrust.

ENGINE START LEVER . . . . . CUTOFF

APU (if available) . . . . . START & ON BUS

PACK SWITCH (affected side). . . . . .OFF

[Causes operating pack to regulate to high flow in flight with flaps up.]

FUEL . . . . . .BALANCE

TRANSPONDER MODE SELECTOR (TCAS equipped airplanes) . . . . . TA

[Prevents climb commands which can exceed single engine performance capability.]

If wing anti–ice is required:

ISOLATION VALVE SWITCH . . . . . AUTO

Plan to land at the nearest suitable airport.

Accomplish the ONE ENGINE INOPERATIVE LANDING checklist.

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ENGINE FIRE, SEVERE DAMAGE OR SEPARATION

Condition: **Fire is detected in the associated engine; severe damage which may be associated with airframe vibration and/or abnormal engine indications has occurred; or the engine has separated.**

AUTOTHROTTLE (if engaged) . . . . . .DISENGAGE

[Allows thrust lever to remain where manually positioned.]

THRUST LEVER . . . . . CLOSE

[Assists in recognition of the afftected engine.]

ENGINE START LEVER . . . . . CUTOFF

ENGINE FIRE WARNING SWITCH . . . . . .PULL

To manually unlock the engine fire warning switch, press the override and pull.

If the engine fire warning switch or ENG OVERHEAT light remains illuminated:

ENGINE FIRE WARNING SWITCH . . . . . .ROTATE L or R

Rotate to the stop and hold for one second.

If after 30 seconds the engine fire warning switch or ENG OVERHEAT light remains illuminated:

ENGINE FIRE WARNING SWITCH . . . . . ROTATE TO REMAINING BOTTLE

Rotate to the opposite stop and hold for one second.

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If high airframe vibration occurs and continues after engine is shut down:

Without delay, reduce airspeed and descend to a safe altitude which results in an acceptable vibration level. If high vibration returns and further airspeed reduction and descent are not practicable, increasing airspeed may reduce vibration.

ISOLATION VALVE SWITCH . . . . . CLOSE

PACK SWITCH (affected side) . . . . . OFF

[Causes operating pack to regulate to high flow in flight with flaps up.]

APU BLEED AIR SWITCH . . . . . OFF

APU (if available) . . . . . START & ON BUS

FUEL . . . . . BALANCE

TRANSPONDER MODE SELECTOR (TCAS equipped airplanes) . . . . . TA

[Prevents climb commands which can exceed single engine performance capability.]

If wing anti–ice is required:

ISOLATION VALVE SWITCH (after fire has been extinguished) . . . . . AUTO

Plan to land at the nearest suitable airport.

Accomplish the ONE ENGINE INOPERATIVE LANDING checklist when appropriate.

ENGINE FIRE/OVERHEAT DETECTOR FAULT

Condition: **The FAULT light illuminated indicates both loops on an engine have failed.**

The fire detection system in one or both engines is inoperative.

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ENGINE HIGH OIL TEMPERATURE

Condition: **Engine oil temperature is in the yellow band, or at or above the red radial.**

If temperature is at or above the red radial:

Accomplish the ENGINE FAILURE/SHUTDOWN checklist.

If temperature is in the yellow band:

AUTOTHROTTLE (if engaged) . . . . . .DISENGAGE

[Allows thrust lever to remain where manually positioned.]

THRUST LEVER (affected engine) . . . . . RETARD

Retard until engine oil temperature is within normal operating range or thrust lever is closed.

If temperature is in the yellow band for more than 15 minutes:

Accomplish the ENGINE FAILURE/SHUTDOWN checklist.

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ENGINE LIMIT/SURGE/STALL

Condition: **One or more of the following conditions:**

- Engine RPM or EGT indications are abnormal, approaching or exceeding limits
- No response to thrust lever movement
- Abnormal engine noises.

AUTOTHROTTLE (if engaged) . . . . . .DISENGAGE

[Allows thrust lever to remain where manually positioned.]

THRUST LEVER . . . . . RETARD

Retard until indications remain within appropriate limits or thrust lever is closed.

If indications are abnormal or EGT continues to increase:

ENGINE START LEVER . . . . . CUTOFF

APU (If available) . . . . . START & ON BUS

PACK SWITCH (affected side) . . . . . .OFF

[Causes operating pack to regulate to high flow in flight with flaps up.]

FUEL . . . . . .BALANCE

TRANSPONDER MODE SELECTOR
(TCAS equipped airplanes) . . . . . TA

[Prevents climb commands which can exceed single engine performance capability.]

If wing anti-ice is required:

ISOLATION VALVE SWITCH . . . . . AUTO

Plan to land at the nearest suitable airport.

Accomplish the ONE ENGINE INOPERATIVE LANDING checklist.

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If indications are stabilized and EGT decreases:

THRUST LEVER . . . . . .ADVANCE

Advance slowly. Check that RPM and EGT follow thrust lever movement.

Operate engine normally or at a reduced thrust setting which is surge and stall free.

ENGINE LOW OIL PRESSURE

Condition: **Engine oil pressure is in the yellow band with takeoff thrust set, LOW OIL PRESSURE light illuminated, or engine oil pressure is at or below the red radial.**

If engine oil pressure is in the yellow band with takeoff thrust set:

DO NOT TAKEOFF.

If engine oil pressure is at or below the red radial:

Accomplish the ENGINE FAILURE/SHUTDOWN checklist.

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ENGINE OIL FILTER BYPASS

Condition: **The engine OIL FILTER BYPASS light illuminated indicates an impending bypass of the scavenge oil filter.**

AUTOTHROTTLE (if engaged) . . . . . .DISENGAGE

[Allows thrust lever to remain where manually positioned.]

THRUST LEVER . . . . . RETARD

Retard until the OIL FILTER BYPASS light extinguishes or thrust lever is closed.

If the OIL FILTER BYPASS light extinguishes:

Operate the engine at reduced thrust to keep the light extinguished.

If the OIL FILTER BYPASS light remains illuminated:

Accomplish the ENGINE FAILURE/SHUTDOWN checklist.

ENGINE OVERHEAT

Condition: **The ENG OVERHEAT light illuminated indicates an overheat is detected on the related engine.**

AUTOTHROTTLE (if engaged) . . . . . .DISENGAGE

[Allows thrust lever to remain where manually positioned.]

THRUST LEVER . . . . . CLOSE

If the ENG OVERHEAT light remains illuminated:

Accomplish the ENGINE FIRE, SEVERE DAMAGE OR SEPARATION checklist.

If the ENG OVERHEAT light extinguishes:

Operate the engine at reduced thrust to keep the light extinguished.

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ENGINE TAILPIPE FIRE

Condition: **Tailpipe fire is reported with no engine fire warning.**

ENGINE START LEVER. . . . . CUTOFF

If bleed air is available:

PACK SWITCHES . . . . . OFF

[Allows maximum bleed air for engine motoring.]

ISOLATION VALVE SWITCH . . . . . AUTO

ENGINE No. 1 and No. 2 BLEED AIR SWITCHES . . . . . VERIFY ON

If APU is operating:

APU BLEED AIR SWITCH. . . . . ON

If engine start switch is OFF/CONT:

Allow N2 to decrease below 20%.

ENGINE START SWITCH. . . . . GRD

Motor engine until tailpipe fire is reported extinguished.

ENGINE START SWITCH. . . . . OFF

TOWER AND FLIGHT ATTENDANTS . . . . . NOTIFY

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HIGH ENGINE VIBRATION

Condition: **VIB levels are in excess of 4.0 units accompanied by airframe vibrations.**

If not in icing conditions:

AUTOTHROTTLE (if engaged) . . . . . .DISENGAGE

[Allows thrust lever to remain where manually positioned.]

THRUST LEVER . . . . . RETARD

Retard to maintain VIB below 4 units.

Note: If the VIB indication does not decrease when the thrust lever is retarded, check other engine indications. If other engine indications are normal, no further action is required.

If in icing conditions:

During descent or holding accomplish the following on one engine at a time at approximately 15 minute intervals:

ENGINE START SWITCH . . . . . FLT

THRUST . . . . . ADJUST

Adjust thrust to 45% N1. After approximately five seconds, advance thrust lever slowly to a minimum of 80% N1.

If vibration does not decrease:

Accomplish the procedure for "If not in icing conditions."

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INFLIGHT ENGINE START

Condition: **Engine start is desired after a shutdown with no fire or apparent damage.**

Complete the ENGINE FAILURE AND SHUTDOWN checklist before attempting an inflight engine start.

INFLIGHT START ENVELOPE . . . . . CHECK

CAUTION: Starter assist should be used if N2 is below 15%.

THRUST LEVER . . . . . CLOSE

ENGINE START LEVER . . . . . CUTOFF

If starter assist is required:

PACK SWITCH (affected side) . . . . . OFF

DUCT PRESSURE . . . . . MINIMUM 30 PSI

If required, advance the thrust lever to increase duct pressure.

IGNITION SELECT SWITCH . . . . . BOTH

ENGINE START SWITCH . . . . . GRD/FLT

ENGINE START LEVER . . . . . IDLE DETENT

Position engine start lever to IDLE detent at a minimum of 15% N2.

If no increase in EGT is observed within 30 seconds:

ENGINE START LEVER . . . . . CUTOFF

ENGINE START SWITCH . . . . . OFF

After engine start:

ELECTRICAL . . . . . GENERATOR ON

PACK SWITCH . . . . . AUTO

ENGINE START SWITCH . . . . . AS REQUIRED

APU . . . . . AS REQUIRED

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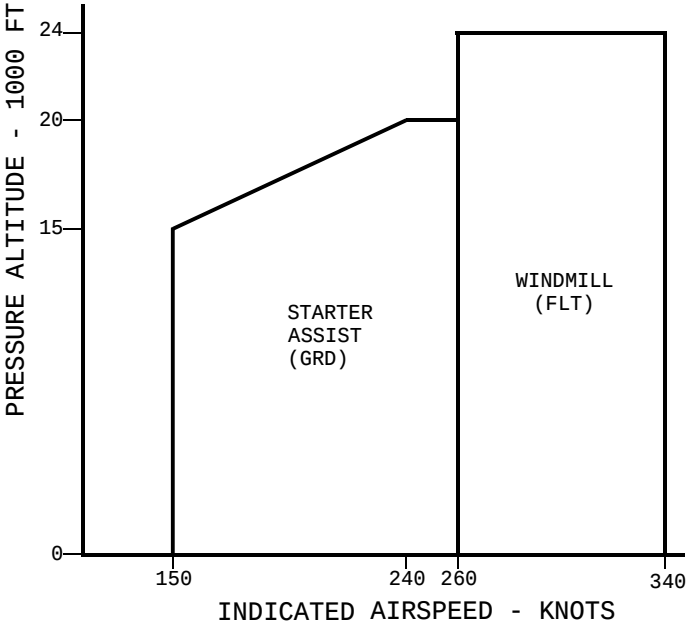
After engine start: (continued)

TRANSPONDER MODE SELECTOR
(TCAS equipped airplanes) . . . . . TA/RA

INFLIGHT START ENVELOPE



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LOSS OF THRUST ON BOTH ENGINES

Condition: **Loss of all thrust on both engines or loss of thrust lever response from both engines.**

ENGINE START SWITCHES . . . . . FLT

ENGINE START LEVERS . . . . . CUTOFF

EGT decreasing:

ENGINE START LEVERS . . . . . .IDLE DETENT

If EGT reaches 930°C, repeat the above steps.

Note: In moderate to heavy rain it may take up to 3 minutes to accelerate to idle.

APU (if available) . . . . . START & ON BUS

Do not wait for successful engine start(s) prior to starting APU.

[The APU has demonstrated the capability to provide electrical and pneumatic power up to 20,000 feet.]

If neither restart is successful and N2 is below 15%:

WING ANTI–ICE SWITCH . . . . . .OFF

PACK SWITCHES . . . . . .OFF

APU BLEED AIR SWITCH . . . . . .ON

IGNITION SELECT SWITCH . . . . . BOTH

EITHER ENGINE START SWITCH . . . . . GRD

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When engine parameters have stabilized:

APU BLEED AIR SWITCH . . . . . OFF

ENGINE START SWITCH . . . . . FLT

THRUST LEVER . . . . . ADVANCE

ENGINE GENERATOR SWITCH . . . . . ON

PACK SWITCH . . . . . AUTO

Accomplish the INFLIGHT ENGINE START checklist to start the other engine.

If neither IRS attitude display recovers after a generator bus is restored:

IRS MODE SELECTOR SWITCHES . . . . . ATT

Maintain straight and level, constant airspeed flight until attitude displays recover (approximately 30 seconds).

MAGNETIC HEADING . . . . . ENTER

APU . . . . . AS REQUIRED

LOW IDLE

Condition: **The LOW IDLE light illuminated indicates the related engine is below minimum required RPM.**

AUTOTHROTTLE (if engaged) . . . . . DISENGAGE

[Allows thrust lever to remain where manually positioned.]

THRUST LEVERS . . . . . ADVANCE THRUST LEVERS UNTIL LIGHT EXTINGUISHES

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ONE ENGINE INOPERATIVE LANDING

Condition: **Landing must be accomplished with one engine inoperative.**

Plan a flaps 15 landing.

Set VREF 15.

Maintain VREF 15 + 5 knots minimum on final approach. Apply normal wind correction but do not go below VREF 15 + 5 knots to assure adequate speed for go-around.

---------- DEFERRED ITEMS ----------

==> DESCENT – APPROACH

ANTI–ICE . . . . . AS REQUIRED

Use engine anti–ice on the operative engine only.

AIR CONDITIONING AND PRESSURIZATION . . . . . SET

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If additional go–around thrust is desired, below 10,000 feet configure the pressurization system for a no engine bleed landing:

ISOLATION VALVE SWITCH . . . . . CLOSE

ENGINE No. 1 BLEED AIR SWITCH . . . . . .OFF

APU BLEED AIR SWITCH . . . . . .ON

Do not open the APU bleed valve if the engine fire warning switch remains illuminated.

LEFT PACK SWITCH . . . . . AUTO

ENGINE No. 2 BLEED AIR SWITCH . . . . . .OFF

ALTIMETERS AND INSTRUMENTS . . . . . SET & X–CHECKED

N1 AND IAS BUGS . . . . . .CHECKED & SET, VREF 15

GROUND PROXIMITY FLAP INHIBIT SWITCH . . . . . .FLAP INHIBIT

GO–AROUND PROCEDURE . . . . . REVIEW

Accomplish normal go–around procedure except:

- **Use flaps 1**
- **Maintain VREF 15 + 5 knots to flap retraction altitude**
- **Limit bank angle to 15 degrees until reaching VREF 15 + 15 knots or minimum maneuver speed**
- **Accelerate to flaps 1 maneuvering speed prior to flap retraction.**

---------- DEFERRED ITEMS ----------

==> LANDING

ENGINE START SWITCH (operating engine) . . . . . .ON

RECALL. . . . . CHECKED

SPEED BRAKE . . . . . ARMED, GREEN LIGHT

LANDING GEAR . . . . . .DOWN, 3 GREEN

FLAPS . . . . . 15, GREEN LIGHT

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POWER MANAGEMENT CONTROL (PMC) INOPERATIVE

Condition: **The power management control INOP light illuminated indicates the PMC is inoperative or OFF.**

THRUST LEVERS (if required) . . . . . ADJUST

Flight may continue with the autothrottle ON.

In manual operation, adjust the thrust levers as required.

Observe engine limits.

REVERSER

Condition: **The REVERSER light illuminated indicates a fault is detected in the related engine reverser system.**

Note: Additional system failures may cause inflight deployment.

Expect normal reverser operation after landing.

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REVERSER UNLOCKED (IN FLIGHT)

Condition: **The REVERSER UNLOCKED light illuminated indicates that either of the two reverser sleeves has mechanically unlocked or that the REVERSER UNLOCKED light is giving a false indication.**

Note: Only multiple failures could allow the engine to go into reverse thrust.

Note: Unstowed reverser sleeves produce buffet, yaw, roll and increased airplane drag.

FORWARD THRUST LEVER (affected engine) . . . . . CHECK

CAUTION: Do not actuate the reverse thrust lever.

If the engine responds to forward thrust lever movement and no buffet or yaw exists:

Operate the engine normally.

If the engine does not respond to forward thrust lever movement or buffet or yaw exists:

Accomplish the ENGINE FAILURE/ SHUTDOWN checklist.

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START VALVE OPEN

Condition: **The START VALVE OPEN light illuminated indicates the start valve has opened or remains open after engine start.**

ENGINE START SWITCH . . . . . OFF

If START VALVE OPEN light remains illuminated:

ISOLATION VALVE SWITCH . . . . . CLOSE

PACK SWITCH (affected side) . . . . . OFF

[Causes operating pack to regulate to high flow in flight with flaps up.]

ENGINE BLEED AIR SWITCH (affected engine) . . . . . OFF

APU BLEED AIR SWITCH (engine No. 1 only) . . . . . OFF

If during ground operations:

GROUND AIR SOURCE (if in use) . . . . . DISCONNECT

ENGINE START LEVER . . . . . CUTOFF

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VOLCANIC ASH

Condition: **Static discharge around the windshield, bright glow in the engine inlets, smoke or dust on the flight deck or an acrid odor indicates the airplane is in volcanic ash.**

Exit volcanic ash as quickly as possible. Consider a 180 degree turn.

OXYGEN MASKS AND SMOKE GOGGLES (if required) . . . . . ON

CREW COMMUNICATIONS (if required) . . . . . ESTABLISH

AUTOTHROTTLE (if engaged) . . . . . DISENGAGE

[Allows thrust lever to remain where manually positioned.]

THRUST LEVERS . . . . . CLOSE

Conditions permitting, operate at idle thrust.

[Reduces possible engine damage and/or flameout by decreasing EGT.]

ENGINE START SWITCHES . . . . . FLT

PACK SWITCHES . . . . . HIGH

WING ANTI–ICE . . . . . ON

[Increases bleed air extraction to improve engine stall margin.]

ENGINE ANTI–ICE . . . . . ON

[Increases bleed air extraction to improve engine stall margin.]

APU (if available) . . . . . START

[Provides backup electrical and pneumatic source, if required.]

Note: Encountering volcanic ash can lead to abnormal systems reactions such as:

- **Engine malfunctions, increasing EGT, engine stall or flameout.**
- **Decrease or loss of airspeed indications.**
- **Equipment cooling OFF light.**

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If engines have flamed out or stalled, or EGT rapidly approaches or exceeds limit:

Accomplish the LOSS OF THRUST ON BOTH ENGINES checklist.

Note: Engines are very slow to accelerate to idle at high altitudes, which may be interpreted as a hung start or an engine malfunction.

Plan to land at the nearest suitable airport.

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AIR CONDITIONING SMOKE/FUMES

Condition: **A concentration of air conditioning smoke/fumes is identified.**

OXYGEN MASKS AND SMOKE GOGGLES (if required) . . . . . ON

CREW COMMUNICATIONS (if required) . . . . . ESTABLISH

RECIRCULATION FAN SWITCH(ES) . . . . . OFF

[Eliminates possible source of smoke/fumes.]

If smoke/fumes stop:

Continue flight with the recirculation fan switch(es) OFF.

If smoke/fumes continue:

ISOLATION VALVE SWITCH . . . . . CLOSE

R PACK SWITCH . . . . . OFF

If smoke/fumes stop:

Continue flight with the R PACK switch OFF and ISOLATION VALVE switch CLOSE.

If smoke/fumes continue:

R PACK SWITCH . . . . . AUTO

L PACK SWITCH . . . . . OFF

If smoke/fumes stop:

Continue flight with the L PACK switch OFF and ISOLATION VALVE switch CLOSE.

If smoke/fumes continue:

L PACK SWITCH . . . . . AUTO

Plan to land at nearest suitable airport.

Accomplish the SMOKE/FUMES REMOVAL checklist, if required.

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APU DET INOP

Condition: **The APU DET INOP light illuminated indicates the APU fire detection loop has failed.**

APU SWITCH . . . . . OFF

CAUTION: Do not operate the APU. An APU fire would not be detected and the APU would continue to run.

APU FIRE

Condition: **Fire is detected in the APU.**

APU FIRE WARNING SWITCH . . . . . PULL & ROTATE

Rotate to the stop and hold for one second. To manually unlock the APU fire warning switch, press the override and pull.

APU SWITCH . . . . . OFF

If the fire warning switch remains illuminated:

In flight:

Plan to land at the nearest suitable airport.

On the ground:

STANDBY POWER SWITCH . . . . . BAT

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CARGO FIRE

Condition: **Fire is detected in the related cargo compartment.**

CARGO FIRE ARM SWITCH (FWD/AFT) . . . . . ARMED

CARGO FIRE DISCHARGE SWITCH . . . . . PUSH

Push and hold for 1 second.

Note: DISCH light may require up to 30 seconds to illuminate.

RECIRCULATION FAN SWITCHES . . . . . OFF

ONE PACK SWITCH . . . . . OFF

Plan to land at the nearest suitable airport.

CARGO FIRE DETECTOR FAULT

Condition: **The DETECTOR FAULT light illuminated indicates both loops in one or both cargo compartments have failed.**

The fire detection system in one or both cargo compartments is inoperative.

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ELECTRICAL SMOKE/FUMES OR FIRE

Condition: **Electrical smoke/fumes/fire is identified.**

OXYGEN MASKS AND SMOKE GOGGLES (if required) . . . . . ON

CREW COMMUNICATIONS (if required) . . . . . ESTABLISH

RECIRCULATION FAN SWITCH(ES) . . . . . OFF

[Removes fan as a possible source of smoke/fumes. Stops recirculation of smoke/fumes and increases fresh air flow.]

If smoke/fumes/fire source is known:

ELECTRICAL POWER (affected equipment) . . . . . REMOVE

If practical, remove power from affected equipment by switch or circuit breaker in flight deck or cabin.

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If smoke/fumes/fire persists or source is unknown:

BUS TRANSFER SWITCH . . . . . OFF

[Prevents unwanted transfer of power.]

GALLEY POWER SWITCH . . . . . OFF

EQUIPMENT COOLING SUPPLY/EXHAUST SWITCHES . . . . . ALTERNATE

[Removes normal fans as a possible source of smoke/fumes.]

CABIN READING & GALLEY ATTENDANT WORK LIGHTS . . . . . ON

Instruct flight attendants to:
- **turn on cabin reading lights**
- **turn on galley attendants work lights.**

[Prepares cabin lighting prior to depowering cabin fluorescent lights.]

CABIN EQUIPMENT . . . . . OFF

Instruct flight attendants to:
- **turn off galley power switches**
- **turn off cabin fluorescent light switches.**

Plan to land at the nearest suitable airport.

Accomplish the SMOKE/FUMES REMOVAL checklist, if required.

ENGINE FIRE/OVERHEAT DETECTOR FAULT

Condition: **The FAULT light illuminated indicates both loops on an engine have failed.**

The fire detection system in one or both engines is inoperative.

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ENGINE FIRE, SEVERE DAMAGE OR SEPARATION

Condition: **Fire is detected in the associated engine; severe damage which may be associated with airframe vibration and/or abnormal engine indications has occurred; or the engine has separated.**

AUTOTHROTTLE (if engaged) . . . . . .DISENGAGE

[Allows thrust lever to remain where manually positioned.]

THRUST LEVER . . . . . CLOSE

[Assists in recognition of the afftected engine.]

ENGINE START LEVER . . . . . CUTOFF

ENGINE FIRE WARNING SWITCH . . . . . .PULL

To manually unlock the engine fire warning switch, press the override and pull.

If the engine fire warning switch or ENG OVERHEAT light remains illuminated:

ENGINE FIRE WARNING SWITCH . . . . . .ROTATE L or R

Rotate to the stop and hold for one second.

If after 30 seconds the engine fire warning switch or ENG OVERHEAT light remains illuminated:

ENGINE FIRE WARNING SWITCH . . . . . ROTATE TO REMAINING BOTTLE

Rotate to the opposite stop and hold for one second.

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If high airframe vibration occurs and continues after engine is shut down:

Without delay, reduce airspeed and descend to a safe altitude which results in an acceptable vibration level. If high vibration returns and further airspeed reduction and descent are not practicable, increasing airspeed may reduce vibration.

ISOLATION VALVE SWITCH . . . . . CLOSE

PACK SWITCH (affected side) . . . . . OFF

[Causes operating pack to regulate to high flow in flight with flaps up.]

APU BLEED AIR SWITCH . . . . . OFF

APU (if available) . . . . . START & ON BUS

FUEL . . . . . BALANCE

TRANSPONDER MODE SELECTOR (TCAS equipped airplanes) . . . . . TA

[Prevents climb commands which can exceed single engine performance capability.]

If wing anti–ice is required:

ISOLATION VALVE SWITCH (after fire has been extinguished) . . . . . AUTO

Plan to land at the nearest suitable airport.

Accomplish the ONE ENGINE INOPERATIVE LANDING checklist when appropriate.

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ENGINE OVERHEAT

Condition: **The ENG OVERHEAT light illuminated indicates an overheat is detected on the related engine.**

AUTOTHROTTLE (if engaged) . . . . . .DISENGAGE

[Allows thrust lever to remain where manually positioned.]

THRUST LEVER . . . . . CLOSE

If the ENG OVERHEAT light remains illuminated:

Accomplish the ENGINE FIRE, SEVERE DAMAGE OR SEPARATION checklist.

If the ENG OVERHEAT light extinguishes:

Operate the engine at reduced thrust to keep the light extinguished.

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ENGINE TAILPIPE FIRE

Condition: **Tailpipe fire is reported with no engine fire warning.**

ENGINE START LEVER. . . . . CUTOFF

If bleed air is available:

PACK SWITCHES . . . . . OFF

[Allows maximum bleed air for engine motoring.]

ISOLATION VALVE SWITCH . . . . . AUTO

ENGINE No. 1 and No. 2
BLEED AIR SWITCHES . . . . . VERIFY ON

If APU is operating:

APU BLEED AIR SWITCH. . . . . ON

If engine start switch is OFF/CONT:

Allow N2 to decrease below 20%.

ENGINE START SWITCH. . . . . GRD

Motor engine until tailpipe fire is reported extinguished.

ENGINE START SWITCH. . . . . OFF

TOWER AND FLIGHT ATTENDANTS . . . . . NOTIFY

LAVATORY SMOKE

Condition: **The SMOKE light illuminated indicates smoke is detected in one or more lavatories.**

Verify that lavatory fire is contained.

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SMOKE/FUMES REMOVAL

Condition: **Smoke/fumes removal is required.**

OXYGEN MASKS AND SMOKE GOGGLES (if required) . . . . . ON

CABIN DOOR . . . . . CLOSE

[Prevents smoke/fumes contamination of/from other compartments.]

CREW COMMUNICATIONS (if required) . . . . . ESTABLISH

If pack(s) are on and smoke/fumes source is confirmed to be on the flight deck or main cabin:

L AND R PACK SWITCHES . . . . . HIGH

RECIRCULATION FAN SWITCH(ES) . . . . . OFF

PRESSURIZATION MODE SELECTOR . . . . . STBY

CABIN ALTITUDE INDICATOR (maximum 10,000 feet) . . . . . INCREASE

[Increases the ventilation rate.]

CABIN RATE SELECTOR . . . . . MAXIMUM INCR

ENGINE No. 1 and No. 2 BLEED AIR SWITCHES . . . . . VERIFY ON

ENGINE THRUST . . . . . MAXIMUM PRACTICAL N1 (minimum 45%)

[Provides maximum cabin ventilation.]

FLIGHT DECK AIR CONDITIONING AND GASPER OUTLETS . . . . . OPEN

CAUTION: Do not open any flight deck window. Keep the cabin door closed.

If smoke/fumes are uncontrollable:

AIRPLANE ALTITUDE . . . . . LOWEST SAFE ALTITUDE or 10,000 FT, whichever is higher

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If smoke/fumes are uncontrollable: (continued)

At 14,000 feet or below:

PRESSURIZATION MODE SELECTOR . . . . . MAN

OUTFLOW VALVE SWITCH . . . . . OPEN

Position outflow valve to full open. This causes the cabin airflow to carry smoke/fumes aft.

If packs are off and smoke/fumes source is confirmed to be on the flight deck:

CAUTION: Window should not be opened unless the source is confirmed to be originating on the flight deck.

NORMAL HOLDING AIRSPEED . . . . . ESTABLISH

[High airspeed may prevent opening the window.]

FIRST OFFICER'S SLIDING WINDOW . . . . . OPEN

WHEEL WELL FIRE

Condition: **A fire is detected in the main wheel well.**

Observe extend limit speed (270K/.82M)

LANDING GEAR LEVER . . . . . .DOWN

Note: Do not use FMC fuel predictions with landing gear extended.

If the landing gear must be retracted for airplane performance, leave the landing gear extended for 20 minutes after the WHEEL WELL fire warning light has extinguished.

LANDING GEAR LEVER (if required)
(235 Kts maximum) . . . . . UP & OFF

Plan to land at the nearest suitable airport.

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ALL FLAPS UP LANDING

Condition: **Leading edge devices and trailing edge flaps cannot be extended.**

Burn off fuel to reduce touchdown speed.

Set VREF 40 + 55 knots.

[If tire placard speed is exceeded, tire failure may occur.]

Check the appropriate Non-Normal Configuration Landing Distance table in the ADVISORY INFORMATION section of the Performance-Inflight chapter.

Maintain flaps up maneuvering speed until on final.

Limit bank angle to 15° below flaps up maneuvering speed.

---------- DEFERRED ITEMS ----------

==> DESCENT - APPROACH

ANTI-ICE . . . . . AS REQUIRED

AIR CONDITIONING &
PRESSURIZATION . . . . . SET

ALTIMETERS &
INSTRUMENTS . . . . . SET & X-CHECKED

N1 & IAS BUGS . . . . . CHECKED & SET, VREF 40 + 55 KNOTS

GROUND PROXIMITY FLAP
INHIBIT SWITCH . . . . . FLAP INHIBIT

GO-AROUND PROCEDURE . . . . . REVIEW

Accomplish normal go-around except:
- **Limit bank angle to 15° below flaps up maneuvering speed.**
- **Accelerate to flaps up maneuvering speed.**

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---------- DEFERRED ITEMS ----------

==> LANDING

ENGINE START SWITCHES . . . . . ON

RECALL . . . . . CHECKED

SPEED BRAKE . . . . . ARMED, GREEN LIGHT

LANDING GEAR . . . . . DOWN, 3 GREEN

FLAPS . . . . . UP, NO LIGHTS

FASTEN BELTS SWITCH . . . . . ON

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ALTERNATE FLAPS OPERATION

Condition: **Flaps fail to extend or retract in response to flap lever selection and no asymmetry is indicated.**

Plan a flaps 15 landing.

Set VREF 15.

Note: Asymmetry protection is not provided when the alternate extension system is used.

Plan for alternate flap extension.

[Alternate extension time to flaps 15 is approximately 2 minutes. When the leading edge devices have been extended by the alternate system, they cannot be retracted unless hydraulic system B is operational and normal flap operation is restored. The drag penalty with the leading edge devices extended may make it impossible to reach an alternate field.]

---------- DEFERRED ITEMS ----------

==> DESCENT - APPROACH

ANTI-ICE . . . . . AS REQUIRED

AIR COND & PRESS . . . . . SET

ALTIMETERS & INSTRUMENTS . . . . . SET & X-CHECKED

N1 & IAS BUGS . . . . . .CHECKED & SET, VREF 15

GROUND PROXIMITY FLAP INHIBIT SWITCH. . . . . .FLAP INHIBIT

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---------- DEFERRED ITEMS ----------

==> LANDING

ALTERNATE FLAPS MASTER SWITCH (230 knots maximum) . . . . . ARM

Set flap lever to desired flap position and extend flaps on maneuvering speed schedule using the alternate flaps position switch.

Note: The LE FLAPS TRANSIT light will remain illuminated until the flaps approach the flaps 10 position.

All leading edge devices go to the FULL EXT position and cannot be retracted unless hydraulic system B is operational and normal flap operation is restored.

ENGINE START SWITCHES . . . . . ON

RECALL . . . . . CHECKED

SPEED BRAKE . . . . . ARMED, GREEN LIGHT

LANDING GEAR . . . . . DOWN, 3 GREEN

FLAPS . . . . . 15, GREEN LIGHT

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ASYMMETRICAL OR NO LEADING EDGE DEVICES

Condition: **The LE FLAPS TRANSIT light illuminated indicates asymmetrical or no leading edge devices.**

Plan a flaps 15 landing.

Set VREF 15 + 5 knots.

Limit bank angle to 15° below flaps up maneuvering speed

Note: Do not use FMC fuel predictions with flaps extended.

---------- DEFERRED ITEMS ----------

==> DESCENT - APPROACH

ANTI-ICE . . . . . AS REQUIRED

AIR CONDITIONING & PRESSURIZATION . . . . . SET

ALTIMETERS & INSTRUMENTS . . . . . SET & X-CHECKED

N1 & IAS BUGS . . . . . CHECKED & SET, VREF 15 + 5 KNOTS

GROUND PROXIMITY FLAP INHIBIT SWITCH . . . . . FLAP INHIBIT

---------- DEFERRED ITEMS ----------

==> LANDING

ENGINE START SWITCHES . . . . . ON

RECALL . . . . . CHECKED

SPEED BRAKE . . . . . ARMED, GREEN LIGHT

LANDING GEAR . . . . . DOWN, 3 GREEN

FLAPS . . . . . 15, AMBER LIGHT

[With asymmetrical or no leading edge devices extended, the amber LE FLAPS TRANSIT Light is illuminated. On speed tape equipped airplanes, operation within the lower yellow airspeed band is normal for this condition.]

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ASYMMETRICAL TRAILING EDGE FLAPS

Condition: **An uncommanded roll occurs when a new flap selection is made and a difference between the left and right flap indication is observed.**

FLAPS . . . . . SET

Move flap lever to the detent nearest the smallest actual flap position.

CAUTION: Do not attempt to move the trailing edge flaps with the alternate flaps switch as there is no asymmetry protection.

If flaps 15 or more:

Set VREF for smallest flap position.

If flaps between 1 and 15:

Set VREF 40 + 30 knots.

Check the appropriate Non-Normal Configuration Landing Distance table in the ADVISORY INFORMATION section of the Performance-Inflight chapter.

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If flaps less than 1:

Use TRAILING EDGE FLAPS UP LANDING checklist.

Note: Do not use FMC fuel predictions with flaps extended.

---------- DEFERRED ITEMS ----------

==> DESCENT - APPROACH

ANTI-ICE . . . . . AS REQUIRED

AIR CONDITIONING &
PRESSURIZATION . . . . . SET

ALTIMETERS &
INSTRUMENTS . . . . . SET & X-CHECKED

N1 & IAS BUGS . . . . . CHECKED & SET,
VREF AS REQUIRED

GROUND PROXIMITY FLAP
INHIBIT SWITCH . . . . . FLAP INHIBIT

---------- DEFERRED ITEMS ----------

==> LANDING

ENGINE START SWITCHES . . . . . ON

RECALL . . . . . CHECKED

SPEED BRAKE . . . . . ARMED, GREEN LIGHT

LANDING GEAR . . . . . DOWN, 3 GREEN

FLAPS . . . . . ____, GREEN LIGHT

AUTO SLAT FAIL

Condition: **The AUTO SLAT FAIL light illuminated indicates failure of the auto slat system.**

No crew action required in flight.

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FEEL DIFFERENTIAL PRESSURE

Condition: **The FEEL DIFF PRESS light illuminated indicates significant differential pressure in the elevator feel computer.**

No crew action required in flight.

FLAP LOAD RELIEF

Condition: **The FLAP LOAD RELIEF light illuminated indicates the flap load relief system has been activated and flaps are retracting from 40 to 30 due to excessive airspeed.**

Check flap position and maintain appropriate airspeed.

FLIGHT CONTROL LOW PRESSURE

Condition: **A FLT CONTROL LOW PRESSURE light illuminated indicates the related hydraulic system pressure to ailerons, elevators and rudder is low or the rudder pressure reducer has failed in the low pressure mode.**

FLIGHT CONTROL SWITCH . . . . . STBY RUD

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JAMMED OR RESTRICTED ELEVATOR OR AILERON

Condition: **Restricted movement of the elevator or aileron/spoiler is experienced.**

AUTOPILOT (if engaged) . . . . . .DISENGAGE

AUTOTHROTTLE (if engaged) . . . . . .DISENGAGE

Verify thrust is symmetrical.

JAMMED OR RESTRICTED SYSTEM . . . . . .OVERPOWER

Use maximum force, including a combined effort of both pilots, if required. A maximum two-pilot effort on the controls will not cause a cable or system failure.

Do not turn off any flight control switches.

If controls are normal:

Accomplish the normal DESCENT-APPROACH and LANDING checklists.

If controls are not normal:

If electric trim is desired, the Stabilizer Trim Override switch must be positioned to OVERRIDE.

Do not make abrupt thrust changes. Extend or retract speedbrake slowly and smoothly.

Limit bank angle to 15°.

Plan to land at the nearest suitable airport.

Plan a flaps 15 landing.

Set VREF 15 + 5.

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If controls are not normal: (continued)

---------- DEFERRED ITEMS ----------

==> **DESCENT - APPROACH**

ANTI-ICE . . . . . AS REQUIRED

AIR CONDITIONING & PRESSURIZATION . . . . . SET

ALTIMETERS & INSTRUMENTS . . . . . SET & X-CHECKED

N1 & IAS BUGS . . . . . CHECKED & SET, VREF 15 + 5 KNOTS

GROUND PROXIMITY FLAP INHIBIT SWITCH . . . . . FLAP INHIBIT

GO-AROUND PROCEDURE . . . . . REVIEW

Accomplish normal go-around procedure. Advance thrust to go-around smoothly and slowly to avoid excessive pitch-up.

---------- DEFERRED ITEMS ----------

==> **LANDING**

ENGINE START SWITCHES . . . . . ON

RECALL . . . . . CHECKED

SPEED BRAKE . . . . . ARMED, GREEN LIGHT

LANDING GEAR . . . . . DOWN, 3 GREEN

FLAPS . . . . . 15, GREEN LIGHT

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JAMMED STABILIZER

Condition: **The stabilizer fails to respond to electric trim inputs.**

MANUAL TRIM . . . . . .APPLY

Use force as needed to cause the disconnect clutch to disengage. Steady pressure on the manual trim handles is required until the desired trim is attained. A maximum two-pilot effort on the trim wheels will not cause a cable or system failure.

Note: Handle should be folded inside stabilizer trim wheel when not in use.

If the trim wheel moves when released:

STABILIZER TRIM AUTOPILOT CUTOUT SWITCH . . . . . CUTOUT

[Autopilot is not available.]

Maintain in-trim airspeed until start of approach.

Plan a flaps 15 landing.

Set VREF 15.

Establish landing configuration early.

Note: Anticipate higher than normal elevator forces during approach and landing.

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---------- DEFERRED ITEMS ----------

==> DESCENT - APPROACH

ANTI-ICE . . . . . AS REQUIRED

AIR CONDITIONING &
PRESSURIZATION . . . . . SET

ALTIMETERS &
INSTRUMENTS . . . . . SET & X-CHECKED

N1 & IAS BUGS . . . . . CHECKED & SET, VREF 15

GROUND PROXIMITY FLAP
INHIBIT SWITCH . . . . . FLAP INHIBIT

---------- DEFERRED ITEMS ----------

==> LANDING

ENGINE START SWITCHES . . . . . ON

RECALL . . . . . CHECKED

SPEED BRAKE . . . . . ARMED, GREEN LIGHT

LANDING GEAR . . . . . DOWN, 3 GREEN

FLAPS . . . . . 15, GREEN LIGHT

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LEADING EDGE FLAPS TRANSIT

Condition: **The LE FLAPS TRANSIT light remains illuminated.**

Note: Do not use FMC fuel predictions with flaps extended.

If the light remains illuminated with the trailing edge flaps extended:

Accomplish the ASYMMETRICAL OR NO LEADING EDGE DEVICES checklist.

If the light remains illuminated with the trailing edge flaps up:

AIRSPEED . . . . . 230 KNOTS MAXIMUM

If roll is encountered:

Accomplish the ASYMMETRICAL OR NO LEADING EDGE DEVICES checklist.

If no roll is encountered:

Recycle flaps to flaps 1 then UP.

If the LE FLAPS TRANSIT light extinguishes:

Proceed normally.

If the LE FLAPS TRANSIT light remains illuminated:

LE DEVICES ANNUNCIATOR PANEL . . . . . CHECK

If any light(s) for only one leading edge device is illuminated:

Limit airspeed to 300 knots (280 knots for turbulent air penetration) or .65 Mach, whichever is lower.

If any light(s) for more than one leading edge device is illuminated:

Limit airspeed to 230 knots.

Accomplish the ASYMMETRICAL OR NO LEADING EDGE DEVICES checklist for landing.

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MACH TRIM FAIL

Condition: **The MACH TRIM FAIL light illuminated indicates failure of the mach trim system.**

Limit airspeed to .74 Mach.

RUNAWAY STABILIZER

Condition: **Continuing rotation of the stabilizer trim wheel in a manner not appropriate for flight conditions.**

CONTROL COLUMN. . . . . HOLD FIRMLY

AUTOPILOT (if engaged). . . . . .DISENGAGE

Do not re-engage the autopilot.

Control airplane pitch attitude manually with control column and main electric trim as required.

If runaway continues:

STABILIZER TRIM CUTOUT SWITCHES . . . . . CUTOUT

If runaway continues:

STABILIZER TRIM WHEEL . . . . . .GRASP & HOLD

STABILIZER . . . . . .TRIM MANUALLY

Anticipate trim requirements.

Complete the normal DESCENT-APPROACH and LANDING checklists.

Establish proper airspeed and in-trim condition early on final approach.

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SPEED BRAKE

Condition: **The flashing SPEED BRAKE light illuminated indicates the speed brakes are up while in the landing configuration.**

SPEED BRAKE
LEVER . . . . . ARMED/DOWN DETENT (AS REQUIRED)

SPEED BRAKE DO NOT ARM

Condition: **The SPEED BRAKE DO NOT ARM light illuminated indicates a fault in the automatic speed brake system.**

Note: Speed brakes may be used in flight.

Complete the normal DESCENT-APPROACH and LANDING checklists except do not arm speed brakes for landing.

Manually deploy the speed brakes immediately upon touchdown.

SPEED TRIM FAIL

Condition: **The SPEED TRIM FAIL light illuminated indicates failure of the speed trim system.**

No crew action required in flight.

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STABILIZER OUT OF TRIM

Condition: **The STAB OUT OF TRIM light illuminated indicates the autopilot is not trimming the stabilizer properly.**

Note: Momentary illumination of the STAB OUT OF TRIM light during large changes in trim requirements is normal.

If the stabilizer is not trimming:

CONTROL COLUMN . . . . . HOLD FIRMLY

AUTOPILOT . . . . . .DISENGAGE

STABILIZER TRIM . . . . . AS REQUIRED

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TRAILING EDGE FLAPS UP LANDING

Condition: **The flap position indicator remains in the UP position, or less than 1, when flaps 1 is selected.**

Burn off fuel to reduce touchdown speed.

Set VREF 40 + 40 knots.

Check the appropriate Non-Normal Configuration Landing Distance table in the ADVISORY INFORMATION section of the Performance-Inflight chapter.

Maintain 210 knots until on final.

Limit bank angle to 15° below 210 knots.

---------- DEFERRED ITEMS ----------

==> DESCENT - APPROACH

ANTI-ICE . . . . . AS REQUIRED

AIR CONDITIONING &
PRESSURIZATION . . . . . SET

ALTIMETERS &
INSTRUMENTS . . . . . SET & X-CHECKED

N1 & IAS BUGS . . . . . CHECKED & SET, VREF 40 + 40 KNOTS

[A nuisance stick shaker may occur when slowing to VREF 40 + 40 at high gross weights and/or bank angles greater than 15°.]

GROUND PROXIMITY FLAP
INHIBIT SWITCH . . . . . FLAP INHIBIT

GO-AROUND PROCEDURE . . . . . REVIEW

Accomplish normal go-around except:
- **Limit bank angle to 15 ° below 210 knots.**
- **Accelerate to 210 knots.**
- **Do not exceed 230 knots with leading edge flaps extended.**

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---------- DEFERRED ITEMS ----------

==> LANDING

ALTERNATE FLAPS MASTER SWITCH . . . . . ARM

ALTERNATE FLAPS POSITION SWITCH (230 knots maximum) . . . . . .MOMENTARY DOWN

Verify leading edge devices annunciator indicates FULL EXT.

Note: The LE FLAPS TRANSIT light will remain illuminated.

ENGINE START SWITCHES. . . . . .ON

RECALL. . . . . CHECKED

SPEED BRAKE . . . . . ARMED, GREEN LIGHT

LANDING GEAR . . . . . .DOWN, 3 GREEN

FLAPS . . . . . .UP, AMBER LIGHT

[On speed tape equipped airplanes, operation within the lower yellow airspeed band is normal for this condition.]

FASTEN BELTS SWITCH . . . . . .ON

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UNCOMMANDED RUDDER

Condition: **Uncommanded rudder pedal displacement or pedal kicks.**

AUTOPILOT (if engaged) . . . . . DISENGAGE

Maintain control of the airplane with all available flight controls. If roll is uncontrollable, immediately reduce pitch/angle of attack and increase airspeed. Do not attempt to maintain altitude until control is recovered.

AUTOTHROTTLE (if engaged) . . . . . DISENGAGE

Verify thrust is symmetrical.

YAW DAMPER SWITCH . . . . . OFF

RUDDER TRIM . . . . . CENTER

RUDDER PEDALS . . . . . FREE & CENTER

Use maximum force including a combined effort of both pilots, if required, to free and center the rudder pedals.

If rudder pedal position or movement is not normal and the condition is the not the result of rudder trim:

SYSTEM B FLIGHT CONTROL SWITCH . . . . . STBY RUD

A slight rudder deflection may remain, but continued rudder pedal pressure may help maintain an in-trim conditon.

Sufficient directional control is available on landing using differential braking and nose wheel steering.

Crosswind capability may be reduced.

Do not use auto brakes.

Consider checking rudder freedom of movement at a safe altitude using slow rudder inputs while in the landing configuration and at approach airspeed.

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If condition was the result of rudder trim or environmental factors:

YAW DAMPER SWITCH . . . . . ON

Accomplish the normal DESCENT - APPROACH and LANDING checklists.

UNCOMMANDED YAW OR ROLL

Condition: **Uncommanded yaw or roll occurs in flight.**

AUTOPILOT (if engaged) . . . . . DISENGAGE

Maintain control of the airplane with all available flight controls. If roll is uncontrollable, immediately reduce pitch/angle of attack and increase airspeed. Do not attempt to maintain altitude until control is recovered.

AUTOTHROTTLE (if engaged) . . . . . DISENGAGE

Verify thrust is symmetrical.

YAW DAMPER SWITCH . . . . . OFF

If it is confirmed that the autopilot or autothrottle is not the cause of the uncommanded yaw or roll, the autopilot and autothrottle may be re-engaged at the pilot's discretion.

YAW DAMPER

Condition: **The YAW DAMPER light illuminated indicates the yaw damper is disengaged.**

YAW DAMPER SWITCH . . . . . OFF THEN ON

If light remains illuminated:

YAW DAMPER SWITCH . . . . . OFF

Do not exceed flaps 30 if crosswind exceeds 30 knots.

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AIRSPEED UNRELIABLE

Condition: **Pitch attitude not consistent with existing phase of flight, altitude, thrust, and weight, or noise and/or low frequency buffeting.**

Crosscheck ground speed and winds provided by the IRS and FMC to determine airspeed accuracy if indicated airspeed is questionable.

Note: Erroneous or unreliable airspeed indications may be caused by blocked or frozen pitot-static system(s), or a severely damaged or missing radome.

AIRPLANE ATTITUDE/THRUST . . . . . ADJUST

Maintain airplane control. Attitude and thrust information is provided in the Performance-Inflight section.

PITOT STATIC HEAT . . . . . CHECK ON

MACH/AIRSPEED INDICATORS . . . . . CROSS CHECK

FLIGHT RECORDER OFF

Condition: **The flight recorder OFF light illuminated indicates that the recorder is not operating.**

No crew action required.

SG FAIL

Condition: **The SG FAIL alert or blanking of both EADI and EHSI displays indicates a failure of the related symbol generator.**

EFI TRANSFER SWITCH . . . . . AS REQUIRED

Do not engage either autopilot.

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FMC/CDU ALERTING MESSAGE

Condition: **The FMC alert light and MSG light illuminated indicate an FMC/CDU alerting message exists.**

Take action as required by the message.

FMC FAIL - SINGLE FMC AIRPLANES

Condition: **The FMC alert light and FAIL lights illuminated indicate an FMC failure.**

Without an operating FMC, LNAV and VNAV are not available. AN/CDU LNAV operation is available (as installed.)

Note: When preparing for approach use the N1 manual set knobs to set N1 bugs.

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FMC FAIL - DUAL FMC AIRPLANES

Condition: **Illumination of the FMC alert light accompanied by:**

- loss of FMC information on both CDU's and loss of FMC data on the captains's navigation display map mode indicates left FMC failure
- illumination of the FMC message light, loss of FMC data on the first officer's navigation display map mode, and a SINGLE FMC OPERATION scratchpad message indicates right FMC failure
- loss of FMC information on the CDU and loss of FMC data on both navigation display map modes indicates dual FMC failure.

If only the left or right FMC has failed:

FMC SOURCE
SELECT SWITCH . . . . . .BOTH ON L or BOTH ON R

Select operating FMC.

If DUAL FMC OP RESTORED message appears:

FMC SOURCE SELECT SWITCH . . . . . .NORMAL

If dual FMC failure has occurred:

LNAV and VNAV are not available. Resume conventional navigation. AN/CDU LNAV operation is available (as installed.)

Note: When preparing for approach, use manual N1 set knobs to set the N1 bugs.

INSTRUMENT SWITCH

Condition: **The INSTR SWITCH light illuminated indicates either the electronic flight instrument or IRS transfer switch has been moved from the NORMAL position.**

No crew action required in flight.

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IRS DC FAIL

Condition: **An IRS DC FAIL light illuminated indicates the related IRS DC power has failed.**

If all other IRS lights are extinguished, operate normally.

Note: With both IRS DC FAIL lights illuminated, the switched hot battery bus is not powered or the battery is nearly discharged.

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IRS FAULT

Condition: **An IRS FAULT light illuminated indicates the related IRS system has detected a fault. On the ground, the IRS FAULT light accompanied by an ALIGN light may indicate the entered present position is incorrect.**

On the ground:

If the ALIGN light is illuminated:

IRS MODE SELECTOR . . . . . OFF

[The FAULT light extinguishes immediately and the ALIGN light extinguishes after approximately 30 seconds.]

After the ALIGN light extinguishes:

IRS MODE SELECTOR. . . . . NAV

PRESENT POSITION . . . . . ENTER

If the ALIGN light illuminates again, re-enter present position.

If the FAULT light illuminates again, notify maintenance.

In flight:

Note: The IRS ATT and/or NAV mode(s) may be inoperative.

IRS MODE SELECTOR (affected IRS) . . . . . ATT

Maintain straight and level, constant airspeed flight until attitude displays recover (approximately 30 seconds.)

If the FAULT light extinguishes:

MAGNETIC HEADING . . . . . ENTER

Enter updated heading periodically on the POS INIT page or on the overhead IRS display unit by selecting HDG/STS.

Do not use autopilot approach mode.

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In flight: (continued)

If the FAULT light remains illuminated:

IRS TRANSFER
SWITCH . . . . . .BOTH ON L or BOTH ON R

Note: Do not engage either autopilot.

IRS ON DC

Condition: **An IRS ON DC light illuminated indicates the related IRS is operating from the switched hot battery bus.**

Power to the right IRS is removed after 5 minutes.

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CROSSFEED SELECTOR INOPERATIVE

Condition: **The crossfeed VALVE OPEN light remaining illuminated bright blue indicates the crossfeed valve position disagrees with the crossfeed selector position.**

Valve Closed:

Flight conditions permitting, vary thrust to maintain fuel balance. If unable to maintain acceptable balance, land as soon as possible.

Valve Open:

Maintain fuel balance with selective use of fuel pumps.

FUEL FILTER BYPASS

Condition: **A fuel FILTER BYPASS light illuminated indicates impending fuel filter bypass due to a contaminated filter.**

Note: Erratic engine operation and flameout may occur due to fuel contamination.

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FUEL PUMP LOW PRESSURE

Condition: **A fuel pump LOW PRESSURE light illuminated indicates the related fuel pump output pressure is low.**

Note: Fuel pump LOW PRESSURE lights may flicker when tank quantity is low and the airplane is in turbulent air or during climb or descent.

If one main tank fuel pump LOW PRESSURE light is illuminated:

MAIN TANK FUEL PUMP SWITCH . . . . . OFF

[Sufficient fuel pressure is available for normal operation.]

If both main tank fuel pump LOW PRESSURE lights are illuminated:

Note: At high altitude, thrust deterioration or engine flameout may occur.

If one center tank fuel pump LOW PRESSURE light is illuminated:

CROSSFEED SELECTOR . . . . . OPEN

[Prevents fuel imbalance.]

CENTER TANK FUEL PUMP SWITCH (affected side) . . . . . OFF

When the other center tank fuel pump LOW PRESSURE light illuminates:

CROSSFEED SELECTOR . . . . . CLOSE

REMAINING CENTER TANK FUEL PUMP SWITCH . . . . . OFF

If both center tank fuel pump LOW PRESSURE lights are illuminated:

BOTH CENTER TANK FUEL PUMP SWITCHES . . . . . OFF

Center tank fuel is unusable. Main tank fuel may not be sufficient for the planned flight.

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FUEL QUANTITY INDICATOR INOPERATIVE

Condition: **The related fuel quantity indicator is blank, displays ERRxx, zeros or dashes.**

(U5 FMC's)

If a fuel quantity indicator fails:

VNAV . . . . . DISENGAGE

[FMC gross weight calculations are not provided with an inoperative fuel quantity indicator.]

If fuel quantity in FMC is not correct:

Do not use FMC speed and altitude information.

Use manually calculated gross weight and performance information from the QRH.

Do not use minimum maneuver speed or buffet margin information.

Use the Flap Maneuver Speed table in the Performance- Inflight section.

Stick shaker, VMO/MMO and minimum flap retraction speed displays are unaffected.

CAUTION: If the FMC is used for manual entry of gross weight to calculate reference speeds or fuel remaining at destination, the information will not remain accurate.

(U7.5 FMC's)

If dashes are displayed on the FUEL quantity line on the FMC PERF INIT page:

FMC FUEL WEIGHT . . . . . ENTER

Enter and periodically update manually calculated fuel weight on the FMC PERF INIT page.

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If dashes are not displayed on the FUEL quantity line on the FMC PERF INIT page:

Do not use FMC speed and altitude information.

Use manually calculated gross weight and performance information from the QRH.

Do not use minimum maneuver speed or buffet margin information.

Use the Flap Maneuver Speed table in the Performance-Inflight section.

Stick shaker, VMO/MMO and minimum flap retraction speed displays are unaffected.

CAUTION: If the FMC is used for manual entry of gross weight to calculate reference speeds or fuel remaining at destination, the information will not remain accurate.

FUEL TEMP LOW

Condition: **Fuel temperature is approaching minimum.**

When fuel temperature is approaching fuel temperature limit (3 degrees C (5 degrees F) above the fuel freeze point or - 45 degrees C (- 49 degrees F) whichever is higher):

Increase speed, change altitude and/or deviate to a warmer air mass to achieve a TAT equal to or higher than the fuel temperature limit.

TAT will increase approximately 0.5 to 0.7 degrees C for each .01 Mach increase in speed. In extreme conditions, it may be necessary to descend as low as FL250.

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INADVERTENT TRANSFER OF FUEL INTO CENTER TANK

Condition: **An inadvertent increase in quantity in the center tank and a decrease in quantity in either main tank can result in excessive main tank imbalance.**

FUEL PUMP SWITCHES (decreasing main tank) . . . . . OFF

Note: At high altitude, thrust deterioration or engine flameout may occur.

CENTER TANK FUEL PUMP SWITCH (affected side) . . . . . ON

MAIN TANK FUEL PUMP SWITCHES . . . . . ON

When the center tank fuel pump LOW PRESSURE light illuminates:

CENTER TANK FUEL PUMP SWITCH (affected side) . . . . . OFF

Repeat the above steps as necessary.

MINIMUM FUEL OPERATION

Condition: **A minimum fuel condition exists when the quantity in any main tank is 1000 pounds (453 kilograms) or less.**

MAIN TANK FUEL PUMP SWITCHES . . . . . ALL ON

CROSSFEED SELECTOR . . . . . OPEN

[Ensures remaining fuel available to both engines.]

Apply thrust changes slowly and smoothly. If a climb is required, maintain the minimum pitch attitude required for safe flight.

[Minimizes the possibility of uncovering the fuel pumps.]

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HYDRAULIC PUMP LOW PRESSURE

Condition: **A hydraulic pump LOW PRESSURE light illuminated indicates output pressure of the related pump is low.**

HYDRAULIC PUMP SWITCH . . . . . OFF

[Intermittent illumination of the hydraulic pump LOW PRESSURE light may be the result of single electric pump operation and a high demand on the hydraulic system.]

HYDRAULIC PUMP OVERHEAT

Condition: **A hydraulic pump OVERHEAT light illuminated indicates a fluid or motor overheat in the related electric motor-driven pump.**

ELECTRIC HYDRAULIC PUMP SWITCH . . . . . OFF

[Intermittent illumination of the hydraulic pump LOW PRESSURE light may be the result of single electric pump operation and a high demand on the hydraulic system.]

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LOSS OF SYSTEM A

Condition: **Both system A hydraulic pump LOW PRESSURE lights illuminated accompanied by low system A pressure indicates loss of system A. Other indications include:**

- system A flight control LOW PRESSURE light illuminated
- FEEL DIFF PRESS light illuminated when flaps are up.

SYSTEM A FLIGHT CONTROL SWITCH . . . . . STBY RUD

SYSTEM A HYDRAULIC PUMP SWITCHES . . . . . OFF

Note: Inoperative items:
- **ground spoilers**
- **inboard flight spoilers**
- **autopilot A**
- **normal nose wheel steering**
- **alternate brakes.**

Note: Engine No. 1 thrust reverser has standby pressure.

Plan for manual gear extension.

[When the gear has been lowered manually, it cannot be retracted. The drag penalty with gear extended may make it impossible to reach an alternate field.]

NOSE WHEEL STEERING SWITCH . . . . . ALT

---------- DEFERRED ITEMS ----------

==> DESCENT - APPROACH

ANTI-ICE . . . . . AS REQUIRED

AIR CONDITIONING & PRESSURIZATION . . . . . SET

ALTIMETERS & INSTRUMENTS . . . . . SET & X-CHECKED

N1 & IAS BUGS . . . . . CHECKED & SET

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---------- DEFERRED ITEMS ----------

==> LANDING

LANDING GEAR LEVER . . . . . OFF

MANUAL GEAR EXTENSION HANDLES . . . . . PULL

[The uplock is released when the handle is pulled to its limit. The related red landing gear indicator light illuminates, indicating uplock release.]

Wait 15 seconds after the last manual gear extension handle is pulled:

LANDING GEAR LEVER . . . . . DOWN

ENGINE START SWITCHES . . . . . ON

RECALL . . . . . CHECKED

SPEED BRAKE . . . . . ARMED, GREEN LIGHT

LANDING GEAR . . . . . DOWN, 3 GREEN

FLAPS . . . . . _____, GREEN LIGHT

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LOSS OF SYSTEM B

Condition: **Both system B hydraulic pump LOW PRESSURE lights illuminated accompanied by low system B pressure indicates loss of system B. Other indications include:**

- system B flight control LOW PRESSURE light illuminated
- FEEL DIFF PRESS light illuminated when flaps are up.

SYSTEM B FLIGHT CONTROL SWITCH . . . . . STBY RUD

SYSTEM B HYDRAULIC PUMP SWITCHES . . . . . OFF

Note: Inoperative items:
- **outboard flight spoilers**
- **autopilot B**
- **yaw damper**
- **alternate nose wheel steering**
- **normal brakes.**

Note: Alternate brakes are available. Engine No. 2 thrust reverser and leading edge flaps and slats have standby pressure. Trailing edge flaps have alternate electrical power.

Plan for flaps 15 landing

Set VREF 15

Plan for alternate flap extension.

[Alternate extension time to flaps 15 is approximately 2 minutes. When the leading edge devices have been extended by the alternate system, they cannot be retracted unless hydraulic system B is operational and normal flap operation is restored. The drag penalty with the leading edge devices extended may make it impossible to reach an alternate field.]

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---------- DEFERRED ITEMS ----------

==> DESCENT - APPROACH

AUTO BRAKE . . . . . OFF

ANTI-ICE . . . . . AS REQUIRED

AIR CONDITIONING &
PRESSURIZATION . . . . . SET

ALTIMETERS &
INSTRUMENTS . . . . . SET & X-CHECKED

N1 & IAS BUGS . . . . . CHECKED & SET, VREF 15

GROUND PROXIMITY FLAP
INHIBIT SWITCH . . . . . FLAP INHIBIT

ALTERNATE FLAPS MASTER
SWITCH (230 knots maximum) . . . . . ARM

Set flap lever to desired flap position and extend flaps on maneuvering speed schedule using the alternate flaps position switch.

Note: The LE FLAPS TRANSIT light will remain illuminated until the flaps approach the flaps 10 position.

---------- DEFERRED ITEMS ----------

==> LANDING

ENGINE START SWITCHES . . . . . ON

RECALL . . . . . CHECKED

SPEED BRAKE . . . . . ARMED, GREEN LIGHT

LANDING GEAR . . . . . DOWN, 3 GREEN

FLAPS . . . . . 15, GREEN LIGHT

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MANUAL REVERSION

Condition: **Both system A & B hydraulic pump LOW PRESSURE lights illuminated accompanied by low system A & B pressure indicates loss of both system A & B. Other indications include:**

- system A & B flight control LOW PRESSURE lights illuminated.

SYSTEM A AND B
FLIGHT CONTROL SWITCHES . . . . . STBY RUD

SYSTEM A and B
HYDRAULIC PUMP SWITCHES . . . . . OFF

Note: Inoperative items:
- **ground spoilers**
- **all flight spoilers**
- **nose wheel steering**
- **autopilot A & B**
- **yaw damper**

Note: Thrust reversers and leading edge flaps and slats have standby pressure. Trailing edge flaps have alternate electrical power. Inboard and outboard brakes have accumulator pressure only.

Plan for flaps 15 landing

Set VREF 15

Plan for manual gear and alternate flap extension.

[When the gear has been lowered manually, it cannot be retracted. Alternate extension time to flaps 15 is approximately 2 minutes. When the leading edge devices are extended by the alternate system, they cannot be retracted unless hydraulic system B is operational and normal flap operation is restored. The drag penalty with gear and/or leading edge devices extended may make it impossible to reach an alternate field.]

Note: The crosswind capability of the airplane is greatly reduced.

On touchdown, apply steady brake pressure without modulating the brakes.

Do not attempt to taxi the airplane after stopping.

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---------- DEFERRED ITEMS ----------

==> DESCENT - APPROACH

AUTO BRAKE . . . . . OFF

ANTI-ICE . . . . . AS REQUIRED

AIR CONDITIONING &
PRESSURIZATION . . . . . SET

ALTIMETERS &
INSTRUMENTS . . . . . SET & X-CHECKED

N1 & IAS BUGS . . . . . CHECKED & SET, VREF 15

GROUND PROXIMITY FLAP
INHIBIT SWITCH . . . . . FLAP INHIBIT

GO-AROUND PROCEDURE . . . . . REVIEW

Accomplish normal go-around procedure except:
- **Advance thrust to go-around smoothly and slowly to avoid excessive pitch-up**
- **Prepare to trim**
- **Limit bank angle to 15 degrees until 150 knots (or speed tape minimum maneuver speed)**

ALTERNATE FLAPS MASTER
SWITCH (230 knots maximum) . . . . . ARM

Set flap lever to desired flap position and extend flaps on maneuvering speed schedule using the alternate flaps position switch.

Note: The LE FLAPS TRANSIT light will remain illuminated until the flaps approach the flaps 10 position.

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---------- DEFERRED ITEMS ----------

==> LANDING

LANDING GEAR LEVER . . . . . OFF

MANUAL GEAR EXTENSION HANDLES . . . . . PULL

[The uplock is released when the handle is pulled to its limit. The related red landing gear indicator light illuminates, indicating uplock release.]

Wait 15 seconds after the last manual gear extension handle is pulled:

LANDING GEAR LEVER . . . . . DOWN

ENGINE START SWITCHES. . . . . ON

RECALL . . . . . CHECKED

LANDING GEAR . . . . . DOWN, 3 GREEN

FLAPS . . . . . 15, GREEN LIGHT

STANDBY HYDRAULIC LOW PRESSURE

Condition: **The standby hydraulic LOW PRESSURE light illuminated indicates output pressure of the standby pump is low.**

Note: With a loss of hydraulic system A and B, the rudder is inoperative.

STANDBY HYDRAULIC LOW QUANTITY

Condition: **The standby hydraulic LOW QUANTITY light illuminated indicates low quantity in the standby hydraulic reservoir.**

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ANTISKID INOPERATIVE

Condition: **The ANTISKID INOP light illuminated indicates a system fault is detected by the antiskid system.**

AUTO BRAKE SELECT SWITCH . . . . . OFF

Brake with caution.

AUTO BRAKE DISARM

Condition: **The AUTO BRAKE DISARM light illuminated indicates the auto brake has disconnected after being set.**

On ground:

AUTO BRAKE SELECT SWITCH . . . . . OFF

If the AUTO BRAKE DISARM light remains illuminated:

DO NOT TAKEOFF.

Inflight:

AUTO BRAKE SELECT SWITCH . . . . . OFF THEN RESELECT

If the AUTO BRAKE DISARM Light re-illuminates:

AUTO BRAKE SELECT SWITCH . . . . . OFF

Use manual brakes.

BRAKE PRESSURE INDICATOR ZERO PSI

Condition: **The brake accumulator has no nitrogen precharge.**

Accumulator braking is not available.

Note: If hydraulic system indications are normal, brake operation is unaffected.

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GEAR LEVER WILL NOT MOVE UP AFTER TAKEOFF

Condition: **The landing gear lever cannot be placed to the UP position in the normal manner due to one or more of the following:**

- failure of the landing gear lever lock solenoid
- failure of the air/ground system

Note: Do not use FMC fuel predictions.

LANDING GEAR LEVER . . . . . DOWN

If the takeoff configuration warning remains silent after the flaps are fully retracted and the thrust levers are beyond the vertical position:

Note: This condition indicates a failure of the landing gear lever lock solenoid.

LANDING GEAR OVERRIDE TRIGGER . . . . . PULL

LANDING GEAR LEVER . . . . . UP & OFF

If the takeoff configuration warning sounds when flaps are fully retracted:

Note: This condition indicates a failure of the air/ground system.

LANDING GEAR AIR/GND RELAY AND LIGHTS C/B (P6) . . . . . PULL

Plan to land at the nearest suitable airport.

CAUTION: Do not operate the speed brakes in flight.

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If the takeoff configuration warning sounds when flaps are fully retracted: (continued)

---------- DEFERRED ITEMS ----------

==> DESCENT – APPROACH

ANTI–ICE . . . . . AS REQUIRED

AIR CONDITIONING & PRESSURIZATION . . . . . SET

ALTIMETER & INSTRUMENTS . . . . . SET & X-CHECKED

N1 & IAS BUGS . . . . . CHECKED & SET, VREF

---------- DEFERRED ITEMS ----------

==> LANDING

ENGINE START SWITCHES . . . . . ON

RECALL . . . . . CHECKED

SPEED BRAKE . . . . . DOWN DETENT

LANDING GEAR LEVER . . . . . DOWN

LANDING GEAR AIR/GND RELAY AND LIGHTS C/B (P6) . . . . . RESET

LANDING GEAR . . . . . DOWN, 3 GREEN

LANDING GEAR AIR/GND RELAY AND LIGHTS C/B (P6) . . . . . PULL

FLAPS . . . . . _____, GREEN LIGHT

Manually deploy the speed brakes immediately upon touchdown.

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LANDING GEAR LEVER JAMMED IN THE UP POSITION

Condition: **The landing gear lever will not move from the UP position, or all landing gear remain up after attempting manual gear extension.**

---------- DEFERRED ITEMS ----------

==> DESCENT – APPROACH

TOTAL FUEL QUANTITY . . . . . CHECK

The following checklist should not be accomplished with less than 3,800 kgs of total fuel quantity available. Land with gear up or land with any available landing gear if less than 3,800 kgs total fuel quantity is available.

SYSTEM A FLIGHT CONTROL SWITCH. . . . . STBY RUD

SYSTEM A HYDRAULIC PUMP SWITCHES . . . . . OFF

Cycle the speed brake lever until system A hydraulic pressure is below 500 psi.

CAUTION: Do not repressurize hydraulic system A in flight or on the ground and do not use alternate nose wheel steering because the landing gear may retract.

Note: Inoperative items:

- **ground spoilers**
- **inboard flight spoilers**
- **autopilot A**
- **normal nose wheel steering**
- **alternate brakes.**

Note: Engine No. 1 thrust reverser has standby pressure.

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MANUAL GEAR EXTENSION HANDLES . . . . . PULL

Continue to cycle the speed brakes to relieve the hydraulic pressure generated as the landing gear extends. Up to 4 minutes may be required until each gear is extended.

Note: With the landing gear lever UP or OFF, the red gear indicator lights will be illuminated.

If all landing gear are not verified down and locked:

Accomplish the PARTIAL OR GEAR UP LANDING checklist.

Plan for a flaps 15 landing.

Set VREF 15 + 10 knots.

[Provides tail clearance during landing since hydraulic system A flight spoilers will be unlocked and may float.]

ANTI–ICE . . . . . AS REQUIRED

AIR CONDITIONING & PRESSURIZATION . . . . . SET

ALTIMETERS & INSTRUMENTS . . . . . SET & X-CHECKED

N1 AND IAS BUGS . . . . . CHECKED & SET
VREF 15 + 10

GROUND PROXIMITY FLAP INHIBIT SWITCH . . . . . FLAP INHIBIT

GROUND PROXIMITY GEAR INHIBIT SWITCH . . . . . GEAR INHIBIT

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---------- DEFERRED ITEMS ----------

==> LANDING

ENGINE START SWITCHES . . . . . ON

RECALL . . . . . CHECKED

SPEED BRAKE . . . . . ARMED, GREEN LIGHT

LANDING GEAR . . . . . DOWN, 3 GREEN

FLAPS . . . . . 15, GREEN LIGHT

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MANUAL GEAR EXTENSION

Condition: **All landing gear do not indicate down and locked when the landing gear lever is placed in the DOWN position.**

LANDING GEAR LEVER . . . . . OFF

MANUAL GEAR EXTENSION HANDLES . . . . . PULL

[The uplock is released when the handle is pulled to its limit. The related red landing gear indicator light illuminates, indicating uplock release.]

Wait 15 seconds after the last MANUAL GEAR EXTENSION HANDLE is pulled:

LANDING GEAR LEVER . . . . . DOWN

If all landing gear indicate down and locked:

Land normally.

If a green landing gear indicator light still fails to illuminate:

WHEEL WELL LIGHT SWITCH. . . . . ON

GEAR DOWN LOCK VISUAL INDICATOR(S) . . . . . CHECK

[Verify main landing gear mechanical down lock indicators are aligned and nose landing gear arrow heads are in contact.]

If the gear down lock visual indicator(s) verify gear down and locked:

Land normally.

If all landing gear are not verified down and locked:

Accomplish the PARTIAL OR GEAR UP LANDING checklist.

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PARTIAL OR GEAR UP LANDING

Condition: **All landing gear do not indicate down and locked after attempting manual gear extension.**

Brief crew and passengers on emergency landing and evacuation procedures.

Burn off fuel to reduce touchdown speed.

Plan a flaps 40 landing.

Set VREF 40.

---------- DEFERRED ITEMS ----------

==> DESCENT – APPROACH

ANTI–ICE . . . . . AS REQUIRED

AIR CONDITIONING &
PRESSURIZATION . . . . . SET

ALTIMETERS &
INSTRUMENTS . . . . . SET & X-CHECKED

N1 & IAS BUGS . . . . . .CHECKED & SET, VREF 40

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---------- DEFERRED ITEMS ----------

==> Prior to approach:

AURAL WARN C/B (P6) . . . . . PULL

[Prevents warning horn with gear retracted and landing flaps selected.]

AUTO SPEED BRAKE C/B (P6) . . . . . PULL

GROUND PROXIMITY GEAR INHIBIT SWITCH . . . . . GEAR INHIBIT

STANDBY POWER SWITCH . . . . . BAT

AUTO BRAKE SELECT SWITCH . . . . . OFF

AVAILABLE GEAR . . . . . EXTEND IF DESIRED

ENGINE BLEED AIR SWITCHES . . . . . OFF

[Ensures the airplane is depressurized at touchdown.]

APU SWITCH . . . . . OFF

LANDING PROCEDURE . . . . . REVIEW

- **Position fuel pump switches OFF just prior to flare.**
- **After stop, accomplish the PASSENGER EVACUATION checklist.**

Note: Do not raise the speed brakes unless stopping distance is critical.

---------- DEFERRED ITEMS ----------

==> LANDING

ENGINE START SWITCHES . . . . . ON

RECALL . . . . . CHECKED

SPEED BRAKE . . . . . DOWN DETENT

LANDING GEAR . . . . . _____ DOWN GREEN LIGHT(S)

FLAPS . . . . . 40, GREEN LIGHT

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WHEEL WELL FIRE

Condition: **A fire is detected in the main wheel well.**

Observe extend limit speed (270K/.82M)

LANDING GEAR LEVER . . . . . DOWN

Note: Do not use FMC fuel predictions with landing gear extended.

If the landing gear must be retracted for airplane performance, leave the landing gear extended for 20 minutes after the WHEEL WELL fire warning light has extinguished.

LANDING GEAR LEVER (if required)
(235 Kts maximum) . . . . . UP & OFF

Plan to land at the nearest suitable airport.

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CONFIGURATION WARNING NNC.15.1
GROUND PROXIMITY ALERT NNC.15.1
OVERSPEED NNC.15.2

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ALTITUDE ALERT

Condition: **The ALT ALERT indication indicates that the airplane is approaching or deviating from a selected altitude.**

Reset selected altitude (if necessary).

Maintain appropriate altitude.

CONFIGURATION WARNING

Condition: **An intermittent warning horn sounds when advancing thrust levers to takeoff, or a steady warning horn sounds inflight.**

Assure proper airplane configuration.

GROUND PROXIMITY ALERT

Condition: **The GPWS provides warnings and/or alerts for any of the following potentially hazardous flight conditions:**

- excessive descent rate
- excessive terrain closure rate
- altitude loss after takeoff or go–around
- unsafe terrain clearance when not in the landing configuration
- excessive deviation below an ILS glideslope
- bank angles and descent below the selected minimum altitudes.

Correct the flight path or the airplane configuration.

Note: If an alert occurs when flying under daylight VMC conditions, and positive visual verification is made that no hazard exists, the alert may be regarded as cautionary and the approach may be continued.

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OVERSPEED

Condition: **The mach/airspeed warning clacker is activated when the maximum operating speed of Vmo/Mmo is exceeded.**

Reduce thrust and, if required, adjust attitude to reduce airspeed to less than Vmo/Mmo.

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Approach to Stall Recovery NNM.1.1

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Terrain Avoidance NNM.1.3

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Approach to Stall Recovery

The following is immediately accomplished at the first indication of stall buffet or stick shaker.

Pilot Flying	Pilot Not Flying
Advance thrust levers to maximum thrust, smoothly adjusting pitch attitude* to avoid ground contact or obstacles. Level the wings (do not change flaps or landing gear configuration).	Verify maximum thrust. Monitor altitude and airspeed. Call out any trend toward terrain contact.
When ground contact is no longer a factor, adjust pitch attitude to accelerate while minimizing altitude loss. Return to speed appropriate for the configuration.	

Note: *At high altitudes it may be necessary to decrease pitch attitude below the horizon to achieve acceleration.

Rejected Takeoff

The decision to reject a takeoff rests solely with the captain and must be made in time so that stopping action can begin by V1. If the decision is made to reject the takeoff, the captain should clearly announce “REJECT,” commence the stopping action and assume control of the airplane if the first officer is making the takeoff. The first officer should not abandon control of the airplane until the captain makes a positive input on the controls.

Prior to 80 knots, the takeoff should be rejected for system failure(s), unusual noise or vibration, tire failure, abnormally slow acceleration, engine failure, engine fire, unsafe takeoff configuration warning, or if the airplane is unsafe or unable to fly.

Above 80 knots, reject for engine failure, fire or if the airplane is unsafe or unable to fly.

During the takeoff, the crew member recognizing the malfunction will call it out clearly and precisely.

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Captain	First Officer
Without delay: Simultaneously close the thrust levers (disengage the autothrottles, if required) and apply maximum manual wheel brakes or verify operation of RTO autobrakes. If RTO autobrakes is selected, monitor system performance and apply manual wheel brakes if the AUTO BRAKE DISARM light illuminates or deceleration is not adequate. Raise SPEED BRAKE lever. Apply maximum reverse thrust consistent with conditions. Continue maximum braking until certain the airplane will stop on the runway.	Verify actions as follows: Thrust levers closed. Autothrottles disengaged. Maximum brakes applied. Verify SPEED BRAKE lever UP and call "SPEEDBRAKES UP." If SPEED BRAKE lever is not UP, call "SPEEDBRAKES NOT UP." Reverse thrust applied. Call out omitted action items.
Field length permitting: Initiate movement of the reverse thrust levers to reach the reverse idle detent by taxi speed.	Call out 60 knots. Communicate the reject decision to the Control Tower and appropriate crew members as soon as practical.

When the airplane is stopped, perform procedures as required.

Review Brake Cooling Schedule for brake cooling time and precautions (refer to Inflight Performance Chapter).

Consider the following:

The possibility of wheel fuse plugs melting.
The need to clear the runway.
The requirement for remote parking.
Alerting fire equipment.
Not setting the parking brake unless passenger evacuation is necessary.
Advising the ground crew of the hot brake hazard.
Advising passengers of the need to remain seated or evacuate.
Completion of Non-Normal checklist (if appropriate) for conditions which caused the RTO.

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Terrain Avoidance

Ground Proximity Caution

Accomplish the following maneuver for any of these aural alerts:

- CAUTION TERRAIN
- TERRAIN AHEAD (JAA)
- SINK RATE
- TERRAIN, TERRAIN
- DON’T SINK
- TOO LOW FLAPS
- TOO LOW GEAR
- TOO LOW TERRAIN
- GLIDESLOPE
- BANK ANGLE

Pilot Flying	Pilot Not Flying
Correct the flight path or the airplane configuration.	

The below glideslope deviation alert may be cancelled or inhibited for:

- localizer or backcourse approach
- circling approach from an ILS
- when conditions require a deliberate approach below glideslope
- unreliable glideslope signal.

Note: If a terrain caution occurs when flying under daylight VMC, and positive visual verification is made that no obstacle or terrain hazard exists, the alert may be regarded as cautionary and the approach may be continued.

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Ground Proximity Warning

Accomplish the following maneuver for any of these conditions:

- Activation of the “PULL UP” or “TERRAIN TERRAIN PULL UP” warning.
- Activation of the “TERRAIN AHEAD PULL UP” (JAA) warning.
- Other situations resulting in unacceptable flight toward terrain.

Pilot Flying	Pilot Not Flying
Disconnect autopilot. Disconnect autothrottle. Aggressively apply maximum* thrust. Simultaneously roll wings level and rotate to an initial pitch attitude of 20.° Retract speedbrakes If terrain remains a threat, continue rotation up to the pitch limit indicator (if available) or stick shaker or initial buffet.	Assure maximum* thrust. Verify all required actions have been completed and call out any omissions.
Do not change gear or flap configuration until terrain separation is assured. Monitor radio altimeter for sustained or increasing terrain separation. When clear of terrain, slowly decrease pitch attitude and accelerate.	Monitor vertical speed and altitude (radio altitude for terrain clearance and barometric altitude for a minimum safe altitude.) Call out any trend toward terrain contact.

Note: Aft control column force increases as the airspeed decreases. In all cases, the pitch attitude that results in intermittent stick shaker or initial buffet is the upper pitch attitude limit. Flight at intermittent stick shaker may be required to obtain a positive terrain separation. Smooth, steady control will avoid a pitch attitude overshoot and stall.

Note: Do not use flight director commands.

Note: *Maximum thrust means “maximum certified thrust.” On engines without electronic thrust limiting capability, overboost or “firewalling the thrust levers” should only be considered during emergency situations when all other available actions have been taken and terrain contact is imminent.

Note: If positive visual verification is made that no obstacle or terrain hazard exists when flying under daylight VMC conditions prior to a terrain or obstacle warning, the alert may be regarded as cautionary and the approach may be continued.

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Traffic Avoidance

Immediately accomplish the following by recall whenever a TCAS traffic advisory (TA) or resolution advisory (RA) occurs.

CAUTION: Once an RA has been issued, safe separation could be compromised if current vertical speed is changed, except as necessary to comply with the RA. This is because TCAS II-to-TCAS II coordination may be in progress with the intruder aircraft, and any change in vertical speed that does not comply with the RA may negate the effectiveness of the other aircraft's compliance with the RA.

Note: If stick shaker or initial buffet occurs during the maneuver, immediately accomplish the APPROACH TO STALL RECOVERY procedure.

Note: If high speed buffet occurs during the maneuver, relax pitch force as necessary to reduce buffet, but continue the maneuver.

Note: Do not use flight director commands until clear of conflict.

For TA:

Pilot Flying	Pilot Not Flying
Look for traffic using traffic display as a guide. Call out any conflicting traffic	
If traffic is sighted, maneuver as required.	

For RA, except a climb in landing configuration:

WARNING: A DESCEND (fly down) RA issued below 1000 feet AGL should not be followed.

Pilot Flying	Pilot Not Flying
If maneuvering is required, disengage the autopilot and autothrottle. Smoothly adjust pitch and thrust to satisfy the RA command. Follow the planned lateral flight path unless visual contact with the conflicting traffic requires other action.	
Attempt to establish visual contact. Call out any conflicting traffic.	

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For a climb RA in landing configuration:

Pilot Flying	Pilot Not Flying
Disengage the autopilot and autothrottle. Advance thrust levers forward to ensure maximum thrust is attained and call for FLAPS 15. Smoothly adjust pitch to satisfy the RA command. Follow the planned lateral flight path unless visual contact with the conflicting traffic requires other action.	Verify maximum thrust set. Position flap lever to 15 detent.
After positive rate of climb established, call for GEAR UP.	Position gear lever up.
Attempt to establish visual contact. Call out any conflicting traffic.	

Upset Recovery

An upset can generally be defined as unintentionally exceeding the following conditions:

- Pitch attitude greater than 25 degrees nose up, or
- Pitch attitude greater than 10 degrees nose down, or
- Bank angle greater than 45 degrees, or
- Within above parameters but flying at airspeeds inappropriate for the conditions.

The following techniques represent a logical progression for recovering the airplane. The sequence of actions is for guidance only and represents a series of options to be considered and used depending on the situation. Not all actions may be necessary once recovery is under way. If needed, use pitch trim sparingly. Careful use of rudder to aid roll control should be considered only if roll control is ineffective and the airplane is not stalled.

These techniques assume that the airplane is not stalled. A stall condition can exist at any attitude and may be recognized by continuous stick shaker activation accompanied by one or more of the following:

- Buffeting which could be heavy at times
- Lack of pitch authority and/or roll control
- Inability to arrest descent rate.

If the airplane is stalled, recovery from the stall must be accomplished first by applying and maintaining nose down elevator until stall recovery is complete and stick shaker activation ceases.

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Nose High Recovery

Pilot Flying	Pilot Not Flying
• Recognize and confirm the situation	
• Disconnect autopilot and autothrottle • Apply as much as full nose-down elevator • * Apply appropriate nose down stabilizer trim • Reduce thrust • * Roll (adjust bank angle) to obtain a nose down pitch rate Complete the recovery: • When approaching the horizon roll to wings level • Check airspeed and adjust thrust • Establish pitch attitude	• Call out attitude, airspeed and altitude throughout the recovery • Verify all required actions have been completed and call out any omissions.

Nose Low Recovery

Pilot Flying	Pilot Not Flying
• Recognize and confirm the situation	
• Disconnect autopilot and autothrottle • Recover from stall, if required • * Roll in shortest direction to wings level (unload and roll if bank angle is more than 90 degrees) Recover to level flight: • Apply nose up elevator • *Apply nose up trim, if required • Adjust thrust and drag as required.	• Call out attitude, airspeed and altitude throughout the recovery • Verify all required actions have been completed and call out any omissions.

WARNING: * Excessive use of pitch trim or rudder may aggravate an upset situation or may result in loss of control and/or high structural loads.

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Windshear

The following is immediately accomplished by recall if windshear conditions are encountered. The following conditions are indications of windshear encounters:

- activation of the “WINDSHEAR” warning
- inadvertent windshear encounter or other situation resulting in unacceptable flight path deviations

Note: In general, unacceptable flight path deviations are recognized as uncontrolled changes from normal steady state flight conditions below 1000 feet AGL, in excess of any of the following:

- 15 knots indicated airspeed
- 500 fpm vertical speed
- 5° pitch attitude
- 1 dot displacement from the glideslope
- unusual thrust lever position for a significant period of time.

Pilot Flying	Pilot Not Flying
MANUAL FLIGHT • Disconnect autopilot • Press either TO/GA switch • Aggressively apply maximum* thrust • Disconnect autothrottle • Simultaneously roll wings level and rotate toward an initial pitch attitude of 15 ° • Retract speedbrakes • Follow flight director TO/GA guidance (if available). AUTOMATIC FLIGHT • Press either TO/GA switch** • Verify TO/GA mode annunciation • Verify thrust advances to GA power • Retract speedbrakes • Monitor system performance***.	• Assure maximum* thrust • Verify all required actions have been completed and call out any omissions.

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Pilot Flying	Pilot Not Flying
• Do not change flap or gear configuration until windshear is no longer a factor • Monitor vertical speed and altitude • Do not attempt to regain lost airspeed until windshear is no longer a factor.	• Monitor vertical speed and altitude • Call out any trend toward terrain contact, descending flight path, or significant airspeed changes.

*Maximum thrust means “maximum certified thrust.” On engines without electronic thrust limiting capability, overboost or “firewalling the thrust levers” should only be considered during emergency situations when all other available actions have been taken and terrain contact is imminent.

** If TO/GA is not available, disconnect autopilot and autothrottle and fly manually.

Note: Aft control column force increases as the airspeed decreases. In all cases, the pitch attitude that results in intermittent stick shaker or initial buffet is the upper pitch attitude limit. Flight at intermittent stick shaker may be required to obtain a positive terrain separation. Smooth, steady control will avoid a pitch attitude overshoot and stall.

WARNING: * Severe windshear may exceed the performance of the AFDS. The pilot flying must be prepared to disconnect the autopilot and autothrottle and fly manually.**

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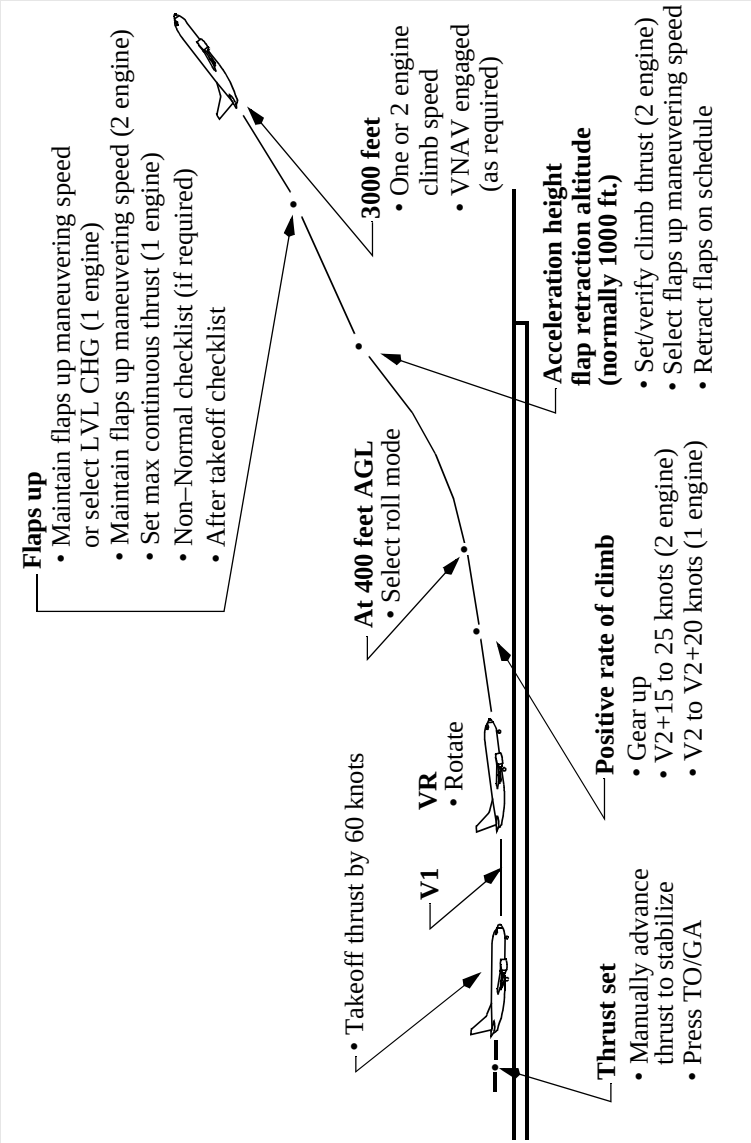
Takeoff



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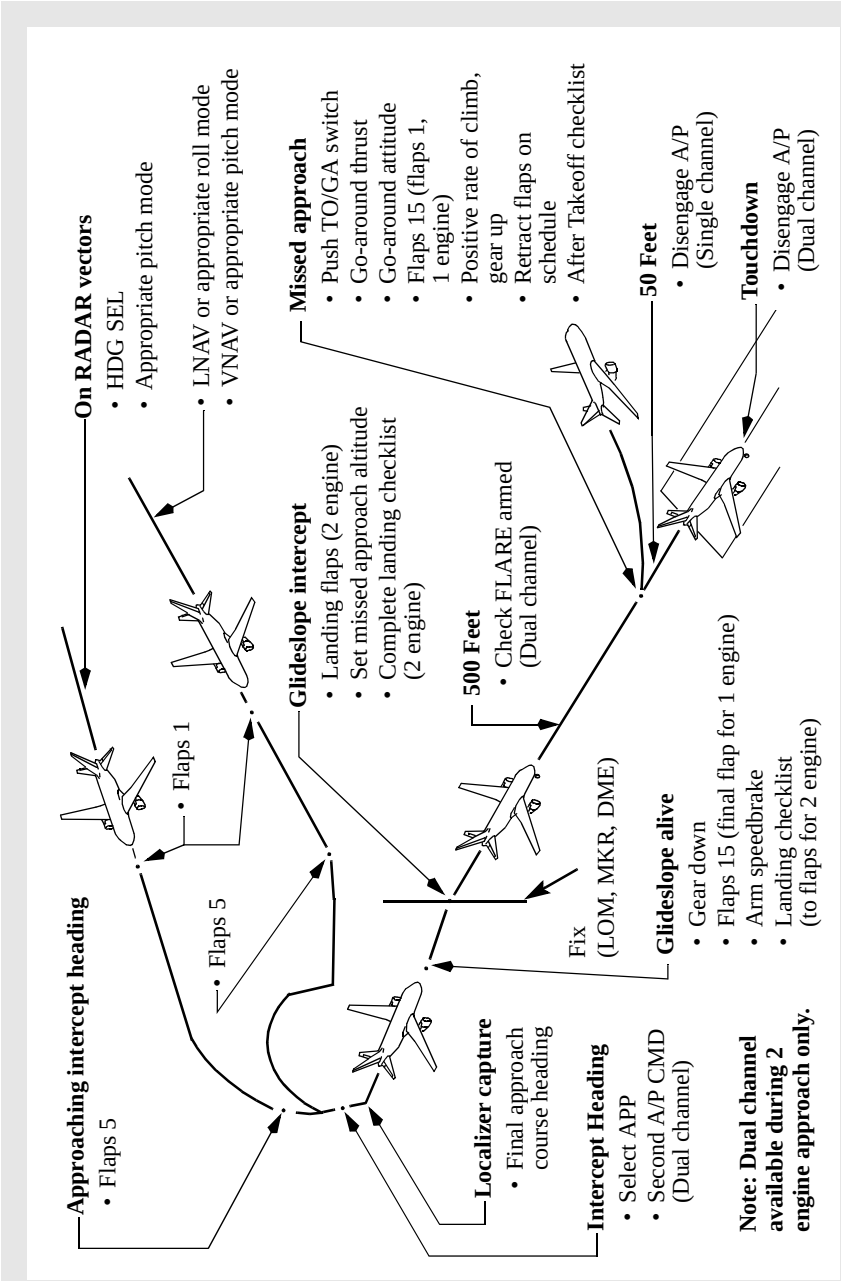
ILS Approach



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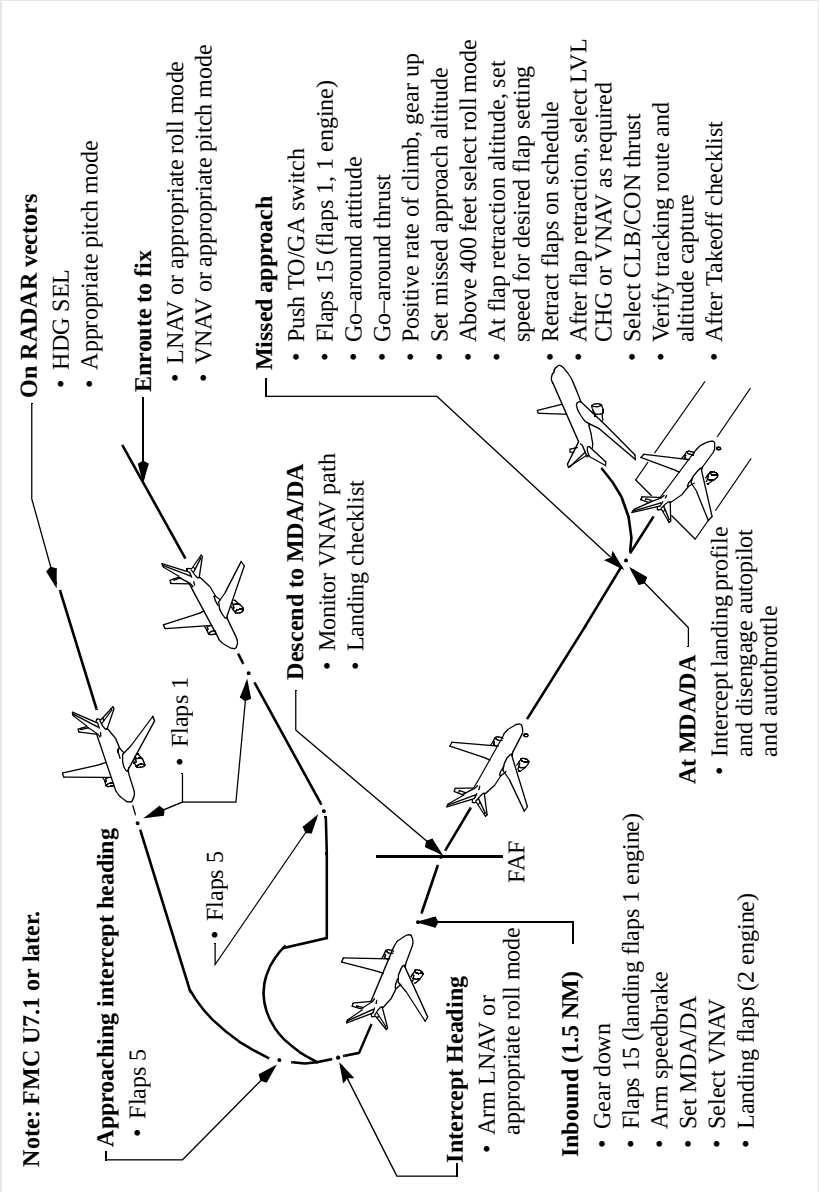
Instrument Approach using VNAV



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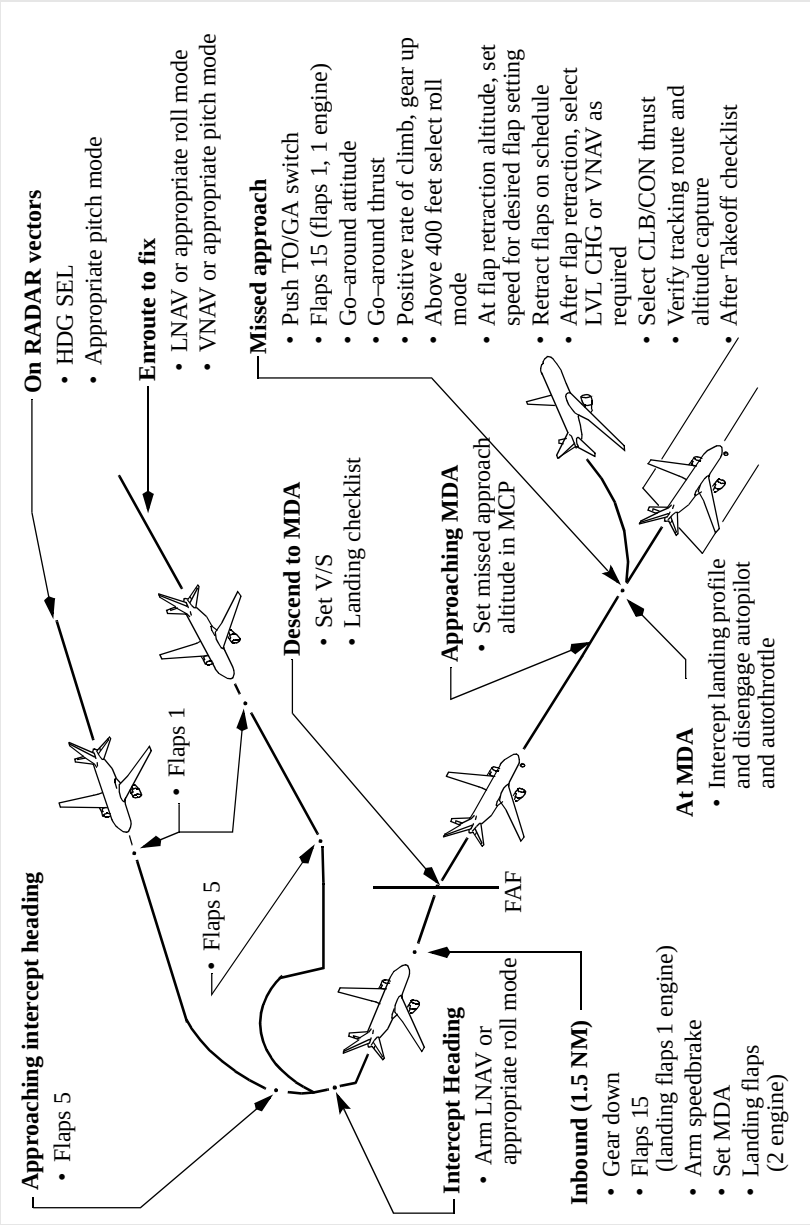
Instrument Approach using V/S



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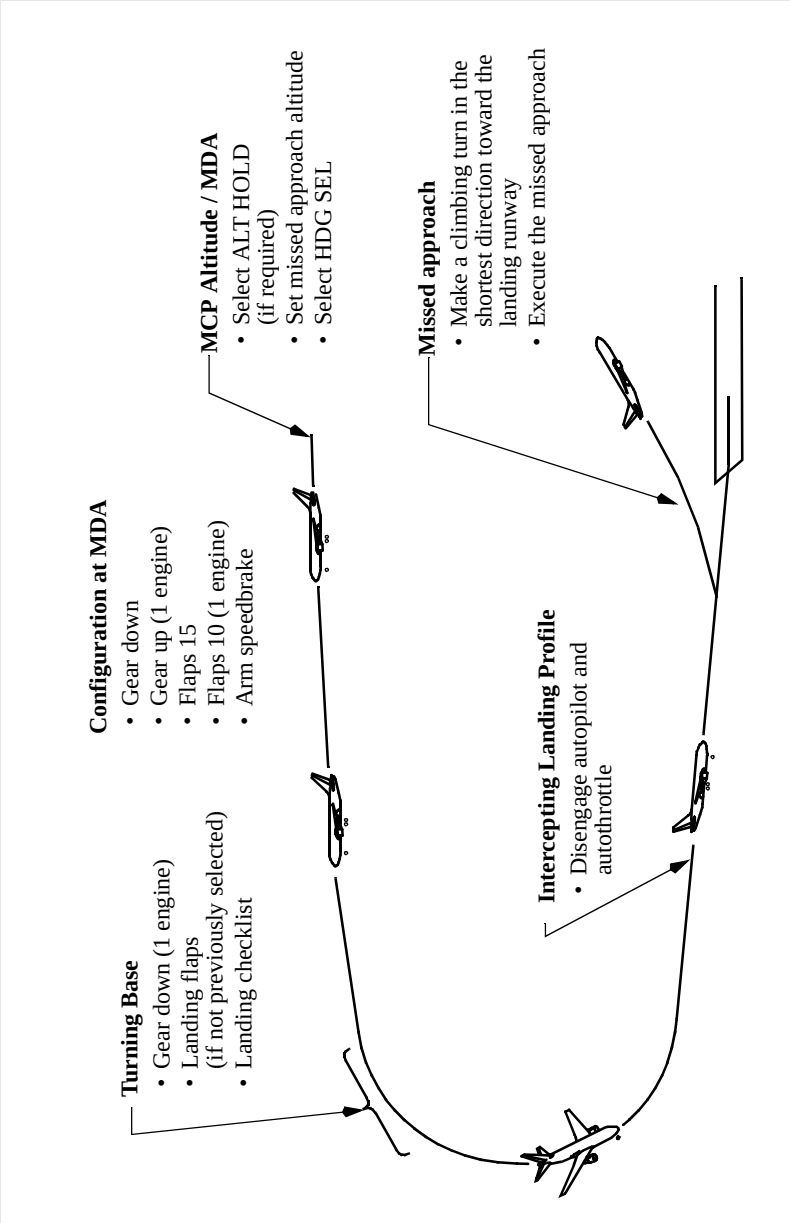
Circling Approach



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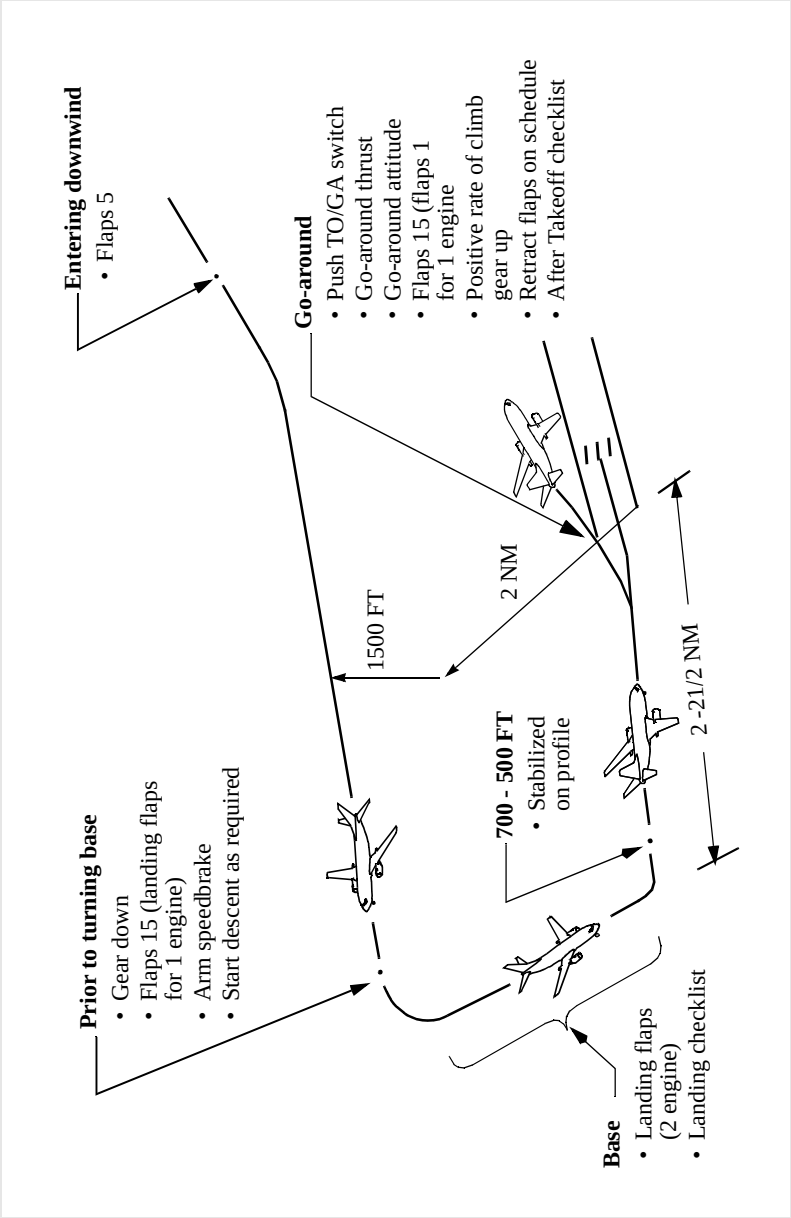
Visual Traffic Pattern



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ABORTED ENGINE START NNC.0.1,NNC.7.1
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D

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E

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G

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J

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L

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M

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T

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U

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V

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W

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Y

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Z

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